



Assetto Corsa EVO

Game Version : **0.7.1**

Assets Type : **Cars**

Subtype : **Car Physics Assets**



Preface

This document is an independent, community-driven technical reference for the vehicle physics assets of **Assetto Corsa EVO**. It is **not** an official publication of Kunos Simulazioni, nor is it endorsed, reviewed, or maintained by the studio. Assetto Corsa EVO and related marks remain the property of their respective owners.

The material herein was compiled through careful inspection of shipped physics assets, comparative analysis across vehicle configurations, and reasoned interpretation of field names, value ranges, and observed behavior. Where the engine's internal implementation is not publicly specified, units, roles and relationships are **inferred**. They should therefore be treated as working hypotheses - useful for modding, research, and discussion - rather than as authoritative studio specifications.

Readers should expect that some entries may contain inaccuracies, incomplete coverage, or interpretations that later evidence revises. Game updates, newly examined assets, and community verification may all require corrections. This reference is published so that knowledge of otherwise opaque systems can be shared, challenged, and improved openly.

Use this guide as a structured starting point. Cross-check critical parameters against in-game behaviour and primary asset data before relying on any single conclusion. Contributions that refine descriptions, confirm units, or document previously unmapped fields are welcome and strengthen the resource for everyone.

Reporting Corrections

If you find an error, an incomplete field, or a more convincing interpretation of a unit or description, please report it on GitHub using the **Documentation correction** issue template :

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1. Brake System [.brakesystem]

A. Description

Vehicle-wide braking authority: how hard the car can stop, how that force is split front/rear, and how electronic helpers (EBB, steer-brake, hybrid brake blending) reshape that split in real time.

Per-corner pad/disc thermal and friction behaviour lives in linked `.BRAKES` compounds — this asset points at them and sets the global hydraulic picture.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Peak stopping torque, static bias, handbrake	<code>.BRAKESYSTEM</code>	—
Pad/disc heat, fade, wear $\mu(T)$	Paths → <code>.BRAKES</code>	Asset 2. Brakes
ABS / TC slip maps	—	<code>.CARELECTRONICS</code>
Setup bias overrides	Cockpit bias fields	<code>.CARSETUP</code> / limits

Car Data (or equivalent) load one `.BRAKESYSTEM` per vehicle. Without it, the sim has no global brake torque or bias baseline.

II. What you are really tuning

- Raw decelerating power** — *Total Torque* is the ceiling when the pedal is fully down (examples: Delta ~2800 Nm, GTAm ~4100 Nm, 296 GT3 ~4300 Nm, SF-25 ~5500 Nm). Raise it and the car stops harder if the tyres and compounds can take it; otherwise you just lock earlier.
- Static platform** — *Front Bias* is the default axle split before electronics. Street/GT cars often sit around 0.65–0.78 (or 65–78 depending on authoring scale — check whether the car stores a 0–1 ratio or a percent-like float). Too far forward → early front lock / understeer on turn-in; too far rear → snap oversteer under heavy braking.
- Driver authority** — *Has Cockpit Bias* + *Bias Step* expose in-car balance clicks (true on 296 GT3 / SF-25; false on GTAm / Delta). Step is the increment per press.
- Handbrake** — *Hand Brake Torque* is a separate channel (often rear-biased). Rally/AWD cars may use a large value (Delta 1300 Nm); pure race cars may leave it near zero (SF-25).
- Compound links** — *Front/Rear Compound Path* attach `.BRAKES` assets. GT3 examples point at shared racing pads; many road cars leave them None until compounds are wired.
- Live redistribution** — Controllers rewrite bias (or torque) from telemetry through staged LUT pipelines. That is where modern “smart” braking and hybrid brake-by-wire live.

III. Architecture

1 - MECHANICAL BASELINE

Fields 1-7: torque, bias, handbrake, cockpit adjusters, compound paths. This is enough for a simple car with EBB off.

2 - CONTROLLERS (SHARE STAGE PATTERN)

Four controller slots reuse the same stage recipe (*Name + Stages[]*):

Block	Typical job
Controller EBB	Single EBB pipeline (legacy / simpler cars)
Controllers EBB [x]	Multiple named EBB pipelines (e.g. SF-25 migration maps <i>Mig0</i> , <i>Mig2</i> ...)
Steer Brake Controller	Micro brake / yaw assist from steer or yaw-delta inputs
Torque Controller EBB	Scales total brake torque (lockup protection, region blending)

Each **stage** is one closed loop :

1. Read an *Input Var* (Brake, LatG, LoadSpreadLF, SlipAngleRearMAX, ErsCoastTorque, ...).
2. Map it through a *LUT* (.CURVE).
3. Combine with previous stages via *Add* or *Multiply*.
4. Smooth with *Filter Gain*, camp with *Up/Down Limit*.
5. *Current Value* is runtime; *Const Value* is a static fallback when input is **Const** or unused.

3- EBB MODE SWITCH

Mode	Meaning
ebbDisabled	Static bias only (common in simpler examples even when controller trees exist)
ebbInternal	Engine built-in distribution logic
ebbDynamicControllerAbsolute	Custom stages drive absolute balance
ebbDynamicControllerRelative	Custom stages offset the base bias

EBB Front Multiplier scales how aggressively EBB can push front authority. **EBB Min Speed** gates the system below a speed threshold (0 = always eligible).

IV. How to read the examples

1 - ALFA GIULIA GTAM / LANCIA DELTA

Mechanical baseline filled; a Controller EBB tree exists (load transfer, slip angle, pedal), but **EBB Mode = ebbDisabled**, so those stages are dormant unless mode is switched. Useful as templates for enabling dynamic bias later.

2 - FERRARI 296 GT3

Strong total torque, cockpit bias on, shared GT3 .BRAKES pads front/rear, controllers empty, EBB disabled. Classic “race car with adjustable static bias + proper compounds.”

3 - FERRARI 296 GT3

Highest torque, fine cockpit steps, no handbrake. Several **Controllers EBB** named migration maps combine **Brake** with **ErsCoastTorque** LUTs under kers/ebb/ — hybrid brake blending between hydraulic and regen drag. This is the reference pattern for ERS-aware brake systems.

V. Practical notes

- Controllers present ≠ controllers active. Always check **EBB Mode**.
- Bias units are not always written the same way across cars (ratio vs percent-like). Compare against known good cars of the same authoring style before copying values blindly.
- Compound paths (*None*) mean thermal/fade behaviour is incomplete even if torque numbers look race-ready.
- Steer-brake and torque-EBB blocks are powerful; leave them *None* until you have LUTs and a clear stability goal.
- Schema typo: **Troque Controller EBB** = Torque Controller EBB.

VI. Related assets

- **2. Brakes [.brakes]** — pad/disc thermal & friction curves referenced by compound paths
- **9. Car Electronics [.carelectronics]** — ABS/TC (wheel slip intervention, not bias)
- **5-7. Car Setup / Limits / Units** — garage bias and display mapping when cockpit adjust is enabled

B. Schema

```
| 1. Total Torque : float
| 2. Front Bias : float
| 3. Hand Brake Torque : float
| 4. Has Cockpit Bias : boolean
| 5. Bias Step : float
| 6. Front Compound Path [x] : float | can have multiple Front Compound
Path
| 7. Rear Compound path [x] : float | can have multiple Rear Compound
Path
| 8. Controller EBB : object with an array of stages within
| | 8a. Name : string
```

- 8b. Stages [x] : object | Controller EBB can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 9. Controllers EBB [x] : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Controllers EBB [x] can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 10. Steer Brake Controller : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Steer Brake Controller can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 11. Torque Controller EBB : object with an array of stages within
 - 8a. Name : string
 - 8b. Stages [x] : object | Torque Controller EBB can have multiple stages
 - 8b1. Input Var : enum
 - 8b2. Combinator Mode : enum
 - 8b3. Lut : string - path
 - 8b4. Filter Gain : float
 - 8b5. Up Limit : float
 - 8b6. Down Limit : float
 - 8b7. Current Value : float
 - 8b8. Const Value : float
- 12. EBB Mode : enum
- 13. EBB Front Multiplier : float
- 14. EBB Min Speed : float

Enum List - Brake System

Enum	Values
Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
CombinatorMode	UndefinedMode, Add, Mult
EBB Mode	ebbDisabled, ebbInternal, ebbDynamicControllerAbsolute, ebbDynamicControllerRelative

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Total Torque	Nm (Newton-meters)	Defines the maximum total braking torque available for the entire vehicle.
2.	Front Bias	% (Percentage / Distribution ratio)	Sets the percentage of total braking torque allocated to the front axle (e.g., 0.60 = 60%).
3.	Hand Brake Torque	Nm (Newton-meters)	Specifies the maximum braking torque applied by pulling the mechanical handbrake.
4.	Has Cockpit Bias	None (Boolean : True / False)	Toggles whether the driver can adjust the brake balance manually from inside the cockpit while driving.
5.	Bias Step	% (Percentage increment)	The step size by which the brake bias changes per click when adjusted (e.g., 0.005 for 0.5% increments).
6.	Front Compound Path	None (File path)	Points to the physics/lookup file defining the friction and thermal properties of the front brake pads.
7.	Rear Compound Path	None (File path)	Points to the physics/lookup file defining the friction and thermal properties of the rear brake pads.
8a.	Name	None (String)	An internal label or identifier for the specific brake controller or modifier stage.
8b1.	Input Var	None (Telemetry enum)	The telemetry variable used as the trigger input for this controller (e.g., <i>SteerDEG</i> , <i>Brake</i> , <i>Speed</i>).

ID	Name	Unit of Measurement	Description
8b2.	Combinator Mode	None (Math enum : Add / Mult)	Determines how this stage's output combines with the baseline value (Addition or Multiplication).
8b3.	Lut	None (.curve file path)	References a Look-Up Table curve file that maps the raw input value to a specific output factor.
8b4.	Filter Gain	Coefficient (Smoothing multiplier)	Controls the input signal smoothing filter; acts as a dampener to prevent sudden spikes in controller application.
8b5.	Up Limit	Depends on the input variable	The upper bounding limit for the input signal, clamping values that exceed this threshold.
8b6.	Down Limit	Depends on the input variable	The lower bounding limit for the input signal, clamping values that fall below this threshold.
8b7.	Current Value	Depends on the input variable	The current, real-time value processed by the controller during simulation.
8b8.	Const Value	Depends on the input variable	A fallback fallback constant value used if no dynamic input curve or telemetry is active.
12.	EBB Mode	None (Mode enum)	Selects the operational logic profile for the Electronic Brake Balance system.
13.	EBB Front Multiplier	Coefficient (Scaling factor)	Scaling factor that dynamically adjusts the front brake bias authority under EBB intervention.
14.	EBB Min Speed	Km/h or m/s * (Spped threshold)	The minimum vehicle speed below which the Electronic Brake Balance system deactivates.

D. Example data

I. Chosen Cars for Example

- Alfa Romeo Giulia GTAm (slug : ks_alfa_romeo_giulia_gtam)
- Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)
- Ferrari 296 GT3 (slug : ks_ferrari_296_gt3)
- Ferrari SF25 (slug : ks_ferrari_sf_25)

II. Example

Alfa Romeo Giulia GTAm

- 1. Total Torque : 4100.00000
- 2. Front Bias : 0.75000
- 3. Hand Brake Torque : 200.00000
- 4. Has Cockpit Bias : false
- 5. Bias Step : 0.50000
- 6. Front Compound Path : None
- 7. Rear Compound path : None
- 8. Controller EBB
 - 8a. Name : None
 - 8b. Stages 1
 - 8b1. Input Var : LoadSpreadLF
 - 8b2. Combinator Mode : Add
 - 8b3. Lut :
 - content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_0.curve
 - 8b4. Filter Gain : 0.01000
 - 8b5. Up Limit : 1.00000
 - 8b6. Down Limit : 0.00000
 - 8b7. Current Value : 0.00000
 - 8b8. Const Value : 0.00000
 - 8b. Stages 2
 - 8b1. Input Var : SlipAngleRearMAX
 - 8b2. Combinator Mode : Mult
 - 8b3. Lut :
 - content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_1.curve
 - 8b4. Filter Gain : 0.95000
 - 8b5. Up Limit : 1.20000
 - 8b6. Down Limit : 0.00000
 - 8b7. Current Value : 0.00000
 - 8b8. Const Value : 0.00000
 - 8b. Stages 3
 - 8b1. Input Var : Brake
 - 8b2. Combinator Mode : Mult
 - 8b3. Lut :
 - content\cars\ks_alfa_romeo_giulia_gtam\data\ctrl_ebbCONTROLLER_2.curve
 - 8b4. Filter Gain : 0.95000
 - 8b5. Up Limit : 1.20000
 - 8b6. Down Limit : 0.00000
 - 8b7. Current Value : 0.00000
 - 8b8. Const Value : 0.00000
- 9. Controllers EBB [x] : None
- 10. Steer Brake Controller : None
- 11. Torque Controller EBB : None
- 12. EBB Mode : ebbDisabled
- 13. EBB Front Multiplier 1.10000
- 14. EBB Min Speed : 0.00000

Lancia Delta HF Integrale EVO II (slug : ks_lancia_delta_hf_integrale_evo_ii)

- 1. Total Torque : 2800.00000
- 2. Front Bias : 0.78000
- 3. Hand Brake Torque : 1300.00000
- 4. Has Cockpit Bias : false
- 5. Bias Step : 0.00000
- 6. Front Compound Path : None

- 7. Rear Compound path : None
- 8. Controller EBB
 - 8a. Name : None
 - 8b. Stages 1
 - 8b1. Input Var : LoadSpreadLF
 - 8b2. Combinator Mode : Add
 - 8b3. Lut :

content\cars\ks_lancia_delta_hf_integrale_evo_ii\data\ctrl_ebbCONTROLLER

0.curve

- 8b4. Filter Gain : 0.01000
- 8b5. Up Limit : 1.00000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 8b. Stages 2
 - 8b1. Input Var : Brake
 - 8b2. Combinator Mode : Mult
 - 8b3. Lut :

content\cars\ks_lancia_delta_hf_integrale_evo_ii\data\ctrl_ebbCONTROLLER

1.curve

- 8b4. Filter Gain : 0.95000
- 8b5. Up Limit : 1.20000
- 8b6. Down Limit : 0.00000
- 8b7. Current Value : 0.00000
- 8b8. Const Value : 0.00000
- 9. Controllers EBB [x] : None
- 10. Steer Brake Controller : None
- 11. Torque Controller EBB : None
- 12. EBB Mode : ebbDisabled
- 13. EBB Front Multiplier 1.20000
- 14. EBB Min Speed : 0.00000

Ferrari 296 GT3

- 1. Total Torque : 4300.00000
- 2. Front Bias : 65.00000
- 3. Hand Brake Torque : 200.00000
- 4. Has Cockpit Bias : true
- 5. Bias Step : 0.20000
- 6. Front Compound Path 1 :

content\cars\common_phsx\brakes\racing\racing_GT3_pad2.brakes

- 7. Rear Compound path 2 :

content\cars\common_phsx\brakes\racing\racing_GT3_pad2.brakes

- 8. Controllers EBB [x] : None
- 9. Controller EBB : None
- 10. Steer Brake Controller : None
- 11. Torque Controller EBB : None
- 12. EBB Mode : ebbDisabled
- 13. EBB Front Multiplier 0.00000
- 14. EBB Min Speed : 0.00000

Ferrari SF25


```

- 1. Total Torque : 5500.00000
- 2. Front Bias : 54.00000
- 3. Hand Brake Torque : 0.00000
- 4. Has Cockpit Bias : true
- 5. Bias Step : 0.10000
- 6. Front Compound Path : None
- 7. Rear Compound path : None
- 8. Controller EBB : None
- 9. Controllers EBB 1
  | 8a. Name : Mig0
  | 8b. Stages 1
  |   | 8b1. Input Var : Brake
  |   | 8b2. Combinator Mode : Add
  |   | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_0.c
urve
  |   | 8b4. Filter Gain : 0.90000
  |   | 8b5. Up Limit : 10.00000
  |   | 8b6. Down Limit : 0.00000
  |   | 8b7. Current Value : 0.00000
  |   | 8b8. Const Value : 0.00000
  | 8b. Stages 2
  |   | 8b1. Input Var : ErsCoastTorque
  |   | 8b2. Combinator Mode : Add
  |   | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
  |   | 8b4. Filter Gain : 0.90000
  |   | 8b5. Up Limit : 10.00000
  |   | 8b6. Down Limit : 0.00000
  |   | 8b7. Current Value : 0.00000
  |   | 8b8. Const Value : 0.00000
- 8. Controllers EBB 2
  | 8a. Name : Mig2
  | 8b. Stages 1
  |   | 8b1. Input Var : Brake
  |   | 8b2. Combinator Mode : Add
  |   | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_2.c
urve
  |   | 8b4. Filter Gain : 0.90000
  |   | 8b5. Up Limit : 10.00000
  |   | 8b6. Down Limit : 0.00000
  |   | 8b7. Current Value : 0.00000
  |   | 8b8. Const Value : 0.00000
  | 8b. Stages 2
  |   | 8b1. Input Var : ErsCoastTorque
  |   | 8b2. Combinator Mode : Add
  |   | 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
  |   | 8b4. Filter Gain : 0.90000
  |   | 8b5. Up Limit : 10.00000
  |   | 8b6. Down Limit : 0.00000
  |   | 8b7. Current Value : 0.00000
  |   | 8b8. Const Value : 0.00000

```

```

- 8. Controllers EBB 3
  - 8a. Name : Mig4
  - 8b. Stages 1
    - 8b1. Input Var : Brake
    - 8b2. Combinator Mode : Add
    - 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_4.c
urve
    - 8b4. Filter Gain : 0.90000
    - 8b5. Up Limit : 10.00000
    - 8b6. Down Limit : 0.00000
    - 8b7. Current Value : 0.00000
    - 8b8. Const Value : 0.00000
  - 8b. Stages 2
    - 8b1. Input Var : ErsCoastTorque
    - 8b2. Combinator Mode : Add
    - 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
    - 8b4. Filter Gain : 0.90000
    - 8b5. Up Limit : 10.00000
    - 8b6. Down Limit : 0.00000
    - 8b7. Current Value : 0.00000
    - 8b8. Const Value : 0.00000
- 8. Controllers EBB 4
  - 8a. Name : Mig6
  - 8b. Stages 1
    - 8b1. Input Var : Brake
    - 8b2. Combinator Mode : Add
    - 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_6.c
urve
    - 8b4. Filter Gain : 0.90000
    - 8b5. Up Limit : 10.00000
    - 8b6. Down Limit : 0.00000
    - 8b7. Current Value : 0.00000
    - 8b8. Const Value : 0.00000
  - 8b. Stages 2
    - 8b1. Input Var : ErsCoastTorque
    - 8b2. Combinator Mode : Add
    - 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve
    - 8b4. Filter Gain : 0.90000
    - 8b5. Up Limit : 10.00000
    - 8b6. Down Limit : 0.00000
    - 8b7. Current Value : 0.00000
    - 8b8. Const Value : 0.00000
- 8. Controllers EBB 5
  - 8a. Name : Mig8
  - 8b. Stages 1
    - 8b1. Input Var : Brake
    - 8b2. Combinator Mode : Add
    - 8b3. Lut :
content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_migration_8.c
urve

```

- | | | 8b4. Filter Gain : 0.90000
- | | | 8b5. Up Limit : 10.00000
- | | | 8b6. Down Limit : 0.00000
- | | | 8b7. Current Value : 0.00000
- | | | 8b8. Const Value : 0.00000
- | | 8b. Stages 2
 - | | 8b1. Input Var : ErsCoastTorque
 - | | 8b2. Combinator Mode : Add
 - | | 8b3. Lut :

content\cars\ks_ferrari_sf_25\data\kers\ebb\ebb_controller_KERStorque.cu
rve

- | | | 8b4. Filter Gain : 0.90000
- | | | 8b5. Up Limit : 10.00000
- | | | 8b6. Down Limit : 0.00000
- | | | 8b7. Current Value : 0.00000
- | | | 8b8. Const Value : 0.00000
- | 10. Steer Brake Controller : None
- | 11. Torque Controller EBB : None
- | 12. EBB Mode : ebbDynamicControllerRelative
- | 13. EBB Front Multiplier 0.00000
- | 14. EBB Min Speed : 0.00000

2. Brakes [.brakes]

A. Description

Hardware-level brake physics for pads and discs: how friction builds with temperature, how heat enters and leaves the assembly, and how material wears away over a stint.

The `.BRAKESYSTEM` asset commands *how much* brake torque to ask for and how to split it. This asset decides whether that torque is still available after heat, fade, rain, and wear have done their work.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Pad/disc thermal mass, cooling, fade	<code>.BRAKES</code>	—
Wear thickness and μ loss per mm	<code>.BRAKES</code>	—
Friction vs temperature curve	<code>.BRAKES</code> → Perf Curve	
Peak torque, bias, EBB, handbrake	—	1. Brake System
ABS / TC	—	<code>.CARELECTRONICS</code>
Brake duct / cooling setup clicks	Cooling responds here	<code>.CARSETUP</code> aero / ducts

Referenced from Brake System via **Front / Rear Compound Path**. One file can serve a single axle (vintage front/rear split) or the whole car (shared GT3 pad compound).

II. What you are really tuning

- Cooling balance** — *Cool Transfer* is the stationary ambient cooling rate. *Cool Speed Factor* scales cooling with vehicle speed (air through ducts). *Rain Cool Factor* boosts dissipation when wet. Together they set whether brakes stabilise or cook over a lap.
- Torque conversion** — *Torque K* scales how efficiently commanded hydraulic effort becomes stopping torque at the disc. Same Brake System torque with a lower *Torque K* feels weaker or softer at the pedal.
- Thermal mass and conduction** — *Thermal Capacity* (surface) and *Core Thermal Capacity* (bulk disc) set how fast temperature rises. *Thermal Conductivity* + *Conduction Thickness* move heat from the hot friction face into the core. *Emissivity* and *Surface* add radiation and exposed area for high-temp cooling.
- Wear life** — *Disk / Pad Thickness* are brand-new thicknesses (mm). *Disk / Pad Consumption Rate* burn material under load/heat. *T Reference Wear* is the temperature above which wear accelerates. Endurance compounds (GT3 pad2) show much lower consumption than vintage road pads.

5. **Wear → performance loss** — As thickness drops, *Perf Decrease M M*, *Mu Reduction M M*, *Area Reduction M M*, and *Gamma Correction M M* degrade bite, friction, contact area, and pedal feel. These are the endurance “brakes get longer” knobs.
6. **Temperature window** — *Perf Curve* maps disc temperature (°C) → friction efficiency (1.0 = full target μ). Shape of that curve is the compound’s personality: cold bite, sweet spot, and fade cliff.

III. Architecture

1 - THERMAL MODELLING (SCHEMA 1-10)

Cooling coefficients, emissivity, exposed surface, surface/core heat capacity, conductivity, and conduction depth. This block decides lap-to-lap temperature stability.

2 - DIMENSIONS AND WEAR (SCHEMA 11-19)

Consumption rates, initial thicknesses, per-millimetre performance penalties, and the wear temperature reference. Fields labelled **M M** mean “per millimetre of material lost.”

3 - PERFORMANCE CURVE (SCHEMA 20)

External `.CURVE` LUT: X = temperature (°C), Y = friction efficiency modifier. The Brake System asks for torque; this curve says how much of the pad’s potential is actually online at the current temperature.

IV. How to read the examples

1 - VINTAGE ROAD FRONT / REAR (SPLIT AXLE)

Two separate `.BRAKES` files: one for the front axle, one for the rear.

The thermal and wear numbers are almost the same on both (capacity, thickness, consumption). What differs is the **Perf Curve** path: front uses `TCURVE_VINTAGE_FRONT.CURVE`, rear uses `TCURVE_VINTAGE_REAR.CURVE`. That is the split-axle pattern: each axle can have its own temperature-to-friction behaviour, even when the hardware specs match.

On the vintage curve, cold bite is already high (about 0.90 at 0 °C), so the brakes work with little warm-up. Peak efficiency (about 1.0) sits around 600 °C, then fades gently toward about 0.80 at 1200 °C. Wear rates are relatively high (about 0.18 disc / 0.20 pad) compared with race endurance pads.

2 - RACING GT3 PAD 2 (SHARED COMPOUND)

One master `.BRAKES` file reused front and rear from Brake System compound paths. Larger *Surface* (0.40 vs 0.20), much lower consumption (about 0.0105 disc / 0.021 pad), earlier *T Reference Wear* (500 °C). *Perf Curve* is denser and stays high through a wide mid band, then drops harder past about 900-1000 °C: a race pad that wants heat but lasts longer in a stint. Cars often swap compounds by pointing both axles at another shared file (for example pad1 vs pad2), rather than authoring separate front/rear hardware.

V. Practical notes

- Brake System torque without a real `.BRAKES` path (*None*) means incomplete fade/wear: stopping power will not behave like a race car under heat.
- Split front/rear files when hardware or thermal load differs; share one compound file when the car swaps pads as a package (GT3).
- Read the Perf Curve before chasing *Total Torque*: a cold-weak race pad needs warm-up; a street curve that is already at 0.9 cold will feel strong immediately.
- High `COOL SPEED FACTOR` couples strongly to duct / ride-height aero setup: brake temps are not only a `.BRAKES` problem.
- Wear fields are slow variables: they matter for endurance and long sessions more than a single qualifying lap.

VI. Related assets

- **1. Brake System [`.brakesystem`]** — torque, bias, compound paths that load this file
- **5. Car Setup [`.carsetup`]** — ducts / cooling-related setup that change effective cooling
- **19. Tyre [`.tyre`]** — separate thermal world; brake heat can couple via tyre thermal transfer factors when modelled

B. Schema

- 1. Cool Transfer : float
- 2. Torque K : float
- 3. Cool Speed Factor : float
- 4. Rain Cool Factor : float
- 5. Emissivity : float
- 6. Surface : float
- 7. Thermal Capacity : float
- 8. Core Thermal Capacity : float
- 9. Thermal Conductivity : float
- 10. Conduction Thickness : float
- 11. Disk Consumption Rate : float
- 12. Pad Consumption Rate : float
- 13. Disk Thickness : float
- 14. Pad Thickness : float
- 15. Perf Decrease M M : float
- 16. Gamma Correction M M : float
- 17. Mu Reduction M M : float
- 18. Area Reduction M M : float
- 19. T Reference Wear : float
- 20. Perf Curve : string – path

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
2.	Torque Conversion / K	Dimensionless Factor	Mechanical torque conversion multiplier; scales how efficiently hydraulic line pressure translates into raw stopping torque.
3.	Cool Speed Factor	Coefficient	Airflow cooling multiplier; dictates the linear or non-linear scaling of heat dissipation as the vehicle speed increases.
4.	Rain Cool Factor	Dimensionless Ratio	Wet-weather cooling modifier; amplifies the global heat dissipation rate to simulate rain and track water spray striking the brake assembly.
5.	Emissivity	Dimensionless (0.0 - 1.0)	Thermal radiation efficiency (ϵ); defines how effectively the brake material radiates infrared heat, becoming highly critical at glowing, high-temperature states.
6.	Surface	m^2 (Square meters)	Total physical exposed surface area of the brake disc/assembly; larger surfaces naturally facilitate faster convective cooling.
7.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the thin outer friction surface layer; determines how quickly the contact zone heats up under friction.
8.	Core Thermal Capacity	J/K or J/°C	Heat storage capacity of the internal "core" mass of the brake disc; serves as the main thermal reservoir absorbing energy from the surface.
9.	Thermal Conductivity	W/(m-K) or Coefficient	Internal heat transfer rate; controls how fast heat energy moves from the hot outer friction surface into the cooler internal core.
10.	Conduction Thickness	m (Meters) or mm	Physical distance/depth representing the boundary layer for internal heat transfer between the surface friction node and the core mass.
11.	Disk Consumption Rate	Wear Coefficient	Material wear rate of the brake disc/rotor, tracking physical thickness reduction relative to temperature and kinetic energy absorption cycles.
12.	Pad Consumption Rate	Wear Coefficient	Material wear rate of the brake pads, tracking friction compound depletion relative to temperature and kinetic energy absorption cycles.

ID	Name	Unit of Measurement	Description
13.	Disk Thickness	mm (Millimeters)	The initial, brand-new physical structural thickness of the brake disc/rotor.
14.	Pad Thickness	mm (Millimeters)	The initial, brand-new physical thickness of the wearable friction material layer on the brake pad.
15.	Perf Decrease M M	% / mm (mm^{-1})	Global percentage drop in overall braking efficiency and biting performance for every single millimeter of total pad/disk material lost to wear.
16.	Gamma Correction M M	Dimensionless Exponent	Non-linear scaling factor; progressively adjusts how material wear non-linearly alters pedal feedback, compliance, and friction behavior over time.
17.	Mu Reduction M M	$\Delta\mu / mm$	The literal linear reduction of the raw friction coefficient (μ) applied as the pads and discs become thinner.
18.	Area Reduction M M	m^2 / mm or $\% / mm$	The progressive reduction of the effective pad-to-disc contact surface area as the pad material shaves down.
19.	T Reference Wear	°C (Degress Celsius)	Threshold reference temperature; defines the thermal boundary above which component wear rates begin to spike exponentially.
20.	Perf Curve	None (File path)	File path pointing to an external .CURVE look-up table mapping raw operating temperature (X-axis in °C) to friction coefficient efficiency (Y-axis).

D. Example data

I. Chosen Brakes for Example

- Vintage Road Front (slug : vintage_road_front) / Vintage Road Rear (vintage_road_rear)
- Racing GT3 [Pad 2] (slug : racing_gt3_pad2)

II. Example

Vintage Road [Front]

```

1. Cool Transfer : 1.30000
2. Torque K : 0.70000

```


- 3. Cool Speed Factor : 1.50000
- 4. Rain Cool Factor : 0.80000
- 5. Emissivity : 0.70000
- 6. Surface : 0.20000
- 7. Thermal Capacity : 100.00000
- 8. Core Thermal Capacity : 1600.00000
- 9. Thermal Conductivity : 250.00000
- 10. Conduction Thickness : 0.00500
- 11. Disk Consumption Rate : 0.18000
- 12. Pad Consumption Rate : 0.19800
- 13. Disk Thickness : 32.00000
- 14. Pad Thickness : 29.00000
- 15. Perf Decrease M M : 15.00000
- 16. Gamma Correction M M : 1.50000
- 17. Mu Reduction M M : 0.01000
- 18. Area Reduction M M : 0.16000
- 19. T Reference Wear : 600.00000
- 20. Perf Curve :

content\cars\common_phsx\brakes\vintage\tcurve_vintage_front.curve

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	0.000	0.900
P1	100.000	0.950
P2	500.000	0.980
P3	600.000	1.000
P4	800.000	0.980
P5	900.000	0.950
P6	1000.000	0.850
P7	1200.000	0.800

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

Vintage Road [Rear]

- 1. Cool Transfer : 1.30000
- 2. Torque K : 0.70000
- 3. Cool Speed Factor : 1.50000
- 4. Rain Cool Factor : 0.80000
- 5. Emissivity : 0.70000
- 6. Surface : 0.20000
- 7. Thermal Capacity : 100.00000
- 8. Core Thermal Capacity : 1600.00000
- 9. Thermal Conductivity : 250.00000
- 10. Conduction Thickness : 0.00500
- 11. Disk Consumption Rate : 0.18000
- 12. Pad Consumption Rate : 0.19800
- 13. Disk Thickness : 32.00000
- 14. Pad Thickness : 29.00000

- 15. Perf Decrease M M : 15.00000
- 16. Gamma Correction M M : 1.50000
- 17. Mu Reduction M M : 0.01000
- 18. Area Reduction M M : 0.16000
- 19. T Reference Wear : 600.00000
- 20. Perf Curve :

content\cars\common_phsx\brakes\vintage\tcurve_vintage_rear.curve

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P0	0.000	0.900
P1	100.000	0.950
P2	500.000	0.980
P3	600.000	1.000
P4	800.000	0.980
P5	900.000	0.950
P6	1000.000	0.850
P7	1200.000	0.800

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

Racing GT3 [Pad 2]

- 1. Cool Transfer : 0.90000
- 2. Torque K : 0.70000
- 3. Cool Speed Factor : 1.60000
- 4. Rain Cool Factor : 0.80000
- 5. Emissivity : 0.70000
- 6. Surface : 0.40000
- 7. Thermal Capacity : 100.00000
- 8. Core Thermal Capacity : 1600.00000
- 9. Thermal Conductivity : 250.00000
- 10. Conduction Thickness : 0.00500
- 11. Disk Consumption Rate : 0.01050
- 12. Pad Consumption Rate : 0.02100
- 13. Disk Thickness : 32.00000
- 14. Pad Thickness : 29.00000
- 15. Perf Decrease M M : 13.00000
- 16. Gamma Correction M M : 1.00000
- 17. Mu Reduction M M : 0.01000
- 18. Area Reduction M M : 0.08000
- 19. T Reference Wear : 500.00000
- 20. Perf Curve :

content\cars\common_phsx\brakes\racing\tcurve_racing_GT3_pad2

	°C Temperature (X)	Friction Coefficient Modifier (Y)
P1	-0.905	0.839
P2	39.477	0.914
P3	88.596	0.952
P4	121.867	0.967
P5	156.821	0.974
P6	203.872	0.978
P7	252.036	0.980
P8	303.284	0.980
P9	365.345	0.980
P10	380.803	0.979
P11	443.598	0.977
P12	526.801	0.973
P13	599.594	0.966
P14	722.684	0.945
P15	781.584	0.936
P16	849.036	0.923
P17	908.993	0.898
P18	972.698	0.854
P19	1047.645	0.734
P20	1245.218	0.567

Friction Coefficient Modifier = Performance Efficiency where 1.000 = 100%

3. Car Data [.car]

A. Description

The vehicle master file: mass, inertia footprint, chassis geometry, fuel payload, and the path list that wires every other physics asset into one car.

If Brake System / Brakes are the stopping hardware, Car Data is the carcass and the bill of materials. Without it, the sim has nowhere to hang engine, gearbox, suspension, tyres, or aero.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Total mass, fuel mass, tank position	.CAR General	—
Wheelbase, track, CG location, ride refs	.CAR Suspensions	Kinematics detail in .SUSPENSION
Chassis torsion / damping	.CAR General	—
Paths to engine, gearbox, clutch, drivetrain, brakes	.CAR path fields	Individual assets 4, 13, 14, 10, 1...
Tyre compound lists	Front / Rear Tyre Compounds	Grip/thermal in .TYRE
Aero maps, DRS, ERS hooks	Aero / Drs / Ers blocks	.WING, .SURFACE3D, electronics
Garage presets and limits	Setup / Performance Modes paths	.CARSETUP, limits, units

One Car Data file per vehicle configuration. Other assets are modules this file references.

II. What you are really tuning

- Mass and fuel** — *Total Mass* is the baseline vehicle mass used for $F=ma$ (examples: Renault 5 about 910 kg, R8 GT3 about 1355 kg, 296 GTB about 1750 kg). *Fuel / Max Fuel* plus *Kg Per Liter* and *Tank Position* make mass and CG move as the tank empties — endurance balance drift lives here, not in the tyre file.
- Plan footprint** — *Wheel Base*, *Track Front / Track Rear*, and *Longitudinal Cg Location* set the geometric DNA. Long wheelbase + mid CG (296 about 0.40 from the front reference) feels planted; short wheelbase + forward CG (R5 about 0.64) feels darty and nose-heavy. Narrow track (R5 about 1.34 m) rolls and rotates differently from a GT3 track (about 1.67 m).
- Vertical references** — *Base Y Front / Base Y Rear*, pickup heights, and *Minimum Height / Check Rules* anchor ride height and regulatory floors (R8 enables rule checks with a minimum height value; road cars often leave checks off).
- Chassis stiffness** — *Torsional Stiffness* and *Torsional Damping* model the body as a flexible beam between axles. Soft shells (R5 about 11000) twist and muddy platform control; race tubs (R8 about 40000) stay flatter so ARBs and springs do the real work.

5. **Module wiring** — Path fields (*Engine Path, Gearbox Path, Clutch Path, Drivetrain Path, Brakes Path*, coilover / suspension paths, electronics, setup presets) decide which concrete assets this car loads. Wrong path = silent fallback or broken car, even if the numbers in General look fine.
6. **Driver controls and aids hooks** — *Steering System* (steer lock, rack behaviour), *Electronics / Electronics Path, Controls*, tyre compound arrays, *Aero / Drs / Ers*, and *Performance Modes* expose how the car is driven and which packages can be swapped.

III. Architecture

1 - IDENTITY AND GENERAL (SCHEMA 1-3)

Screen names, mass, fuel economy block, body box, pickup heights, rule flags, torsional pair, optional body mesh offset. *General Path* can point at a shared `.GENERAL` override when used.

2 - SUSPENSIONS HUB (SCHEMA 4)

Not the full kinematic hardpoints — those live in `.SUSPENSION / .COILOVER`. Here you get axle geometry (wheelbase, tracks, CG, base Y), damage thresholds, paths to front/rear coilover and suspension, optional heavy springs, ARB / flex-bar stiffness (and their dynamic controllers), and damper cockpit flags.

3 - POWERTRAIN AND BRAKE PATHS (SCHEMA 5-9)

Pointers to `.DRIVETRAIN`, `.GEARBOX`, `.CLUTCH`, `.CARENGINE`, and the brake system asset. This is the assembly list for longitudinal physics.

4 - DRIVER, COLLISION, TIRES (SCHEMA 10-16)

Steering system, electronics (inline and/or path), controls, box colliders, front/rear tyre compound lists.

5 - AERO AND HYBRID (SCHEMA 17-19)

Aero package (downforce elements, surface3d maps, wings path), DRS connections, ERS / hybrid hooks when present.

6 - SETUPS, MESH, AI, MODES (SCHEMA 20-28)

Setup limits path, collider/body mesh, stock / AI / wet setups, performance mode packages, AI car data, and residual fields (including *mm* in the schema tree).

IV. How to read the examples

1 - FERRARI 296 GTB (ROAD MID-ENGINE)

Heavy total mass (1750 kg), 2.60 m wheelbase, wide tracks (about 1.72 m), longitudinal CG about 0.40 (mass biased rearward of geometric mid). Torsional stiffness 30000. Large tank (max 120 L) with a defined tank position — fuel burn will move balance. Full path set to its own coilover, suspension, drivetrain, gearbox, clutch, engine, brakes. Road-car style: rules check off.

2 - AUDI R8 LMS GT3 EVO II (RACE GT3)

Lighter race mass (1355 kg), longer wheelbase (2.70 m), CG about 0.42, tracks about 1.67 m. Stiffer tub (40000) and much stiffer ARBs (front about 56800, rear about 39900). *Check Rules* true with a minimum height set — BoP / regulatory awareness. Efficiency non-zero. Same modular path pattern, aimed at shared race compounds and setups.

3 - RENAULT 5 GT TURBO (LIGHT FWD HOT HATCH)

Light (910 kg), short wheelbase (2.41 m), narrow tracks (about 1.34 / 1.32 m), longitudinal CG about 0.64 (nose-heavy). Soft torsional stiffness (11000) and soft ARBs; flex bar rear present. Small tank (max 50 L). High steer lock (630 in the example) versus GT cars. Shows how Car Data alone already predicts a nervous, front-driven road car before you open the suspension files.

V. Practical notes

- Fix mass, CG, wheelbase, and tracks before chasing setup springs: wrong Car Data geometry cannot be “tuned away” in the garage.
- Longitudinal CG location is a fraction along the wheelbase — compare cars from the same authoring style before copying.
- Fuel tank position matters for stint balance; a rear tank that empties forward is a different car at the end of a race.
- Path fields are load-bearing: a typo in *Engine Path* or suspension path breaks the car even if General looks perfect.
- *Check Rules* / *Minimum Height* are easy to miss on race cars and can fight your ride-height setup.
- Schema label in the source TOC is `.CAR`; on disk the asset is still the Car Data package that references the other extensions.

VI. Related assets

- **15. General** [`.generalcar`] — optional shared general block via *General Path*
- **11 / 17. Coilover** / **Suspension** — loaded from Suspensions paths
- **4 / 13 / 14 / 10. Engine** / **Drivetrain** / **Gearbox** / **Clutch** — powertrain paths
- **1. Brake System** [`.brakesystem`] — *Brakes Path*
- **19 / 16 / 20. Tyre** / **Surface 3D** / **Wing** — compounds and aero hooks
- **5-7. Car Setup** / **Limits** / **Units** — garage presets and limits paths

B. Dependency map

Wiring chart for what `.CAR` (**Car Data**) loads, what those assets load next, and what only *overlays* or *selects* without owning a path.

Legend

Link type	Meaning
Load	Path string (or path list) — A opens B
Optional load	Path often <i>None</i> ; when set, A opens B
Overlay	Session / UI values on top of hardware already loaded
Select	Index or enum into maps / lists defined elsewhere
Override	Tuning part (or similar) redirects paths that .CAR would use

I. Direct loads from Car Data

Car Data field / block (typical)	Loads	Description
General (inline)	—	Mass, fuel, torsion, etc. live here when no external general file
General Path	<u>15. General</u>	Optional shared general block — usually <i>None</i> in shipped cars
Coilover Front / Rear Path	<u>11. Coilover</u>	Spring-damper units
Front / Rear Suspension Path	<u>17. Suspension</u>	Kinematic hardpoints
Drivetrain Path	<u>13. Drivetrain</u>	Drive layout + diffs / AWD
Gearbox Path	<u>14. Gearbox</u>	Ratios + shift behaviour
Clutch Path	<u>10. Clutch</u>	Clamp + autoclutch
Engine Path	<u>4. Car Engine</u>	Torque maps, controllers, turbo list
Brakes Path	<u>1. Brake System</u>	Global brake torque / bias
Electronics / Electronics Path	<u>9. Car Electronics</u>	TC / ABS / EDL / ESP maps
Front / Rear Type Compounds [x]	<u>19. Tyre</u>	Compound list per axle
Aero → Front / Rear Lift, Drag	<u>16. Surface 3D</u>	CX / CZ platform maps
Aero → Wings Path [x]	<u>20. Wing</u>	Discrete aero elements
Setup / AI / wet setup paths	<u>5. Car Setup</u>	Garage presets
Setup Limits path	<u>6. Car Setup Limits</u>	Min / Max / step
(Via shared / car tooling)	<u>7. Car Setup Units</u>	Display units (often <i>common_phsx</i>)

II. Second hop — what loaded assets open next

From	Link	To	Notes
<u>1. Brake System</u>	Load	<u>2. Brakes</u>	Front / rear compound paths
<u>4. Car Engine</u>	Load	<u>18. Turbo</u>	Turbos To Load (0...n)
<u>4. Car Engine</u>	Load	.CURVE	Power / coast / throttle / controller LUTs
<u>11. Coilover</u>	Optional load	<u>12. Damper Curves</u>	Damper.Lut List
<u>12. Damper Curves</u>	Load	.CURVE	Each Damper Lut[x]
<u>10. Clutch</u>	Optional load	.CURVE	Autoclutch up/down profiles, clutch curve
<u>14. Gearbox</u>	Optional load	.CURVE	Autoblip profile
<u>13. Drivetrain</u>	Optional load	.CURVE	Lock / AWD clutch controller LUTs
<u>9. Car Electronics</u>	—	—	Maps are inline; no child physics asset
<u>19. Tyre</u>	Load	.CURVE	Wear + thermal performance curves
<u>20. Wing</u>	Load	.CURVE	AOA CL/CD; optional ground-height multi
<u>16. Surface 3D</u>	Optional load	paths / CSV	Downforce h/dh, import tooling — often None
<u>1. Brake System</u>	Load	.CURVE	EBB / controller stages when present
<u>8. Tuning Parts</u>	Override	Any domain path above	Swap engine, drivetrain, setup limits, coilovers, ...

III. Overlay and select layers (not path owners)

Asset	Link type	Acts on	What it does
<u>5. Car Setup</u>	Overlay	Coilover / suspension feel	Wheel rates, dampers, bump stops, helpers
<u>5. Car Setup</u>	Overlay	Tyre	Pressures, camber, toe, caster
<u>5. Car Setup</u>	Select	<u>9. Electronics</u>	TC / ABS / ESC map indices
<u>5. Car Setup</u>	Select	<u>4. Engine</u> / <u>18. Turbo</u>	Engine map, boost level when adjustable
<u>5. Car Setup</u>	Overlay	<u>13. Drivetrain</u>	Diff power / coast / preload

Asset	Link type	Acts on	What it does
<u>5. Car Setup</u>	Overlay	<u>20. Wing</u> + ride height → <u>16. Surface 3D</u>	Wing angles, target heights
<u>5. Car Setup</u>	Select	<u>19. Tyre</u> lists on .CAR	Compound index
<u>6. Setup Limits</u>	Overlay	Setup UI	Legal min / max / step only
<u>7. Setup Units</u>	Overlay	Setup UI	Labels (bar, °, Ns/m, ...) — not SI truth
<u>9. Electronics</u>	Override	Paths that .CAR uses	Variant packs (drift, Cup aids, LSD, ...)

IV. By concern

1 - STOP

Step	Asset	Role
1	.CAR → <u>1. Brake System</u>	Peak torque, bias, controllers
2	→ <u>2. Brakes</u>	Pad / disc thermal & friction
3	.CAR → <u>9. Electronics</u>	ABS maps
4	Setup	Bias / ducts overlays; ABS index
5	<u>19. Tyre</u>	Longitudinal grip that can take the torque

2 - GO

Step	Asset	Role
1	.CAR → <u>4. Engine</u>	Torque source
2	<u>18. Turbo</u>	Boost / lag when listed
3	.CAR → <u>10. Clutch</u>	Couple to gearbox
4	.CAR → <u>14. Gearbox</u>	Ratios / shifts
5	.CAR → <u>13. Drivetrain</u>	Diffs / AWD to wheels
6	Setup / electronics	Diff locks, TC index, boost level

3 - PLATFORM

Step	Asset	Role
1	.CAR geometry (WB, track, CG)	Axle layout
2	→ <u>17. Suspension</u>	Hardpoints / motion
3	→ <u>11. Coilover</u>	Springs / dampers
4	→ <u>12. Damper Curves</u>	Optional LUT bank
5	Setup / limits / units	Garage rates, clicks, labels

4 - AERO

Step	Asset	Role
1	.CAR Aero → <u>16. Surface 3D</u>	Platform CX / CZ vs height
2	.CAR Aero → <u>20. Wing</u>	BODY / FRONT / REAR / diffuser elements
3	Setup	Ride heights feed Surface 3D axes; wing angles feed Wing
4	Tuning part aero package	Optional path / embedded aero swap

V. Shared libraries (common phsx)

Kind	Typical home	Referenced by	Risk
Tyres	common_phsx\tyres\...	Many .CAR compound lists	One edit hits many cars
Damper packs	common_phsx\dampers\...	Coilover Lut List	One edit hits many cars
Setup units	Shared setup_units	Many cars	UI label drift if forked carelessly
Curves	Next to tyre / wing / damper packs	Parent asset LUT fields	Broken path = silent wrong click / polar

VI. Start here

Intent	Open first	Then
Change peak power / maps	<u>4. Engine</u>	<u>18. Turbo</u> if FI
Change gearing	<u>14. Gearbox</u>	Matching <u>10. Clutch</u> if swap pack
Change LSD / AWD	<u>13. Drivetrain</u>	Setup diff locks / limits

Intent	Open first	Then
Change springs / dampers	<u>11. Coilover</u>	<u>12. Damper Curves</u> if LUT; setup overlays
Change roll centres / camber gain	<u>17. Suspension</u>	Setup alignment only overlays
Change ABS / TC personality	<u>9. Electronics</u>	Setup index + <u>6. Limits</u>
Strip aids (Cup style)	<u>8. Tuning Parts</u> + electronics / limits	Do not only zero setup indices
Change grip / heat / wear	<u>19. Tyre</u>	Setup pressure / camber
Change aero balance	<u>16. Surface 3D</u> + <u>20. Wing</u>	Setup heights / wing angles
Change stopping power	<u>1. Brake System</u>	<u>2. Brakes</u> compounds
Ship a variant (drift, upgrade)	<u>8. Tuning Parts</u>	Retarget paths; keep setup/limits coherent
Change dry mass / fuel / torsion	<u>3. Car Data</u> General (or 15. General if Path used)	—

VII. Practical notes

- Always start from **.CAR paths** before editing a leaf asset — wrong file in *common_phsx* is a common footgun.
- Setup cannot invent hardware: no wing path → wing angle does nothing useful ; empty tyre list → compound index is empty.
- *General Path* / damper Lut / clutch curves / AWD controller curves are **optional** — *None* is normal, not always a bug.
- Tuning parts **override** the graph; read the active part set before trusting stock **.CAR** paths.
- See also per-asset §1 tables in 01...20 for finer “handled here / elsewhere” splits.

C. Schema

- 1. Screen Name : **string**
- 2. General : **object**
 - 2a. Screen Name : **string**
 - 2b. Total Mass : **float**
 - 2c. Tank Position : **x, y, z float**
 - 2d. Fuel : **float**
 - 2e. Max Fuel : **float**
 - 2f. Efficiency : **float**
 - 2g. Kg Per Liter : **float**
 - 2h. Body Box Sizes : **x, y, z float**
 - 2i. Pickup Front Height : **float**
 - 2j. Pickup Rear Height : **float**
 - 2k. Check Rules : **boolean**
 - 2l. Minimum Height : **float**
 - 2m. Torsional Stiffness : **float**
 - 2n. Torsional Damping : **float**
 - 2o. Body Mesh Offset : **object**

- 2o1. Position : x, y, z float
 - 2o2. Rotation : x, y, z, float
 - 2o3. Scale : x, y, z float
- 3. General Path : string – path
- 4. Suspensions : object
 - 4a. Wheel Base : float
 - 4b. Longitudinal Cg Location : float
 - 4c. Base Y Front : float
 - 4d. Base Y Rear : float
 - 4e. Track Front : float
 - 4f. Track Rear : float
 - 4g. Damage : object
 - 4g1. Min Velocity : float
 - 4g2. Gain : float
 - 4g3. Max Damage : float
 - 4g4. Debug Log : boolean
 - 4h. Coilover Front path : string – path
 - 4i. Coilover Rear Path : string – path
 - 4j. Front Suspension Path : string – path
 - 4k. Rear Suspension Path : string – path
 - 4l. Heavy Springs [x] : object | Can have multiple Heavy Springs,
where x can be 1, 2, 3, ...
 - 4l1. Spring Rate : float
 - 4l2. Progressive Sprint Rate : float
 - 4l3. Bump Stop Up : object
 - 4l3a. Range : float
 - 4l3b. Reference : float
 - 4l3c. Force : float
 - 4l3d. Gamma : float
 - 4l3e. Length : float
 - 4l3f. Damping : float
 - 4l4. Bump Stop Down : object
 - 4l4a. Range : float
 - 4l4b. Reference : float
 - 4l4c. Force : float
 - 4l4d. Gamma : float
 - 4l4e. Length : float
 - 4l4f. Damping : float
 - 4l5. Collar Position : float
 - 4l6. Damper : object
 - 4l6a. Fast : object
 - 4l6a1. Bump : float
 - 4l6a2. Rebound : float
 - 4l6b. Slow : object
 - 4l6b1. Bump : float
 - 4l6b2. Rebound : float
 - 4l6c. Fast Threshold Bump : float
 - 4l6d. Fast Threshold Rebound : float
 - 4l6e. Cooling Surface : float
 - 4l6f. Nominal Force : float
 - 4l6g. Min Stress Fatigue : float
 - 4l6h. Max Stress Fatigue : float
 - 4l6i. Thermal Capacity : float
 - 4l6j. Heat Transfer Coefficient : float
 - 4l6k. Lut List : string – path
 - 4l6l. Damper Lut Scale : string – path

```

| | | - 4l7. Helper K : float
| | | - 4l8. Helper Range : float
| | | - 4l9. Rod Controllers : object with an array of stages within
| | |   | 4l9a. Name : string
| | |   | 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
| | |   | 4l9b1. Input Var : enum
| | |   | 4l9b2. Combinator Mode : enum
| | |   | 4l9b3. Lut : string - path
| | |   | 4l9b4. Filter Gain : float
| | |   | 4l9b5. Up Limit : float
| | |   | 4l9b6. Down Limit : float
| | |   | 4l9b7. Current Value : float
| | |   | 4l9b8. Const Value : float
| | | - 4m. Arb Front : object
| | |   | 4m1. Stiffness : float
| | |   | 4m2. Controller : object
| | |   | 4l9a. Name : string
| | |   | 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
| | |   | 4l9b1. Input Var : enum
| | |   | 4l9b2. Combinator Mode : enum
| | |   | 4l9b3. Lut : string - path
| | |   | 4l9b4. Filter Gain : float
| | |   | 4l9b5. Up Limit : float
| | |   | 4l9b6. Down Limit : float
| | |   | 4l9b7. Current Value : float
| | |   | 4l9b8. Const Value : float
| | | - 4n. Arb Rear : object
| | |   | 4n1. Stiffness : float
| | |   | 4n2. Controller : object
| | |   | 4l9a. Name : string
| | |   | 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
| | |   | 4l9b1. Input Var : enum
| | |   | 4l9b2. Combinator Mode : enum
| | |   | 4l9b3. Lut : string - path
| | |   | 4l9b4. Filter Gain : float
| | |   | 4l9b5. Up Limit : float
| | |   | 4l9b6. Down Limit : float
| | |   | 4l9b7. Current Value : float
| | |   | 4l9b8. Const Value : float
| | | - 4o. Flex Bar Front : object
| | |   | 4o1. Stiffness : float
| | |   | 4o2. Controller : object
| | |   | 4l9a. Name : string
| | |   | 4l9b. Stages [x] : float | Rod Controllers can have multiple
stages
| | |   | 4l9b1. Input Var : enum
| | |   | 4l9b2. Combinator Mode : enum
| | |   | 4l9b3. Lut : string - path
| | |   | 4l9b4. Filter Gain : float
| | |   | 4l9b5. Up Limit : float
| | |   | 4l9b6. Down Limit : float
| | |   | 4l9b7. Current Value : float
| | |   | 4l9b8. Const Value : float

```

- 4p. Flex Bar Rear : object
 - 4p1. Stiffness : float
 - 4p2. Controller : object with an array of stages within
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string - path
 - 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
- 4q. Dampers Controller : object
 - 4q1. Wheel Gain : float
 - 4q2. Heave Gain : float
 - 4q3. Pitch Gain : float
 - 4q4. Roll Gain : float
 - 4q5. Front : object
 - 4q5a. Base : object
 - 4q5a1. Type : enum
 - 4q5b. Heave : object
 - 4q5a1. Type : enum
 - 4q5c. Roll : object
 - 4q5a1. Type : enum
 - 4q5d. Pitch : object
 - 4q5a1. Type : enum
 - 4q5e. Max : object
 - 4q5a1. Type : enum
 - 4q5f. Min : object
 - 4q5a1. Type : enum
 - 4q6. Rear : object
 - 4q5a. Base : object
 - 4q5a1. Type : enum
 - 4q5b. Heave : object
 - 4q5a1. Type : enum
 - 4q5c. Roll : object
 - 4q5a1. Type : enum
 - 4q5d. Pitch : object
 - 4q5a1. Type : enum
 - 4q5e. Max : object
 - 4q5a1. Type : enum
 - 4q5f. Min : object
 - 4q5a1. Type : enum
- 4r. Has Dampers Cockpit Settings : boolean
- 5. Drivetrain Path : string - path
- 6. Gearbox Path : string - path
- 7. Clutch Path : string - path
- 8. Engine Path : string - path
- 9. Brakes Path : string - path
- 10. Steering System : object
 - 10a. Four W S Controllers : object with an array of stages within
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages

- 4l9b1. Input Var : **enum**
- 4l9b2. Combinator Mode : **enum**
- 4l9b3. Lut : **string – path**
- 4l9b4. Filter Gain : **float**
- 4l9b5. Up Limit : **float**
- 4l9b6. Down Limit : **float**
- 4l9b7. Current Value : **float**
- 4l9b8. Const Value : **float**
- 11. Electronics : **object**
 - 11a. T C : **object**
 - 11a1. Has T C2 : **boolean**
 - 11a2. Frequency Hz : **float**
 - 11a3. Min Speed Kmh : **float**
 - 11a4. Gear Change Time : **float**
 - 11a5. Min Cut Level : **float**
 - 11a6. Max Cut Level : **float**
 - 11a7. Settings [x] : **object** | T C can have multiple Settings
 - 11a7a. Min Slip Ratio : **float**
 - 11a7b. Max Slip Ratio : **float**
 - 11a7c. Ref Slip Angle Deg : **float**
 - 11a7d. Engine Cut Level : **float**
 - 11a7e. Angular A C Cgain : **float**
 - 11a7f. Oversteer Gain : **float**
 - 11a7g. Slip Angle Activation Deg : **float**
 - 11b. A B S : **object**
 - 11b1. Settings [x] : **object** | A B S can have multiple Settings
 - 11b1a. Min Slip Ratio : **float**
 - 11b1b. Max Slip Ratio : **float**
 - 11b1c. Ref Slip Angle Deg : **float**
 - 11b1d. Cut Level : **float**
 - 11b1f. Max Torque Variation : **float**
 - Frequency : **float**
 - Channels : **integer**
 - Min Speed Kmh : **float**
 - 11c. E D L : **object**
 - 11c1. Active : **boolean**
 - 11c2. Brake Torque Power : **float**
 - 11c3. Brake Torque Coast : **float**
 - 11c4. Dead Zone Coast : **float**
 - 11c5. Dead Zone Power : **float**
 - 11c6. Max Spin Power : **float**
 - 11c7. Max Spin Coaster : **float**
 - 11c8. Min Speed : **float**
 - 11d. E S P : **object**
 - 11d1. Frequency Hz : **float**
 - 11d2. Min Speed Kmh : **float**
 - 11d3. Settings [x] : **object** | E S P can have multiple settings
 - 11d3a. Gain : **float**
 - 11d3b. Steer Gain : **float**
 - 11d3c. Min Steer Gain : **float**
 - 11d3d. Steer Gain Max Speed : **float**
 - 11d3e. Oversteer Gain : **float**
 - 11d3f. Understeer Gain : **float**
 - 11d3g. Max Slip Ratio : **float**
 - 11d3h. Dead Zone : **float**
 - 11d3i. Filter Gain : **float**

- 11d3j. Brake Perc : float
 - 11d3k. Brake Perc Activation : float
 - 12. Electronics Path : string - path
 - 13. Controls : object
 - 13a. Ff Mult : float
 - 13b. Steer Lock : float
 - 13c. Steer Ratio : float
 - 13d. Linear Steer Rod Ratio : float
 - 13e. Steer Assist : float
 - 14. Box Colliders [x] : object | can have multiple Box Colliders
 - 14a. Center : x, y, z float
 - 14b. Size : x, y, z float
 - 14c. Pitch Rotation Deg : float
 - 15. Front Tyre Compounds [x] : string - path | can have multiple Front Tyre Compounds path
 - 16. Rear Tyre Compounds [x] : string - path | can have multiple Rear Tyre Compounds path
 - 17. Aero : object
 - 17a. Slip Gain Multiple : float
 - 17b. Speed Factor Mult : float
 - 17c. Downforces [x] : object | can have multiple Downforces
 - 17c1. Position : x, y, z float
 - 17c2. Cl Gain : float
 - 17c3. Cd Gain : float
 - 17c4. Yaw Gain : float
 - 17c5. Drag Per Cool Transfer : float
 - 17c6. Damage C L [x] : string | can have multiple Damage C L
 - 17c7. Damage C D [x] : string | can have multiple Damage C D
 - 17c8. Downforce Controllers [x] : object | can have multiple Downforce Controllers
 - 17c8a. Combinator Mode : enum
 - 17c8b. Input : enum
 - 17c8c. Filter : float
 - 17c8d. Up Limit : float
 - 17c8e. Down Limit : float
 - 17c8f. Lut : string - path
 - 17c9. Lift Per Front Angle : float
 - 17c10. Lift Per Rear Angle : float
 - 17c11. Drag Per Front Angle : float
 - 17c12. Drag Per Rear Angle : float
 - 17c13. Default Front Angle : float
 - 17c14. Default Rear Angle : float
 - 17d. Front Lift : string - path
 - 17e. Rear Lift : string - path
 - 17f. Drag : string - path
 - 17g. Wings Path [x] : string - path | can have multiple Wings Path
 - 18. Drs : object
 - 18a. Ignore Zones : boolean
 - 18b. Limit G : float
 - 18c. Wing Connections [x] : object | can have multiple Wing Connections
 - 18c1. Mode : enum
 - 18c2. Connected Wing : integer
 - 18c3. Effect : float
 - 18c4. Angle : float
 - 19. Ers : object

- 19a. Torque Lut : string – path
- 19b. Coast Lut : string – path
- 19c. Battery Charge Kj : float
- 19d. Has Button Override : boolean
- 19e. Max Kj Per Lap : float
- 19f. Max Charge Kj Per Lap : float
- 19g. Heat Charge Perc : float
- 19h. Heat Power Kw : float
- 19i. Default Power Controller Index : integer
- 19j. Power Controllers Front [x] : object | can have multiple Power

Controllers Front

- 4l9a. Name : string
- 4l9b. Stages [x] : object | Rod Controllers can have multiple

stages

- 4l9b1. Input Var : enum
- 4l9b2. Combinator Mode : enum
- 4l9b3. Lut : string – path
- 4l9b4. Filter Gain : float
- 4l9b5. Up Limit : float
- 4l9b6. Down Limit : float
- 4l9b7. Current Value : float
- 4l9b8. Const Value : float
- 19k. Power Controllers Rear [x] : object | can have multiple Power

Controllers Rear

- 4l9a. Name : string
- 4l9b. Stages [x] : object | Rod Controllers can have multiple

stages

- 4l9b1. Input Var : enum
- 4l9b2. Combinator Mode : enum
- 4l9b3. Lut : string – path
- 4l9b4. Filter Gain : float
- 4l9b5. Up Limit : float
- 4l9b6. Down Limit : float
- 4l9b7. Current Value : float
- 4l9b8. Const Value : float
- 19l. Brake Rear Correction : float
- 19m. Has Cockpit Controls : boolean
- 19n. Cockpit Controls : object
 - 19n1. Delivery Profile : boolean
 - 19n2. Mgu H Mode : boolean
 - 19n3. Recovery : boolean
- 19o. Has Front Motors : boolean
- 19p. Front Motors : object
 - 19p1. Torque Lut : string – path
 - 19p2 Discharge Time : float
 - 19p3. Torque Vectoring Bias : float
- 20. Setup Limits : string path
- 21. Collider Mesh : string path
- 22. Body Mesh Offset : object
 - 22a. Position : x, y, z float
 - 22b. Rotation: x, y, z float
 - 22c. Scale : x, y, z float
- 23. Stock Setup : string path
- 24. Ai Setup : string path
- 25. Wet Setup : string path

| 26. Performance Modes [x] : object | can have multiple Performance Modes

- | 26a. Performance Mode Name : string
- | 26b. Electronics Settings : object
 - | 26b1. Tc1 : float
 - | 26b2. Tc2 : float
 - | 26b3. Abs : float
 - | 26b4. Esc : float
 - | 26b5. Ebb : float
 - | 26b6. Engine Map : float
 - | 26b7. Telemetry laps To Record : float
 - | 26b8. Turbo Boost Lv : float
 - | 26b9. Ers Deployment Map : float
 - | 26b10. Ers Recharge Lv : float
 - | 26b11. Ers Heat Charging : float
- | 26c. Brakes Settings : object
 - | 26c1. Front Bias : float
 - | 26c2. Torque Multiplier : float
 - | 26c3. Brake Ducts [x] : float | can have multiple Brake Ducts
- | 26d. Damper Settings [x] : object | can have multiple Damper

Settings

- | 26d1. Slow Bump : float
- | 26d2. Fast Bump : float
- | 26d3. Slow Rebound : float
- | 26d4. Fast Rebound : float
- | 26e. Differential Data : object
 - | 26e1. Type : enum
 - | 26e2. Power : float
 - | 26e3. Coast : float
 - | 26e4. Preload : float
 - | 26e5. Front Share : float
 - | 26e6. Torque Bias Ratio Power : float
 - | 26e7. Torque Bias Ratio Coast : float
 - | 26e8. Thermal Capacity : float
 - | 26e9. Surface : float
 - | 26e10. Heat Transfer Coeff : float
 - | 26e11. Wear Factor : float
 - | 26e12. Friction Reduction With T : float
 - | 26e13. Friction Ref T : float
- | 26f. Four W D Differentials : object
 - | 26f1. Front Diff : object
 - | 26e1. Type : enum
 - | 26e2. Power : float
 - | 26e3. Coast : float
 - | 26e4. Preload : float
 - | 26e5. Front Share : float
 - | 26e6. Torque Bias Ratio Power : float
 - | 26e7. Torque Bias Ratio Coast : float
 - | 26e8. Thermal Capacity : float
 - | 26e9. Surface : float
 - | 26e10. Heat Transfer Coeff : float
 - | 26e11. Wear Factor : float
 - | 26e12. Friction Reduction With T : float
 - | 26e13. Friction Ref T : float
 - | 26f2. Center Diff : object
 - | 26e1. Type : enum

- 26e2. Power : float
- 26e3. Coast : float
- 26e4. Preload : float
- 26e5. Front Share : float
- 26e6. Torque Bias Ratio Power : float
- 26e7. Torque Bias Ratio Coast : float
- 26e8. Thermal Capacity : float
- 26e9. Surface : float
- 26e10. Heat Transfer Coeff : float
- 26e11. Wear Factor : float
- 26e12. Friction Reduction With T : float
- 26e13. Friction Ref T : float
- 26f3. Rear Diff : object
 - 26e1. Type : enum
 - 26e2. Power : float
 - 26e3. Coast : float
 - 26e4. Preload : float
 - 26e5. Front Share : float
 - 26e6. Torque Bias Ratio Power : float
 - 26e7. Torque Bias Ratio Coast : float
 - 26e8. Thermal Capacity : float
 - 26e9. Surface : float
 - 26e10. Heat Transfer Coeff : float
 - 26e11. Wear Factor : float
 - 26e12. Friction Reduction With T : float
 - 26e13. Friction Ref T : float
- 26g. Front Lock Controllers : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string - path
 - 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
- 26h. Center Lock Controllers : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum
 - 4l9b3. Lut : string - path
 - 4l9b4. Filter Gain : float
 - 4l9b5. Up Limit : float
 - 4l9b6. Down Limit : float
 - 4l9b7. Current Value : float
 - 4l9b8. Const Value : float
- 26i. Rear Lock Controllers : object
 - 4l9a. Name : string
 - 4l9b. Stages [x] : object | Rod Controllers can have multiple stages
 - 4l9b1. Input Var : enum
 - 4l9b2. Combinator Mode : enum

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| | | 4l9b3. Lut : string - path
| | | 4l9b4. Filter Gain : float
| | | 4l9b5. Up Limit : float
| | | 4l9b6. Down Limit : float
| | | 4l9b7. Current Value : float
| | | 4l9b8. Const Value : float
| | 26j. Awd Clutches [x] : object | can have multiple Awd Clutches
| | | 26j1. Position : integer
| | | 26j2. Preload : float
| | 26k. Turbo Controllers [x] : object with an array of stages within |
can have multiple Turbo Controllers
| | | 4l9a. Name : string
| | | 4l9b. Stages [x] : object | Rod Controllers can have multiple
stages
| | | 4l9b1. Input Var : enum
| | | 4l9b2. Combinator Mode : enum
| | | 4l9b3. Lut : string - path
| | | 4l9b4. Filter Gain : float
| | | 4l9b5. Up Limit : float
| | | 4l9b6. Down Limit : float
| | | 4l9b7. Current Value : float
| | | 4l9b8. Const Value : float
| | 26l. Turbo Settings : object
| | | 26l1. Boost Lv : float
| 27. Ai Car Data : string - path
| 28. mm : integer

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Enum list - Car Engine

Id	Enum	Values
4l9b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
4l9b2	CombinatorMode	UndefinedMode, Add, Mult
4q5a1	Type	<None>, Poly3, Poly5, Piece Wise, Damper Lut Data
18c1	Mode	UseEffect, UseAngle
26e1	Type	LSD, Spool, Torsen, EpicyclicTorsen, EpicyclicLSD, TorqueVectoring

D. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
2a.	Screen Name	None (String)	Human-readable name shown for the General block (mass, fuel, body envelope) in tooling interfaces.
2b.	Total Mass	kg (Kilograms)	Baseline vehicle mass including driver and fluids as configured; fundamental input for $F=ma$ weight transfer and inertia calculations.
2c.	Tank Position	m (Meters, X / Y / Z)	3D coordinates of the fuel tank centroid relative to the vehicle reference frame; governs how CG and weight bias shift as fuel burns.
2d.	Fuel	L (Liters)	Current fuel volume at session start; dynamically reduces mass and alters balance during the stint.
2e.	Max Fuel	L (Liters)	Maximum fuel tank capacity; defines the upper bound for fuel load and endurance strategy calculations.
2f.	Efficiency	Dimensionless ratio	Fuel consumption efficiency modifier applied to engine fuel burn calculations.
2g.	Kg Per Liter	kg/L (Kilograms per Liter)	Fuel density conversion factor; translates volumetric fuel (liters) into mass (kg) for physics weight updates.
2h.	Body Box Sizes	m (Meters, X / Y / Z)	Axis-aligned bounding box dimensions of the chassis body envelope; used for regulatory checks and collision approximation.
2i.	Pickup Front Height	m (Meters)	Vertical pickup/reference height at the front axle; aligns suspension hardpoints and ground contact geometry.
2j.	Pickup Rear Height	m (Meters)	Vertical pickup/reference height at the rear axle; aligns suspension hardpoints and ground contact geometry.
2k.	Check Rules	None (Boolean : True / False)	Enables automated regulatory rule validation (e.g., minimum ride height) against series constraints.
2l.	Minimum Height	m (Meters)	Regulatory minimum ride-height threshold enforced when Check Rules is active.
2m.	Torsional Stiffness	Nm/rad or N/m (Torsional spring rate)	Chassis torsional spring rate; resists twist between front and rear axles under asymmetric loading.

ID	Name	Unit of Measurement	Description
2n.	Torsional Damping	Nm-s/rad (Torsional damping)	Chassis torsional damping coefficient; dissipates oscillatory twist energy in the body structure.
2o1.	Position	m (Meters, X / Y / Z)	Translation offset applied to the visual/collision body mesh relative to the physics origin (General > Body Mesh Offset).
2o2.	Rotation	deg or rad (X / Y / Z)	Euler rotation offset applied to the body mesh for visual/collision alignment (General > Body Mesh Offset).
2o3.	Scale	Dimensionless (X / Y / Z)	Non-uniform scale factor applied to the body mesh offset block for model alignment.
3.	General Path	None (File path)	External reference to a .GENERALCAR asset overriding or extending global mass/inertia parameters.
4a.	Wheel Base	m (Meters)	Longitudinal distance between front and rear axle centerlines; stabilizes high-speed behavior and affects rotation agility.
4b.	Longitudinal Cg Location	Ratio (0.0 - 1.0)	Longitudinal center-of-gravity position along the wheelbase as a fraction from front axle (e.g., 0.45 = slightly front-biased).
4c.	Base Y Front	m (Meters)	Reference vertical suspension hub height at the front axle in the chassis coordinate system.
4d.	Base Y Rear	m (Meters)	Reference vertical suspension hub height at the rear axle in the chassis coordinate system.
4e.	Track Front	m (Meters)	Lateral distance between left and right front tire centerlines; wider track increases roll stability.
4f.	Track Rear	m (Meters)	Lateral distance between left and right rear tire centerlines.
4g1.	Min Velocity	m/s or km/h (Speed threshold)	Minimum impact velocity required before suspension/chassis damage accumulation begins.
4g2.	Gain	Dimensionless coefficient	Damage accumulation rate multiplier per unit of impact energy above the minimum velocity threshold.
4g3.	Max Damage	Ratio (0.0 - 1.0)	Upper cap on cumulative suspension/chassis damage before performance degradation saturates.

ID	Name	Unit of Measurement	Description
4g4.	Debug Log	None (Boolean : True / False)	Enables verbose logging of suspension damage events for development and tuning diagnostics.
4h.	Coilover Front Path	None (File path)	Path to the front axle .coilover asset defining spring rates, damper curves, and bump-stop geometry.
4i.	Coilover Rear Path	None (File path)	Path to the rear axle .COILOVER asset.
4j.	Front Suspension Path	None (File path)	Path to the front .SUSPENSION asset defining kinematics, wishbone geometry, and camber curves.
4k.	Rear Suspension path	None (File path)	Path to the rear .SUSPENSION asset.
4l1.	Spring Rate	N/m (Newtons per meter)	Linear spring stiffness of the heavy-duty auxiliary spring element (used on trucks, off-road, or dual-rate setups).
4l2.	Progressive Spring Rate	N/m ² or N/m (Progressive rate)	Secondary progressive spring rate; stiffness rises non-linearly as compression increases.
4l3a.	Range	m (Meters)	Operational travel range of the upper bump-stop before full engagement (Heavy Springs > Bump Stop Up).
4l3b	Reference	m (Meters)	Reference suspension position at which the upper bump-stop force model begins scaling.
4l3c.	Force	N (Newtons)	Peak resistive force delivered by the upper bump-stop at maximum compression.
4l3d.	Gamma	Dimensionless exponent	Non-linear shaping exponent for upper bump-stop force vs. compression curve.
4l3e.	Length	m (Meters)	Physical contact length of the upper bump-stop element.
4l3f.	Damping	N·s/m (Damping coefficient)	Hydraulic or rubber damping applied during upper bump-stop engagement.
4l4a.	Range	m (Meters)	Operational travel range of the lower (rebound) bump-stop (Heavy Springs > Bump Stop Down).
4l4b.	Reference	m (Meters)	Reference position at which the lower bump-stop force model activates.

ID	Name	Unit of Measurement	Description
4l4c.	Force	N (Newtons)	Peak resistive force delivered by the lower bump-stop at maximum extension.
4l4d.	Gamma	Dimensionless exponent	Non-linear shaping exponent for lower bump-stop force curve.
4l4e.	Length	m (Meters)	Physical contact length of the lower bump-stop element.
4l4f.	Damping	N·s/m (Damping coefficient)	Damping applied during lower bump-stop engagement on rebound.
4l5.	Collar Position	m (Meters)	Adjustable spring perch/collar position controlling static ride height and preload on the heavy spring.
4l6a1.	Bump	N·s/m (Damping rate)	Fast damper channel bump (compression) damping rate within the Heavy Springs embedded damper model.
4l6a2.	Rebound	N·s/m (Damping rate)	Fast damper channel rebound (extension) damping rate.
4l6b1.	Bump	N·s/m (Damping rate)	Slow damper channel bump damping rate for low-velocity suspension movements.
4l6b2.	Rebound	N·s/m (Damping rate)	Slow damper channel rebound damping rate.
4l6c.	Fast Threshold Bump	m/s (Velocity threshold)	Suspension compression velocity above which the fast bump damper circuit engages.
4l6d.	Fast Threshold Rebound	m/s (Velocity threshold)	Suspension extension velocity above which the fast rebound damper circuit engages.
4l6e.	Cooling Surface	m ² (Square meters)	Exposed damper body surface area used for thermal dissipation modeling.
4l6f.	Nominal Force	N (Newtons)	Reference force level for damper nominal operating point and fatigue calculations.
4l6g.	Min Stress Fatigue	Pa or N (Stress threshold)	Lower bound stress level below which damper fatigue accumulation is negligible.
4l6h.	Mas Stress Fatigue	Pa or N (Stress threshold)	Upper bound stress level at which damper fatigue reaches maximum degradation rate.
4l6i.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the damper assembly; controls temperature rise under repeated high-energy cycles.

ID	Name	Unit of Measurement	Description
4I6j.	Heat Transfer Coefficient	W/(m ² ·K) (Coefficient)	Convective heat transfer rate from the damper body to ambient airflow.
4I6k.	Lut List	None (File path)	Path to a damper force-velocity look-up table list defining non-linear damping behavior.
4I6l.	Damper Lut Scale	None (File path)	Scaling curve applied to the damper LUT output for fine-tuning peak forces.
4I7.	Helper K	N/m (Spring rate)	Stiffness of the helper/tender spring assisting the main spring at low compression.
4I8.	Helper Range	m (Meters)	Travel range over which the helper spring is active before the main spring fully engages.
4I9a.	Name	None (String)	Internal identifier for a dynamic controller block. Reused across Rod Controllers, ARB/Flex Bar controllers, Steering, ERS, diff-lock, and turbo controllers.
4I9b1.	Input Var	None (Telemetry enum)	Live telemetry channel used as controller input (e.g., Brake, Gas, Speed, SlipRatio, SteerDEG, ErsChargeLevel).
4I9b2.	Combinator Mode	None (Math enum : Add / Mult)	How this stage output combines with prior stages: additive offset or multiplicative scaling.
4I9b3.	Lut	None (.CURVE file path)	Look-up table mapping the input variable to an output modifier factor.
4I9b4.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter coefficient dampening rapid telemetry spikes for smooth controller response.
4I9b5.	Up Limit	Depends on input variable	Upper clamp on the processed input signal before LUT evaluation.
4I9b6.	Down Limit	Depends on input variable	Lower clamp on the processed input signal before LUT evaluation.
4I9b7.	Current Value	Depends on input variable	Runtime value of the controller stage output during simulation (debug/telemetry).
4I9b8.	Const Value	Depends on input variable	Fallback constant output when no dynamic input or LUT is active.
4m1.	Stiffness	Nm/deg or N/m (Anti-roll bar rate)	Torsional stiffness of the front anti-roll bar resisting body roll in cornering.
4n1.	Stiffness	Nm/deg or N/m (Anti-roll bar rate)	Torsional stiffness of the rear anti-roll bar.

ID	Name	Unit of Measurement	Description
4o1.	Stiffness	Nm/deg or N/m (Flex bar rate)	Stiffness of the front flex/secondary torsion bar (compliance element in complex suspension layouts).
4p1.	Stiffness	Nm/deg or N/m (Flex bar rate)	Stiffness of the rear flex/secondary torsion bar.
4q1.	Wheel Gain	Dimensionless coefficient	Global scaling gain for per-wheel damper authority in the active damper controller.
4q2.	Heave Gain	Dimensionless coefficient	Gain applied to heave-mode (vertical chassis) damper control signals.
4q3.	Pitch Gain	Dimensionless coefficient	Gain applied to pitch-mode damper control (nose dive / squat compensation).
4q4.	Roll Gain	Dimensionless coefficient	Gain applied to roll-mode damper control (lateral weight transfer management).
4q5a1.	Type	None (Enum : none / Poly3 / Poly5 / PieceWise / DamperLutData)	Damper control curve type for each motion mode (Base, Heave, Roll, Pitch, Max, Min) on front/rear axles.
4r.	Has Dampers Cockpit Settings	None (Boolean : True / False)	Allows the driver to adjust damper settings from cockpit controls during a session.
5.	Drivetrain Path	None (File path)	Path to the .DRIVETRAIN asset defining differential type, gear ratios, and torque split logic.
6.	Gearbox Path	None (File path)	Path to the .GEARBOX asset defining gear ratios, shift times, and auto-blip behavior.
7.	Clutch Path	None (File path)	Path to the .CLUTCH asset defining clutch friction, inertia, and autoclutch engagement.
8.	Engine Path	None (File path)	Path to the .CARENGINE asset defining torque curves, turbo, and engine braking.
9.	Brakes Path	None (File path)	Path to the .BRAKESYSTEM asset defining total braking torque, bias, and EBB controllers.
10a.	Four W S Controllers	None (Controller object)	Four-wheel steering dynamic controller array; modulates rear steer angle via staged LUT pipelines (uses 419 stage fields).
11a1.	Has T C2	None (Boolean : True / False)	Enables a secondary traction-control map (TC2) for dual-stage intervention logic.

ID	Name	Unit of Measurement	Description
11a2.	Frequency Hz	Hz (Hertz)	Control loop update frequency of the traction-control system.
11a3.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum vehicle speed below which traction control deactivates.
11a4.	Gear Change Time	s (Seconds)	Temporary TC suppression window duration during gear changes to avoid false interventions.
11a5.	Min Cut Level	Ratio (0.0 - 1.0)	Minimum engine torque cut level applied at the lowest TC intervention step.
11a6.	Max Cut Level	Ratio (0.0 - 1.0)	Maximum engine torque cut level at the most aggressive TC step.
11a7a.	Min Slip Ratio	Dimensionless ratio	Lower slip-ratio threshold at which TC begins monitoring wheel spin for this settings map.
11a7b.	Max Slip Ratio	Dimensionless ratio	Upper slip-ratio threshold triggering full TC intervention for this settings map.
11a7c.	Ref Slip Angle Deg	deg (Degrees)	Reference rear slip angle used to scale TC aggression relative to yaw behavior.
11a7d.	Engine Cut Level	Ratio (0.0 - 1.0)	Engine torque reduction factor applied when slip thresholds are exceeded.
11a7e.	Angular A C Cgain	Dimensionless gain	Yaw-rate / angular acceleration gain for TC correction authority.
11a7f.	Oversteer Gain	Dimensionless gain	Additional TC sensitivity multiplier when oversteer is detected.
11a7g.	Slip Angle Activation Deg	deg (Degrees)	Minimum slip angle required before this TC map becomes active.
11b1a.	Min Slip Ratio	Dimensionless ratio	Lower wheel slip threshold for ABS monitoring on this settings map.
11b1b.	Max Slip Ratio	Dimensionless ratio	Upper slip threshold triggering maximum ABS pressure modulation.
11b1c.	Ref Slip Angle Deg	deg (Degrees)	Reference slip angle for ABS yaw-stability cross-correlation.
11b1d.	Cut Level	Ratio (0.0 - 1.0)	Brake pressure reduction factor applied during ABS pulsing.
11b1f.	Max Torque Variation	Nm or ratio	Maximum allowed brake torque fluctuation per ABS cycle on this map.

ID	Name	Unit of Measurement	Description
11b2.	Frequency	Hz (Hertz)	ABS control loop pulsing frequency.
11b3.	Channels	None (Integer)	Number of independent ABS control channels (typically one per wheel or per axle).
11b4.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum speed below which ABS intervention is disabled.
11c1.	Active	None (Boolean : True / False)	Master enable for Electronic Differential Lock (EDL) brake-based torque transfer.
11c2.	Brake Torque Power	Nm (Newton-meters)	Brake torque applied to the spinning wheel under power (on-throttle) to emulate a locking diff.
11c3.	Brake Torque Coast	Nm (Newton-meters)	Brake torque applied during coasting to synchronize wheel speeds.
11c4.	Dead Zone Coast	Ratio or rad/s	Speed-difference deadband below which coast EDL does not intervene.
11c5.	Dead Zone Power	Ratio or rad/s	Speed-difference deadband below which power EDL does not intervene.
11c6.	Max Spin Power	rad/s or ratio	Maximum allowed wheel spin differential under power before EDL fully engages.
11c7.	Max Spin Coaster	rad/s or ratio	Maximum allowed wheel spin differential on coast before EDL engages.
11c8.	Min Speed	m/s or km/h (Speed threshold)	Minimum vehicle speed required for EDL operation.
11d1.	Frequency Hz	Hz (Hertz)	Electronic Stability Program control loop update frequency.
11d2.	Min Speed Kmh	km/h (Kilometers per hour)	Minimum speed below which ESP deactivates.
11d3a.	Gain	Dimensionless gain	Global ESP intervention authority multiplier for this settings map.
11d3b.	Steer Gain	Dimensionless gain	Steering-angle sensitivity gain for ESP yaw correction.
11d3c.	Min Steer Gain	Dimensionless gain	Minimum steering gain applied at low speeds within ESP logic.
11d3d.	Steer Gain Max Speed	km/h (Kilometers per hour)	Speed at which steering gain reaches its maximum ESP authority.
11d3e.	Oversteer Gain	Dimensionless gain	Corrective gain when rear slip (oversteer) is detected.

ID	Name	Unit of Measurement	Description
11d3f.	Understeer Gain	Dimensionless gain	Corrective gain when front slip (understeer) is detected.
11d3g.	Max Slip Ratio	Dimensionless gain	Maximum wheel slip ratio before ESP applies full brake/engine intervention.
11d3h.	Dead Zone	deg or ratio	Yaw/steer error deadband where ESP remains passive.
11d3i.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter on ESP sensor inputs to prevent oscillatory corrections.
11d3j.	Brake Perc	Ratio (0.0 - 1.0)	Percentage of available brake pressure ESP may apply for stability correction.
11d3k.	Brake Perc Activation	Ratio (0.0 - 1.0)	Slip/yaw threshold fraction at which ESP begins applying brake intervention.
12.	Electronics Path	None (File path)	External reference to a standalone .CARELECTRONICS asset overriding inline TC/ABS/ESP definitions.
13a.	Ff Mult	Dimensionless multiplier	Force-feedback strength multiplier scaling steering wheel torque output to the driver.
13b.	Steer Lock	deg (Degrees)	Maximum steering wheel rotation angle from lock to lock.
13c.	Steer Ratio	Ratio (:1)	Steering rack ratio converting steering wheel angle to front wheel angle.
13d.	Linear Steer Rod Ratio	M/rad or dimensionless	Linear rack displacement per unit steering input; fine-tunes low-speed steering response.
13e.	Steer Assist	Dimensionless multiplier	Power-steering assist strength scaling self-aligning torque feedback.
14a.	Center	m (Meters, X / Y / Z)	Center position of an axis-aligned box collider used for simplified body collision detection.
14b.	Size	m (Meters, X / Y / Z)	Full dimensions (width, height, length) of the box collider.
14c.	Pitch Rotation Deg	deg (Degrees)	Pitch-axis rotation applied to orient the box collider relative to the chassis.
15.	Front Tyre Compounds	None (File path)	Path(s) to front axle .tyre compound assets; multiple entries allow compound selection.
16.	Rear Type Compounds	None (File path)	Path(s) to rear axle .tyre compound assets.

ID	Name	Unit of Measurement	Description
17a.	Slip Gain Multiple	Dimensionless multiplier	Global scaling factor for aerodynamic slip-stream and yaw-sensitivity effects.
17b.	Speed Factor Mult	Dimensionless multiplier	Multiplier applied to speed-dependent aero force calculations (downforce/drag scaling).
17c1.	Position	m (Meters, X / Y / Z)	Application point of an aerodynamic downforce/lift element on the chassis.
17c2.	Cl Gain	Dimensionless coefficient	Lift coefficient gain multiplier for this downforce element.
17c3.	Cd Gain	Dimensionless coefficient	Drag coefficient gain multiplier for this downforce element.
17c4.	Yaw Gain	Dimensionless coefficient	Sensitivity of aero forces to vehicle yaw angle (side-force generation).
17c5.	Drag Per Cool Transfer	Coefficient	Drag penalty factor linked to coolant or brake-duct thermal transfer (aero-thermal coupling).
17c6.	Damage C L	None (String / curve reference)	Damage-state modifier curve for lift coefficient degradation on this aero element.
17c7.	Damage C D	None (String / curve reference)	Damage-state modifier curve for drag coefficient degradation.
17c8a.	Combinator Mode	None (Math enum : Add / Mult)	How downforce controller stage output combines with baseline aero coefficients.
17c8b.	Input	None (Telemetry enum)	Telemetry input driving dynamic downforce modulation (speed, ride height, DRS state, etc.).
17c8c.	Filter	Coefficient (Smoothing multiplier)	Input smoothing filter for downforce controller stages.
17c8d.	Up Limit	Depends on input variable	Upper clamp on downforce controller input.
17c8e.	Down Limit	Depends on input variable	Lower clamp on downforce controller input.
17c8f.	Lut	None (.curve file path)	Look-up table mapping controller input to aero coefficient modifier.
17c9.	Lift Per Front Angle	Coefficient per deg	Rate of lift change per degree of front ride-height / pitch angle.
17c10.	Lift Per Rear Angle	Coefficient per deg	Rate of lift change per degree of rear ride-height / pitch angle.
17c11.	Drag Per Front Angle	Coefficient per deg	Rate of drag change per degree of front body angle.
17c12.	Drag Per Rear Angle	Coefficient per deg	Rate of drag change per degree of rear body angle.

ID	Name	Unit of Measurement	Description
17c13.	Default Front Angle	deg (Degrees)	Baseline front body angle reference for aero angle-dependent calculations.
17c14.	Default Rear Angle	deg (Degrees)	Baseline rear body angle reference for aero angle-dependent calculations.
17d.	Front Lift	None (File path)	Path to a .SURFACE3D front lift aero map defining Cl vs. ride height / speed.
17e.	Rear Lift	None (File path)	Path to a .SURFACE3D rear lift aero map.
17f.	Drag	None (File path)	Path to a .SURFACE3D drag map defining Cd vs. speed and configuration.
17g.	Wings Path	None (File path)	Path(s) to .WING assets defining adjustable aerodynamic devices (splitters, wings, diffusers).
18a.	Ignore Zones	None (Boolean : True / False)	When true, DRS activation ignores track zone restrictions (development override).
18b.	Limit G	G (Lateral G-force)	Maximum lateral G beyond which DRS automatically deactivates for safety.
18c1.	Mode	None (Enum : UseEffect / UseAngle)	DRS wing connection mode: apply a coefficient effect or a physical angle offset.
18c2.	Connected Wing	None (Integer index)	Index of the wing element in the Wings Path array controlled by this DRS connection.
18c3.	Effect	Dimensionless coefficient delta	Aerodynamic coefficient change applied when DRS is open (Mode = UseEffect).
18c4.	Angle	deg (Degrees)	Physical wing angle offset applied when DRS is open (Mode = UseAngle).
19a.	Torque Lut	None (.curve file path)	Look-up table mapping deployment input to electric motor assist torque (kW/Nm).
19b.	Coast Lut	None (.curve file path)	Look-up table defining regenerative braking / coast recovery torque.
19c.	Battery Charge Kj	kJ (Kilojoules)	Total energy storage capacity of the hybrid battery pack.
19d.	Has Button Override	None (Boolean : True / False)	Allows driver button override of automatic ERS deployment strategy.

ID	Name	Unit of Measurement	Description
19e.	Max Kj Per Lap	kJ/lap (Kilojoules per lap)	Regulatory or strategic cap on ERS energy deployment per lap.
19f.	Max Charge Kj Per Lap	kJ/lap (Kilojoules per lap)	Maximum energy that can be recovered into the battery per lap.
19g.	Heat Charge Perc	Ratio (0.0 - 1.0)	Fraction of waste heat energy convertible into battery charge (MGU-H simulation).
19h.	Heat Power Kw	kW (Kilowatts)	Thermal energy recovery power ceiling from exhaust/heat sources.
19i.	Default Power Controller Index	None (Integer)	Index of the default ERS deployment map selected at session start.
19j.	Power Controllers Front	None (Controller array)	Array of front-axle ERS deployment controllers; each entry uses the shared 4I9 stage fields (Name + Stages).
19k.	Power Controllers Rear	None (Controller array)	Array of rear-axle ERS deployment controllers; each entry uses the shared 4I9 stage fields.
19l.	Brake Rear Correction	Ratio or Nm	Rear brake torque correction factor when ERS regenerative braking is active.
19m.	Has Cockpit Controls	None (Boolean : True / False)	Exposes ERS deployment/ recovery controls to the driver in the cockpit UI.
19n1.	Delivery Profile	None (Boolean)	Cockpit toggle for ERS power delivery profile selection.
19n2.	Mgu H Mode	None (Boolean)	Cockpit toggle for MGU-H harvesting mode.
19n3.	Recovery	None (Boolean)	Cockpit toggle for regenerative recovery aggressiveness.
19o.	Has Front Motors	None (Boolean : True / False)	Enables front-axle electric motor(s) for hybrid or torque-vectoring applications.
19p1.	Torque Lut	None (.curve file path)	Torque delivery curve for front-axle electric motor(s).
19p2.	Discharge Time	s (Seconds)	Time constant governing front motor energy discharge rate from the battery.
19p3.	Torque Vectoring Bias	Ratio (-1.0 - 1.0)	Left/right torque distribution bias for front-axle electric torque vectoring.
20.	Setup Limits	None (File path)	Path to .CARSETUPLIMITS asset defining min/max/step bounds for garage setup sliders.

ID	Name	Unit of Measurement	Description
21.	Collider Mesh	None (File path)	Path to the detailed collision mesh used for body contact and scraping detection.
22a.	Position	m (Meters, X / Y / Z)	Root-level body mesh translation offset (distinct from General 2o block).
22b.	Rotation	deg or rad (X / Y / Z)	Root-level body mesh rotation offset.
22c.	Scale	Dimensionless (X / Y / Z)	Root-level body mesh scale factor.
23.	Stock Setup	None (File path)	Path to the default .CARSETUP preset loaded for new sessions.
24.	Ai Setup	None (File path)	Path to the AI-specific .CARSETUP preset used by computer-controlled opponents.
25.	Wet Setup	None (File path)	Path to the wet-weather .CARSETUP preset (tyre pressures, wing angles, etc.).
26a.	Performance Mode Name	None (String)	Label for a performance mode preset (e.g., Qualifying, Race, Wet, Safe).
26b1.	Tc1	None (Map index / level)	Traction Control map level stored in this performance mode.
26b2.	Tc2	None (Map index / level)	Secondary TC map level for dual-TC systems.
26b3.	Abs	None (Map index / level)	ABS intervention map level for this performance mode.
26b4.	Esc	None (Map index / level)	ESP/stability control map level for this performance mode.
26b5.	Ebb	None (Map index / level)	Electronic Brake Balance mode/level for this performance mode.
26b6.	Engine Map	None (Map index)	Engine power map selection (fuel/power trade-off).
26b7.	Telemetry Laps To Record	None (integer / lap number)	Number of laps of telemetry data to retain when this mode is active.
26b8.	Turbo Boost Lv	bar or level index	Turbo boost level preset for this performance mode.
26b9.	Ers Deployment Map	None (Map index)	Turbo boost level preset for this performance mode.
26b10.	Ers Recharge Lv	None (Level index)	ERS energy recovery aggressiveness level.
26b11.	Ers Heat Charging	None (Level index)	MGU-H heat-to-battery charging level.
26c1.	Front Bias	Ratio (0.0 - 1.0)	Brake balance front percentage stored in this performance mode.

ID	Name	Unit of Measurement	Description
26c2.	Torque Multiplier	Dimensionless multiplier	Global braking torque scaling factor for this mode.
26c3.	Brake Ducts	Ratio (0.0 - 1.0)	Brake cooling duct opening level(s); multiple entries for front/rear ducts.
26d1.	Slow Bump	Clicks or N·s/m	Slow-speed compression damper setting for this performance mode.
26d2.	Fast Bump	Clicks or N·s/m	Fast-speed compression damper setting.
26d3.	Slow Rebound	Clicks or N·s/m	Slow-speed rebound damper setting.
26d4.	Fast Rebound	Clicks or N·s/m	Fast-speed rebound damper setting.
26e1.	Type	None (Enum : LSD / Spool / Torsen / EpicyclicTorsen / EpicyclicLSD / TorqueVectoring)	Differential mechanism type governing torque split behavior.
26e2.	Power	Nm or ratio	Differential lock/acceleration sensitivity under power (on-throttle).
26e3.	Coast	Nm or ratio	Differential lock sensitivity on coast (off-throttle).
26e4.	Preload	Nm (Newton-meters)	Static preload torque required before differential plates begin slipping.
26e5.	Front Share	Ratio (0.0 - 1.0)	Fraction of total driveline torque directed to the front axle (AWD/4WD).
26e6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Torque bias ratio between axles or sides under power.
26e7.	Torque Bias Ratio Coast	Ratio	Torque bias ratio between axles or sides on coast.
26e8.	Thermal Capacity	J/K or J/°C	Heat storage capacity of differential components for thermal wear modeling.
26e9.	Surface	m ² (Square meters)	Exposed surface area of differential housing for cooling calculations.
26e10.	Heat Transfer Coef	W/(m ² ·K) (Coefficient)	Heat dissipation rate from differential to ambient airflow.
26e11.	Wear Factor	Dimensionless coefficient	Rate at which differential friction surfaces degrade under load and temperature.
26e12.	Friction Reduction With T	Ratio/°C	Linear friction reduction per degree of temperature rise in the differential.

ID	Name	Unit of Measurement	Description
26e13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature at which nominal differential friction is defined.
26f1.	Front Diff	None (Differential object)	Front axle differential block; contains 26e1–26e13 fields for AWD/4WD layouts.
26f2.	Center Diff	None (Differential object)	Center differential block for torque split between front and rear axles.
26f3.	Rear Diff	None (Differential object)	Rear axle differential block.
26g.	Front Lock Controllers	None (Controller object)	Dynamic controller modulating front differential lock under telemetry conditions.
26h.	Center Lock Controllers	None (Controller object)	Dynamic controller modulating center differential lock.
26i.	Rear Lock Controllers	None (Controller object)	Dynamic controller modulating rear differential lock.
26j1.	Position	None (Integer index)	Index position of an AWD clutch pack in the driveline layout.
26j2.	Preload	Nm (Newton-meters)	Static preload torque on this AWD clutch before slip occurs.
26k.	Turbo Controllers	None (Controller object)	Dynamic controller array modulating turbo boost via staged LUT pipelines (uses 419 stage fields).
26l1.	Boost Lv	bar or level index	Turbo boost level preset within this performance mode.
27.	Ai Car Data	None (File path)	Path to an AI-specific .car data override tuning opponent behavior without altering player physics.
28.	mm	None (Integer)	Internal unit-scaling or precision flag used by the editor toolchain (millimeter reference mode).

E. Example data

I. Chosen Car Data for Example

- Ferrari 296 GTB (slug : ks_ferrari_286_gtb)
- Audi R8 LMS GT3 Evo II (slug : ks_audi_r8_lms_gt3_evo_2)
- Renault 5 GT Turbo (slug : ks_renault_5_gt_turbo)

II. Example

Ferrari 296 GTB

- 1. Screen Name : None
- 2. General
 - 2a. Screen Name : Ferrari 296 GTB
 - 2b. Total Mass : 1750.00000
 - 2c. Tank Position : 0.00000, -0.16715, -0.32949
 - 2d. Fuel : 60.00000
 - 2e. Max Fuel : 120.00000
 - 2f. Efficiency : 0.00000
 - 2g. Kg Per Liter : 0.75500
 - 2h. Body Box Sizes : 2.00000, 0.70000, 4.22149
 - 2i. Pickup Front Height : -0.35200
 - 2j. Pickup Rear Height : -0.35200
 - 2k. Check Rules : false
 - 2l. Minimum Height : 0.00000
 - 2m. Torsional Stiffness : 30000.00000
 - 2n. Torsional Damping : 500.00000
 - 2o. Body Mesh Offset
 - 2o1. Position : 0.000, 0.000, 0.000
 - 2o2. Rotation : 0.000, 0.000, 0.000
 - 2o3. Scale : 0.000, 0.000, 0.000
- 3. General Path : None
- 4. Suspensions
 - 4a. Wheel Base : 2.60000
 - 4b. Longitudinal Cg Location : 0.39500
 - 4c. Base Y Front : -0.11000
 - 4d. Base Y Rear : -0.08200
 - 4e. Track Front : 1.72600
 - 4f. Track Rear : 1.71000
 - 4g. Damage
 - 4g1. Min Velocity : 40.00000
 - 4g2. Gain : 0.00040
 - 4g3. Max Damage : 0.05000
 - 4g4. Debug Log : true
 - 4h. Coilover Front path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.coilover
 - 4i. Coilover Rear Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_rear.coilover
 - 4j. Front Suspension Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_front.suspension
 - 4k. Rear Suspension Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb_rear.suspension
 - 4l. Heavy Springs : None
 - 4m. Arb Front
 - 4m1. Stiffness : 26000.00000
 - 4m2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
 - 4n. Arb Rear
 - 4n1. Stiffness : 20000.00000
 - 4n2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
 - 4o. Flex Bar Front
 - 4o1. Stiffness : 0.00000

- 4o2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 4p. Flex Bar Rear
 - 4p1. Stiffness 0.00000
 - 4p2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 4q. Dampers Controller : None
- 4r. Has Dampers Cockpit Settings : false
- 5. Drivetrain Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.drivetrain
- 6. Gearbox Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.gearbox
- 7. Clutch Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.clutch
- 8. Engine Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.carengine
- 9. Brakes Path :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.brakesystem
- 10. Steering System
 - 10a. Four W S Controllers
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 11. Electronics
 - 11a. T C
 - 11a1. Has T C2 : false
 - 11a2. Frequency Hz : 333.00000
 - 11a3. Min Speed Kmh : 35.00000
 - 11a4. Gear Change Time : 0.0800
 - 11a5. Min Cut Level : 8.00000
 - 11a6. Max Cut Level : 0.90000
 - 11a7. Settings 1
 - 11a7a. Min Slip Ratio : 0.00000
 - 11a7b. Max Slip Ratio : 0.00000
 - 11a7c. Ref Slip Angle Deg : 0.00000
 - 11a7d. Engine Cut Level : 0.00000
 - 11a7e. Angular A C Cgain : 0.00000
 - 11a7f. Oversteer Gain : 0.00000
 - 11a7g. Slip Angle Activation Deg : 0.00000
 - 11a7. Settings 2
 - 11a7a. Min Slip Ratio : 0.09000
 - 11a7b. Max Slip Ratio : 0.35000
 - 11a7c. Ref Slip Angle Deg : 15.00000
 - 11a7d. Engine Cut Level : 1.80000
 - 11a7e. Angular A C Cgain : 4.50000
 - 11a7f. Oversteer Gain : 3.00000
 - 11a7g. Slip Angle Activation Deg : 6.00000
 - 11a7. Settings 3
 - 11a7a. Min Slip Ratio : 0.07000
 - 11a7b. Max Slip Ratio : 0.20000
 - 11a7c. Ref Slip Angle Deg : 12.00000
 - 11a7d. Engine Cut Level : 1.50000
 - 11a7e. Angular A C Cgain : 6.00000
 - 11a7f. Oversteer Gain : 5.00000
 - 11a7g. Slip Angle Activation Deg : 5.00000

- 11a7. Settings 4
 - 11a7a. Min Slip Ratio : 0.04500
 - 11a7b. Max Slip Ratio : 0.15000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 1.0000
 - 11a7e. Angular A C Cgain : 10.00000
 - 11a7f. Oversteer Gain : 7.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000

- 11b. A B S

- 11b1. Settings 1
 - 11b1a. Min Slip Ratio : -1.00000
 - 11b1b. Max Slip Ratio : -1.00000
 - 11b1c. Ref Slip Angle Deg : 0.00000
 - 11b1d. Cut Level : 0.00000
 - 11b1f. Max Torque Variation : 0.70000
- 11b1. Settings 2
 - 11b1a. Min Slip Ratio : 0.07000
 - 11b1b. Max Slip Ratio : 0.12000
 - 11b1c. Ref Slip Angle Deg : 9.00000
 - 11b1d. Cut Level : 0.05000
 - 11b1f. Max Torque Variation : 1.00000

- Frequency : 40.00000
- Channels : 4
- Min Speed Kmh : 0.00000

- 11c. E D L : None

- 11d. E S P

- 11d1. Frequency Hz : 100.00000
- 11d2. Min Speed Kmh : 20.00000
- 11d3. Settings 1
 - 11d3a. Gain : 0.00000
 - 11d3b. Steer Gain : 0.00000
 - 11d3c. Min Steer Gain : 0.00000
 - 11d3d. Steer Gain Max Speed : 0.00000
 - 11d3e. Oversteer Gain : 0.00000
 - 11d3f. Understeer Gain : 0.00000
 - 11d3g. Max Slip Ratio : 0.00000
 - 11d3h. Dead Zone : 0.00000
 - 11d3i. Filter Gain : 0.00000
 - 11d3j. Brake Perc : 0.00000
 - 11d3k. Brake Perc Activation : 0.00000

- 11d3. Settings 2
 - 11d3a. Gain : 0.50000
 - 11d3b. Steer Gain : 0.50000
 - 11d3c. Min Steer Gain : 0.50000
 - 11d3d. Steer Gain Max Speed : 0.00000
 - 11d3e. Oversteer Gain : 0.50000
 - 11d3f. Understeer Gain : 0.50000
 - 11d3g. Max Slip Ratio : 0.19000
 - 11d3h. Dead Zone : 0.30000
 - 11d3i. Filter Gain : 0.95000
 - 11d3j. Brake Perc : 0.10000
 - 11d3k. Brake Perc Activation : 0.90000

- 12. Electronics Path : None

- 13. Controls

- 13a. Ff Mult : 1.10000

- 13b. Steer Lock : 342.00000
- 13c. Steer Ratio : 12.00000
- 13d. Linear Steer Rod Ratio : 0.00200
- 13e. Steer Assist : 1.00000
- 14. Box Colliders 1
 - 14a. Center : 0.00000, -0.26000, 0.33000
 - 14b. Size : 1.78000, 0.15000, 3.94000
 - 14c. Pitch Rotation Deg : 0.200000
- 15. Front Tyre Compounds 1 :
 - content\cars\common_phsx\tyres\hypercar\hypercar_245_35_20.tyre
- 16. Rear Tyre Compounds 1 :
 - content\cars\common_phsx\tyres\hypercar\hypercar_305_35_20.tyre
- 17. Aero
 - 17a. Slip Gain Multiple : 1.00000
 - 17b. Speed Factor Mult : 2.00000
 - 17c. Downforces : None
 - 17d. Front Lift : None
 - 17e. Rear Lift : None
 - 17f. Drag : None
 - 17g. Wings Path 1 :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb0.wing
 - 17g. Wings Path 2 :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb1.wing
 - 17g. Wings Path 3 :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb2.wing
 - 17g. Wings Path 4 :
 - content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb3.wing
- 18. Drs : None
- 19. Ers
 - 19a. Torque Lut :
 - content\cars\ks_ferrari_296_gtb\data\kers\KERS_TORQUE.curve
 - 19b. Coast Lut :
 - content\cars\ks_ferrari_296_gtb\data\kers\KERS_TORQUE.curve
 - 19c. Battery Charge Kj : 26820.00000
 - 19d. Has Button Override : false
 - 19e. Max Kj Per Lap : 0.00000
 - 19f. Max Charge Kj Per Lap : 0.00000
 - 19g. Heat Charge Perc : 0.00000
 - 19h. Heat Power Kw : 0.00000
 - 19i. Default Power Controller Index : 0
 - 19j. Power Controllers Front : None
 - 19k. Power Controllers Rear 1
 - 4l9a. Name : MAP Q
 - 4l9b. Stages 1
 - 4l9b1. Input Var : Gas
 - 4l9b2. Combinator Mode : Add
 - 4l9b3. Lut :
 - content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_Q_CONTROLLER_GAS.curve
 - 4l9b4. Filter Gain : 0.96000
 - 4l9b5. Up Limit : 1.00000
 - 4l9b6. Down Limit : -1.00000
 - 4l9b7. Current Value : 0.00000
 - 4l9b8. Const Value : 0.00000
 - 4l9b. Stages 2
 - 4l9b1. Input Var : Gear

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| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 3
| | | | 4l9b1. Input Var : SlipRatioRearAVG
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 4
| | | | 4l9b1. Input Var : RPMS
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 5
| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHR0TTLE_coast.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 6
| | | | 4l9b1. Input Var : Brake
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 19k. Power Controllers Rear 2
| | | | 4l9a. Name : MAP P
| | | | 4l9b. Stages 1

```



```

| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_P_CONTROLLER_G
AS.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 2
| | | | 4l9b1. Input Var : Gear
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 3
| | | | 4l9b1. Input Var : SlipRatioRearAVG
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 4
| | | | 4l9b1. Input Var : RPMS
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 5
| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 6
| | | | 4l9b1. Input Var : Brake

```

```

| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
- 19k. Power Controllers Rear 3
| | | | 4l9a. Name : MAP H
| | | | 4l9b. Stages 1
| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kers_H_CONTROLLER_G
AS.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
- 4l9b. Stages 2
| | | | 4l9b1. Input Var : Gear
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_GEAR
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
- 4l9b. Stages 3
| | | | 4l9b1. Input Var : SlipRatioRearAVG
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_SLIP
.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
- 4l9b. Stages 4
| | | | 4l9b1. Input Var : RPMS
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_RPM.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
- 4l9b. Stages 5

```

```

| | | | 4l9b1. Input Var : Gas
| | | | 4l9b2. Combinator Mode : Add
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersTHROTTLE_coast.
curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 4l9b. Stages 6
| | | | 4l9b1. Input Var : Brake
| | | | 4l9b2. Combinator Mode : Mult
| | | | 4l9b3. Lut :
content\cars\ks_ferrari_296_gtb\data\kers\controller_kersCONTROLLER_BRAK
E.curve
| | | | 4l9b4. Filter Gain : 0.96000
| | | | 4l9b5. Up Limit : 1.00000
| | | | 4l9b6. Down Limit : -1.00000
| | | | 4l9b7. Current Value : 0.00000
| | | | 4l9b8. Const Value : 0.00000
| | | | 19l. Brake Rear Correction : 5.00000
| | | | 19m. Has Cockpit Controls : false
| | | | 19n. Cockpit Controls
| | | | | 19n1. Delivery Profile : false
| | | | | 19n2. Mgu H Mode : false
| | | | | 19n3. Recovery : false
| | | | 19o. Has Front Motors : false
| | | | 19p. Front Motors
| | | | | 19p1. Torque Lut : None
| | | | | 19p2 Discharge Time : 0.00000
| | | | | 19p3. Torque Vectoring Bias : 0.00000
| | | | 20. Setup Limits :
content\cars\ks_ferrari_296_gtb\data\setup\limits_f296gtb.carsetuplimits
| | | | 21. Collider Mesh :
content\cars\ks_ferrari_296_gtb\collider\ferrari_296_gtb_collider.mesh
| | | | 22 Body Mesh Offset
| | | | | 22a. Position : 0.000, -0.482, 0.247
| | | | | 22b. Rotation: 0.100, 0.000, 0.000
| | | | | 22c. Scale : 1.000, 1.000, 1.000
| | | | 23. Stock Setup : content/cars/
ks_ferrari_296_gtb\data\setup\f296gtb_stock.carsetup
| | | | 24. Ai Setup : None
| | | | 25. Wet Setup : None
| | | | 26. Performance Modes 1
| | | | | 26a. Performance Mode Name : QUAL
| | | | | 26b. Electronics Settings
| | | | | | 26b1. Tc1 : 2.00000
| | | | | | 26b2. Tc2 : 1.00000
| | | | | | 26b3. Abs : 1.00000
| | | | | | 26b4. Esc : 0.00000
| | | | | | 26b5. Ebb : 0.000
| | | | | | 26b6. Engine Map : 0.00000
| | | | | | 26b7. Telemetry laps To Record : 0.00000
| | | | | | 26b8. Turbo Boost Lv : 0.00000
| | | | | | 26b9. Ers Deployment Map : 0.000

```

- | 26b10. Ers Recharge Lv : 100.000
- | 26b11. Ers Heat Charging : 0.000
- 26c. Brakes Settings : None
- 26d. Damper Settings : None
- 26e. Differential Data : None
- 26f. Four W D Differentials : None
- 26g. Front Lock Controllers : None
- 26h. Center Lock Controllers : None
- 26i. Rear Lock Controllers : None
- 26j. Awd Clutches : None
- 26k. Turbo Controllers : None
- 26l. Turbo Settings : None
- 26. Performance Modes 2
- | 26a. Performance Mode Name : PERF
- 26b. Electronics Settings
 - | 26b1. Tc1 : 2.00000
 - | 26b2. Tc2 : 1.00000
 - | 26b3. Abs : 1.00000
 - | 26b4. Esc : 1.00000
 - | 26b5. Ebb : 0.000
 - | 26b6. Engine Map : 0.00000
 - | 26b7. Telemetry laps To Record : 0.00000
 - | 26b8. Turbo Boost Lv : 0.00000
 - | 26b9. Ers Deployment Map : 1.000
 - | 26b10. Ers Recharge Lv : 100.000
 - | 26b11. Ers Heat Charging : 0.000
- 26c. Brakes Settings : None
- 26d. Damper Settings : None
- 26e. Differential Data : None
- 26f. Four W D Differentials : None
- 26g. Front Lock Controllers : None
- 26h. Center Lock Controllers : None
- 26i. Rear Lock Controllers : None
- 26j. Awd Clutches : None
- 26k. Turbo Controllers : None
- 26l. Turbo Settings : None
- 26. Performance Modes 3
- | 26a. Performance Mode Name : HYBRID
- 26b. Electronics Settings
 - | 26b1. Tc1 : 2.00000
 - | 26b2. Tc2 : 1.00000
 - | 26b3. Abs : 1.00000
 - | 26b4. Esc : 1.00000
 - | 26b5. Ebb : 0.000
 - | 26b6. Engine Map : 1.00000
 - | 26b7. Telemetry laps To Record : 0.00000
 - | 26b8. Turbo Boost Lv : 0.00000
 - | 26b9. Ers Deployment Map : 2.000
 - | 26b10. Ers Recharge Lv : 100.000
 - | 26b11. Ers Heat Charging : 0.000
- 26c. Brakes Settings : None
- 26d. Damper Settings : None
- 26e. Differential Data : None
- 26f. Four W D Differentials : None
- 26g. Front Lock Controllers : None
- 26h. Center Lock Controllers : None

- 26i. Rear Lock Controllers : None
- 26j. Awd Clutches : None
- 26k. Turbo Controllers : None
- 26l. Turbo Settings : None
- 27. Ai Car Data :
content\cars\ks_ferrari_296_gtb\data\ks_ferrari_296_gtb.aicardata
- 28. mm : 1

Audi R8 LMS GT3 Evo II

- 1. Screen Name : Audi R8 LMS GT3 Evo II
- 2. General
 - 2a. Screen Name : Audi R8 LMS GT3 Evo II
 - 2b. Total Mass : 1355
 - 2c. Tank Position : 0.00000, -0.02900, -0.25600
 - 2d. Fuel : 0.00000
 - 2e. Max Fuel : 120.00000
 - 2f. Efficiency : 0.40000
 - 2g. Kg Per Liter : 0.75500
 - 2h. Body Box Sizes : 1.98000, 1.19000, 4.46000
 - 2i. Pickup Front Height : -0.34900
 - 2j. Pickup Rear Height : -0.31800
 - 2k. Check Rules : true
 - 2l. Minimum Height : 450.00000
 - 2m. Torsional Stiffness : 40000.00000
 - 2n. Torsional Damping : 400.00000
 - 2o. Body Mesh Offset
 - 2o1. Position : 0.000, 0.000, 0.000
 - 2o2. Rotation : 0.000, 0.000, 0.000
 - 2o3. Scale : 0.000, 0.000, 0.000
- 3. General Path : None
- 4. Suspensions
 - 4a. Wheel Base : 2.70000
 - 4b. Longitudinal Cg Location : 0.42000
 - 4c. Base Y Front : -0.03500
 - 4d. Base Y Rear : -0.03500
 - 4e. Track Front : 1.66700
 - 4f. Track Rear : 1.67000
 - 4g. Damage
 - 4g1. Min Velocity : 40.00000
 - 4g2. Gain : 0.00040
 - 4g3. Max Damage : 0.05000
 - 4g4. Debug Log : true
 - 4h. Coilover Front path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_front.coilover
 - 4i. Coilover Rear Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_rear.coilover
 - 4j. Front Suspension Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_front.suspension

```

| | 4k. Rear Suspension Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2_rear
.suspension
| | | 4l. Heavy Springs : None
| | | 4m. Arb Front
| | | | 4m1. Stiffness : 56800.00000
| | | | 4m2. Controller
| | | | | 4l9a. Name : None
| | | | | 4l9b. Stages : None
| | | 4n. Arb Rear
| | | | 4n1. Stiffness : 39900.00000
| | | | 4n2. Controller
| | | | | 4l9a. Name : None
| | | | | 4l9b. Stages : None
| | | 4o. Flex Bar Front
| | | | 4o1. Stiffness : 0.00000
| | | | 4o2. Controller
| | | | | 4l9a. Name : None
| | | | | 4l9b. Stages : None
| | | 4p. Flex Bar Rear
| | | | 4p1. Stiffness : 0.00000
| | | | 4p2. Controller
| | | | | 4l9a. Name : None
| | | | | 4l9b. Stages : None
| | | 4q. Dampers Controller : None
| | | 4r. Has Dampers Cockpit Settings : false
| | 5. Drivetrain Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.driv
etrain
| | 6. Gearbox Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.gear
box
| | 7. Clutch Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.clut
ch
| | 8. Engine Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.care
ngine
| | 9. Brakes Path :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.brak
esystem
| | 10. Steering System
| | | 10a. Four W S Controllers
| | | | 4l9a. Name : None
| | | | 4l9b. Stages : None
| | 11. Electronics
| | | 11a. T C
| | | | 11a1. Has T C2 : true
| | | | 11a2. Frequency Hz : 333.00000
| | | | 11a3. Min Speed Kmh : 30.00000
| | | | 11a4. Gear Change Time : 0.07000
| | | | 11a5. Min Cut Level : 0.00000
| | | | 11a6. Max Cut Level : 6.00000
| | | | 11a7. Settings 1
| | | | | 11a7a. Min Slip Ratio : 0.00000
| | | | | 11a7b. Max Slip Ratio : 0.00000

```

- 11a7c. Ref Slip Angle Deg : 0.00000
- 11a7d. Engine Cut Level : 0.00000
- 11a7e. Angular A C Cgain : 0.00000
- 11a7f. Oversteer Gain : 0.00000
- 11a7g. Slip Angle Activation Deg : 0.00000
- 11a7. Settings 2
 - 11a7a. Min Slip Ratio : 0.18000
 - 11a7b. Max Slip Ratio : 0.35000
 - 11a7c. Ref Slip Angle Deg : 15.00000
 - 11a7d. Engine Cut Level : 1.50000
 - 11a7e. Angular A C Cgain : 1.00000
 - 11a7f. Oversteer Gain : 1.00000
 - 11a7g. Slip Angle Activation Deg : 5.00000
- 11a7. Settings 3
 - 11a7a. Min Slip Ratio : 0.15000
 - 11a7b. Max Slip Ratio : 0.30000
 - 11a7c. Ref Slip Angle Deg : 13.00000
 - 11a7d. Engine Cut Level : 1.50000
 - 11a7e. Angular A C Cgain : 1.50000
 - 11a7f. Oversteer Gain : 1.50000
 - 11a7g. Slip Angle Activation Deg : 5.00000
- 11a7. Settings 4
 - 11a7a. Min Slip Ratio : 0.12000
 - 11a7b. Max Slip Ratio : 0.28000
 - 11a7c. Ref Slip Angle Deg : 11.00000
 - 11a7d. Engine Cut Level : 1.25000
 - 11a7e. Angular A C Cgain : 2.00000
 - 11a7f. Oversteer Gain : 2.00000
 - 11a7g. Slip Angle Activation Deg : 4.50000
- 11a7. Settings 5
 - 11a7a. Min Slip Ratio : 0.10000
 - 11a7b. Max Slip Ratio : 0.25000
 - 11a7c. Ref Slip Angle Deg : 10.00000
 - 11a7d. Engine Cut Level : 1.00000
 - 11a7e. Angular A C Cgain : 2.50000
 - 11a7f. Oversteer Gain : 2.50000
 - 11a7g. Slip Angle Activation Deg : 4.00000
- 11a7. Settings 6
 - 11a7a. Min Slip Ratio : 0.08000
 - 11a7b. Max Slip Ratio : 0.22000
 - 11a7c. Ref Slip Angle Deg : 8.50000
 - 11a7d. Engine Cut Level : 1.00000
 - 11a7e. Angular A C Cgain : 3.00000
 - 11a7f. Oversteer Gain : 3.00000
 - 11a7g. Slip Angle Activation Deg : 4.00000
- 11a7. Settings 7
 - 11a7a. Min Slip Ratio : 0.07000
 - 11a7b. Max Slip Ratio : 0.22000
 - 11a7c. Ref Slip Angle Deg : 7.50000
 - 11a7d. Engine Cut Level : 1.00000
 - 11a7e. Angular A C Cgain : 3.50000
 - 11a7f. Oversteer Gain : 3.50000
 - 11a7g. Slip Angle Activation Deg : 3.50000
- 11a7. Settings 8
 - 11a7a. Min Slip Ratio : 0.06000
 - 11a7b. Max Slip Ratio : 0.22000

- 11a7c. Ref Slip Angle Deg : 7.00000
- 11a7d. Engine Cut Level : 0.50000
- 11a7e. Angular A C Cgain : 4.00000
- 11a7f. Oversteer Gain : 4.00000
- 11a7g. Slip Angle Activation Deg : 3.00000
- 11a7. Settings 9
 - 11a7a. Min Slip Ratio : 0.05000
 - 11a7b. Max Slip Ratio : 0.19000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 0.45000
 - 11a7e. Angular A C Cgain : 5.00000
 - 11a7f. Oversteer Gain : 5.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11a7. Settings 10
 - 11a7a. Min Slip Ratio : 0.05000
 - 11a7b. Max Slip Ratio : 0.17000
 - 11a7c. Ref Slip Angle Deg : 7.00000
 - 11a7d. Engine Cut Level : 0.40000
 - 11a7e. Angular A C Cgain : 7.50000
 - 11a7f. Oversteer Gain : 6.00000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11a7. Settings 11
 - 11a7a. Min Slip Ratio : 0.04000
 - 11a7b. Max Slip Ratio : 0.15000
 - 11a7c. Ref Slip Angle Deg : 6.00000
 - 11a7d. Engine Cut Level : 0.35000
 - 11a7e. Angular A C Cgain : 8.00000
 - 11a7f. Oversteer Gain : 6.50000
 - 11a7g. Slip Angle Activation Deg : 2.50000
- 11a7. Settings 12
 - 11a7a. Min Slip Ratio : 0.03000
 - 11a7b. Max Slip Ratio : 0.15000
 - 11a7c. Ref Slip Angle Deg : 6.00000
 - 11a7d. Engine Cut Level : 0.30000
 - 11a7e. Angular A C Cgain : 8.00000
 - 11a7f. Oversteer Gain : 7.00000
 - 11a7g. Slip Angle Activation Deg : 2.50000
- 11a7. Settings 13
 - 11a7a. Min Slip Ratio : 0.03000
 - 11a7b. Max Slip Ratio : 0.14000
 - 11a7c. Ref Slip Angle Deg : 5.00000
 - 11a7d. Engine Cut Level : 0.30000
 - 11a7e. Angular A C Cgain : 8.50000
 - 11a7f. Oversteer Gain : 7.50000
 - 11a7g. Slip Angle Activation Deg : 3.00000
- 11b. A B S
 - 11b1. Settings 1
 - 11b1a. Min Slip Ratio : -1.00000
 - 11b1b. Max Slip Ratio : -1.00000
 - 11b1c. Ref Slip Angle Deg : 0.00000
 - 11b1d. Cut Level : 0.00000
 - 11b1f. Max Torque Variation : 0.00000
 - 11b1. Settings 2
 - 11b1a. Min Slip Ratio : 0.12000
 - 11b1b. Max Slip Ratio : 0.14000
 - 11b1c. Ref Slip Angle Deg : 7.00000

- 11b1d. Cut Level : 0.20000
- 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 3
 - 11b1a. Min Slip Ratio : 0.11000
 - 11b1b. Max Slip Ratio : 0.14000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 4
 - 11b1a. Min Slip Ratio : 0.10000
 - 11b1b. Max Slip Ratio : 0.12000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 5
 - 11b1a. Min Slip Ratio : 0.08000
 - 11b1b. Max Slip Ratio : 0.10000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 6
 - 11b1a. Min Slip Ratio : 0.07000
 - 11b1b. Max Slip Ratio : 0.08000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 7
 - 11b1a. Min Slip Ratio : 0.06000
 - 11b1b. Max Slip Ratio : 0.08000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 8
 - 11b1a. Min Slip Ratio : 0.05000
 - 11b1b. Max Slip Ratio : 0.07000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 9
 - 11b1a. Min Slip Ratio : 0.05000
 - 11b1b. Max Slip Ratio : 0.06000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 10
 - 11b1a. Min Slip Ratio : 0.04000
 - 11b1b. Max Slip Ratio : 0.05000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
- 11b1. Settings 11
 - 11b1a. Min Slip Ratio : 0.02500
 - 11b1b. Max Slip Ratio : 0.03500
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000

- 11b1. Settings 12
 - 11b1a. Min Slip Ratio : 0.01000
 - 11b1b. Max Slip Ratio : 0.02000
 - 11b1c. Ref Slip Angle Deg : 7.00000
 - 11b1d. Cut Level : 0.20000
 - 11b1f. Max Torque Variation : 1.00000
 - Frequency : 40.00000
 - Channels : 4
 - Min Speed Kmh : 20.00000
- 11c. E D L : None
- 11d. E S P
 - 11d1. Frequency Hz : 0.00000
 - 11d2. Min Speed Kmh : 0.00000
 - 11d3. Settings : None
- 12. Electronics Path : None
- 13. Controls
 - 13a. Ff Mult : 1.50000
 - 13b. Steer Lock : 240.00000
 - 13c. Steer Ratio : 14.30000
 - 13d. Linear Steer Rod Ratio : 0.00235
 - 13e. Steer Assist : 1.00000
- 14. Box Colliders 1
 - 14a. Center : 0.00000, -0.23500, 0.30000
 - 14b. Size : 1.78000, 0.20000, 4.07000
 - 14c. Pitch Rotation Deg : 0.65000
- 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\racing_slicks\slick_325_680_18.tyre
- 16. Rear Tyre Compounds 1 :
ontent\cars\common_phsx\tyres\racing_slicks\slick_325_705_18.tyre
- 17. Aero
 - 17a. Slip Gain Multiple : 1.00000
 - 17b. Speed Factor Mult : 2.00000
 - 17c. Downforces 1
 - 17c1. Position : 0.82140, -0.34900, 1.57000
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000
 - 17c4. Yaw Gain : 0.00000
 - 17c5. Drag Per Cool Transfer : 0.01500
 - 17c6. Damage C L 1 : 0.01000
 - 17c6. Damage C L 2 : 0.00500
 - 17c6. Damage C L 3 : 0.00300
 - 17c6. Damage C L 4 : 0.00300
 - 17c7. Damage C D 1 : 0.01500
 - 17c7. Damage C D 2 : 0.00500
 - 17c7. Damage C D 3 : 0.00300
 - 17c7. Damage C D 4 : 0.00300
 - 17c8. Downforce Controllers : None
 - 17c9. Lift Per Front Angle : 0.00000
 - 17c10. Lift Per Rear Angle : 0.00000
 - 17c11. Drag Per Front Angle : 0.00000
 - 17c12. Drag Per Rear Angle : 0.00000
 - 17c13. Default Front Angle : 0.00000
 - 17c14. Default Rear Angle : 0.00000
 - 17c. Downforces 2
 - 17c1. Position : 0.82140, -0.34900, 1.57000
 - 17c2. Cl Gain : 1.00000

- 17c3. Cd Gain : 1.00000
- 17c4. Yaw Gain : 0.00000
- 17c5. Drag Per Cool Transfer : 0.01500
- 17c6. Damage C L 1 : 0.01000
- 17c6. Damage C L 2 : 0.00500
- 17c6. Damage C L 3 : 0.00300
- 17c6. Damage C L 4 : 0.00300
- 17c7. Damage C D 1 : 0.01500
- 17c7. Damage C D 2 : 0.00500
- 17c7. Damage C D 3 : 0.00300
- 17c7. Damage C D 4 : 0.00300
- 17c8. Downforce Controllers : None
- 17c9. Lift Per Front Angle : 0.00000
- 17c10. Lift Per Rear Angle : 0.00000
- 17c11. Drag Per Front Angle : 0.00000
- 17c12. Drag Per Rear Angle : 0.00000
- 17c13. Default Front Angle : 0.00000
- 17c14. Default Rear Angle : 0.00000
- 17c. Downforces 3
 - 17c1. Position : 0.82140, -0.31800, -1.13500
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000
 - 17c4. Yaw Gain : -0.10000
 - 17c5. Drag Per Cool Transfer : 0.01000
 - 17c6. Damage C L 1 : 0.00500
 - 17c6. Damage C L 2 : 0.01000
 - 17c6. Damage C L 3 : 0.00300
 - 17c6. Damage C L 4 : 0.00300
 - 17c7. Damage C D 1 : 0.00500
 - 17c7. Damage C D 2 : 0.01500
 - 17c7. Damage C D 3 : 0.00300
 - 17c7. Damage C D 4 : 0.00300
 - 17c8. Downforce Controllers : None
 - 17c9. Lift Per Front Angle : 0.00000
 - 17c10. Lift Per Rear Angle : 0.00000
 - 17c11. Drag Per Front Angle : 0.00000
 - 17c12. Drag Per Rear Angle : 0.00000
 - 17c13. Default Front Angle : 0.00000
 - 17c14. Default Rear Angle : 0.00000
- 17c. Downforces 4
 - 17c1. Position : 0.82140, -0.31800, -1.13500
 - 17c2. Cl Gain : 1.00000
 - 17c3. Cd Gain : 1.00000
 - 17c4. Yaw Gain : -0.10000
 - 17c5. Drag Per Cool Transfer : 0.01000
 - 17c6. Damage C L 1 : 0.00500
 - 17c6. Damage C L 2 : 0.01000
 - 17c6. Damage C L 3 : 0.00300
 - 17c6. Damage C L 4 : 0.00300
 - 17c7. Damage C D 1 : 0.00500
 - 17c7. Damage C D 2 : 0.01500
 - 17c7. Damage C D 3 : 0.00300
 - 17c7. Damage C D 4 : 0.00300
 - 17c8. Downforce Controllers : None
 - 17c9. Lift Per Front Angle : 0.00000
 - 17c10. Lift Per Rear Angle : 0.00000

- | | | 17c11. Drag Per Front Angle : 0.00000
- | | | 17c12. Drag Per Rear Angle : 0.00000
- | | | 17c13. Default Front Angle : 0.00000
- | | | 17c14. Default Rear Angle : 0.00000
- | | 17d. Front Lift :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\frontczmap.surface3d
- | | 17e. Rear Lift :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\rearczmap.surface3d
- | | 17f. Drag :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\cxmap.surface3d
- | | 17g. Wings Path 1 :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_rear_modifier.wing
- | | 17g. Wings Path 2 :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_front_modifier.wing
- | 18. Drs : None
- | 19. Ers : None
- | 20. Setup Limits :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\aero\Audi_R8_GT3_rear_wing_front_modifier.wing
- | 21. Collider Mesh :
content\cars\ks_audi_r8_lms_gt3_evo_2\collider\audi_r8_gt3_evo2_collider.mesh
- | 22. Body Mesh Offset
 - | | 22a. Position : 0.000, -0.419, 0.230
 - | | 22b. Rotation: 0.650, 0.000, 0.000
 - | | 22c. Scale : 1.000, 1.000, 1.000
- | 23. Stock Setup :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\setup\ks_audi_r8_lms_gt3_evo_2_safe.carsetup
- | 24. Ai Setup : None
- | 25. Wet Setup :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\setup\ks_audi_r8_lms_gt3_evo_2_wet.carsetup
- | 26. Performance Modes : None
- | 27. Ai Car Data :
content\cars\ks_audi_r8_lms_gt3_evo_2\data\ks_audi_r8_lms_gt3_evo_2.aicardata
- | 28. mm : 1

Renault 5 GT Turbo

- | 1. Screen Name : None
- | 2. General
 - | | 2a. Screen Name : Renault 5 GT Turbo
 - | | 2b. Total Mass : 910.00000
 - | | 2c. Tank Position : 0.00000, -0.20000, -1.56750
 - | | 2d. Fuel : 30.00000
 - | | 2e. Max Fuel : 50.00000
 - | | 2f. Efficiency : 0.00000
 - | | 2g. Kg Per Liter : 0.75500
 - | | 2h. Body Box Sizes : 1.60000, 1.10000, 3.30000
 - | | 2i. Pickup Front Height : -0.49000

- 2j. Pickup Rear Height : -0.48000
- 2k. Check Rules : false
- 2l. Minimum Height : 0.00000
- 2m. Torsional Stiffness : 11000.00000
- 2n. Torsional Damping : 50.00000
- 2o. Body Mesh Offset
 - 2o1. Position : 0.000, 0.000, 0.000
 - 2o2. Rotation : 0.000, 0.000, 0.000
 - 2o3. Scale : 0.000, 0.000, 0.000
- 3. General Path : None
- 4. Suspensions :
 - 4a. Wheel Base : 2.40700
 - 4b. Longitudinal Cg Location : 0.64000
 - 4c. Base Y Front : -0.42000
 - 4d. Base Y Rear : -0.43000
 - 4e. Track Front : 1.34000
 - 4f. Track Rear : 1.31500
 - 4g. Damage
 - 4g1. Min Velocity : 40.00000
 - 4g2. Gain : 0.00040
 - 4g3. Max Damage : 0.05000
 - 4g4. Debug Log : true
 - 4h. Coilover Front path :
 - content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_front.coil
 - over
 - 4i. Coilover Rear Path :
 - content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_rear.coil
 - over
 - 4j. Front Suspension Path :
 - content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_front.susp
 - ension
 - 4k. Rear Suspension Path :
 - content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo_rear.coil
 - over
- 4l. Heavy Springs : None
- 4m. Arb Front
 - 4m1. Stiffness : 8000.00000
 - 4m2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 4n. Arb Rear
 - 4n1. Stiffness : 6000.00000
 - 4n2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 4o. Flex Bar Front
 - 4o1. Stiffness : 0.00000
 - 4o2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 4p. Flex Bar Rear
 - 4p1. Stiffness : 10000.00000
 - 4p2. Controller
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 4q. Dampers Controller : None

- | L 4r. Has Dampers Cockpit Settings : false
- 5. Drivetrain Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.drivetrain
- 6. Gearbox Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.gearbox
- 7. Clutch Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.clutch
- 8. Engine Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.carengine
- 9. Brakes Path :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_turbo.brakesystem
- 10. Steering System
 - 10a. Four W S Controllers
 - 4l9a. Name : None
 - 4l9b. Stages : None
- 11. Electronics
 - 11a. T C
 - 11a1. Has T C2 : false
 - 11a2. Frequency Hz : 150.00000
 - 11a3. Min Speed Kmh : 20.00000
 - 11a4. Gear Change Time : 0.08000
 - 11a5. Min Cut Level : 10.00000
 - 11a6. Max Cut Level : 1.00000
 - 11a7. Settings 1 :
 - 11a7a. Min Slip Ratio : -1.00000
 - 11a7b. Max Slip Ratio : -1.00000
 - 11a7c. Ref Slip Angle Deg : 0.00000
 - 11a7d. Engine Cut Level : 0.00000
 - 11a7e. Angular A C Cgain : 0.00000
 - 11a7f. Oversteer Gain : 0.00000
 - 11a7g. Slip Angle Activation Deg : 0.00000
 - 11b. A B S
 - 11b1. Settings 1
 - 11b1a. Min Slip Ratio : -1.00000
 - 11b1b. Max Slip Ratio : -1.00000
 - 11b1c. Ref Slip Angle Deg : 0.00000
 - 11b1d. Cut Level : 0.00000
 - 11b1f. Max Torque Variation : 0.70000
 - 11b1. Settings 2
 - 11b1a. Min Slip Ratio : 0.06500
 - 11b1b. Max Slip Ratio : 0.12000
 - 11b1c. Ref Slip Angle Deg : 8.00000
 - 11b1d. Cut Level : 0.40000
 - 11b1f. Max Torque Variation : 0.75000
 - Frequency : 40.00000
 - Channels : 4
 - Min Speed Kmh : 0.00000
 - 11c. E D L
 - 11c1. Active : true
 - 11c2. Brake Torque Power : 40.00000
 - 11c3. Brake Torque Coast : 200.00000
 - 11c4. Dead Zone Coast : 0.05000
 - 11c5. Dead Zone Power : 0.10000
 - 11c6. Max Spin Power : 0.40000
 - 11c7. Max Spin Coaster : 0.20000
 - 11c8. Min Speed : 0.00000

- 11d. E S P
 - 11d1. Frequency Hz : 0.00000
 - 11d2. Min Speed Kmh : 0.00000
 - 11d3. Settings : None
- 12. Electronics Path : None
- 13. Controls
 - 13a. Ff Mult : 2.90000
 - 13b. Steer Lock : 630.00000
 - 13c. Steer Ratio : -19.60000
 - 13d. Linear Steer Rod Ratio : 0.00220
 - 13e. Steer Assist : 1.00000
- 14. Box Colliders 1
 - 14a. Center : 0.00000, -0.24000, -0.29000
 - 14b. Size : 1.40000, 0.10000, 3.23000
 - 14c. Pitch Rotation Deg : 0.00000
- 15. Front Tyre Compounds 1 :
content\cars\common_phsx\tyres\road\road_195_55_13.tyre
- 16. Rear Tyre Compounds 1 :
content\cars\common_phsx\tyres\road\road_195_55_13.tyre
- 17. Aero
 - 17a. Slip Gain Multiple : 1.00000
 - 17b. Speed Factor Mult : 2.00000
 - 17c. Downforces : None
 - 17d. Front Lift : None
 - 17e. Rear Lift : None
 - 17f. Drag : None
 - 17g. Wings Path 1 :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo0.wing
- 18. Drs : None
- 19. Ers : None
- 20. Setup Limits :
content\cars\ks_renault_5_gt_turbo\data\setup\renault_5_gt_turbo.carsetuplimits
- 21. Collider Mesh :
content\cars\ks_renault_5_gt_turbo\collider\renault_5_gt_turbo_collider.mesh
- 22. Body Mesh Offset
 - 22a. Position : 0.000, -0.632, -0.340
 - 22b. Rotation : 0.700, 0.000, 0.000
 - 22c. Scale : 1.000, 1.000, 1.000
- 23. Stock Setup :
content\cars\ks_renault_5_gt_turbo\data\setup\renault_5_gt_turbo.carsetup
- 24. Ai Setup : None
- 25. Wet Setup : None
- 26. Performance Modes : None
- 27. Ai Car Data :
content\cars\ks_renault_5_gt_turbo\data\ks_renault_5_gt_turbo.aicardata
- 28. mm : 1

4. Car Engine [.carengine]

A. Description

Propulsion core: how the power unit makes torque, how fast it revs, how the throttle feels, and how boost or hybrid overlays reshape that delivery.

Car Data only points here via *Engine Path*. This file owns the energy side of longitudinal dynamics — acceleration, engine braking, map strategy, and forced-induction hooks.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Torque vs RPM, coast/engine brake curve	.CARENGINE	—
Idle / limiter, inertia, throttle shaping	.CARENGINE	—
Engine maps (power / fuel / throttle / feel)	.CARENGINE Maps	Setup map index
Turbo / compressor load and boost control	Controllers + .TURBO paths	18. Turbo
Battery pack for motor / hybrid assist	Battery Data block	Car Data ERS hooks
Gear ratios, clutch, final drive	—	Gearbox / clutch / drivetrain

Referenced from Car Data **Engine Path**. Without a valid .CARENGINE, the car has no torque source.

II. What you are really tuning

- Power identity**— *Engine Type* selects *combustion* or *motor*. *Power Curve* is the RPM to torque LUT (the shape everyone feels out of corners). *Coast Curve* is negative torque on a closed throttle — engine braking and entry stability for driven axles.
- Rotational character**— *Inertia* is flywheel / rotating assembly mass. Low values (Alpine motor about 0.06) snap RPM up and down; high values (Camaro about 0.45) feel heavy and slow to build revs. SF-25 may leave inertia unset when handled elsewhere.
- Rev window** — *Minimum* is idle (0 on the EV Alpine; about 600-900 on classic ICE). *Limiter* is the hard ceiling (Alpine about 15200, Camaro about 6600, Datsun about 7000). *Limiter Cycles* softens how the cut feels before a hard stop.
- Throttle feel**— Global and per-map throttle response curves map pedal to effective opening. *Throttle Lag Up / Dn* smooth tip-in and lift. Camaro uses noticeable lag (about 0.8 up / 0.9 down); race maps often leave lag at 0 and swap **linear** vs **aggressive** curves instead.
- Maps** — *Maps[]* are selectable power modes: *Power Mult*, *Consumption Mult*, optional throttle curve, lag, and map type (*gamma / cervone*). SF-25 stacks many maps (full power down to about 0.72 power with matching fuel cuts) — strategy and drivability live here more than in the raw power curve.

6. **Boost and hybrid extras** — Turbo / wastegate controller stages (same LUT pipeline pattern as brakes EBB), *Max Turbo Boost*, *Bov Threshold*, and *Turbos To Load* point at `.TURBO` assets (Camaro loads a compressor turbo). *Battery Data* holds pack capacity and thermal limits for motor / hybrid use. *Start ECU Assist* is launch / anti-stall logic (RPM window, clutch slip target, gains).

III. Architecture

1 - CORE PROPULSION (SCHEMA 1-4)

Engine type, inertia, power curve path, coast curve path.

2 - MAPS AND REV LIMITS (SCHEMA 5-8)

Map array with per-map throttle behaviour; idle, limiter, limiter cycles.

3 - GLOBAL THROTTLE AND START (SCHEMA 9-15)

Fallback throttle curve and lag, rev choking, ignition time, starter torque, Start ECU Assist (launch control style fields).

4 - FORCED INDUCTION (SCHEMA 16-20)

Turbo Controllers and Waste Gate Controllers (shared stage fields: Input Var, Combinator, Lut, Filter, limits), max boost, BOV threshold, turbos-to-load paths.

5 - BATTERY (SCHEMA 21)

Capacity, charge/discharge efficiency, temperature loss, convection, thermal capacity, temp clamps — relevant when Engine Type is motor or hybrid assist is active.

IV. How to read the examples

1 - ALPINE A290 B (ELECTRIC MOTOR)

ENGINE TYPE : MOTOR, very low inertia, idle at 0, high limiter (about 15200). No maps, no turbos, no boost. Power + coast + throttle curves only. Clean EV template: instant response, no spool, no multi-map strategy.

2 - FERRARI SF-25 (RACE ICE + STRATEGY MAPS)

combustion, rich map list mixing power multipliers (1.00 down to about 0.72) with consumption multipliers and linear/aggressive throttle curves. Coast uses a smooth curve. Shows how modern race engines separate **peak curve** from **selectable maps** for fuel and drivability.

3 - CHEVROLET CAMARO ZL1 (BOOSTED MUSCLE)

High inertia (0.45), idle 900, limiter 6600, strong throttle lag, Start ECU Assist with clutch enabled. Turbo controller driven by *RPMS* into a boost LUT; *Turbos To Load* points at a compressor `.TURBO`; *Max Turbo Boost* about 0.6. Battery block present but zeroed. Forced-induction ICE with launch assist.

4 - DATSUN 240Z FAIRLADY (CLASSIC NA)

Combustion, moderate inertia (0.18), idle 600, limiter 7000, map type *cervone*, no turbo load.
Straightforward naturally aspirated road/race engine without boost plumbing.

V. Practical notes

- Shape the **Power Curve** before stacking maps: maps only scale and re-throttle what the curve already is.
- Coast curve is half of entry balance on driven wheels — do not leave a placeholder if engine braking matters.
- Controllers present does not mean boost is active: check *Turbos To Load* and *Max Turbo Boost*, and read the linked *.TURBO* lag/gamma fields.
- *motor* vs *combustion* changes which blocks matter (battery / zero idle vs turbo / idle RPM).
- Throttle lag and aggressive maps can mask or exaggerate turbo kick — tune them together with the *.TURBO* asset.
- Schema spellings to expect: *Throttle Rev Chocking*, *Dischage Efficiency*, *Start E C U Assist*, *Speed Range K H M*.

VI. Related assets

- **3. Car Data [.car]** — *Engine Path* loads this file
- **18. Turbo [.turbo]** — spool / wastegate assets listed in *Turbos To Load*
- **14 / 10 / 13. Gearbox / Clutch / Drivetrain** — consume engine torque downstream
- **5. Car Setup [.carsetup]** — engine map / turbo boost level selection in the garage

B. Schema

```
- 1. Engine Type : enum
- 2. Inertia : float
- 3. Power Curve : string - path
- 4. Coast Curve : string - path
- 5. Maps [x] : object | can have multiple Maps
  - 5a. Type : enum
  - 5b. Power Mult : float
  - 5c. Consumption Mult : float
  - 5d. Throttle Response Curve : string - path
  - 5e. Throttle Gain K R P M : float
  - 5f. Throttle Ref R P M Move : float
  - 5g. Throttle Lag Up : float
  - 5h. Throttle Lag Dn : float
  - 5i. Throttling Factor : float
- 6. Minimum : integer
- 7. Limiter : integer
- 8. Limiter Cycles : integer
- 9. Throttle Response Curve : string - path
- 10. Throttle Lag Up : float
- 11. Throttle Lag Dn : float
```

- 12. Throttle Rev Chocking : float
- 13. Ignition Time S : float
- 14. Starter Engine Torque : float
- 15. Start E C U Assist : object
 - 15a. Rpm Range : float
 - 15b. Gain : float
 - 15c. Speed Range K H M : float
 - 15d. Rpm Limiter : float
 - 15e. Limiter Cycles : integer
 - 15f. Use Clutch : boolean
 - 15g. Slip Ratio Target : float
 - 15h. Clutch Gain : float
 - 15i. Gas Gain : float
- 16. Turbo Controllers [x] : object with an array of stages within | can have multiple Turbo Controllers
 - 16a. Name : string
 - 16b. Stages [x] | object | can have multiple Stage
 - 16b1. Input Var : enum
 - 16b2. Combinator Mode : enum
 - 16b3. Lut : string - path
 - 16b4. Filter Gain : float
 - 16b5. Up Limit : float
 - 16b6. Down Limit : float
 - 16b7. Current Value : float
 - 16b8. Const Value : float
- 17. Waste Gate Controllers [x] : object with an array of stages within | can have multiple Waste Gate Controllers
 - 16a. Name : string
 - 16b. Stages [x] | object | can have multiple Stage
 - 16b1. Input Var : enum
 - 16b2. Combinator Mode : enum
 - 16b3. Lut : string - path
 - 16b4. Filter Gain : float
 - 16b5. Up Limit : float
 - 16b6. Down Limit : float
 - 16b7. Current Value : float
 - 16b8. Const Value : float
- 18. Max Turbo Boost : float
- 19. Bov Threshold : float
- 20. Turbos To Load [x] : string - path | can have multiple Turbos To Load
- 21. Battery Data : object
 - 21a. Capacity Kwh : float
 - 21b. Dischage Efficiency : float
 - 21c. Charge Efficiency : float
 - 21d. Temp Eff Loss Per Deg : float
 - 21e. Convection K : float
 - 21f. Convection Forced K : float
 - 21g. Thermal Capacity : float
 - 21h. Max Temp : float
 - 21i. Min Temp : float

Enum - Car Engine

1	Engine Type	motor, combustion
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5a	Type	gamma, cervone
16b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
16b2	CombinatorMode	UndefinedMode, Add, Mult

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Engine Type	None (Enum : motor / combustion)	Selects the propulsion model: motor for electric-only powertrains, combustion for internal-combustion (ICE) or hybrid ICE behavior.
2.	Inertia	kg·m ² (Kilogram square meters)	Rotational inertia of the engine assembly (crankshaft, flywheel, pistons). Low values yield fast revving (F1); high values create sluggish RPM changes (heavy V8).
3.	Power Curve	None (.CURVE file path)	Primary look-up table mapping engine RPM (X-axis) to torque output in Nm (Y-axis); the foundation of all acceleration and top-speed calculations.
4.	Coast Curve	None (.CURVE file path)	Look-up table defining negative engine torque (engine braking) as a function of RPM when throttle is fully released (0%).
5.	Maps	None (Object array)	Array of selectable engine maps (power modes); each entry defines power/consumption multipliers and per-map throttle behavior.
5a.	Type	None (Enum : gamma / cervone)	Engine map blending/interpolation profile type governing how map parameters transition between operating points.
5b.	Power Mult	Dimensionless multiplier	Scalar applied to the base power/torque curve output for this engine map (e.g., 0.82 = 82% power for economy mode).
5d.	Throttle Response Curve	None (.CURVE file path)	Per-map look-up table mapping pedal position to effective throttle plate opening (linear, aggressive, progressive, wet).

ID	Name	Unit of Measurement	Description
5e.	Throttle Gain K RPM	Dimensionless gain	RPM-dependent gain scaling throttle response sensitivity; modulates how aggressively the engine reacts at different rev ranges.
5f.	Throttle Ref RPM Move	RPM (Revolutions per minute)	Reference RPM delta threshold triggering throttle gain adjustments during rapid RPM changes.
5g.	Throttle Lap Up	S (Seconds) or ratio	Time constant or smoothing factor for throttle opening response; higher values soften pedal tip-in.
5h.	Throttle Lag Dn	S (Seconds) or ratio	Time constant or smoothing factor for throttle closing response; higher values delay lift-off engine braking onset.
5i.	Throttling Factor	Dimensionless coefficient	Global throttle shaping factor applied within this engine map for fine-tuning drivability.
6.	Minimum	RPM (Revolutions per minute)	Idle RPM — the baseline engine speed maintained at zero throttle input.
7.	Limiter	RPM (Revolutions per minute)	Maximum engine RPM before the rev limiter cuts fuel/ignition to protect the engine.
8.	Limiter Cycles	None (Integer)	Number of soft-cut cycles the rev limiter performs before hard limiting; affects limiter feel and sound.
9.	Throttle Response Curve	None (.CURVE file path)	Global (non-map-specific) throttle pedal-to-plate response curve used when no per-map override is active.
10.	Throttle Lag Up	s (Seconds) or ratio	Global throttle opening lag/smoothing applied across all maps unless overridden per map.
11.	Throttle Lag Dn	s (Seconds) or ratio	Global throttle closing lag/smoothing applied across all maps unless overridden per map.
12.	Throttle Rev Chocking	Dimensionless coefficient	Rev-choking factor limiting RPM rise under partial throttle; simulates intake restriction or anti-lag behavior.
13.	Ignition Time S	s (Seconds) or ratio	Ignition timing delay or startup ignition sequence duration; affects initial fire-up and low-RPM response.
14.	Starter Engine Torque	Nm (Newton-meters)	Cranking torque delivered by the starter motor to bring the engine from rest to idle RPM.

ID	Name	Unit of Measurement	Description
15a.	Rpm Range	RPM (Revolutions per minute)	RPM window within which the start/launch assist system actively modulates torque and clutch slip.
15b.	Gain	Dimensionless gain	Authority multiplier of the start ECU assist torque and slip control loop.
15c.	Speed Range KMH	km/h (Kilometers per hour)	Vehicle speed range over which launch assist remains active before disengaging.
15d.	Rpm Limiter	RPM (Revolutions per minute)	Temporary RPM ceiling enforced during launch assist to prevent over-rev on clutch dump.
15e.	Limiter Cycles	None (Integer)	Number of rev-limiter cycles permitted during launch assist before hard cut.
15f.	Use Clutch	None (Boolean : True / False)	Enables clutch-slip-based launch control logic instead of pure throttle modulation.
15g.	Slip Ratio Target	Dimensionless ratio	Target wheel slip ratio maintained by launch assist for optimal traction off the line.
15h.	Clutch Gain	Dimensionless gain	Clutch engagement modulation gain within the launch assist controller.
15i.	Gas Gain	Dimensionless gain	Throttle input scaling gain applied during launch assist for traction management.
16.	Turbo Controllers	None (Controller array)	Array of dynamic turbo boost controllers; each entry contains a Name and Stages pipeline (fields 16a, 16b1–16b8).
16a.	Name	None (String)	Internal identifier for a turbo or wastegate controller block.
16b1.	Input Var	None (Telemetry enum)	Telemetry channel driving the controller stage (commonly RPMS, Gas, or Boost pressure).
16b2.	Combinator Mode	None (Math enum : Add / Mult)	How this stage output combines with prior stages: additive offset or multiplicative scaling.
16b3.	Lut	None (.CURVE file path)	Look-up table mapping the input variable to a turbo boost or wastegate modifier.
16b4.	Filter Gain	Coefficient (Smoothing multiplier)	Low-pass filter coefficient smoothing rapid input spikes for stable boost control.
16b5.	Up Limit	Depends on input variable	Upper clamp on the processed controller input signal.

ID	Name	Unit of Measurement	Description
16b6.	Down Limit	Depends on input variable	Lower clamp on the processed controller input signal.
16b7.	Current Value	Depends on input variable	Runtime output value of the controller stage during simulation.
16b8.	Const Value	Depends on input variable	Fallback constant output when no dynamic input or LUT is active.
17.	Waste Gate Controllers	None (Controller array)	Array of wastegate dynamic controllers; reuses the same 16a Name and 16b1–16b8 stage field structure as Turbo Controllers.
18.	Max Turbo Boost	bar (bar gauge pressure)	Maximum allowable turbocharger boost pressure ceiling; hard limit for forced-induction output.
19.	Bov Threshold	bar or ratio	Blow-off valve (BOV) activation threshold; pressure differential at which excess boost is vented.
20.	Turbos To Load	None (.TURBO file path)	Path(s) to .turbo assets defining turbocharger spool dynamics, compressor maps, and wastegate physics.
21a.	Capacity Kwh	kWh (Kilowatt-hours)	Total electrical energy storage capacity of the traction battery pack.
21b.	Discharge Efficiency	Ratio (0.0 - 1.0)	Energy conversion efficiency from battery to motor during discharge (e.g., 0.80 = 80%).
21c.	Charge Efficiency	Ratio (0.0 - 1.0)	Energy conversion efficiency from regenerative braking or charging back into the battery.
21d.	Temp Eff Loss Per Deg	Ratio/°C (Per degree Celsius)	Efficiency loss per degree of battery temperature deviation from optimal operating point.
21e.	Convection K	W/(m ² ·K) or coefficient	Natural convection heat transfer coefficient for passive battery cooling.
21f.	Convection Forced K	W/(m ² ·K) or coefficient	Forced convection coefficient when active cooling (airflow or liquid) is applied.
21g.	Thermal Capacity	J/K or J/°C	Heat storage capacity of the battery pack; controls temperature rise rate under sustained load.
21h.	Max Temp	°C (Degrees Celsius)	Maximum safe operating temperature before power derating or thermal shutdown.

ID	Name	Unit of Measurement	Description
21i	Min Temp	°C (Degrees Celsius)	Minimum operating temperature; below this, efficiency loss or power limiting may apply.

D. Example data

I. Chosen Car Engine for Example

- Alpine A290 b (slug : ks_alpine_a290_b)
- Ferrari SF 25 (slug : ks_ferrari_sf_25)
- Chevrolet Camaro ZL1 (slug : ks_renault_5_gt_turbo)
- Datsun 240z Fairlady (slug : ks_datsun_240z_fairlady)

II. Example

Alpine A290 b

```

1. Engine Type : motor
2. Inertia : 0.06000
3. Power Curve : content\cars\ks_alpine_a290_b\data\HEADER_POWER.curve
4. Coast Curve : content\cars\ks_alpine_a290_b\data\coast.curve
5. Maps : None
6. Minimum : 0
7. Limiter : 15200
8. Limiter Cycles : 20
9. Throttle Response Curve :
content\cars\ks_alpine_a290_b\data\throttle.curve
10. Throttle Lag Up : 0.00000
11. Throttle Lag Dn : 0.00000
12. Throttle Rev Chocking : 0.00000
13. Ignition Time S : 0.00000
14. Starter Engine Torque : 0.00000
15. Start E C U Assist
  15a. Rpm Range : 0.00000
  15b. Gain : 0.00000
  15c. Speed Range K H M : 0.00000
  15d. Rpm Limiter : 0.00000
  15e. Limiter Cycles : 0
  15f. Use Clutch : false
  15g. Slip Ratio Target : 0.00000
  15h. Clutch Gain : 0.00000
  15i. Gas Gain : 0.00000
16. Turbo Controllers : None
17. Waste Gate Controllers : None
18. Max Turbo Boost : 0.00000
19. Bov Threshold : 0.00000
20. Turbos To Load : None
21. Battery Data
  21a. Capacity Kwh : 52.000

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- 21b. Dischage Efficiency : 0.800
- 21c. Charge Efficiency : 0.600
- 21d. Temp Eff Loss Per Deg : 0.01000
- 21e. Convection K : 0.00100
- 21f. Convection Forced K : 0.00050
- 21g. Thermal Capacity : 100.000
- 21h. Max Temp : 35.000
- 21i. Min Temp : 15.000

Ferrari SF 25

- 1. Engine Type : combustion
- 2. Inertia : None
- 3. Power Curve : content\cars\ks_ferrari_sf_25\data\HEADER_POWER.curve
- 4. Coast Curve : content\cars\ks_ferrari_sf_25\data\coast_smooth.curve
- 5. Maps 1
 - 5a. Type : gamma
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 0.60000
 - 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.00000
 - 5h. Throttle Lag Dn : 0.00000
 - 5i. Throttling Factor : 0.00000
- 5. Maps 2
 - 5a. Type : gamma
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 0.60000
 - 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.00000
 - 5h. Throttle Lag Dn : 0.00000
 - 5i. Throttling Factor : 0.00000
- 5. Maps 3
 - 5a. Type : gamma
 - 5b. Power Mult : 0.89000
 - 5c. Consumption Mult : 0.56000
 - 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.00000
 - 5h. Throttle Lag Dn : 0.00000
 - 5i. Throttling Factor : 0.00000
- 5. Maps 4
 - 5a. Type : gamma
 - 5b. Power Mult : 0.89000
 - 5c. Consumption Mult : 0.56000
 - 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve

- | 5e. Throttle Gain K R P M : 0.00000
- | 5f. Throttle Ref R P M Move : 0.00000
- | 5g. Throttle Lag Up : 0.00000
- | 5h. Throttle Lag Dn : 0.00000
- | 5i. Throttling Factor : 0.00000
- 5. Maps 5
 - | 5a. Type : gamma
 - | 5b. Power Mult : 0.82000
 - | 5c. Consumption Mult : 0.55000
 - | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
 - | 5e. Throttle Gain K R P M : 0.00000
 - | 5f. Throttle Ref R P M Move : 0.00000
 - | 5g. Throttle Lag Up : 0.00000
 - | 5h. Throttle Lag Dn : 0.00000
 - | 5i. Throttling Factor : 0.00000
- 5. Maps 6
 - | 5a. Type : gamma
 - | 5b. Power Mult : 0.82000
 - | 5c. Consumption Mult : 0.55000
 - | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
 - | 5e. Throttle Gain K R P M : 0.00000
 - | 5f. Throttle Ref R P M Move : 0.00000
 - | 5g. Throttle Lag Up : 0.00000
 - | 5h. Throttle Lag Dn : 0.00000
 - | 5i. Throttling Factor : 0.00000
- 5. Maps 7
 - | 5a. Type : gamma
 - | 5b. Power Mult : 0.72000
 - | 5c. Consumption Mult : 0.50000
 - | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
 - | 5e. Throttle Gain K R P M : 0.00000
 - | 5f. Throttle Ref R P M Move : 0.00000
 - | 5g. Throttle Lag Up : 0.00000
 - | 5h. Throttle Lag Dn : 0.00000
 - | 5i. Throttling Factor : 0.00000
- 5. Maps 8
 - | 5a. Type : gamma
 - | 5b. Power Mult : 0.72000
 - | 5c. Consumption Mult : 0.50000
 - | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
 - | 5e. Throttle Gain K R P M : 0.00000
 - | 5f. Throttle Ref R P M Move : 0.00000
 - | 5g. Throttle Lag Up : 0.00000
 - | 5h. Throttle Lag Dn : 0.00000
 - | 5i. Throttling Factor : 0.00000
- 5. Maps 9
 - | 5a. Type : gamma
 - | 5b. Power Mult : 0.95000
 - | 5c. Consumption Mult : 0.58500
 - | 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
 - | 5e. Throttle Gain K R P M : 0.00000

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| 5f. Throttle Ref R P M Move : 0.00000
| 5g. Throttle Lag Up : 0.00000
| 5h. Throttle Lag Dn : 0.00000
| 5i. Throttling Factor : 0.00000
- 5. Maps 10
| 5a. Type : gamma
| 5b. Power Mult : 0.95000
| 5c. Consumption Mult : 0.58500
| 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_aggressive.curve
| 5e. Throttle Gain K R P M : 0.00000
| 5f. Throttle Ref R P M Move : 0.00000
| 5g. Throttle Lag Up : 0.00000
| 5h. Throttle Lag Dn : 0.00000
| 5i. Throttling Factor : 0.00000
- 5. Maps 11
| 5a. Type : gamma
| 5b. Power Mult : 1.00000
| 5c. Consumption Mult : 0.60000
| 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_progressive.curve
| 5e. Throttle Gain K R P M : 0.00000
| 5f. Throttle Ref R P M Move : 0.00000
| 5g. Throttle Lag Up : 0.00000
| 5h. Throttle Lag Dn : 0.00000
| 5i. Throttling Factor : 0.00000
- 5. Maps 12
| 5a. Type : gamma
| 5b. Power Mult : 1.00000
| 5c. Consumption Mult : 0.60000
| 5d. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_wet.curve
| 5e. Throttle Gain K R P M : 0.00000
| 5f. Throttle Ref R P M Move : 0.00000
| 5g. Throttle Lag Up : 0.00000
| 5h. Throttle Lag Dn : 0.00000
| 5i. Throttling Factor : 0.00000
- 6. Minimum : 2000
- 7. Limiter : 15000
- 8. Limiter Cycles : 60
- 9. Throttle Response Curve :
content\cars\ks_ferrari_sf_25\data\throttle_linear.curve
| 10. Throttle Lag Up : 0.00000
| 11. Throttle Lag Dn : 0.00000
| 12. Throttle Rev Chocking : 0.00000
| 13. Ignition Time S : 0.00000
| 14. Starter Engine Torque : 40.00000
- 15. Start E C U Assist
| 15a. Rpm Range : 0.00000
| 15b. Gain : 0.00000
| 15c. Speed Range K H M : 0.00000
| 15d. Rpm Limiter : 0.00000
| 15e. Limiter Cycles : 0
| 15f. Use Clutch : false
| 15g. Slip Ratio Target : 0.00000
| 15h. Clutch Gain : 0.00000
```

- L 15i. Gas Gain : 0.00000
- 16. Turbo Controllers 1
 - 16a. Name : None
 - 16b. Stages 1
 - 16b1. Input Var : RPMS
 - 16b2. Combinator Mode : Add
 - 16b3. Lut :
- content\cars\ks_ferrari_sf_25\data\ctrl_turbo0CONTROLLER_0.curve
 - 16b4. Filter Gain : 0.99000
 - 16b5. Up Limit : 10000.00000
 - 16b6. Down Limit : 0.00000
 - 16b7. Current Value : 0.00000
 - L 16b8. Const Value : 0.00000
- 17. Waste Gate Controllers : None
- 18. Max Turbo Boost : 2.20000
- 19. Bov Threshold : 0.00000
- 20. Turbos To Load 1 :
- content\cars\ks_ferrari_sf_25\data\ks_ferrari_sf_250.turbo
- 21. Battery Data
 - 21a. Capacity Kwh : 0.000
 - 21b. Discharge Efficiency : 0.000
 - 21c. Charge Efficiency : 0.000
 - 21d. Temp Eff Loss Per Deg : 0.00000
 - 21e. Convection K : 0.00000
 - 21f. Convection Forced K : 0.00000
 - 21g. Thermal Capacity : 0.000
 - 21h. Max Temp : 0.000
 - L 21i. Min Temp : 0.000

Chevrolet Camaro ZL1

- 1. Engine Type : combustion
- 2. Inertia : 0.45000
- 3. Power Curve :
- content\cars\ks_chevrolet_camaro_zl1\data\HEADER_POWER.curve
- 4. Coast Curve : content\cars\ks_chevrolet_camaro_zl1\data\coast.curve
- 5. Maps 1
 - 5a. Type : gamma
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 1.00000
 - 5d. Throttle Response Curve :
- content\cars\ks_chevrolet_camaro_zl1\data\throttlemap_response.curve
 - 5e. Throttle Gain K R P M : 0.00000
 - 5f. Throttle Ref R P M Move : 0.00000
 - 5g. Throttle Lag Up : 0.80000
 - 5h. Throttle Lag Dn : 0.92000
 - L 5i. Throttling Factor : 0.00000
- 6. Minimum : 900
- 7. Limiter : 6600
- 8. Limiter Cycles : 30
- 9. Throttle Response Curve :
- content\cars\ks_chevrolet_camaro_zl1\data\throttlemap_response.curve
- 10. Throttle Lag Up : 0.80000
- 11. Throttle Lag Dn : 0.90000

- 12. Throttle Rev Chocking : 0.00000
- 13. Ignition Time S : 0.00000
- 14. Starter Engine Torque : 60.00000
- 15. Start E C U Assist
 - 15a. Rpm Range : 1500.00000
 - 15b. Gain : 0.10000
 - 15c. Speed Range K H M : 120.00000
 - 15d. Rpm Limiter : 3800.00000
 - 15e. Limiter Cycles : 0
 - 15f. Use Clutch : true
 - 15g. Slip Ratio Target : 0.14000
 - 15h. Clutch Gain : 5.00000
 - 15i. Gas Gain : 1.75000
- 16. Turbo Controllers 1
 - 16a. Name : None
 - 16b. Stages 1
 - 16b1. Input Var : RPMS
 - 16b2. Combinator Mode : Add
 - 16b3. Lut :
 - content\cars\ks_chevrolet_camaro_zl1\data\rpms_boost.curve
 - 16b4. Filter Gain : 0.00000
 - 16b5. Up Limit : 1.20000
 - 16b6. Down Limit : 0.00000
 - 16b7. Current Value : 0.00000
 - 16b8. Const Value : 0.00000
- 17. Waste Gate Controllers : None
- 18. Max Turbo Boost : 0.60000
- 19. Bov Threshold : 0.00000
- 20. Turbos To Load 1 :
 - content\cars\ks_chevrolet_camaro_zl1\data\ks_chevrolet_camaro_zl1_compressor.turbo
- 21. Battery Data
 - 21a. Capacity Kwh : 0.000
 - 21b. Dischage Efficiency : 0.000
 - 21c. Charge Efficiency : 0.000
 - 21d. Temp Eff Loss Per Deg : 0.00000
 - 21e. Convection K : 0.00000
 - 21f. Convection Forced K : 0.00000
 - 21g. Thermal Capacity : 0.000
 - 21h. Max Temp : 0.000
 - 21i. Min Temp : 0.000

Datsun 240z Fairlady

- 1. Engine Type : combustion
- 2. Inertia : 0.18000
- 3. Power Curve :
 - content\cars\ks_datsun_240z_fairlady\data\HEADER_POWER.curve
- 4. Coast Curve :
 - content\cars\ks_datsun_240z_fairlady\data\ks_datsun_240z_fairlady_coast1.curve
- 5. Maps 1
 - 5a. Type : cervone
 - 5b. Power Mult : 1.00000
 - 5c. Consumption Mult : 1.00000

```
| | 5d. Throttle Response Curve :
content\cars\ks_datsun_240z_fairlady\data\throttle.curve
| | 5e. Throttle Gain K R P M : 0.00000
| | 5f. Throttle Ref R P M Move : 0.00000
| | 5g. Throttle Lag Up : 0.00000
| | 5h. Throttle Lag Dn : 0.00000
| | 5i. Throttling Factor : 0.00000
| 6. Minimum : 600
| 7. Limiter : 7000
| 8. Limiter Cycles : 8
| 9. Throttle Response Curve :
content\cars\ks_datsun_240z_fairlady\data\throttle.curve
| 10. Throttle Lag Up : 0.00000
| 11. Throttle Lag Dn : 0.00000
| 12. Throttle Rev Chocking : 0.00000
| 13. Ignition Time S : 0.00000
| 14. Starter Engine Torque : 30.00000
| 15. Start E C U Assist
| | 15a. Rpm Range : 0.00000
| | 15b. Gain : 0.00000
| | 15c. Speed Range K H M : 0.00000
| | 15d. Rpm Limiter : 0.00000
| | 15e. Limiter Cycles : 0
| | 15f. Use Clutch : false
| | 15g. Slip Ratio Target : 0.00000
| | 15h. Clutch Gain : 0.00000
| | 15i. Gas Gain : 0.00000
| 16. Turbo Controllers : None
| 17. Waste Gate Controllers : None
| 18. Max Turbo Boost : 0.00000
| 19. Bov Threshold : 0.00000
| 20. Turbos To Load : None
| 21. Battery Data
| | 21a. Capacity Kwh : 0.000
| | 21b. Dischage Efficiency : 0.000
| | 21c. Charge Efficiency : 0.000
| | 21d. Temp Eff Loss Per Deg : 0.00000
| | 21e. Convection K : 0.00000
| | 21f. Convection Forced K : 0.00000
| | 21g. Thermal Capacity : 0.000
| | 21h. Max Temp : 0.000
| | 21i. Min Temp : 0.000
```

7. Car Setup [.carsetup]

A. Description

Pit-garage snapshot: the adjustable values layered on top of fixed Car Data geometry and asset paths — springs, dampers, alignment, aero targets, diff, electronics maps, and fuel load for a given session or preset.

Car Data defines what the car *is*. Car Setup defines how it is *configured today*. Limits and units (assets 6–7) decide what the UI will allow and how numbers are displayed.

I. Role in the stack

Concern	Handled here	Handled elsewhere
ARB, bias, ducts, diff locks for this preset	.CARSETUP	Hardware ceilings in other assets
Per-corner wheel rate, bump stops, helpers	.CARSETUP Suspensions	Base coilover / suspension kinematics
Damper clicks / rates	.CARSETUP Dampers	Damper curves asset when used
Pressures, camber, toe, caser, compound index	.CARSETUP Alignements	Tyre compound physics in .tyre
TC / ABS / ESC / EBB / engine map / ERS / boost	.CARSETUP Electronics	Maps defined in electronics / engine
Ride height targets, wing angles, collars	.CARSETUP Aero	Wing / surface3d aero maps
Legal min/max / steps	—	<u>6. Car Setup Limits</u>
Display units (bar, clicks, deg)	—	<u>7. Car Setup Units</u>

Loaded from Car Data stock / AI / wet setup paths, or from tuning-part / performance-mode packages. *Import Setup* can inherit another .CARSETUP as a base.

II. What you are really tuning

- Mechanical balance** — *Arbs* (typically index 1 front, 2 rear) shift roll stiffness front/rear — primary understeer/oversteer lever without touching springs. *Steer Ratio* changes rack feel. Brakes block sets *Front Bias*, optional *Torque Multiplier*, and *Brake Ducts*. Differential *Power / Coast / Preload* shape exit drive and entry stability (often zeroed on open or non-adjustable diffs; race cars fill them in — 488 uses coast and preload).
- Springs and stops** — Per-corner *Suspensions*[: *Wheel Rate* is the main spring (or wheel-rate) number. Bump stops up/down (*Range + Rate*) catch end travel. Helper springs (*Helper Rate / Range*) fill gaps when used. Quattro / Junior sit around mid road-race rates; 488 Challenge jumps to much higher wheel rates (about 150000-160000 in the example).
- Dampers** — Per-corner slow/fast bump and rebound. Schema skips *4d* and goes to *4e. Fast Rebound*. Values may be authored as physical coefficients or as click-like numbers depending on the car (Quattro slow bump thousands; 488 slow bump single digits) — always read Limits/Units for that vehicle.

4. **Alignment and tyres** — Per-corner pressure, camber, toe, caster, plus derived *Static Camber / Toe Out Linear*, and *Compound* index into Car Data tyre lists. Race presets push more negative camber (488 about -3.5 deg) than road examples (about -1.0 to -1.3).
5. **Electronics package** — TC1/TC2, ABS, ESC, EBB, engine map, turbo boost level, ERS maps/levels, telemetry laps. Safe race presets often raise TC/ABS (488 safe_1 uses 3/3); older or mechanical presets may leave aids at 0.
6. **Aero and fuel** — Collar positions, front/rear target ride heights, front/rear wing angles (488 rear wing 12 deg in the example; Quattro/Junior wings at 0). *Fuel Strategy* is session fuel load. Metadata: *Final State Name, Version, Is Setup Shared*.

III. Architecture

1 - IMPORT AND MECHANICAL BALANCE (SCHEMA 1-2)

Optional import path; ARBs, steer ratio, brakes sub-block, differential sub-block.

2 - PER-CORNER STRUCTURE (SCHEMA 3-5)

Indexed suspensions, dampers, and alignments (usually 1-4 = LF, RF, LR, RR). This is where left/right and front/rear asymmetry lives.

3 - ELECTRONICS (SCHEMA 6)

Driver-aid and powertrain map selectors for this preset.

4 - AERO AND FUEL (SCHEMA 7-8)

Platform and wing targets; fuel quantity for the run.

5 - PRESET IDENTITY (SCHEMA 9-11)

Final state name/path, version integer, shared flag for multiplayer or garage sharing.

IV. How to read the examples

1 - AUDI SPORT QUATTRO

Mechanical road/rally style: ARBs 25000 / 20000, brake bias 72, diff locks at 0. Wheel rates about 50000 front / lower rear in the dump, cold pressures 28, mild camber (-1.0). ABS on, TC off. Wings at 0, fuel 30. A balanced base preset without heavy race aero.

2 - ALFA ROMEO JUNIOR

Stiffer ARBs (about 40000 / 38000), bias 80, pressures 26-27, camber about -1.3. TC and ABS both 1. Very low fuel in the dump (3) — short-run or display preset. Still road-car aero angles at 0.

3 - FERRARI 488 CHALLENGE ECO (PRESET SAFE 1)

Race Challenge setup: ARB split 34000 / 17000, bias 64, coast lock 0.3 and preload 10. High wheel rates, aggressive camber (-3.5), lower pressures (about 23.5-24), rear wing 12 deg. TC 3 / ABS 3 — clearly a safer electronics package on a stiff aero platform. Named mech/visual preset pair in *Final State Name*.

V. Practical notes

- Setup cannot invent hardware: if Car Data has no wing path or the tyre list is empty, wing angle and compound indices do nothing useful.
- Always pair with **Setup Limits** before copying numbers between cars — min/max/step and click scales differ.
- Damper and spring magnitudes are not universal SI across all examples; treat them as car-local until Units/Limits say otherwise.
- Schema spelling: **Alignements** (not Alignments); damper ID jumps **4c to 4e** (no 4d).
- Import Setup plus performance-mode packages can stack overrides — check what actually loaded in-session.
- Electronics indices must match maps defined in engine / electronics assets, or the garage selection is an empty switch.

VI. Related assets

- **3. Car Data [.car]** — stock / AI / wet setup paths that load this file
- **6 / 7. Car Setup Limits / Units** — legal range and display mapping
- **11 / 17 / 19 / 20. Coilover / Suspension / Tyre / Wing** — physics underneath the garage numbers
- **8. Car Tuning Parts [.tuningpart]** — can embed or swap full setup packages
- **9 / 4 / 18. Electronics / Engine / Turbo** — maps selected by Electronics fields

B. Schema

```
1. Import Setup : string - path
2. Mechanical Balance : object
  2a. Arbs [x] : float | can have multiple Arbs
  2b. Steer Ratio : float
  2c. Brakes : object
    2c1. Front Bias : float
    2c2. Torque Multiplier : float
    2c3. Brake Ducts [x] : float | can have multiple Brake Ducts
  2d. Differential : object
    2d1. Power : float
    2d2. Coast : float
    2d3. Preload : float
3. Suspensions [x] : object
  3a. Wheel Rate : float
  3b. Bump Stop Up : object
    3b1. Range : float
    3b2. Rate : float
```

- 3c. Bump Stop Down : **object**
 - 3c1. Range : **float**
 - 3c2. Rate : **float**
- 3d. Helper Rate : **float**
- 3e. Helper Range : **float**
- 4. Dampers [x] : **object** | can have multiple Dampers
 - 4a. Slow Bump : **float**
 - 4b. Fast Bump : **float**
 - 4c. Slow Rebound : **float**
 - 4e. Fast Rebound : **float**
- 5. Alignments [x] : **object** | can have multiple Alignments
 - 5a. Pressure : **float**
 - 5b. Camber : **float**
 - 5c. Toe : **float**
 - 5d. Caster : **float**
 - 5e. Static Camber : **float**
 - 5f. Toe Out Linear : **float**
 - 5g. Compound : **float**
- 6. Electronics : **object**
 - 6a. Tc1 : **float**
 - 6b. Tc2 : **float**
 - 6c. Abs : **float**
 - 6d. Esc : **float**
 - 6e. Ebb : **float**
 - 6f. Engine Map : **float**
 - 6g. Telemetry Laps to Record : **float**
 - 6h. Turbo Boost Lv : **float**
 - 6i. Ers Deployment Map : **float**
 - 6j. Ers Recharge Lv : **float**
 - 6k. Ers Heat Charging : **float**
- 7. Aero : **object**
 - 7a. Collar Positions Mm [x] : **float** | can have multiple Collar Positions Mm
 - 7b. Front Target Height : **float**
 - 7c. Rear Target Height : **float**
 - 7d. Front Wing Angle : **float**
 - 7e. Rear Wing Angle : **float**
- 8. Fuel Strategy : **object**
 - 8a. Fuel : **float**
- 9. Final State Name : **string – path**
- 10. Version : **integer**
- 11. Is Setup Shared : **boolean**

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Import Setup	None (File path)	Optional path to a base .CARSETUP preset from which this setup inherits or overlays values.
2.	Mechanical Balance	None (Object)	Parent block grouping chassis balance adjustments: anti-roll bars, steering ratio, brakes, and differential.

ID	Name	Unit of Measurement	Description
2a.	Arbs	Nm/deg or N/m (Anti-roll bar stiffness)	Anti-roll bar stiffness values; index 1 = front axle, index 2 = rear axle. Primary tool for shifting roll stiffness balance.
2b.	Steer Ratio	Ratio (:1) or clicks	Adjustable steering rack ratio override applied at setup level; modifies low-speed agility and steering wheel sensitivity.
2c.	Brakes	None (Object)	Brake setup sub-block within Mechanical Balance; contains bias, torque scaling, and duct settings.
2c1.	Front Bias	% (Percentage)	Brake balance percentage sent to the front axle (e.g., 72 = 72% front / 28% rear).
2c2.	Torque Multiplier	% or dimensionless multiplier	Global braking torque scaling factor for this setup (100 = 100% of base brake system torque).
2c3.	Brake Ducts	Ratio (0.0 - 1.0) or clicks	Brake cooling duct opening level(s); multiple entries typically map to front/rear duct positions.
2d.	Differential	None (Object)	Differential setup sub-block within Mechanical Balance; governs power, coast, and preload lock.
2d1.	Power	% or Nm (Differential lock)	Differential locking intensity under acceleration (on-throttle); higher values increase drive-wheel coupling on corner exit.
2d2.	Coast	% or Nm (Differential lock)	Differential locking intensity on coast/deceleration; stabilizes the rear under braking and lift-off.
2d3.	Preload	Nm (Newton-meters)	Static preload torque locking the differential plates at zero throttle; resists free differential action mid-corner.
3.	Suspensions	None (Object array, per wheel)	Per-corner suspension block; indices 1–4 map to LF, RF, LR, RR wheels respectively.
3a.	Wheel Rate	N/m (Newtons per meter)	Effective wheel-rate spring stiffness at this corner; controls ride frequency, body roll, and aero platform stability.
3b.	Bump Stop Up	None (Object)	Upper bump-stop definition within each Suspensions entry; limits compression travel beyond spring range.
3b1.	Range	m (Meters)	Travel range before the upper bump-stop engages (Suspensions > Bump Stop Up).

ID	Name	Unit of Measurement	Description
3b2.	Rate	N/m or N (Bump-stop stiffness)	Stiffness/force rate of the upper bump-stop resisting compression beyond the range threshold.
3c.	Bump Stop Down	None (Object)	Lower bump-stop definition within each Suspensions entry; limits droop/extension travel.
3c1.	Range	m (Meters)	Travel range before the lower (rebound) bump-stop engages (Suspensions > Bump Stop Down).
3c2.	Rate	N/m or N (Bump-stop stiffness)	Stiffness/force rate of the lower bump-stop resisting droop/extension.
3d.	Helper Rate	N/m (Spring rate)	Stiffness of the helper/tender spring assisting the main coil at low compression.
3e.	Helper Range	m (Meters)	Travel range over which the helper spring is active before the main spring fully carries load.
4.	Dampers	None (Object array, per wheel)	Per-corner damper settings; indices 1–4 map to LF, RF, LR, RR. Values may be raw N-s/m or UI clicks depending on carsetupunits.
4a.	Slow Bump	N-s/m or clicks	Slow-shaft-speed compression damping; controls body roll and pitch from driver inputs (braking, cornering).
4b.	Fast Bump	N-s/m or clicks	Fast-shaft-speed compression damping; absorbs kerb strikes, bumps, and high-frequency track irregularities.
4c.	Slow Rebound	N-s/m or clicks	Slow-shaft-speed extension damping; regulates how the chassis settles after weight transfer releases.
4e.	Fast Rebound	N-s/m or clicks	Fast-shaft-speed extension damping; controls how quickly the tyre returns to the track surface after a kerb hit.
5.	Alignments	None (Object array, per wheel)	Per-corner tyre pressure and alignment geometry; indices 1–4 map to LF, RF, LR, RR.
5a.	Pressure	PSI or bar (Cold tyre pressure)	Cold inflation pressure before leaving the pits; determines hot operating pressure and tyre carcass temperature window.

ID	Name	Unit of Measurement	Description
5b.	Camber	deg (Degrees)	Adjustable camber angle (negative = top of wheel leans inward); maximizes contact patch during body roll in corners.
5c.	Toe	deg (Degrees)	Toe angle relative to vehicle centerline; toe-out front increases turn-in, toe-in rear stabilizes straight-line stability.
5d.	Caster	deg (Degrees)	Steering axis caster angle; influences straight-line stability, self-centering torque, and camber gain during steering.
5e.	Static Camber	deg (Degrees)	Computed/static camber at ride height including kinematic offset; read-only or derived alignment reference.
5f.	Toe Out Linear	deg/m or rad/m (Toe curve)	Linear toe change rate as a function of suspension travel or steering; models compliance and kinematic toe curve.
5g.	Compound	None (Compound index)	Tyre compound selection index referencing the available front/rear tyre compound paths in Car Data.
6.	Electronics	None (Object)	Driver-aid and powertrain map presets stored in this setup (TC, ABS, ESC, engine map, ERS, turbo).
6a.	Tc1	None (Map level / index)	Traction Control map level (primary TC); references TC settings defined in Car Data electronics.
6b.	Tc2	None (Map level / index)	Secondary Traction Control map level for dual-TC systems.
6c.	Abs	None (Map level / index)	ABS intervention map level; controls slip thresholds and pulsing aggressiveness.
6d.	Esc	None (Map level / index)	Electronic Stability Program map level; governs yaw/brake intervention authority.
6e.	Ebb	None (Map level / index)	Electronic Brake Balance mode or level for dynamic bias adjustment.
6f.	Engine Map	None (Map index)	Engine power map selection referencing engine map entries in the .CARENGINE asset.
6g.	Telemetry Laps To Record	None (Integer)	Number of laps of onboard telemetry to retain when this setup is active.
6h.	Turbo Boost Lv	bar or level index	Turbo boost level preset for forced-induction engines.

ID	Name	Unit of Measurement	Description
6i.	Ers Deployment Map	None (Map index)	Hybrid ERS power deployment map selection.
6j.	Ers Recharge Lv	None (Level index)	ERS regenerative recovery aggressiveness level.
6k.	Ers Heat Charging	None (Level index)	MGU-H heat-to-battery charging level for hybrid systems.
7.	Aero	None (Object)	Aerodynamic setup block: ride heights, spring collar positions, and adjustable wing angles.
7a.	Collar Positions Mm	mm (millimeters)	Spring perch/collar position per corner controlling static ride height; indices 1–4 = LF, RF, LR, RR.
7b.	Front Target Height	mm (millimeters)	Target front ride height reference for aero platform setup and regulatory ride-height checks.
7c.	Rear Target Height	mm (millimeters)	Target rear ride height; combined with front height defines rake (nose-up/nose-down aero attitude).
7d.	Front Wing Angle	deg (Degrees)	Mechanical angle of the front aero device (splitter/wing); increases downforce and drag when raised.
7e.	Rear Wing Angle	deg (Degrees)	Mechanical angle of the rear wing element; primary high-speed cornering stability vs. straight-line speed trade-off.
8.	Fuel Strategy	None (Object)	Fuel load planning block for this setup preset.
8a.	Fuel	L (Liters)	Fuel load strategy for this setup preset; affects weight, CG position, and stint length planning.
9.	Final State Name	None (String / preset identifier)	Composite preset identifier linking mechanical and visual setup states (e.g., mech + visual preset names).
10.	Version	None (Integer)	Setup file format version for backward compatibility with the editor and simulation loader.
11.	Is Setup Shared	None (Boolean : True / False)	Indicates whether this setup preset is shared/exportable across sessions or locked to a single vehicle profile.

D. Example data

I. Chosen Car Engine for Example

- Audi Sport Quattro (slug : ks_audi_sport_quattro)
- Alfa Romeo Junior (slug : ks_alfa_romeo_junior)
- Ferrari 488 Challenge Evo (slug : ks_ferrari_488_challenge_evo [preset : safe_1])

II. Example

Audi Sport Quattro

- 1. Import Setup : None
- 2. Mechanical Balance
 - 2a. Arbs 1 : 25000.00000
 - 2a. Arbs 2 : 20000.00000
 - 2b. Steer Ratio : 18.00000
 - 2c. Brakes
 - 2c1. Front Bias : 72.00000
 - 2c2. Torque Multiplier : 100.00000
 - 2c3. Brake Ducts : None
 - 2d. Differential
 - 2d1. Power : 0.00000
 - 2d2. Coast : 0.00000
 - 2d3. Preload : 0.00000
- 3. Suspensions 1
 - 3a. Wheel Rate : 50000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.02645
 - 3b2. Rate : 650.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.11855
 - 3c2. Rate : 650.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 2
 - 3a. Wheel Rate : 50000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.02645
 - 3b2. Rate : 650.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.11855
 - 3c2. Rate : 650.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 3
 - 3a. Wheel Rate : 42750.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.04487
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.04513
 - 3c2. Rate : 400.00000

- 3. Suspensions 4
 - 3a. Wheel Rate : 42750.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.04487
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.04513
 - 3c2. Rate : 400.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 4. Dampers 1
 - 4a. Slow Bump : 6000.00000
 - 4b. Fast Bump : 1500.00000
 - 4c. Slow Rebound : 8000.00000
 - 4e. Fast Rebound : 2300.00000
- 4. Dampers 2
 - 4a. Slow Bump : 6000.00000
 - 4b. Fast Bump : 1500.00000
 - 4c. Slow Rebound : 8000.00000
 - 4e. Fast Rebound : 2300.00000
- 4. Dampers 3
 - 4a. Slow Bump : 5000.00000
 - 4b. Fast Bump : 1200.00000
 - 4c. Slow Rebound : 7000.00000
 - 4e. Fast Rebound : 1900.00000
- 4. Dampers 4
 - 4a. Slow Bump : 5000.00000
 - 4b. Fast Bump : 1200.00000
 - 4c. Slow Rebound : 7000.00000
 - 4e. Fast Rebound : 1900.00000
- 5. Alignements 1
 - 5a. Pressure : 28.00000
 - 5b. Camber : -1.00000
 - 5c. Toe : -0.03000
 - 5d. Caster : -0.04510
 - 5e. Static Camber : -1.69155
 - 5f. Toe Out Linear : 0.00098
 - 5g. Compound : 0.00000
- 5. Alignements 2
 - 5a. Pressure : 28.00000
 - 5b. Camber : -1.00000
 - 5c. Toe : -0.03000
 - 5d. Caster : -0.04510
 - 5e. Static Camber : -1.76592
 - 5f. Toe Out Linear : 0.00095
 - 5g. Compound : 0.00000
- 5. Alignements 3
 - 5a. Pressure : 28.00000
 - 5b. Camber : -0.60000
 - 5c. Toe : 0.05000
 - 5d. Caster : 0.00000
 - 5e. Static Camber : -0.66797
 - 5f. Toe Out Linear : 0.00022
 - 5g. Compound : 0.00000
- 5. Alignements 4
 - 5a. Pressure : 28.00000

- 5b. Camber : -0.60000
- 5c. Toe : 0.05000
- 5d. Caster : 0.00000
- 5e. Static Camber : -0.75145
- 5f. Toe Out Linear : 0.00022
- 5g. Compound : 0.00000
- 6. Electronics
 - 6a. Tc1 : 0.00000
 - 6b. Tc2 : 0.00000
 - 6c. Abs : 1.00000
 - 6d. Esc : 0.00000
 - 6e. Ebb : 0.000
 - 6f. Engine Map : 0.00000
 - 6g. Telemetry Laps to Record : 0.00000
 - 6h. Turbo Boost Lv : 0.00000
 - 6i. Ers Deployment Map : 0.000
 - 6j. Ers Recharge Lv : 0.000
 - 6k. Ers Heat Charging : 0.000
- 7. Aero
 - 7a. Collar Positions Mm 1 : 112.59116
 - 7a. Collar Positions Mm 2 : 107.20738
 - 7a. Collar Positions Mm 3 : 76.21446
 - 7a. Collar Positions Mm 4 : 75.33346
 - 7b. Front Target Height : 200.00000
 - 7c. Rear Target Height : 200.00000
 - 7d. Front Wing Angle : 0.00000
 - 7e. Rear Wing Angle : 0.00000
- 8. Fuel Strategy
 - 8a. Fuel : 30.00000
- 9. Final State Name :
ks_audi_sport_quattro_preset_sq_mech_1_preset_sq_visual_1
- 10. Version : None
- 11. Is Setup Shared : false

Alfa Romeo Junior

- 1. Import Setup
- 2. Mechanical Balance
 - 2a. Arbs 1 : 40000.00000
 - 2a. Arbs 1 : 38000.00000
 - 2b. Steer Ratio : -14.60000
 - 2c. Brakes
 - 2c1. Front Bias : 80.00000
 - 2c2. Torque Multiplier : 100.00000
 - 2c3. Brake Ducts : None
 - 2d. Differential
 - 2d1. Power : 0.00000
 - 2d2. Coast : 0.00000
 - 2d3. Preload : 0.00000
- 3. Suspensions 1
 - 3a. Wheel Rate : 49000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.01194
 - 3b2. Rate : 400.00000

- 3c. Bump Stop Down
 - 3c1. Range : 0.07306
 - 3c2. Rate : 400.00000
- 3d. Helper Rate : 0.00000
- 3e. Helper Range : 0.00000
- 3. Suspensions 2
 - 3a. Wheel Rate : 49000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.01194
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.07306
 - 3c2. Rate : 400.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 3
 - 3a. Wheel Rate : 45000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : -0.00872
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.0372
 - 3c2. Rate : 500.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 4
 - 3a. Wheel Rate : 45000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : -0.00872
 - 3b2. Rate : 400.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.09372
 - 3c2. Rate : 500.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 4. Dampers 1
 - 4a. Slow Bump : 8200.00000
 - 4b. Fast Bump : 3000.00000
 - 4c. Slow Rebound : 10500.00000
 - 4e. Fast Rebound : 3300.00000
- 4. Dampers 2
 - 4a. Slow Bump : 8200.00000
 - 4b. Fast Bump : 3000.00000
 - 4c. Slow Rebound : 10500.00000
 - 4e. Fast Rebound : 3300.00000
- 4. Dampers 3
 - 4a. Slow Bump : 5000.00000
 - 4b. Fast Bump : 3500.00000
 - 4c. Slow Rebound : 6500.00000
 - 4e. Fast Rebound : 3300.00000
- 4. Dampers 4
 - 4a. Slow Bump : 5000.00000
 - 4b. Fast Bump : 3500.00000
 - 4c. Slow Rebound : 6500.00000
 - 4e. Fast Rebound : 3300.00000
- 5. Alignements 1

- 5a. Pressure : 26.00000
- 5b. Camber : -1.30000
- 5c. Toe : -0.05000
- 5d. Caster : -0.04980
- 5e. Static Camber : -1.57347
- 5f. Toe Out Linear : 0.00018
- 5g. Compound : 0.00000
- 5. Alignements 2
 - 5a. Pressure : 26.00000
 - 5b. Camber : -1.30000
 - 5c. Toe : -0.05000
 - 5d. Caster : -0.04980
 - 5e. Static Camber : -1.56779
 - 5f. Toe Out Linear : 0.00016
 - 5g. Compound : 0.00000
- 5. Alignements 3
 - 5a. Pressure : 27.00000
 - 5b. Camber : -1.40000
 - 5c. Toe : 0.05000
 - 5d. Caster : -0.12000
 - 5e. Static Camber : -1.39969
 - 5f. Toe Out Linear : -0.00142
 - 5g. Compound : 0.00000
- 5. Alignements 4
 - 5a. Pressure : 27.00000
 - 5b. Camber : -1.40000
 - 5c. Toe : 0.05000
 - 5d. Caster : -0.12000
 - 5e. Static Camber : -1.39560
 - 5f. Toe Out Linear : -0.00143
 - 5g. Compound : 0.00000
- 6. Electronics
 - 6a. Tc1 : 1.00000
 - 6b. Tc2 : 1.00000
 - 6c. Abs : 1.00000
 - 6d. Esc : 0.00000
 - 6e. Ebb : 0.000
 - 6f. Engine Map : 0.00000
 - 6g. Telemetry Laps to Record : 0.00000
 - 6h. Turbo Boost Lv : 0.00000
 - 6i. Ers Deployment Map : 0.000
 - 6j. Ers Recharge Lv : 0.000
 - 6k. Ers Heat Charging : 0.000
- 7. Aero
 - 7a. Collar Positions Mm 1 : 102.44676
 - 7a. Collar Positions Mm 2 : 102.44756
 - 7a. Collar Positions Mm 3 : 105.31527
 - 7a. Collar Positions Mm 4 : 105.31618
 - 7b. Front Target Height : 185.00000
 - 7c. Rear Target Height : 190.00000
 - 7d. Front Wing Angle : 0.00000
 - 7e. Rear Wing Angle : 0.00000
- 8. Fuel Strategy
 - 8a. Fuel : 3.00000
- 9. Final State Name :

ks_alfa_romeo_junior_preset_mln_mech_1_preset_mln_visual_1

- 10. Version : 0
- 11. Is Setup Shared : false

Ferrari 488 Challenge Evo [preset : safe_1]

- 1. Import Setup : None
- 2. Mechanical Balance
 - 2a. Arbs 1 : 34000.00000
 - 2a. Arbs 2 : 17000.00000
 - 2b. Steer Ratio : 15.00000
 - 2c. Brakes
 - 2c1. Front Bias : 64.00000
 - 2c2. Torque Multiplier : 100.00000
 - 2c3. Brake Ducts : None
 - 2d. Differential
 - 2d1. Power : 0.00000
 - 2d2. Coast : 0.30000
 - 2d3. Preload : 10.00000
- 3. Suspensions 1
 - 3a. Wheel Rate : 160000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.02748
 - 3b2. Rate : 500.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.01252
 - 3c2. Rate : 300.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 2
 - 3a. Wheel Rate : 160000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.02748
 - 3b2. Rate : 500.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.01252
 - 3c2. Rate : 300.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 3
 - 3a. Wheel Rate : 150000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.01958
 - 3b2. Rate : 300.00000
 - 3c. Bump Stop Down
 - 3c1. Range : 0.04042
 - 3c2. Rate : 300.00000
 - 3d. Helper Rate : 0.00000
 - 3e. Helper Range : 0.00000
- 3. Suspensions 4
 - 3a. Wheel Rate : 150000.00000
 - 3b. Bump Stop Up
 - 3b1. Range : 0.01958
 - 3b2. Rate : 300.00000

- 3c. Bump Stop Down
 - 3c1. Range : 0.04042
 - 3c2. Rate : 300.00000
- 3d. Helper Rate : 0.00000
- 3e. Helper Range : 0.00000
- 4. Dampers 1
 - 4a. Slow Bump : 8.00000
 - 4b. Fast Bump : 5000.00000
 - 4c. Slow Rebound : 9.00000
 - 4e. Fast Rebound : 5000.00000
- 4. Dampers 2
 - 4a. Slow Bump : 8.00000
 - 4b. Fast Bump : 5000.00000
 - 4c. Slow Rebound : 9.00000
 - 4e. Fast Rebound : 5000.00000
- 4. Dampers 3
 - 4a. Slow Bump : 9.00000
 - 4b. Fast Bump : 5000.00000
 - 4c. Slow Rebound : 10.00000
 - 4e. Fast Rebound : 5000.00000
- 4. Dampers 4
 - 4a. Slow Bump : 9.00000
 - 4b. Fast Bump : 5000.00000
 - 4c. Slow Rebound : 10.00000
 - 4e. Fast Rebound : 5000.00000
- 5. Alignements 1
 - 5a. Pressure : 24.00000
 - 5b. Camber : -3.50000
 - 5c. Toe : -0.10000
 - 5d. Caster : -0.07000
 - 5e. Static Camber : -3.09347
 - 5f. Toe Out Linear : 0.00055
 - 5g. Compound : 0.00000
- 5. Alignements 2
 - 5a. Pressure : 24.00000
 - 5b. Camber : -3.50000
 - 5c. Toe : -0.10000
 - 5d. Caster : -0.07000
 - 5e. Static Camber : -3.09532
 - 5f. Toe Out Linear : 0.00056
 - 5g. Compound : 0.00000
- 5. Alignements 3
 - 5a. Pressure : 23.50000
 - 5b. Camber : -3.00000
 - 5c. Toe : 0.10000
 - 5d. Caster : 0.00000
 - 5e. Static Camber : -3.21468
 - 5f. Toe Out Linear : -0.00023
 - 5g. Compound : 0.00000
- 5. Alignements 4
 - 5a. Pressure : 23.50000
 - 5b. Camber : -3.00000
 - 5c. Toe : 0.10000
 - 5d. Caster : 0.00000
 - 5e. Static Camber : -3.21772
 - 5f. Toe Out Linear : -0.00023

- | L 5g. Compound : 0.00000
- 6. Electronics
 - | 6a. Tc1 : 3.00000
 - | 6b. Tc2 : 3.00000
 - | 6c. Abs : 3.00000
 - | 6d. Esc : 0.00000
 - | 6e. Ebb : 0.000
 - | 6f. Engine Map : 0.00000
 - | 6g. Telemetry Laps to Record : 5.00000
 - | 6h. Turbo Boost Lv : 1.00000
 - | 6i. Ers Deployment Map : 0.000
 - | 6j. Ers Recharge Lv : 0.000
 - | L 6k. Ers Heat Charging : 0.000
- 7. Aero
 - | 7a. Collar Positions Mm 1 : 0.88076
 - | 7a. Collar Positions Mm 2 : 0.88185
 - | 7a. Collar Positions Mm 3 : 29.68013
 - | 7a. Collar Positions Mm 4 : 29.68015
 - | 7b. Front Target Height : 75.00000
 - | 7c. Rear Target Height : 90.00000
 - | 7d. Front Wing Angle : 0.00000
 - | L 7e. Rear Wing Angle : 12.00000
- 8. Fuel Strategy
 - | L 8a. Fuel : 30.00000
- 9. Final State Name :
ks_ferrari_488_challenge_evo_preset_f488ce_mech_1_preset_f488ce_visual_1
- 10. Version : 0
- 11. Is Setup Shared : false

6. Car Setup Limits [.carsetuplimits]

A. Description

Garage rulebook for .CARSETUP: which parameters exist as sliders, how far they can move, in what steps, and how the UI should treat them.

Car Setup stores the *current* values. Setup Limits define the *allowed envelope* and interface behaviour. Without limits, the garage has no legal min/max/step model for that car.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Min / max / step per setup field	.CARSETUPLIMITS	—
Locked vs adjustable (<i>Is Modifiable</i>)	.CARSETUPLIMITS	—
Hide raw value, sign, boolean treatment	.CARSETUPLIMITS	—
Optional LUT for non-linear click mapping	.CARSETUPLIMITS	Curves on disk
Display unit string on a limit object	Often here	Canonical labels in <u>7. Car Setup Units</u>
Actual spring/damper/aero numbers in use	—	<u>5. Car Setup</u>
Physics hardware underneath	—	Coilover, tyre, wing, etc.

Usually referenced from Car Data *Setup Limits* path. Tree layout mirrors Car Setup (mechanical balance, suspensions, dampers, alignments, electronics, aero, fuel).

II. What you are really tuning

1. **The limit object (shared anatomy)** —Almost every tunable reuses the same nested fields (schema IDs reused as *2a1...2a10* under each parent):

Field	Role
Step	Garage increment per click
Min / Max	Absolute allowed range
Lut	Optional curve remapping clicks to physics
Is Modifiable	false = locked / greyed out
Hide Value	Hide numeric readout
Is Negative	Sign / direction convention for UI

Field	Role
Unit	Label string (may be None if Units asset owns it)
Fractional Digits	Decimal precision on screen
Treat As Boolean	Present as on/off instead of a continuous slider

When *Min == Max* and *Is Modifiable* is false, the parameter is frozen at a factory value.

2. **What you lock vs open** — Road cars often freeze ARBs, springs, dampers, wings, and diff — only pressures or mild alignment stay open. Motorsport cars open most flags but keep **narrow** min/max windows so the platform stays in its aero/kinematic sweet spot. BOP can tighten windows (raise minimum wing, clamp ride height) without rewriting core Car Data mass or engine curves.
3. **Categories (same map as setup)** — Mechanical balance (ARBs, steer ratio, brake bias/ducts, diff), per-corner suspensions and dampers, alignments (pressures, camber, toe...), electronics map indices, aero (collars, heights, wing angles), fuel strategy, plus *Use Single Compound* for tyre UI behaviour.

III. Architecture

1 - BINDING (SCHEMA 1)

Car Data path (often None when the limits file is already paired by Car Data's own Setup Limits field).

2 - MECHANICAL BALANCE (SCHEMA 2)

Limit objects for ARBs *[x]*, steer ratio, brakes (bias, torque multiplier, ducts), differential (power, coast, preload). Same subtree shape as setup, but each leaf is a limit object, not a float.

3 - PER-CORNER AND ELECTRONICS (SCHEMA 3-6)

Suspensions, dampers, alignments arrays; electronics selectors (TC, ABS, maps...). Indexing matches setup corners (typically 1-4).

4 - AERO, FUEL, COMPOUND MODE (SCHEMA 7-9)

Aero limit blocks, fuel limit, *Use Single Compound* flag for whether front/rear compounds are forced to stay linked in the UI.

IV. How to read the examples

1 - BMW M4 CSL (LOCKED PRODUCTION-STYLE ENVELOPE)

ARBs show *Is Modifiable* : false with Min equal to Max (front stuck at 30000, rear at 11000 in the dump). That is the stock profile: the garage exposes little or no ARB travel because the car is not meant to be a full race adjuster. Read other false flags the same way — frozen factory identity.

2 - LAMBORGHINI COUNTACH (OPEN ADJUSTABLE ENVELOPE)

ARBs *Is Modifiable* : *true* with real windows (front about 28000-68000 step 4000; rear about 5000-50000 step 5000). The car can be explored across a wide mechanical range. Compare min/max width and step size to judge how “race engineer” vs “coarse kit” the garage feels.

V. Practical notes

- A setup value outside min/max should be treated as illegal or clamped — fix the setup or the limits, do not assume the sim will honour out-of-range garage numbers.
- Copying setup numbers between cars without copying limits is how you get greyed sliders or impossible clicks.
- *Step* and *LUT* together define the feel of the menu: linear physics steps vs non-linear click tables.
- Schema reuses limit-field IDs (*2a1...2a10*) under many parents — that is intentional templating, not a documentation error.
- Setup file spelling is **Alignments**; limits schema may say **Alignments** — same corner block.
- Fuel under Fuel Strategy is **8a** in the corrected schema tree (paired with setup’s fuel field).
- For display unit policy, still check **7. Car Setup Units**; limit *Unit* strings can be *None*.

VI. Related assets

- **5. Car Setup [.carsetup]** — values constrained by these limits
- **7. Car Setup Units [.carsetupunits]** — unit labels / localisation for the same tree
- **3. Car Data [.car]** — *Setup Limits* path that loads this file
- **8. Car Tuning Parts [.tuningparts]** — performance modes may ship with their own setup/limit packages

B. Schema

```
1. Car Data : string - path
2. Mechanical Balance : object
  2a. Arbs [x] : object | can have multiple Arbs
    2a1. Step : float
    2a2. Min : float
    2a3. Max : float
    2a4. Lut : string - path
    2a5. Is Modifiable : boolean
    2a6. Hide Value : boolean
    2a7. Is Negative : boolean
    2a8. Unit : string
    2a9. Fractional Digits : integer
    2a10. Treat As Boolean : boolean
  2b. Steer Ratio : object
    2a1. Step : float
    2a2. Min : float
    2a3. Max : float
    2a4. Lut : string - path
```

- 2a5. Is Modifiable : **boolean**
- 2a6. Hide Value : **boolean**
- 2a7. Is Negative : **boolean**
- 2a8. Unit : **string**
- 2a9. Fractional Digits : **integer**
- 2a10. Treat As Boolean : **boolean**
- 2c. Brakes : **object**
 - 2c1. Front Bias : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string** – path
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
 - 2c2. Torque Multiplier : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string** – path
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
 - 2c3. Brake Ducts [x] : **object** | can have multiple Brake Ducts
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string** – path
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
- 2d. Differential : **object**
 - 2d1. Power : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**
 - 2a4. Lut : **string** – path
 - 2a5. Is Modifiable : **boolean**
 - 2a6. Hide Value : **boolean**
 - 2a7. Is Negative : **boolean**
 - 2a8. Unit : **string**
 - 2a9. Fractional Digits : **integer**
 - 2a10. Treat As Boolean : **boolean**
 - 2d2. Coast : **object**
 - 2a1. Step : **float**
 - 2a2. Min : **float**
 - 2a3. Max : **float**

- 2a4. Lut : string - path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer
- 2a10. Treat As Boolean : boolean
- 2d3. Preload : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 3. Suspensions [x] : object | can have multiple Suspensions
 - 3a. Wheel Rate : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 3b. Bump Stop Up : object
 - 3b1. Range : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 3b2. Rate : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 3c. Bump Stop Down : object
 - 3b1. Range : object
 - 2a1. Step : float

- 2a2. Min : float
- 2a3. Max : float
- 2a4. Lut : string - path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer
- 2a10. Treat As Boolean : boolean
- 3b2. Range : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 3d. Helper Rate : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 3e. Helper Range : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 4. Dampers [x] : object | can have multiple Dampers
 - 4a. Slow Bump : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 4b. Fast Bump : object
 - 2a1. Step : float

- 2a2. Min : float
- 2a3. Max : float
- 2a4. Lut : string - path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer
- 2a10. Treat As Boolean : boolean
- 4c. Slow Rebound : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 4d. Fast Rebound : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 5. Alignments [x] : object | can have multiple Alignments
 - 5a. Pressure : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 5b. Camber : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
 - 5c. Toe : object
 - 2a1. Step : float

- 2a2. Min : float
- 2a3. Max : float
- 2a4. Lut : string - path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer
- 2a10. Treat As Boolean : boolean
- 5d. Caster : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 5e. Static Camber : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 5f. Toe Out Linear : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 5g. Compound : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 6. Electronics : object
 - 6a. Tc1 : object
 - 2a1. Step : float

- 2a2. Min : float
- 2a3. Max : float
- 2a4. Lut : string - path
- 2a5. Is Modifiable : boolean
- 2a6. Hide Value : boolean
- 2a7. Is Negative : boolean
- 2a8. Unit : string
- 2a9. Fractional Digits : integer
- 2a10. Treat As Boolean : boolean
- 6b. Tc2 : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 6c. Abs : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 6d. Esc : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 6e. Ebb : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 6f. Engine Map : object
 - 2a1. Step : float
 - 2a2. Min : float

- | 2a3. Max : float
- | 2a4. Lut : string - path
- | 2a5. Is Modifiable : boolean
- | 2a6. Hide Value : boolean
- | 2a7. Is Negative : boolean
- | 2a8. Unit : string
- | 2a9. Fractional Digits : integer
- | 2a10. Treat As Boolean : boolean
- 6g. Telemetry Laps To Record : object
 - | 2a1. Step : float
 - | 2a2. Min : float
 - | 2a3. Max : float
 - | 2a4. Lut : string - path
 - | 2a5. Is Modifiable : boolean
 - | 2a6. Hide Value : boolean
 - | 2a7. Is Negative : boolean
 - | 2a8. Unit : string
 - | 2a9. Fractional Digits : integer
 - | 2a10. Treat As Boolean : boolean
- 6h. Turbo Boost Lv : object
 - | 2a1. Step : float
 - | 2a2. Min : float
 - | 2a3. Max : float
 - | 2a4. Lut : string - path
 - | 2a5. Is Modifiable : boolean
 - | 2a6. Hide Value : boolean
 - | 2a7. Is Negative : boolean
 - | 2a8. Unit : string
 - | 2a9. Fractional Digits : integer
 - | 2a10. Treat As Boolean : boolean
- 6i. Ers Deployment Map : object
 - | 2a1. Step : float
 - | 2a2. Min : float
 - | 2a3. Max : float
 - | 2a4. Lut : string - path
 - | 2a5. Is Modifiable : boolean
 - | 2a6. Hide Value : boolean
 - | 2a7. Is Negative : boolean
 - | 2a8. Unit : string
 - | 2a9. Fractional Digits : integer
 - | 2a10. Treat As Boolean : boolean
- 6j. Ers Recharge Lv : object
 - | 2a1. Step : float
 - | 2a2. Min : float
 - | 2a3. Max : float
 - | 2a4. Lut : string - path
 - | 2a5. Is Modifiable : boolean
 - | 2a6. Hide Value : boolean
 - | 2a7. Is Negative : boolean
 - | 2a8. Unit : string
 - | 2a9. Fractional Digits : integer
 - | 2a10. Treat As Boolean : boolean
- 6k. Ers Heat Charging : object
 - | 2a1. Step : float
 - | 2a2. Min : float
 - | 2a3. Max : float

- 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 7. Aero : object
 - 7a. Collar Positions [x] : object | can have multiple Collar Positions
- 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 7b. Front Target Height : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 7c. Rear Target Height : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 7d. Front Wing Angle : object
 - 2a1. Step : float
 - 2a2. Min : float
 - 2a3. Max : float
 - 2a4. Lut : string - path
 - 2a5. Is Modifiable : boolean
 - 2a6. Hide Value : boolean
 - 2a7. Is Negative : boolean
 - 2a8. Unit : string
 - 2a9. Fractional Digits : integer
 - 2a10. Treat As Boolean : boolean
- 7e. Rear Wing Angle : object
 - 2a1. Step : float
 - 2a2. Min : float

```

    2a3. Max : float
    2a4. Lut : string - path
    2a5. Is Modifiable : boolean
    2a6. Hide Value : boolean
    2a7. Is Negative : boolean
    2a8. Unit : string
    2a9. Fractional Digits : integer
    2a10. Treat As Boolean : boolean
- 8. Fuel Strategy : object
    - 8a. Fuel : object
        - 2a1. Step : float
        - 2a2. Min : float
        - 2a3. Max : float
        - 2a4. Lut : string - path
        - 2a5. Is Modifiable : boolean
        - 2a6. Hide Value : boolean
        - 2a7. Is Negative : boolean
        - 2a8. Unit : string
        - 2a9. Fractional Digits : integer
        - 2a10. Treat As Boolean : boolean
- 9. Use Single Compound : boolean

```

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Car Data	None (File path)	Path to the parent .car asset this limits file is bound to; links regulatory bounds to the correct vehicle.
2.	Mechanical Balance	None (Object)	Parent block for chassis balance limit definitions: ARBs, steering, brakes, and differential.
2a.	Arbs	None (Limit object array)	Per-axle anti-roll bar limit blocks; index 1 = front, 2 = rear. Each entry contains fields 2a1–2a10.
2a1.	Step	Depends on parameter (physics unit or clicks)	Garage increment applied per adjustment click; defines the resolution of slider/stepper controls.
2a2.	Min	Depends on parameter	Absolute minimum allowable physics value for the linked setup parameter.
2a3.	Max	Depends on parameter	Absolute maximum allowable physics value for the linked setup parameter.
2a4.	Lut	None (.CURVE file path)	Optional look-up table mapping linear UI steps to non-linear physical output values.
2a5.	Is Modifiable	None (Boolean : True / False)	When false, the garage slider is locked/greyed out; parameter cannot be changed by the user.

ID	Name	Unit of Measurement	Description
2a6.	Hide Value	None (Boolean : True / False)	When true, obscures the raw numeric value in the UI (proprietary team data simulation).
2a7.	Is Negative	None (Boolean : True / False)	UI sign convention flag; allows negative values to display/be edited correctly (camber, toe).
2a8.	Unit	None (String)	Display unit label rendered in the garage UI (e.g., mm, deg, PSI, bar, Nm, clicks).
2a9.	Fractional Digits	None (Integer)	Number of decimal places shown in the garage UI for this parameter.
2a10.	Treat As Boolean	None (Boolean : True / False)	Renders the parameter as an on/off toggle instead of a numeric slider when true.
2b.	Steer Ratio	None (Limit object)	Limit definition for the adjustable steering ratio parameter in Car Setup.
2c.	Brakes	None (Object)	Brake setup limits sub-block within Mechanical Balance.
2c1.	Front Bias	None (Limit object)	Limit definition for brake balance front percentage.
2c2.	Torque Multiplier	None (Limit object)	Limit definition for global braking torque scaling.
2c3.	Brake Ducts	None (Limit object array)	Per-duct brake cooling limit blocks; multiple entries for front/rear duct positions.
2d.	Differential	None (Object)	Differential setup limits sub-block within Mechanical Balance.
2d1.	Power	None (Limit object)	Limit definition for on-throttle differential lock intensity.
2d2.	Coast	None (Limit object)	Limit definition for coast/off-throttle differential lock intensity.
2d3.	Preload	None (Limit object)	Limit definition for static differential preload torque.
3.	Suspensions	None (Object array, per wheel)	Per-corner suspension limit blocks; indices 1–4 = LF, RF, LR, RR.
3a.	Wheel Rate	None (Limit object)	Limit definition for per-corner wheel-rate spring stiffness.
3b.	Bump Stop Up	None (Object)	Upper bump-stop limit sub-block within each Suspensions entry.
3b1.	Range	None (Limit object)	Bump-stop travel range limit. Under 3b Bump Stop Up; ID reused under 3c Bump Stop Down.

ID	Name	Unit of Measurement	Description
3b2.	Rate	None (Limit object)	Bump-stop stiffness/force rate limit. Under 3b Bump Stop Up; ID reused under 3c Bump Stop Down.
3c.	Bump Stop Down	None (Object)	Lower bump-stop limit sub-block; contains 3b1 (Range) and 3b2 (Rate) limit objects per schema.
3d.	Helper Rate	None (Limit object)	Limit definition for helper/tender spring stiffness.
3e.	Helper Range	None (Limit object)	Limit definition for helper spring active travel range.
4.	Dampers	None (Object array, per wheel)	Per-corner damper limit blocks; indices 1–4 = LF, RF, LR, RR.
4a.	Slow Bump	None (Limit object)	Limit definition for slow-speed compression damper clicks/values.
4b.	Fast Bump	None (Limit object)	Limit definition for fast-speed compression damper clicks/values.
4c.	Slow Rebound	None (Limit object)	Limit definition for slow-speed rebound damper clicks/values.
4d.	Fast Rebound	None (Limit object)	Limit definition for fast-speed rebound damper clicks/values (4d here; Car Setup uses 4e).
5.	Alignments	None (Object array, per wheel)	Per-corner alignment and tyre limit blocks; indices 1–4 = LF, RF, LR, RR.
5a.	Pressure	None (Limit object)	Limit definition for cold tyre inflation pressure.
5b.	Camber	None (Limit object)	Limit definition for adjustable camber angle.
5c.	Toe	None (Limit object)	Limit definition for toe angle.
5d.	Caster	None (Limit object)	Limit definition for caster angle.
5e.	Static Camber	None (Limit object)	Limit definition for computed/static camber display or override bounds.
5f.	Toe Out Linear	None (Limit object)	Limit definition for linear toe compliance curve coefficient.
5g.	Compound	None (Limit object)	Limit definition for tyre compound selection index bounds.
6.	Electronics	None (Object)	Driver-aid and powertrain map limit definitions stored in the garage UI.
6a.	Tc1	None (Limit object)	Limit definition for primary Traction Control map level selection.

ID	Name	Unit of Measurement	Description
6b.	Tc2	None (Limit object)	Limit definition for secondary TC map level.
6c.	Abs	None (Limit object)	Limit definition for ABS map level selection.
6d.	Esc	None (Limit object)	Limit definition for ESP/stability control map level.
6e.	Ebb	None (Limit object)	Limit definition for Electronic Brake Balance level.
6f.	Engine Map	None (Limit object)	Limit definition for engine power map index selection.
6g.	Telemetry Laps To Record	None (Limit object)	Limit definition for onboard telemetry lap recording count.
6h.	Turbo Boost Lv	None (Limit object)	Limit definition for turbo boost level preset.
6i.	Ers Deployment Map	None (Limit object)	Limit definition for ERS deployment map index.
6j.	Ers Recharge Lv	None (Limit object)	Limit definition for ERS recharge aggressiveness level.
6k.	Ers Heat Charging	None (Limit object)	Limit definition for MGU-H heat charging level.
7.	Aero	None (Object)	Aerodynamic setup limit block: ride heights, collar positions, and wing angles.
7a.	Collar Positions	None (Limit object array)	Per-corner spring collar/ride-height limit blocks under 7. Aero; indices 1–4 = LF, RF, LR, RR.
7b.	Front Target Height	None (Limit object)	Limit definition for front target ride height.
7c.	Rear Target Height	None (Limit object)	Limit definition for rear target ride height.
7d.	Front Wing Angle	None (Limit object)	Limit definition for front aero device angle adjustment range.
7e.	Rear Wing Angle	None (Limit object)	Limit definition for rear wing angle adjustment range.
8.	Fuel Strategy	None (Object)	Fuel load planning limit block.
8a.	Fuel	None (Limit object)	Fuel load volume limit under 8. Fuel Strategy.
9.	Use Single Compound	None (Boolean : True / False)	Forces all four corners to share a single tyre compound selection when true.

D. Example data

I. Chosen Car Engine for Example

- BMW M4 CSL (slug : ks_bmw_m4_csl)
- Lamborghini Countach (slug : ks_lamborghini_countach)

II. Example

BMW M4 CSL

```
- 1. Car Data : None
- 2. Mechanical Balance
  - 2a. Arbs 1
    - 2a1. Step : 1.00000
    - 2a2. Min : 30000.00000
    - 2a3. Max : 30000.00000
    - 2a4. Lut : None
    - 2a5. Is Modifiable : false
    - 2a6. Hide Value : false
    - 2a7. Is Negative : false
    - 2a8. Unit : None
    - 2a9. Fractional Digits : 0
    - 2a10. Treat As Boolean : false
  - 2a. Arbs 2
    - 2a1. Step : 1.00000
    - 2a2. Min : 11000.00000
    - 2a3. Max : 11000.00000
    - 2a4. Lut : None
    - 2a5. Is Modifiable : false
    - 2a6. Hide Value : false
    - 2a7. Is Negative : false
    - 2a8. Unit : None
    - 2a9. Fractional Digits : 0
    - 2a10. Treat As Boolean : false
  - 2b. Steer Ratio
    - 2a1. Step : 1.00000
    - 2a2. Min : 15.00000
    - 2a3. Max : 15.00000
    - 2a4. Lut : None
    - 2a5. Is Modifiable : false
    - 2a6. Hide Value : false
    - 2a7. Is Negative : false
    - 2a8. Unit : None
    - 2a9. Fractional Digits : None
    - 2a10. Treat As Boolean : 0
  - 2c. Brakes
    - 2c1. Front Bias
      - 2a1. Step : 1.00000
      - 2a2. Min : 0.00000
      - 2a3. Max : 100.00000
      - 2a4. Lut : None
      - 2a5. Is Modifiable : false
```

- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : None
- 2a10. Treat As Boolean : false
- 2c2. Torque Multiplier
 - 2a1. Step : 1.00000
 - 2a2. Min : 100.00000
 - 2a3. Max : 100.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

- 2d. Differential

- 2d1. Power
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.25000
 - 2a3. Max : 0.25000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

- 2d2. Coast
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.55000
 - 2a3. Max : 0.55000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

- 2d3. Preload
 - 2a1. Step : 1.00000
 - 2a2. Min : 1.00000
 - 2a3. Max : 1.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

- 3. Suspensions 1

- 3a. Wheel Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 47000.00000
 - 2a3. Max : 47000.00000
 - 2a4. Lut : None

- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b. Bump Stop Up
 - 3b1. Range
 - 2a1. Step : 0.01000
 - 2a2. Min : -0.09036
 - 2a3. Max : 0.15964
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 600.00000
 - 2a3. Max : 600.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.07036
 - 2a3. Max : 0.07036
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 1000.00000
 - 2a3. Max : 1000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 5000.00000
 - 2a3. Max : 5000.00000

- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3e. Helper Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.03000
 - 2a3. Max : 0.03000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3. Suspensions 2
 - 3a. Wheel Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 47000.00000
 - 2a3. Max : 47000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b. Bump Stop Up
 - 3b1. Range
 - 2a1. Step : 0.01000
 - 2a2. Min : -0.09036
 - 2a3. Max : 0.15964
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 600.00000
 - 2a3. Max : 600.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 1.00000

- 2a2. Min : 0.07036
- 2a3. Max : 0.07036
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 1000.00000
 - 2a3. Max : 1000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 5000.00000
 - 2a3. Max : 5000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3e. Helper Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.03000
 - 2a3. Max : 0.03000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3. Suspensions 3
 - 3a. Wheel Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 33000.00000
 - 2a3. Max : 33000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b. Bump Stop Up
 - 3b1. Range

- 2a1. Step : 1.00000
- 2a2. Min : 0.13069
- 2a3. Max : 0.13069
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 500.00000
 - 2a3. Max : 500.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.10431
 - 2a3. Max : 0.10431
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 1000.00000
 - 2a3. Max : 1000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 5000.00000
 - 2a3. Max : 5000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3e. Helper Range

- 2a1. Step : 1.00000
- 2a2. Min : 0.03000
- 2a3. Max : 0.03000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 3. Suspensions 4
 - 3a. Wheel Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 33000.00000
 - 2a3. Max : 33000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b. Bump Stop Up
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.13069
 - 2a3. Max : 0.13069
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 1.00000
 - 2a2. Min : 500.00000
 - 2a3. Max : 500.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.10431
 - 2a3. Max : 0.10431
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

3b2. Rate

└ 2a1. Step : 1.00000

└ 2a2. Min : 1000.00000

└ 2a3. Max : 1000.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

3d. Helper Rate

└ 2a1. Step : 1.00000

└ 2a2. Min : 5000.00000

└ 2a3. Max : 5000.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

3e. Helper Range

└ 2a1. Step : 1.00000

└ 2a2. Min : 0.03000

└ 2a3. Max : 0.03000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

4. Dampers 1

4a. Slow Bump

└ 2a1. Step : 1.00000

└ 2a2. Min : 5500.00000

└ 2a3. Max : 5500.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

4b. Fast Bump

└ 2a1. Step : 1.00000

└ 2a2. Min : 1500.00000

└ 2a3. Max : 1500.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

4c. Slow Rebound

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 7500.00000

└─ 2a3. Max : 7500.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

4d. Fast Rebound

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 4750.00000

└─ 2a3. Max : 4750.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

4. Dampers 2

4a. Slow Bump

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 5500.00000

└─ 2a3. Max : 5500.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

4b. Fast Bump

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 1500.00000

└─ 2a3. Max : 1500.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

4c. Slow Rebound

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 7500.00000

└─ 2a3. Max : 7500.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

- 4d. Fast Rebound

└ 2a1. Step : 1.00000

└ 2a2. Min : 4750.00000

└ 2a3. Max : 4750.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

- 4. Dampers 3

- 4a. Slow Bump

└ 2a1. Step : 1.00000

└ 2a2. Min : 4500.00000

└ 2a3. Max : 4500.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

- 4b. Fast Bump

└ 2a1. Step : 1.00000

└ 2a2. Min : 2000.00000

└ 2a3. Max : 2000.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

- 4c. Slow Rebound

└ 2a1. Step : 1.00000

└ 2a2. Min : 6500.00000

└ 2a3. Max : 6500.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

└ 2a10. Treat As Boolean : false

- 4d. Fast Rebound

└ 2a1. Step : 1.00000

└ 2a2. Min : 4250.00000

└ 2a3. Max : 4250.00000

└ 2a4. Lut : None

└ 2a5. Is Modifiable : false

└ 2a6. Hide Value : false

└ 2a7. Is Negative : false

└ 2a8. Unit : None

└ 2a9. Fractional Digits : 0

- | L 2a10. Treat As Boolean : false
- 4. Dampers 4
 - 4a. Slow Bump
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 4500.00000
 - | 2a3. Max : 4500.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | L 2a10. Treat As Boolean : false
 - 4b. Fast Bump
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 2000.00000
 - | 2a3. Max : 2000.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | L 2a10. Treat As Boolean : false
 - 4c. Slow Rebound
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 6500.00000
 - | 2a3. Max : 6500.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | L 2a10. Treat As Boolean : false
 - 4d. Fast Rebound
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 4250.00000
 - | 2a3. Max : 4250.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | L 2a10. Treat As Boolean : false
- 5. Alignments 1
 - 5a. Pressure
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 20.00000
 - | 2a3. Max : 35.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None

- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false
- 5b. Camber
 - | 2a1. Step : 0.10000
 - | 2a2. Min : -2.50000
 - | 2a3. Max : -1.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5c. Toe
 - | 2a1. Step : 0.01000
 - | 2a2. Min : -0.15000
 - | 2a3. Max : 0.20000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5d. Caster
 - | 2a1. Step : 0.01000
 - | 2a2. Min : -0.09195
 - | 2a3. Max : 0.02195
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5e. Static Camber
 - | 2a1. Step : 1.00000
 - | 2a2. Min : -1.50000
 - | 2a3. Max : -1.50000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5f. Toe Out Linear
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 0.00050
 - | 2a3. Max : 0.00050
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5g. Compound

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 0.00000

└─ 2a3. Max : 2.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : true

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5. Alignments 2

5a. Pressure

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 20.00000

└─ 2a3. Max : 35.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : true

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5b. Camber

└─ 2a1. Step : 0.10000

└─ 2a2. Min : -2.50000

└─ 2a3. Max : -1.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : true

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5c. Toe

└─ 2a1. Step : 0.01000

└─ 2a2. Min : -0.15000

└─ 2a3. Max : 0.20000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : true

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5d. Caster

└─ 2a1. Step : 0.01000

└─ 2a2. Min : -0.09195

└─ 2a3. Max : 0.02195

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

- L 2a10. Treat As Boolean : false
- 5e. Static Camber
 - | 2a1. Step : 1.00000
 - | 2a2. Min : -1.50000
 - | 2a3. Max : -1.50000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5f. Toe Out Linear
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 0.00050
 - | 2a3. Max : 0.00050
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5g. Compound
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 0.00000
 - | 2a3. Max : 2.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 5. Alignments 3
 - 5a. Pressure
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 20.00000
 - | 2a3. Max : 35.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
 - 5b. Camber
 - | 2a1. Step : 0.10000
 - | 2a2. Min : -3.00000
 - | 2a3. Max : -0.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : true
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5c. Toe

└─ 2a1. Step : 0.01000

└─ 2a2. Min : -0.15000

└─ 2a3. Max : 0.20000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : true

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5d. Caster

└─ 2a1. Step : 0.00000

└─ 2a2. Min : 0.00000

└─ 2a3. Max : 0.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5e. Static Camber

└─ 2a1. Step : 1.00000

└─ 2a2. Min : -2.00000

└─ 2a3. Max : -2.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5f. Toe Out Linear

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 0.00020

└─ 2a3. Max : 0.00020

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : false

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5g. Compound

└─ 2a1. Step : 1.00000

└─ 2a2. Min : 0.00000

└─ 2a3. Max : 2.00000

└─ 2a4. Lut : None

└─ 2a5. Is Modifiable : true

└─ 2a6. Hide Value : false

└─ 2a7. Is Negative : false

└─ 2a8. Unit : None

└─ 2a9. Fractional Digits : 0

└─ 2a10. Treat As Boolean : false

5. Alignments 4

5a. Pressure

- 2a1. Step : 1.00000
- 2a2. Min : 20.00000
- 2a3. Max : 35.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5b. Camber

- 2a1. Step : 0.10000
- 2a2. Min : -3.00000
- 2a3. Max : -0.50000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5c. Toe

- 2a1. Step : 0.01000
- 2a2. Min : -0.15000
- 2a3. Max : -0.20000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5d. Caster

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5e. Static Camber

- 2a1. Step : 1.00000
- 2a2. Min : -2.00000
- 2a3. Max : -2.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5f. Toe Out Linear

- 2a1. Step : 1.00000
- 2a2. Min : 0.00020
- 2a3. Max : 0.00020
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

5g. Compound

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000
- 2a3. Max : 2.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

6. Electronics

6a. Tc1

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000
- 2a3. Max : 10.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

6b. Tc2

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

6c. Abs

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

6d. Esc

2a1. Step : 1.00000
2a2. Min : 0.00000
2a3. Max : 2.00000
2a4. Lut : None
2a5. Is Modifiable : true
2a6. Hide Value : false
2a7. Is Negative : false
2a8. Unit : None
2a9. Fractional Digits : 0
2a10. Treat As Boolean : false

6e. Ebb

2a1. Step : 0.00000
2a2. Min : 0.00000
2a3. Max : 0.00000
2a4. Lut : None
2a5. Is Modifiable : false
2a6. Hide Value : false
2a7. Is Negative : false
2a8. Unit : None
2a9. Fractional Digits : 0
2a10. Treat As Boolean : false

6f. Engine Map

2a1. Step : 1.00000
2a2. Min : 0.00000
2a3. Max : 0.00000
2a4. Lut : None
2a5. Is Modifiable : false
2a6. Hide Value : false
2a7. Is Negative : false
2a8. Unit : None
2a9. Fractional Digits : 0
2a10. Treat As Boolean : false

6g. Telemetry Laps To Record

2a1. Step : 1.00000
2a2. Min : 0.00000
2a3. Max : 99.00000
2a4. Lut : None
2a5. Is Modifiable : true
2a6. Hide Value : false
2a7. Is Negative : false
2a8. Unit : None
2a9. Fractional Digits : 0
2a10. Treat As Boolean : false

6h. Turbo Boost Lv

2a1. Step : 0.00000
2a2. Min : 0.00000
2a3. Max : 0.00000
2a4. Lut : None
2a5. Is Modifiable : false
2a6. Hide Value : false
2a7. Is Negative : false
2a8. Unit : None
2a9. Fractional Digits : 0
2a10. Treat As Boolean : false

6i. Ers Deployment Map

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 6j. Ers Recharge Lv

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 6k. Ers Heat Charging

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 7. Aero

- 7a. Collar Positions 1

- 2a1. Step : 1.00000
- 2a2. Min : 140.00000
- 2a3. Max : 140.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 7a. Collar Positions 2

- 2a1. Step : 1.00000
- 2a2. Min : 140.00000
- 2a3. Max : 140.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 7a. Collar Positions 3

- 2a1. Step : 1.00000
- 2a2. Min : 70.00000
- 2a3. Max : 70.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 7a. Collar Positions 4
 - 2a1. Step : 1.00000
 - 2a2. Min : 70.00000
 - 2a3. Max : 70.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 7b. Front Target Height
 - 2a1. Step : 1.00000
 - 2a2. Min : 50.00000
 - 2a3. Max : 200.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 7c. Rear Target Height
 - 2a1. Step : 1.00000
 - 2a2. Min : 50.00000
 - 2a3. Max : 200.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 7d. Front Wing Angle
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 7e. Rear Wing Angle
 - 2a1. Step : 0.00000

- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false
- 8. Fuel Strategy
 - 8a. Fuel
 - 2a1. Step : 1.00000
 - 2a2. Min : 2.00000
 - 2a3. Max : 59.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : true
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 9. Use Single Compound : false

Lamborghini Countach

- 1. Car Data : None
- 2. Mechanical Balance
 - 2a. Arbs 1
 - 2a1. Step : 4000.00000
 - 2a2. Min : 28000.00000
 - 2a3. Max : 68000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : true
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 2a. Arbs 2
 - 2a1. Step : 5000.00000
 - 2a2. Min : 5000.00000
 - 2a3. Max : 50000.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : true
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 2b. Steer Ratio
 - 2a1. Step : 1.45000
 - 2a2. Min : -14.50000
 - 2a3. Max : -14.50000

- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : None
- 2a10. Treat As Boolean : 0
- 2c. Brakes
 - 2c1. Front Bias
 - 2a1. Step : 1.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 100.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : None
 - 2a10. Treat As Boolean : false
 - 2c2. Torque Multiplier
 - 2a1. Step : 10.00000
 - 2a2. Min : 100.00000
 - 2a3. Max : 100.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 2d. Differential
 - 2d1. Power
 - 2a1. Step : 0.02000
 - 2a2. Min : 0.20000
 - 2a3. Max : 0.20000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 2d2. Coast
 - 2a1. Step : 0.04000
 - 2a2. Min : 0.40000
 - 2a3. Max : 0.40000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 2d3. Preload
 - 2a1. Step : 1.00000
 - 2a2. Min : 10.00000

- 2a3. Max : 10.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3. Suspensions 1

- 3a. Wheel Rate

- 2a1. Step : 4000.00000
- 2a2. Min : 50000.00000
- 2a3. Max : 100000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b. Bump Stop Up

- 3b1. Range

- 2a1. Step : 0.01200
- 2a2. Min : 0.02933
- 2a3. Max : 0.02933
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3c. Bump Stop Down

- 3b1. Range

- 2a1. Step : 0.00150
- 2a2. Min : 0.10567
- 2a3. Max : 0.010567
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3d. Helper Rate

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3e. Helper Range

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3. Suspensions 2

- 3a. Wheel Rate

- 2a1. Step : 4000.00000
- 2a2. Min : 50000.00000
- 2a3. Max : 100000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b. Bump Stop Up

- 3b1. Range

- 2a1. Step : 0.01200
- 2a2. Min : 0.02933
- 2a3. Max : 0.02933
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3c. Bump Stop Down
 - 3b1. Range
 - 2a1. Step : 0.00150
 - 2a2. Min : 0.10567
 - 2a3. Max : 0.010567
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 3b2. Rate
 - 2a1. Step : 35.00000
 - 2a2. Min : 350.00000
 - 2a3. Max : 350.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3d. Helper Rate
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 3e. Helper Range
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

3. Suspensions 3

3a. Wheel Rate

- 2a1. Step : 4000.00000
- 2a2. Min : 50000.00000
- 2a3. Max : 100000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b. Bump Stop Up

3b1. Range

- 2a1. Step : 0.01200
- 2a2. Min : 0.02911
- 2a3. Max : 0.02911
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3c. Bump Stop Down

3b1. Range

- 2a1. Step : 0.00150
- 2a2. Min : 0.10589
- 2a3. Max : 0.10589
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None

- | | 2a9. Fractional Digits : 0
- | | 2a10. Treat As Boolean : false

- 3d. Helper Rate

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3e. Helper Range

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3. Suspensions 4

- 3a. Wheel Rate

- | 2a1. Step : 4000.00000
- | 2a2. Min : 50000.00000
- | 2a3. Max : 100000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3b. Bump Stop Up

- 3b1. Range

- | 2a1. Step : 0.01200
- | 2a2. Min : 0.02911
- | 2a3. Max : 0.02911
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 3b2. Rate

- | 2a1. Step : 35.00000
- | 2a2. Min : 350.00000
- | 2a3. Max : 350.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3c. Bump Stop Down

- 3b1. Range
 - 2a1. Step : 0.00150
 - 2a2. Min : 0.10589
 - 2a3. Max : 0.10589
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false

- 3b2. Rate

- 2a1. Step : 35.00000
- 2a2. Min : 350.00000
- 2a3. Max : 350.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3d. Helper Rate

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 3e. Helper Range

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 4. Dampers 1

- 4a. Slow Bump

- 2a1. Step : 1000.00000
- 2a2. Min : 10000.00000
- 2a3. Max : 10000.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 2

- 4a. Slow Bump

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 3

- 4a. Slow Bump

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4. Dampers 4

- 4a. Slow Bump

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4b. Fast Bump

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4c. Slow Rebound

- | 2a1. Step : 1000.00000
- | 2a2. Min : 10000.00000
- | 2a3. Max : 10000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 4d. Fast Rebound

- | 2a1. Step : 100.00000
- | 2a2. Min : -2000.00000
- | 2a3. Max : 2000.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 1

- 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : psi
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.20000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5d. Caster

- | 2a1. Step : 0.00166
- | 2a2. Min : -0.02000
- | 2a3. Max : -0.02000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 0.18000
- | 2a2. Min : -1.80000
- | 2a3. Max : -1.80000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 2

- | 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : psi
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.20000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false

- 2a7. Is Negative : false
- 2a8. Unit : Deg
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5d. Caster

- 2a1. Step : 0.00166
- 2a2. Min : -0.02000
- 2a3. Max : -0.02000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5e. Static Camber

- 2a1. Step : 0.18000
- 2a2. Min : -1.80000
- 2a3. Max : -1.80000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- 2a1. Step : 0.00000
- 2a2. Min : 0.00000
- 2a3. Max : 0.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : false
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5g. Compound

- 2a1. Step : 1.00000
- 2a2. Min : 0.00000
- 2a3. Max : 1.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false
- 2a7. Is Negative : false
- 2a8. Unit : None
- 2a9. Fractional Digits : 0
- 2a10. Treat As Boolean : false

- 5. Alignments 3

- 5a. Pressure

- 2a1. Step : 1.00000
- 2a2. Min : 20.00000
- 2a3. Max : 35.00000
- 2a4. Lut : None
- 2a5. Is Modifiable : true
- 2a6. Hide Value : false

- | 2a7. Is Negative : false
- | 2a8. Unit : psi
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.20000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5d. Caster

- | 2a1. Step : 0.00166
- | 2a2. Min : -0.02000
- | 2a3. Max : -0.02000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 0.18000
- | 2a2. Min : -1.80000
- | 2a3. Max : -1.80000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5. Alignments 4

- | 5a. Pressure

- | 2a1. Step : 1.00000
- | 2a2. Min : 20.00000
- | 2a3. Max : 35.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : psi
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5b. Camber

- | 2a1. Step : 0.10000
- | 2a2. Min : -2.50000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5c. Toe

- | 2a1. Step : 0.01000
- | 2a2. Min : -0.20000
- | 2a3. Max : 0.20000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : Deg
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 5d. Caster

- | 2a1. Step : 0.00166
- | 2a2. Min : -0.02000
- | 2a3. Max : -0.02000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5e. Static Camber

- | 2a1. Step : 0.18000
- | 2a2. Min : -1.80000
- | 2a3. Max : -1.80000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5f. Toe Out Linear

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 5g. Compound

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6. Electronics

- | 6a. Tc1

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 1.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- | 6b. Tc2

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false

- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6c. Abs

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6d. Esc

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6e. Ebb

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6f. Engine Map

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

6g. Telemetry Laps To Record

- | 2a1. Step : 1.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 99.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : true
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None

- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6h. Turbo Boost Lv

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6i. Ers Deployment Map

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6j. Ers Recharge Lv

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 6k. Ers Heat Charging

- | 2a1. Step : 0.00000
- | 2a2. Min : 0.00000
- | 2a3. Max : 0.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None
- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false

- 7. Aero

- | 7a. Collar Positions 1

- | 2a1. Step : 7.00000
- | 2a2. Min : 70.00000
- | 2a3. Max : 70.00000
- | 2a4. Lut : None
- | 2a5. Is Modifiable : false
- | 2a6. Hide Value : false
- | 2a7. Is Negative : false
- | 2a8. Unit : None

- | 2a9. Fractional Digits : 0
- | 2a10. Treat As Boolean : false
- 7a. Collar Positions 2
 - | 2a1. Step : 7.00000
 - | 2a2. Min : 70.00000
 - | 2a3. Max : 70.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7a. Collar Positions 3
 - | 2a1. Step : 5.00000
 - | 2a2. Min : 50.00000
 - | 2a3. Max : 50.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7a. Collar Positions 4
 - | 2a1. Step : 5.00000
 - | 2a2. Min : 50.00000
 - | 2a3. Max : 50.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7b. Front Target Height
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 20.00000
 - | 2a3. Max : 180.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0
 - | 2a10. Treat As Boolean : false
- 7c. Rear Target Height
 - | 2a1. Step : 1.00000
 - | 2a2. Min : 20.00000
 - | 2a3. Max : 180.00000
 - | 2a4. Lut : None
 - | 2a5. Is Modifiable : false
 - | 2a6. Hide Value : false
 - | 2a7. Is Negative : false
 - | 2a8. Unit : None
 - | 2a9. Fractional Digits : 0

- 2a10. Treat As Boolean : false
 - 7d. Front Wing Angle
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
 - 7e. Rear Wing Angle
 - 2a1. Step : 0.00000
 - 2a2. Min : 0.00000
 - 2a3. Max : 0.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : false
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 8. Fuel Strategy
 - 8a. Fuel
 - 2a1. Step : 1.00000
 - 2a2. Min : 1.00000
 - 2a3. Max : 120.00000
 - 2a4. Lut : None
 - 2a5. Is Modifiable : true
 - 2a6. Hide Value : false
 - 2a7. Is Negative : false
 - 2a8. Unit : None
 - 2a9. Fractional Digits : 0
 - 2a10. Treat As Boolean : false
- 9. Use Single Compound : false

7. Car Setup Units [.carsetupunits]

A. Description

UI label map for the garage: which display string sits next to each setup field (N/m, PSI, degrees, litres, or None).

This asset does **not** change physics. Car Setup holds values; Setup Limits hold min/max/step; Setup Units hold how those numbers are *named on screen*. Wrong units mislead telemetry and modding even when the sim maths stay correct.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Display unit strings per setup field	.CARSETUPUNITS	—
Current numeric values	—	5. Car Setup
Legal range / step / lock flags	—	6. Car Setup Limits (may also carry a Unit string)
Real SI physics inside assets	—	Coilover, tyre, wing, engine...

Tree layout mirrors Setup / Limits (mechanical balance, suspensions, dampers, alignments, electronics, aero, fuel). Often shipped as a shared *common_phsx* asset (*setup_units*) reused across many cars.

II. What you are really tuning

1. **Readability profile**— Each leaf is a *string* (or None). Typical tokens :

Field	Role
N/m, Ns/m, N, Nm	Force / rate / damper / preload style SI
%	Ratios (bias, torque multiplier)
PSI	Tyre pressure presentation
°	Angles (camber, toe, wing)
m, mm	Lenghts (bump-stop range vs ride height)
L	Fuel volume
None	No unit label (indices, ratios without a symbol, unused fields)

Choosing *Ns/m* vs a click-style label (when used) is the difference between an engineering garage and a driver “count the clicks” garage.

2. **Consistency with Limits**— If Limits say a damper steps in clicks but Units still print *Ns/m*, the UI lies. Keep Units, Limits steps, and the magnitude of Setup values in the same authoring language for that car.

3. **None is intentional**— Electronics maps, compound index, steer ratio, and many diff locks often show None — they are indices or dimensionless selectors, not physical quantities with a unit glyph.

III. Architecture

1 - MECHANICAL BALANCE (SCHEMA 1)

Unit strings for ARBs, steer ratio, brake bias / torque multiplier / ducts, diff power / coast / preload.

2 - SUSPENSIONS AND DAMPERS (SCHEMA 2-3)

Per-corner (or multi-index) wheel rate, bump-stop range/rate, helpers; slow/fast bump and rebound. Schema reuses 2b1/2b2 under both bump-stop up and down — same pattern as Limits ID reuse.

3 - ALIGNMENTS (SCHEMA 4)

Pressure, camber, toe, caster, static camber, toe-out related field, compound. Schema name here is **Alignments** (Setup file often spells **Alignements**).

4 - ELECTRONICS (SCHEMA 5)

TC, ABS, ESC, EBB, engine map, telemetry laps, turbo boost level, ERS fields — almost always None in the shared example.

5 - AERO, FUEL, COMPOUND MODE (SCHEMA 6-8)

Collar positions, ride heights, wing angles (schema jumps 6c to 6e, no 6d), fuel string, *Use Single Compound* label.

IV. How to read the examples

1 - SHARED SETUP UNITS (COMMON PHSX)

Single shared file, not per-car. Engineering-facing defaults:

- ARBs and wheel rates : N/m
- Dampers : Ns/m
- Bump-stop range : m , rate : N
- Helpers : N/m and m
- Brake bias / torque multiplier : %
- Diff preload : Nm (power/coast None)
- Pressures : PSI ; camber/toe/static/toe-out : $^{\circ}$
- Collar positions: m ; target height : mm ; wings : $^{\circ}$
- Fuel : L
- Electronics and many unused labels : *None*

That is the scientific profile: garage text matches SI-style authoring. A click-profile car would swap damper (and sometimes spring) strings toward dimensionless click labels while Limits steps match detent counts.

V. Practical notes

- Changing a Units string does not convert the stored Setup float — it only changes the caption. To change system (PSI vs bar) you need consistent values, limits, and labels together.
- *None* means “no unit glyph,” not “field missing.”
- Schema quirks: missing aero 6d; bump-stop down reuses 2b1/2b2; fuel is **7a** here (Setup/Limits fuel lives under strategy **8a** — parallel trees, different numbering).
- Field name drift: Units may say *Toe Out Camber* while Setup says *Toe Out Linear* — treat as the toe-related alignment label for that slot.
- Shared *common_phsx* units are a baseline; car-specific overrides only appear if that car ships its own `.CARSETUPUNITS`.

VI. Related assets

- **5. Car Setup** `[.carsetup]` — values shown beside these labels
- **6. Car Setup Limits** `[.carsetuplimits]` — range/step; may duplicate Unit strings
- **3. Car Data** `[.car]` — wire which setup/limits/units package the car uses
- Telemetry / HUD consumers — should agree with these strings for driver-facing consistency

B. Schema

```
1. Mechanical Balance : object
  | 1a. Arbs [x] : string | can have multiple Arbs
  | 1b Steer Ratio : string
  | 1c. Brakes : object
  |   | 1c1. Front Bias : string
  |   | 1c2. Torque Multiplier : string
  |   | 1c3. Brake Ducts [x] : string | can have multiple Brake Ducts
  | 1d. Differential : object
  |   | 1d1. Power : string
  |   | 1d2. Coast : string
  |   | 1d3. Preload : string
2. Suspensions [x] : object | can have multiple Suspensions
  | 2a. Wheel Rate : string
  | 2b. Bump Stop Up : object
  |   | 2b1. Range : string
  |   | 2b2. Rate : string
  | 2c. Bump Stop Down : object
  |   | 2b1. Range : string
  |   | 2b2. Rate : string
  | 2d. Helper Rate : string
  | 2e. Helper Range : string
3. Dampers [x] : object | can have multiple Dampers
  | 3a. Slow Bump : string
  | 3b. Fast Bump : string
  | 3c. Slow Rebound : string
  | 3d. Fast Rebound : string
```

- 4. Alignments [x] : object | can have multiple Alignments
 - 4a. Pressure : string
 - 4b. Camber : string
 - 4c. Toe : string
 - 4d. Caster : string
 - 4e. Static Camber : string
 - 4f. Toe Out Camber : string
 - 4g. Compound : string
- 5. Electronics : object
 - 5a. Tc1 : string
 - 5b. Tc2 : string
 - 5c. Abs : string
 - 5d. Esc : string
 - 5e. Ebb : string
 - 5f. Engine Map : string
 - 5g. Telemetry Laps To Record : string
 - 5h. Turbo Boost Lv : string
 - 5i. Ers Deployment Map : string
 - 5j. Ers Recharge Lv : string
 - 5k. Ers Heat Charging : string
- 6. Aero : object
 - 6a. Collar Positions [x] : string | can have multiple Collar Positions
 - 6b. Front Target Height : string
 - 6c. Rear Target Height : string
 - 6e. Front Wing Angle : string
 - 6f. Rear Wing Angle : string
- 7. Fuel Strategy : object
 - 7a. Fuel : string
- 8. Use Single Compound : string

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Mechanical Balance	None (Object)	Parent block defining UI unit strings for chassis balance parameters.
1a.	Arbs	None (String)	Display unit label for anti-roll bar stiffness; index 1 = front, 2 = rear (e.g., N/m).
1b.	Steer Ratio	None (String)	Display unit label for steering ratio (e.g., ratio value or None if dimensionless).
1c.	Brakes	None (Object)	Brake setup unit sub-block within Mechanical Balance.
1c1.	Front Bias	None (String)	Display unit label for brake balance front percentage (e.g., %).
1c2.	Torque Multiplier	None (String)	Display unit label for global braking torque multiplier (e.g., %).

ID	Name	Unit of Measurement	Description
1c3.	Brake Ducts	None (String)	Display unit label for brake duct opening level(s); multiple entries for front/rear.
1d.	Differential	None (Object)	Differential setup unit sub-block within Mechanical Balance.
1d1.	Power	None (String)	Display unit label for on-throttle differential lock (e.g., %, Nm, or None).
1d2.	Coast	None (String)	Display unit label for coast differential lock (e.g., %, Nm, or None).
1d3.	Preload	None (String)	Display unit label for differential preload torque (e.g., Nm).
2.	Suspensions	None (Object array, per wheel)	Per-corner suspension unit block; indices 1–4 (or more) map to wheel positions.
2a.	Wheel Rate	None (String)	Display unit label for wheel-rate spring stiffness (e.g., N/m, N/mm, kg/mm, Hz).
2b.	Bump Stop Up	None (Object)	Upper bump-stop unit sub-block within each Suspensions entry.
2b1.	Range	None (String)	Display unit label for upper bump-stop travel range under 2b Bump Stop Up (e.g., m).
2b2.	Rate	None (String)	Display unit label for upper bump-stop force rate under 2b Bump Stop Up (e.g., N).
2c.	Bump Stop Down	None (Object)	Lower bump-stop unit sub-block; child fields reuse IDs 2b1 and 2b2 per schema.
2d.	Helper Rate	None (String)	Display unit label for helper/tender spring stiffness (e.g., N/m).
2e.	Helper Range	None (String)	Display unit label for helper spring travel range (e.g., m).
3.	Dampers	None (Object array, per wheel)	Per-corner damper unit block; indices 1–4 (or more) map to wheel positions.
3a.	Slow Bump	None (String)	Display unit label for slow-speed compression damping (e.g., Ns/m or clicks).
3b.	Fast Bump	None (String)	Display unit label for fast-speed compression damping (e.g., Ns/m or clicks).
3c.	Slow Rebound	None (String)	Display unit label for slow-speed rebound damping (e.g., Ns/m or clicks).

ID	Name	Unit of Measurement	Description
3d.	Fast Rebound	None (String)	Display unit label for fast-speed rebound damping (e.g., Ns/m or clicks).
4.	Alignments	None (Object array, per wheel)	Per-corner alignment and tyre unit block; indices 1–4 map to wheel positions.
4a.	Pressure	None (String)	Display unit label for cold tyre pressure (e.g., PSI, bar).
4b.	Camber	None (String)	Display unit label for camber angle (e.g., °).
4c.	Toe	None (String)	Display unit label for toe angle (e.g., °).
4d.	Caster	None (String)	Display unit label for caster angle (e.g., ° or None).
4e.	Static Camber	None (String)	Display unit label for computed/ static camber (e.g., °).
4f.	Toe Out Camber	None (String)	Display unit label for toe-out linear/camber curve coefficient (e.g., °).
4g.	Compound	None (String)	Display unit label for tyre compound index (often None / dimensionless).
5.	Electronics	None (Object)	Driver-aid and powertrain map unit labels for garage UI.
5a.	Tc1	None (String)	Display unit label for primary Traction Control map level.
5b.	Tc2	None (String)	Display unit label for secondary TC map level.
5c.	Abs	None (String)	Display unit label for ABS map level.
5d.	Esc	None (String)	Display unit label for ESP/stability control map level.
5e.	Ebb	None (String)	Display unit label for Electronic Brake Balance level.
5f.	Engine Map	None (String)	Display unit label for engine power map index.
5g.	Telemetry Laps To Record	None (String)	Display unit label for telemetry lap recording count.
5h.	Turbo Boost Lv	None (String)	Display unit label for turbo boost level (e.g., bar).
5i.	Ers Deployment Map	None (String)	Display unit label for ERS deployment map index.
5j.	Ers Recharge Lv	None (String)	Display unit label for ERS recharge level.

ID	Name	Unit of Measurement	Description
5k.	Ers Heat Charging	None (String)	Display unit label for MGU-H heat charging level.
6.	Aero	None (Object)	Aerodynamic setup unit block: ride heights, collar positions, and wing angles.
6a.	Collar Positions	None (String)	Display unit label for per-corner spring collar position (e.g., m, mm); indices 1–4 = LF, RF, LR, RR.
6b.	Front Target Height	None (String)	Display unit label for front target ride height (e.g., mm).
6c.	Rear Target Height	None (String)	Display unit label for rear target ride height (e.g., mm).
6e.	Front Wing Angle	None (String)	Display unit label for front aero device angle (e.g., °). Schema skips 6d.
6f.	Rear Wing Angle	None (String)	Display unit label for rear wing angle (e.g., °).
7.	Fuel Strategy	None (Object)	Fuel load planning unit block.
7a.	Fuel	None (String)	Display unit label for fuel load volume (e.g., L).
8.	Use Single Compound	None (String)	Display unit label for the single-compound toggle (often None / boolean).

D. Example data

I. Chosen Car Engine for Example

- Setup Units (slug : setup_units) [common_phsx]

II. Example

Setup Units

```

1. Mechanical Balance
  | 1a. Arbs 1 : N/m
  | 1a. Arbs 2 : N/m
  | 1b Steer Ratio : None
  | 1c. Brakes
  |   | 1c1. Front Bias : %
  |   | 1c2. Torque Multiplier : %
  |   | 1c3. Brake Ducts : None
  | 1d. Differential
  |   | 1d1. Power : None
  |   | 1d2. Coast : None
  |   | 1d3. Preload : Nm

```

- 2. Suspensions 1
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 2
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 3
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 4
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 5
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2d. Helper Rate : N/m
 - 2e. Helper Range : m
- 2. Suspensions 6
 - 2a. Wheel Rate : N/m
 - 2b. Bump Stop Up
 - 2b1. Range : m
 - 2b2. Rate : N
 - 2c. Bump Stop Down

- | | | 2b1. Range : m
- | | | 2b2. Rate : N
- | | 2d. Helper Rate : N/m
- | | 2e. Helper Range : m
- 3. Dampers 1
 - | | 3a. Slow Bump : Ns/m
 - | | 3b. Fast Bump : Ns/m
 - | | 3c. Slow Rebound : Ns/m
 - | | 3d. Fast Rebound : Ns/m
- 3. Dampers 2
 - | | 3a. Slow Bump : Ns/m
 - | | 3b. Fast Bump : Ns/m
 - | | 3c. Slow Rebound : Ns/m
 - | | 3d. Fast Rebound : Ns/m
- 3. Dampers 3
 - | | 3a. Slow Bump : Ns/m
 - | | 3b. Fast Bump : Ns/m
 - | | 3c. Slow Rebound : Ns/m
 - | | 3d. Fast Rebound : Ns/m
- 3. Dampers 4
 - | | 3a. Slow Bump : Ns/m
 - | | 3b. Fast Bump : Ns/m
 - | | 3c. Slow Rebound : Ns/m
 - | | 3d. Fast Rebound : Ns/m
- 3. Dampers 5
 - | | 3a. Slow Bump : Ns/m
 - | | 3b. Fast Bump : Ns/m
 - | | 3c. Slow Rebound : Ns/m
 - | | 3d. Fast Rebound : Ns/m
- 3. Dampers 6
 - | | 3a. Slow Bump : Ns/m
 - | | 3b. Fast Bump : Ns/m
 - | | 3c. Slow Rebound : Ns/m
 - | | 3d. Fast Rebound : Ns/m
- 4. Alignments 1
 - | | 4a. Pressure : PSI
 - | | 4b. Camber : °
 - | | 4c. Toe : °
 - | | 4d. Caster : None
 - | | 4e. Static Camber : °
 - | | 4f. Toe Out Camber : °
 - | | 4g. Compound : None
- 4. Alignments 2
 - | | 4a. Pressure : PSI
 - | | 4b. Camber : °
 - | | 4c. Toe : °
 - | | 4d. Caster : None
 - | | 4e. Static Camber : °
 - | | 4f. Toe Out Camber : °
 - | | 4g. Compound : None
- 4. Alignments 3
 - | | 4a. Pressure : PSI
 - | | 4b. Camber : °
 - | | 4c. Toe : °
 - | | 4d. Caster : None
 - | | 4e. Static Camber : °

- | 4f. Toe Out Camber : °
 - | 4g. Compound : None
- 4. Alignments 4
 - | 4a. Pressure : PSI
 - | 4b. Camber : °
 - | 4c. Toe : °
 - | 4d. Caster : None
 - | 4e. Static Camber : °
 - | 4f. Toe Out Camber : °
 - | 4g. Compound : None
- 5. Electronics
 - | 5a. Tc1 : None
 - | 5b. Tc2 : None
 - | 5c. Abs : None
 - | 5d. Esc : None
 - | 5e. Ebb : None
 - | 5f. Engine Map : None
 - | 5g. Telemetry Laps To Record : None
 - | 5h. Turbo Boost Lv : None
 - | 5i. Ers Deployment Map : None
 - | 5j. Ers Recharge Lv : None
 - | 5k. Ers Heat Charging : None
- 6. Aero
 - | 6a. Collar Positions 1 : m
 - | 6a. Collar Positions 2 : m
 - | 6a. Collar Positions 3 : m
 - | 6a. Collar Positions 4 : m
 - | 6b. Front Target Height : mm
 - | 6c. Rear Target Height : mm
 - | 6e. Front Wing Angle : °
 - | 6f. Rear Wing Angle : °
- 7. Fuel Strategy
 - | 7a. Fuel : L
- 8. Use Single Compound : None

8. Car Tuning Parts [.tuningpart]

A. Description

Modular switchboard for vehicle variants: each part picks a physics domain and either redirects asset paths or embeds override data (engine tune maths, aero package, full performance-mode setups).

Baseline Car Data still defines the stock car. Activating a tuning part hot-swaps or overlays pieces of that stack — drift drivetrain, alternate gearbox, “No ABS / No TC” limits — without duplicating the whole vehicle folder.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Which domain this part overrides	<i>Physics Tuning</i> enum	—
Path redirects to alternate assets	<i>[physics_tuning]</i> object	Target .CARENGINE, .DRIVETRAIN, ...
Engine multiplier / waveform tune	Engine Tube fields	Base curve still in .CARENGINE
Embedded aero package / wing index	Aero Package / Wing fields	.WING, .SURFACE3D
Embedded garage packages	Performance Modes block	Also exist as standalone .CARSETUP
UI category label	<i>Car Part Type</i>	Showroom / upgrade menus
Stock wiring of the car	—	<u>3. Car Data</u>

One car can ship many .TUNINGPART files (Supra drift set, Datsun upgrade catalogue, Cup electronics packs). Selection is combinatorial: engine part + gearbox part + setup limits part, etc.

II. What you are really tuning

1. **Domain selector** — *Physics Tuning* chooses the override family: Engine, Engine Tune, Gearbox, Drivetrain, Clutch, Brakes / Brake System, Electronics, Suspensions, Suspensions Geometry, Wing, Aero Package, Setup, Setup Limits, Performance Modes Tuning, or None.

The object under field **2** is keyed by that choice — Engine parts expose an Engine path block; Setup Limits parts expose a Setup Limits path block. Wrong enum + wrong nested keys = part that does nothing.

2. **Path redirects** — Most mechanical parts are thin wrappers: Path (or front/rear coilover / geometry paths) pointing at an alternate asset. Examples: Supra drift .CARENGINE / .DRIVETRAIN / .GEARBOX / .CARSETUP / .CARSETUPLIMITS; Datsun 5-speed gearbox, L28 engine, LSD drivetrain, G-nose coilovers.
3. **Engine Tune scalars** — When tuning type is Engine Tune (not a full Engine path swap): multiplier, add, and cosine amplitude / period / phase reshape the existing power delivery without replacing the whole .CARENGINE file.
4. **Brakes and suspension packages** — Brake System / Brakes paths (front/rear compounds), coilover front/rear, suspension geometry front/rear. Lets a single menu item restyle the undercarriage or calipers.

5. **Aero and performance modes** — Wing index override; full Aero Package Data (downforce elements, surface3d paths, controllers). Performance Modes Tuning can embed named modes with electronics, brakes, dampers, diffs, lock controllers, AWD clutches, turbo settings — a setup package living inside the tuning part.
6. **Electronics restrictors** — *Physics Tuning : None* plus *Car Part Type : Mechanics_Electronics* (Porsche Cup “No ABS No TC” / “Only ABS”) tags an electronics policy. Companion parts often retarget **Setup Limits** so the garage cannot re-enable stripped aids.

III. Architecture

1 - HEADER (SCHEMA 1 AND 3)

Physics Tuning enum; *Car Part Type* enum/string for UI taxonomy (*Mechanics_Engine*, *Mechanics_Drivetrain*, *Mechanics_Electronics*, exterior categories, or None for pure setup/limits packs).

2 - CONDITIONAL PAYLOAD (SCHEMA 2)

[*physics_tuning*] object whose fields depend on enum 1:

Family	Typical contents
Path-based modules	Engine / Gearbox / Drivetrain / Clutch / Brake System / Setup / Setup Limits <i>Path</i>
Engine Tune	Multiplier, Add, Cos Ampl / Period / Phase
Brakes	Front / Rear compound paths
Suspensions	Coilover front/rear paths
Suspensions Geometry	Geometry path front/rear
Wing	Wing index override
Aero Package	Slip/speed multi, downforces array, lift/drag maps, wings paths
Performance Modes Tuning	Named modes with nested electronics / chassis / driveline / turbo settings

3 - NESTED CONTROLLERS

Inside aero downforces and performance-mode diffs/locks, the familiar stage pipeline reappears (Input, Combinator, Lut, Filter, limits) — same pattern as brake EBB / turbo controllers.

IV. How to read the examples

1 - TOYOTA SUPRA MK IV (DRIFT PACKAGE SET)

Seven parts that rebuild the car for drift: suspensions geometry, drivetrain, engine, coilovers, gearbox, setup, setup limits — each with *Physics Tuning* matching the domain and a path into ...*drift*... assets. Shows the “variant pack” pattern: many small redirects, one coherent alternate identity.

2 - DATSUN 240Z FAIRLADY (CATALOGUE UPGRADES)

Eight parts as optional upgrades: 5-speed gearbox, brake system, clutch, G-nose coilovers, matching setup + limits, L28 engine, LSD drivetrain. Classic aftermarket tree — pick gearbox and LSD independently of engine.

3 - PORSCHE 992 GT3 CUP (ELECTRONICS / BOP STYLE)

“No ABS No TC” and “Only ABS” use *Physics Tuning : None* with *Mechanics_Electronics*. Paired Setup Limits parts force garage rules for those electronics states. Regulation by menu item + limits file, not by rewriting Car Electronics maps in place.

V. Practical notes

- Always read **field 1 first**: it tells you which subtree under field 2 is meaningful.
- A path redirect must point at a real asset of the right type; a gearbox part pointing at a `.BRAKESYSTEM` (OCR noise in some dumps) is a content bug.
- Setup / Setup Limits parts often set *Car Part Type : None* — they are garage packages, not showroom body parts.
- Electronics “None” tuning still needs companion limits if you want the UI locked to the intended aid set.
- Performance Modes inside a tuning part can duplicate what Car Data performance modes already do — check which layer actually loads in-session.
- Schema typos to expect: Front Brias, Brakes rear path typed oddly in the tree dump; enum table OCR wrapping around Physics Tuning values.

VI. Related assets

- **4. Car Data [.car]** — baseline paths that parts override
- **4 / 13 / 14 / 10 / 1. Engine / Drivetrain / Gearbox / Clutch / Brake System** — common redirect targets
- **5 / 6. Car Setup / Setup Limits** — drift and Cup companion packs
- **11 / 17 / 20. Coilover / Suspension / Wing** — suspension and aero redirects
- **9. Car Electronics** — maps still live here; tuning parts often only gate them via limits / None tags

B. Schema

```
| 1. Physics Tuning : enum
| 2. [physics_tuning] : object | object key take the value of 1. Physics
Tuning. Key-Value on this object will depend on 1. selection
| | 2a. (Engine, Gearbox, Drivetrain, Clutch, Brake System, Electronics,
Wing, Setup, Setup Limits) Path : string - path
| | 2b. (Engine Tune) Multiplier : float
| | 2c. (Engine Tune) Add : float
| | 2d. (Engine Tune) Cos Ampl : float
| | 2e. (Engine Tune) Cos Period : float
```

- 2f. (Engine Tune) Cos Phase : float
- 2g. (Brakes) Front Path : string - path
- 2h. (Brakes) Rear Path : float
- 2i. (Suspensions) Coilover Path Front : string - path
- 2j. (Suspensions) Coilover Path Rear : string - path
- 2k. (Suspensions Geometry) Geometry Path Front : string - path
- 2l. (Suspensions Geometry) Geometry Path Rear : string - path
- 2m. (Wing) Wing Index Override : integer
- 2n. (Aero Package) Data : object
 - 2n1. Slip Gain Mult : float
 - 2n2. Speed Factor Mult : float
 - 2n3. Downforces [x] : object | can have multiple Downforces
 - 2n3a. Position : x, y, z float
 - 2n3b. Cl Gain : float
 - 2n3c. Cd Gain : float
 - 2n3d. Yaw Gain : float
 - 2n3e. Drag Per Cool Transfer : float
 - 2n3f. Damage C L [x] : float | can have multiple Damage C L
 - 2n3g. Damage C D [x] : float | can have multiple Damage C D
 - 2n3h. Downforce Controllers [x] : object | can have multiple

Downforce Controllers

- 2n3h1. Combinator Mode : enum
- 2n3h2. Input : enum
- 2n3h3. Filter : float
- 2n3h4. Up Limit : float
- 2n3h5. Down Limit : float
- 2n3h6. Lut : string - path
- 2n3i. Lift Per Front Angle : float
- 2n3j. Lift Per Rear Angle : float
- 2n3k. Drag Per Front Angle : float
- 2n3l. Drag Per Rear Angle : float
- 2n3m. Default Front Angle : float
- 2n3n. Default Rear Angle : float
- 2n4. Front Lift : string - path
- 2n5. Rear Lift : string - path
- 2n6. Drag : string - path
- 2n7. Wings Path [x] : string - path
- 2o. (Performance Modes Tuning) Performance Modes [x] : object | can

have multiple Performance Modes

- 2o1. Performance Mode Name : string
- 2o2. Electronics Settings : object
 - 2o2a. Tc1 : float
 - 2o2b. Tc2 : float
 - 2o2c. Abs : float
 - 2o2d. Esc : float
 - 2o2e. Ebb : float
 - 2o2f. Engine Map : float
 - 2o2g. Telemetry Laps To Record : float
 - 2o2h. Turbo Boost Lv : float
 - 2o2i. Ers Deployment Map : float
 - 2o2j. Ers Recharge Lv : float
 - 2o2k. Ers Heat Charging : float
- 2o3. Brake Settings : object
 - 2o3a. Front Bias : float
 - 2o3b. Torque Multiplier : float
 - 2o3c. Brake Ducts [x] : float | can have multiple Brake Ducts

```
| | | 2o4. Damper Settings [x] : object | can have multiple Damper
Settings
| | | | 2o4a. Slow Bump : float
| | | | 2o4b. Fast Bump : float
| | | | 2o4c. Slow Rebound : float
| | | | 2o4d. Fast Rebound : float
| | | | 2o5. Differential Data : object
| | | | | 2o5a. Type : enum
| | | | | 2o5b. Power : float
| | | | | 2o5c. Coast : float
| | | | | 2o5d. Preload : float
| | | | | 2o5e. Front Share : float
| | | | | 2o5f. Torque Bias Ratio Power : float
| | | | | 2o5g. Torque Bias Ratio Coast : float
| | | | | 2o5h. Thermal Capacity : float
| | | | | 2o5i. Surface : float
| | | | | 2o5j. Heat Transfer Coeff : float
| | | | | 2o5k. Wear Factor : float
| | | | | 2o5l. Friction Reduction With T : float
| | | | | 2o5m. Friction Ref T : float
| | | | 2o6. Four W D Differentials : object
| | | | | 2o6a. Front Diff : object
| | | | | | 2o6a1. Type : enum
| | | | | | 2o6a2. Power : float
| | | | | | 2o6a3. Coast : float
| | | | | | 2o6a4. Preload : float
| | | | | | 2o6a5. Front Share : float
| | | | | | 2o6a6. Torque Bias Ratio Power : float
| | | | | | 2o6a7. Torque Bias Ratio Coast : float
| | | | | | 2o6a8. Thermal Capacity : float
| | | | | | 2o6a9. Surface : float
| | | | | | 2o6a10. Heat Transfer Coeff : float
| | | | | | 2o6a11. Wear Factor : float
| | | | | | 2o6a12. Friction Reduction With T : float
| | | | | | 2o6a13. Friction Ref T : float
| | | | | 2o6b. Center Diff : object
| | | | | | 2o6b1. Type : enum
| | | | | | 2o6b2. Power : float
| | | | | | 2o6b3. Coast : float
| | | | | | 2o6b4. Preload : float
| | | | | | 2o6b5. Front Share : float
| | | | | | 2o6b6. Torque Bias Ratio Power : float
| | | | | | 2o6b7. Torque Bias Ratio Coast : float
| | | | | | 2o6b8. Thermal Capacity : float
| | | | | | 2o6b9. Surface : float
| | | | | | 2o6b10. Heat Transfer Coeff : float
| | | | | | 2o6b11. Wear Factor : float
| | | | | | 2o6b12. Friction Reduction With T : float
| | | | | | 2o6b13. Friction Ref T : float
| | | | 2o6c. Rear Diff : object
| | | | | 2o6c1. Type : enum
| | | | | 2o6c2. Power : float
| | | | | 2o6c3. Coast : float
| | | | | 2o6c4. Preload : float
| | | | | 2o6c5. Front Share : float
| | | | 2o6c6. Torque Bias Ratio Power : float
```

- 2o6c7. Torque Bias Ratio Coast : float
- 2o6c8. Thermal Capacity : float
- 2o6c9. Surface : float
- 2o6c10. Heat Transfer Coeff : float
- 2o6c11. Wear Factor : float
- 2o6c12. Friction Reduction With T : float
- 2o6c13. Friction Ref T : float
- 2o7. Front Lock Controllers : object
 - 2o7a. Name : string
 - 2o7b. Stages [x] : object | can have multiple Stages
 - 2o7b1. Input Var : enum
 - 2o7b2. Combinator Mode : enum
 - 2o7b3. Lut : string – path
 - 2o7b4. Filter Gain : float
 - 2o7b5. Up Limit : float
 - 2o7b6. Down Limit : float
 - 2o7b7. Current Value : float
 - 2o7b8. Const Value : float
- 2o8. Center Lock Controllers : object
 - 2o8a. Name : string
 - 2o8b. Stages [x] : object
 - 2o8b1. Input Var : enum
 - 2o8b2. Combinator Mode : enum
 - 2o8b3. Lut : string – path
 - 2o8b4. Filter Gain : float
 - 2o8b5. Up Limit : float
 - 2o8b6. Down Limit : float
 - 2o8b7. Current Value : float
 - 2o8b8. Const Value : float
- 2o9. Rear Lock Controllers : object
 - 2o9a. Name : string
 - 2o9b. Stages [x] : object
 - 2o9b1. Input Var : enum
 - 2o9b2. Combinator Mode : enum
 - 2o9b3. Lut : string – path
 - 2o9b4. Filter Gain : float
 - 2o9b5. Up Limit : float
 - 2o9b6. Down Limit : float
 - 2o9b7. Current Value : float
 - 2o9b8. Const Value : float
- 2o10. Awd Clutches [x] : object
 - 2o10a. Position : integer
 - 2o10b. Preload : float
 - 2o10c. Controllers : object
 - 2o10c1. Name : string
 - 2o10c2. Stages [x] : object
 - 2o10c2a. Input Var : enum
 - 2o10c2b. Combinator Mode : enum
 - 2o10c2c. Lut : string – path
 - 2o10c2d. Filter Gain : float
 - 2o10c2e. Up Limit : float
 - 2o10c2f. Down Limit : float
 - 2o10c2g. Current Value : float
 - 2o10c2h. Const Value : float
- 2o11. Turbo Controllers [x] : object
 - 2o11a. Name : string

- └ 2o11b. Stages [x] : object
 - └ 2o11b1. Input Var : enum
 - └ 2o11b2. Combinator Mode : enum
 - └ 2o11b3. Lut : string – path
 - └ 2o11b4. Filter Gain : float
 - └ 2o11b5. Up Limit : float
 - └ 2o11b6. Down Limit : float
 - └ 2o11b7. Current Value : float
 - └ 2o11b8. Const Value : float
- └ 2o12. Turbo Settings : object
 - └ 2o12a. Boost Lv : float
- └ 3. Car Part Type : enum

Enum - Car Tuning Pars

1	Physics Tuning	<None>, Engine Tune, Engine, Gearbox, Drivetrain, Clutch, Brakes, Brake System, Electronics, Suspensions, Suspensions Geometry, Wing, Aero Package, Setup, Setup Limits, Performance Modes Tuning
2n3h1	Combinator Mode	UndefinedMode, AddH, MultH, AddYawGain, MultYawGain, AddClGain, MultClGain
2n3h2	Input	UndefinedInput, Brake, Gas, Yaw, LatG, LonG, Steer, Speed, SusTravelLR, SusTravelRR
2o5a	Type	LSD, Spool, Torsen, EpicyclicTorsen, EpicyclicLSD, TorqueVectoring
2o7b1	Input Var	UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque
2o7b2	Combinator Mode	UndefinedMode, Add, Mult

3	Car Part Type	<None>, Exterior_Body, Exterior_FrontBumper, Exterior_FrontBumperLip, Exterior_FrontFogLights, Exterior_FrontBumperCanards, Exterior_FrontLights, Exterior_RearLights, Exterior_RearFogLights, Exterior_AuxLights, Exterior_FrontAirIntakeGrid, Exterior_FrontWindshield, Exterior_Hood, Exterior_HoodAirscoopes, Exterior_SlideFrontBody, Exterior_SlideMirrors, Exterior_LeftDoor, Exterior_RightDoor, Exterior_Skirts, Exterior_SlideRearBody, Exterior_RearDiffuser, Exterior_RearTailgateSpoiler, Exterior_RearSpoiler, Exterior_RoofAirScopes, Exterior_RearWindshield, Exterior_RimFL, Exterior_RimFR, Exterior_RimRL, Exterior_RimRR, Exterior_RimBlurFL, Exterior_RimBlurFR, Exterior_RimBlurRL, Exterior_RimBlurRR, Exterior_TyreFL, Exterior_TyreFR, Exterior_TyreRL, Exterior_TyreRR, Exterior_CaliperFL, Exterior_CaliperFR, Exterior_CaliperRL, Exterior_CaliperRR, Exterior_DiscFL, Exterior_DiscFR, Exterior_DiscRL, Exterior_DiscRR, Exterior_PlateFront, Exterior_PlateRear, Exterior_Exhaust, Exterior_Lumirank, Exterior_FrontSplitter, Exterior_FuelCap, Exterior_Decal, Exterior_Storage, Exterior_Top, Interior_Body, Interior_FrontSeats, Interior_RearSeats, Interior_FireExtinguisher, Interior_SteeringWheelAndHub, Interior_MirrorGadget, Interior_FloorCarpets, Interior_HandbrakeLever, Interior_GearShifterLever, Interior_RollbarOrCage, Interior_CameraDevice, Interior_CockpitLights, Interior_AnalogDashboard, Interior_DigitalDashboard, Interior_SeatbeltOn, Interior_SeatbeltOff, Interior_HarnessOn, Interior_HarnessOff, Interior_SafetyNetOn, Interior_SafetyNetOff, Interior_PedalAccelerator, Interior_PedalBrake, Interior_PedalClutch, Interior_Storage, Interior_CupHolder, Mechanics_SuspensionGeometry_Front, Mechanics_SuspensionGeometry_Rear, Mechanics_SuspensionCoilover_Front, Mechanics_SuspensionCoilover_Rear, Mechanics_Engine, Mechanics_Turbo, Mehanics_Ecu, Mechanics_Pistons, Mechanics_Rod, Mechanics_Cam_Shaft, Mechanics_Crank_Shaft, Mechanics_Head, Mechanics_Injection_System, Mechanics_Radiator, Mechanics_Oil_Radiator, Mechanics_Intercooler, Mechanics_Air_Filter, Mechanics_Intake_Manifold, Mechanics_Exhaust_Manifold, Mechanics_Catalitic_Converter, Mechanics_Center_Pipe, Mechanics_Muffler, Mechanics_Differential, Mechanics_Clutch, Mechanics_Gearbox, Mechanics_Electronics, Mechanics_Drivetrain, Mechanics_Aero_0, Mechanics_Aero_1, Mechanics_Aero_2, Mechanics_Aero_3, Mechanics_Aero_4, Mechanics_Aero_5, Mechanics_Aero_6, Mechanics_Aero_7, Mechanics_Aero_8, Mechanics_Aero_9, Mechanics_Downforce_0, Mechanics_Downforce_1, Mechanics_Downforce_2, Mechanics_Downforce_3, Mechanics_PadsFL, Mechanics_PadsFR, Mechanics_PadsRL, Mechanics_PadsRR, Mechanics_CaliperFL, Mechanics_CaliperFR, Mechanics_CaliperRL, Mechanics_CaliperRR, Mechanics_DiscFL, Mechanics_DiscFR, Mechanics_DiscRL, Mechanics_DiscRR
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C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Physics Tuning	None (Enum)	Primary enum selector declaring which mechanical domain this tuning part overrides (Engine, Gearbox, Drivetrain, Setup Limits, Aero Package, Performance Modes Tuning, etc.).
2.	[physics_tuning]	None (Object)	Dynamic object whose JSON key equals the Physics Tuning enum value; only the branch matching field 1 is populated at runtime.
2a.	Path	None (File path)	File path redirect for component swap tuning parts; active when Physics Tuning is Engine, Gearbox, Drivetrain, Clutch, Brake System, Electronics, Wing, Setup, or Setup Limits.
2b.	Multiplier	Dimensionless float	Engine tune torque multiplier scalar applied on top of the base power curve when Physics Tuning is Engine Tune.
2c.	Add	Nm (Newton-meters)	Flat torque offset added to the engine output curve when Physics Tuning is Engine Tune.
2d.	Cos Ampl	Dimensionless coefficient	Cosine-wave torque modulation amplitude for periodic engine tune effects (Engine Tune mode).
2e.	Cos Period	s (Seconds) or rad	Period of the cosine torque modulation in Engine Tune mode.
2f.	Cos Phase	rad (Radians) or deg	Phase offset of the cosine torque modulation in Engine Tune mode.
2g.	Front Path	None (File path)	Path to an alternative front brake asset (.brakes or related) when Physics Tuning is Brakes.
2h.	Rear Path	Depends on parameter (float)	Path or scalar reference for rear brake configuration when Physics Tuning is Brakes (schema types as float; verify against asset).
2i.	Coilover Path Front	None (File path)	Path to a replacement front .COILOVER asset when Physics Tuning is Suspensions.
2j.	Coilover Path Rear	None (File path)	Path to a replacement rear .COILOVER asset when Physics Tuning is Suspensions.
2k.	Geometry Path Front	None (File path)	Path to a replacement front .SUSPENSION geometry asset when Physics Tuning is Suspensions Geometry.

ID	Name	Unit of Measurement	Description
2l.	Geometry Path Rear	None (File path)	Path to a replacement rear .SUSPENSION geometry asset when Physics Tuning is Suspensions Geometry.
2m.	Wing Index Override	None (Integer)	Integer index selecting which wing aero profile slot to override when Physics Tuning is Wing.
2n.	Data	None (Object)	Root aero package data block when Physics Tuning is Aero Package; contains downforce modifiers and wing path overrides.
2n1.	Slip Gain Mult	Dimensionless multiplier	Global multiplier applied to tyre slip-driven aero gain within the aero package.
2n2.	Speed Factor Mult	Dimensionless multiplier	Multiplier scaling speed-dependent aero factor curves in the package.
2n3.	Downforces [x]	None (Object array)	Per-element downforce modifier block; multiple entries for distinct aero surfaces or zones.
2n3a.	Position	m (Meters, X / Y / Z)	3D position offset (X/Y/Z) of this downforce element relative to the vehicle reference frame.
2n3b.	Cl Gain	Dimensionless coefficient	Lift coefficient (CL) gain multiplier for this downforce element.
2n3c.	Cd Gain	Dimensionless coefficient	Drag coefficient (CD) gain multiplier for this downforce element.
2n3d.	Yaw Gain	Dimensionless coefficient	Yaw-angle sensitivity gain for this downforce element.
2n3e.	Drag Per Cool Transfer	Coefficient	Drag penalty per unit of cooling airflow transfer through this element.
2n3f.	Damage C L [x]	Dimensionless coefficient	Array of lift-damage coefficients reducing CL under aerodynamic damage states.
2n3g.	Damage C D [x]	Dimensionless coefficient	Array of drag-damage coefficients reducing CD efficiency under damage.
2n3h.	Downforce Controllers [x]	None (Object array)	Dynamic downforce controller block; multiple controllers can combine inputs (brake, steer, speed, etc.).
2n3h1.	Combinator Mode	None (Enum)	Enum selecting how this controller combines with others (AddH, MultH, AddClGain, MultClGain, etc.).

ID	Name	Unit of Measurement	Description
2n3h2.	Input	None (Enum)	Input signal source for the controller (Brake, Gas, Yaw, LatG, Speed, SusTravellR, etc.).
2n3h3.	Filter	Coefficient (Smoothing multiplier)	Low-pass filter time constant smoothing the controller input signal.
2n3h4.	Up Limit	Depends on input variable	Upper clamp limit for controller output contribution.
2n3h5.	Down Limit	Depends on input variable	Lower clamp limit for controller output contribution.
2n3h6.	Lut	None (.CURVE file path)	Optional .CURVE LUT mapping filtered input to aero output modifier.
2n3i.	Lift Per Front Angle	Coefficient per deg	Lift sensitivity per degree of front wing angle for this package.
2n3j.	Lift Per Rear Angle	Coefficient per deg	Lift sensitivity per degree of rear wing angle.
2n3k.	Drag Per Front Angle	Coefficient per deg	Drag sensitivity per degree of front wing angle.
2n3l.	Drag Per Rear Angle	Coefficient per deg	Drag sensitivity per degree of rear wing angle.
2n3m.	Default Front Angle	deg (Degrees)	Default front wing angle reference used when no setup override is active.
2n3n.	Default Rear Angle	deg (Degrees)	Default rear wing angle reference used when no setup override is active.
2n4.	Front Lift	None (File path)	Path to front lift aero curve/asset override within the aero package.
2n5.	Rear Lift	None (File path)	Path to rear lift aero curve/asset override within the aero package.
2n6.	Drag	None (File path)	Path to drag aero curve/asset override within the aero package.
2n7.	Wings Path [x]	None (File path)	Path(s) to alternative wing definition assets; multiple entries for multi-element wings.
2o.	Performance Modes [x]	None (Object array)	Performance mode preset block when Physics Tuning is Performance Modes Tuning; multiple modes (Wet, Quali, etc.).
2o1.	Performance Mode Name	None (String)	Display name string for this performance mode shown in UI selection menus.
2o2.	Electronics Settings	None (Object)	Electronics preset sub-block bundled into this performance mode.

ID	Name	Unit of Measurement	Description
2o2a.	Tc1	None (Map index / level)	Traction Control map level preset for this performance mode.
2o2b.	Tc2	None (Map index / level)	Secondary TC map level preset.
2o2c.	Abs	None (Map index / level)	ABS map level preset.
2o2d.	Esc	None (Map index / level)	ESC/ESP map level preset.
2o2e.	Ebb	None (Map index / level)	Electronic Brake Balance level preset.
2o2f.	Engine Map	None (Map index)	Engine power map index preset.
2o2g.	Telemetry Laps To Record	None (Integer)	Telemetry lap recording count preset.
2o2h.	Turbo Boost Lv	bar or level index	Turbo boost level preset.
2o2i.	Ers Deployment Map	None (Map index)	ERS deployment map preset.
2o2j.	Ers Recharge Lv	None (Level index)	ERS recharge level preset.
2o2k.	Ers Heat Charging	None (Level index)	MGU-H heat charging level preset.
2o3.	Brake Settings	None (Object)	Brake setup preset sub-block for this performance mode.
2o3a.	Front Bias	Ratio (0.0 - 1.0)	Brake balance front percentage preset
2o3b.	Torque Multiplier	Dimensionless multiplier	Global braking torque multiplier preset.
2o3c.	Brake Ducts [x]	Ratio (0.0 - 1.0)	Brake duct opening level preset(s); multiple entries for front/rear ducts.
2o4.	Damper Settings [x]	None (Object array)	Per-corner damper preset block; indices map to wheel positions.
2o4a.	Slow Bump	Clicks or N·s/m	Slow-speed compression damping preset for this corner.
2o4b.	Fast Bump	Clicks or N·s/m	Fast-speed compression damping preset.
2o4c.	Slow Rebound	Clicks or N·s/m	Slow-speed rebound damping preset.
2o4d.	Fast Rebound	Clicks or N·s/m	Fast-speed rebound damping preset.
2o5.	Differential Data	None (Object)	Single differential override block within a performance mode.
2o5a.	Type	None (Enum)	Differential type enum (LSD, Spool, Torsen, EpicyclicTorsen, TorqueVectoring, etc.).
2o5b.	Power	Nm or ratio	On-throttle differential lock intensity preset.

ID	Name	Unit of Measurement	Description
2o5c.	Coast	Nm or ratio	Coast/off-throttle differential lock intensity preset.
2o5d.	Preload	Nm (Newton-meters)	Static differential preload torque preset.
2o5e.	Front Share	Ratio (0.0 - 1.0)	Front torque share fraction for torque-vectoring or active diffs.
2o5f.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Torque bias ratio under power.
2o5g.	Torque Bias Ratio Coast	Ratio	Torque bias ratio on coast
2o5h.	Thermal Capacity	J/K or J/°C	Thermal mass of differential friction surfaces for heat model.
2o5i.	Surface	m ² (Square meters)	Effective friction surface area for differential heat generation.
2o5j.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Heat transfer coefficient to ambient/oil for differential cooling.
2o5k.	Wear Factor	Dimensionless coefficient	Wear accumulation rate factor for differential friction surfaces.
2o5l.	Friction Reduction With T	Ratio/°C	Friction reduction factor as a function of differential temperature.
2o5m.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for friction reduction curve.
2o6.	Four W D Differentials	None (Object)	AWD/4WD triple-differential preset container (front, center, rear).
2o6a.	Front Diff	None (Object)	Front axle differential preset sub-block.
2o6a1.	Type	None (Enum)	Differential type enum for the front diff.
2o6a2.	Power	Nm or ratio	On-throttle lock intensity for the front diff.
2o6a3.	Coast	Nm or ratio	Coast lock intensity for the front diff.
2o6a4.	Preload	Nm (Newton-meters)	Preload torque for the front diff.
2o6a5.	Front Share	Ratio (0.0 - 1.0)	Front torque share for the front diff.
2o6a6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Power torque bias ratio for the front diff.
2o6a7.	Torque Bias Ratio Coast	Ratio	Coast torque bias ratio for the front diff.
2o6a8.	Thermal Capacity	J/K or J/°C	Thermal capacity of the front diff friction pack.

ID	Name	Unit of Measurement	Description
2o6a9.	Surface	m ² (Square meters)	Friction surface area for the front diff.
2o6a10.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Heat transfer coefficient for the front diff.
2o6a11.	Wear Factor	Dimensionless coefficient	Wear rate factor for the front diff.
2o6a12.	Friction Reduction With T	Ratio/°C	Temperature-dependent friction reduction for the front diff.
2o6a13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for front diff friction model.
2o6b.	Center Diff	None (Object)	Center differential preset sub-block.
2o6b1.	Type	None (Enum)	Differential type enum for the center diff.
2o6b2.	Power	Nm or ratio	On-throttle lock intensity for the center diff.
2o6b3.	Coast	Nm or ratio	Coast lock intensity for the center diff.
2o6b4.	Preload	Nm (Newton-meters)	Preload torque for the center diff.
2o6b5.	Front Share	Ratio (0.0 - 1.0)	Front torque share for the center diff.
2o6b6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Power torque bias ratio for the center diff.
2o6b7.	Torque Bias Ratio Coast	Ratio	Coast torque bias ratio for the center diff.
2o6b8.	Thermal Capacity	J/K or J/°C	Thermal capacity of the center diff friction pack.
2o6b9.	Surface	m ² (Square meters)	Friction surface area for the center diff.
2o6b10.	Heat Transfer Coeff	W/(m ² ·K) (Coefficient)	Heat transfer coefficient for the center diff.
2o6b11.	Wear Factor	Dimensionless coefficient	Wear rate factor for the center diff.
2o6b12.	Friction Reduction With T	Ratio/°C	Temperature-dependent friction reduction for the center diff.
2o6b13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for center diff friction model.
2o6c.	Rear Diff	None (Object)	Rear axle differential preset sub-block.
2o6c1.	Type	None (Enum)	Differential type enum for the rear diff.
2o6c2.	Power	Nm or ratio	On-throttle lock intensity for the rear diff.

ID	Name	Unit of Measurement	Description
2o6c3.	Coast	Nm or ratio	Coast lock intensity for the rear diff.
2o6c4.	Preload	Nm (Newton-meters)	Preload torque for the rear diff.
2o6c5.	Front Share	Ratio (0.0 - 1.0)	Front torque share for the rear diff.
2o6c6.	Torque Bias Ratio Power	Ratio (e.g., 2.0:1)	Power torque bias ratio for the rear diff.
2o6c7.	Torque Bias Ratio Coast	Ratio	Coast torque bias ratio for the rear diff.
2o6c8.	Thermal Capacity	J/K or J/°C	Thermal capacity of the rear diff friction pack.
2o6c9.	Surface	m² (Square meters)	Friction surface area for the rear diff.
2o6c10.	Heat Transfer Coeff	W/(m²·K) (Coefficient)	Heat transfer coefficient for the rear diff.
2o6c11.	Wear Factor	Dimensionless coefficient	Wear rate factor for the rear diff.
2o6c12.	Friction Reduction With T	Ratio/°C	Temperature-dependent friction reduction for the rear diff.
2o6c13.	Friction Ref T	°C (Degrees Celsius)	Reference temperature for rear diff friction model.
2o7.	Front Lock Controllers	None (Object)	Front axle lock controller definition for active/vectoring diffs.
2o7a.	Name	None (String)	Display/debug name for the front lock controller.
2o7b.	Stages [x]	None (Object array)	Multi-stage controller pipeline; each stage processes an input through a LUT and limits.
2o7b1.	Input Var	None (Telemetry enum)	Input variable enum for this stage (SlipRatio, Steer, LatG, ErsChargeLevel, etc.).
2o7b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for merging stage output (Add or Mult).
2o7b3.	Lut	None (.curve file path)	Look-up table (.curve) mapping filtered input to lock fraction.
2o7b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain smoothing the stage input signal.
2o7b5.	Up Limit	Depends on input variable	Upper output limit for this stage.
2o7b6.	Down Limit	Depends on input variable	Lower output limit for this stage.
2o7b7.	Current Value	Depends on input variable	Runtime current value output of this stage (telemetry/state).
2o7b8.	Const Value	Depends on input variable	Constant fallback input value when Input Var is Const.

ID	Name	Unit of Measurement	Description
2o8.	Center Lock Controllers	None (Object)	Center differential lock controller definition.
2o8a.	Name	None (String)	Display/debug name for the center lock controller.
2o8b.	Stages [x]	None (Object array)	Multi-stage controller pipeline for center diff lock.
2o8b1.	Input Var	None (Telemetry enum)	Input variable enum for center lock stage.
2o8b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode (Add/Mult) for center lock stage.
2o8b3.	Lut	None (.curve file path)	LUT path for center lock stage.
2o8b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for center lock stage input.
2o8b5.	Up Limit	Depends on input variable	Upper output limit for center lock stage.
2o8b6.	Down Limit	Depends on input variable	Lower output limit for center lock stage.
2o8b7.	Current Value	Depends on input variable	Current runtime value for center lock stage.
2o8b8.	Const Value	Depends on input variable	Constant input value for center lock stage.
2o9.	Rear Lock Controllers	None (Object)	Rear axle lock controller definition.
2o9a.	Name	None (String)	Display/debug name for the rear lock controller.
2o9b.	Stages [x]	None (Object array)	Multi-stage controller pipeline for rear diff lock.
2o9b1.	Input Var	None (Telemetry enum)	Input variable enum for rear lock stage.
2o9b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for rear lock stage.
2o9b3.	Lut	None (.curve file path)	LUT path for rear lock stage.
2o9b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for rear lock stage.
2o9b5.	Up Limit	Depends on input variable	Upper output limit for rear lock stage.
2o9b6.	Down Limit	Depends on input variable	Lower output limit for rear lock stage.
2o9b7.	Current Value	Depends on input variable	Current runtime value for rear lock stage.
2o9b8.	Const Value	Depends on input variable	Constant input value for rear lock stage.

ID	Name	Unit of Measurement	Description
2o10	Awd Clutches [x]	None (Object array)	AWD clutch pack definition; multiple clutches for front/rear axle coupling.
2o10a.	Position	None (Integer)	Integer position/index of this clutch in the AWD coupling matrix.
2o10b.	Preload	Nm (Newton-meters)	Static preload torque on this AWD clutch pack.
2o10c.	Controllers	None (Object)	Nested lock controller for this AWD clutch.
2o10c1.	Name	None (String)	Display name for the AWD clutch controller.
2o10c2.	Stages [x]	None (Object array)	Multi-stage pipeline controlling AWD clutch lock.
2o10c2a.	Input Var	None (Telemetry enum)	Input variable enum for AWD clutch stage.
2o10c2b.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for AWD clutch stage.
2o10c2c.	Lut	None (.curve file path)	LUT path for AWD clutch stage.
2o10c2d.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for AWD clutch stage.
2o10c2e.	Up Limit	Depends on input variable	Upper output limit for AWD clutch stage.
2o10c2f.	Down Limit	Depends on input variable	Lower output limit for AWD clutch stage.
2o10c2g.	Current Value	Depends on input variable	Current runtime value for AWD clutch stage.
2o10c2h.	Const Value	Depends on input variable	Constant input for AWD clutch stage.
2o11	Turbo Controllers [x]	None (Object array)	Turbo boost controller block within performance mode tuning.
2o11a.	Name	None (String)	Display name for turbo controller.
2o11b.	Stages [x]	None (Object array)	Multi-stage turbo boost control pipeline.
2o11b1.	Input Var	None (Telemetry enum)	Input variable enum for turbo stage.
2o11b2.	Combinator Mode	None (Math enum : Add / Mult)	Combinator mode for turbo stage.
2o11b3.	Lut	None (.curve file path)	LUT path mapping input to boost modifier.
2o11b4.	Filter Gain	Coefficient (Smoothing multiplier)	Filter gain for turbo stage input.
2o11b5.	Up Limit	Depends on input variable	Upper boost limit for this stage.

ID	Name	Unit of Measurement	Description
2o11b6.	Down Limit	Depends on input variable	Lower boost limit for this stage.
2o11b7.	Current Value	Depends on input variable	Current runtime boost modifier value.
2o11b8.	Const Value	Depends on input variable	Constant input for turbo stage.
2o12.	Turbo Settings	None (Object)	Turbo settings sub-block within performance mode.
2o12a.	Boost Lv	bar or level index	Target turbo boost level preset for this performance mode.
3.	Car Part Type	None (Enum)	Enum categorizing the visual/mechanical part slot this tuning part occupies in the parts menu (Mechanics_Engine, Mechanics_Gearbox, Exterior_*, etc.).

D. Example data

I. Chosen Cars for Example

- Toyota Supra MK IV (slug : ks_toyota_supra_mkiv) [7 tuning parts]
- Datsun 240z Fairlady (slug : ks_datsun_240z_fairlady) [8 tuning parts]
- Porsche 992 GT3 Cup (slug : ks_porsche_992_gt3_cup) [4 tuning parts]

II. Example

Toyota Supra MK IV

1. Drift Front Geometry (file :

ks_toyota_supra_geometry_front_drift_geometry.tuningpart)

```

| 1. Physics Tuning : Suspensions Geometry
| 2. Suspensions Geometry
|   | 2a. Geometry Path Front :
content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_front_drift.
suspension
|   | 2b. Geometry Path Rear :
content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_rear.suspens
ion
| 3. Car Part Type : Mechanics_SuspensionGeometry_Front

```

2. Drift Drivetrain (file : ks_toyota_supra_mkiv_drivetrain_drift.tuningpart)

```

| 1. Physics Tuning : Drivetrain
| 2. Drivetrain
|   | 2a. Path :
content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_drift.drivet
rain

```

└ 3. Car Part Type : Mechanics_Drivetrain

3. Drift Engine (file : ks_toyota_supra_mkiv_engine_drift.tuningpart)

└ 1. Physics Tuning : Engine

└ 2. Engine

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_drift.carengine

└ 3. Car Part Type : Mechanics_Engine

4. Drift Front Suspension (file : ks_toyota_supra_mkiv_front_susp_drift.tuningpart)

└ 1. Physics Tuning : Suspensions

└ 2. Suspensions

└ └ 2a. Coilover Path Front :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_front.coilover

└ └ 2b. Coilover Path Rear :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_rear.coilover

└ 1c. Car Part Type : Mechanics_SuspensionCoilover_Front

5. Drift Gearbox (file : ks_toyota_supra_mkiv_gearbox_drift.tuningpart)

└ 1. Physics Tuning : Gearbox

└ 2. Gearbox

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\ks_toyota_supra_mkiv_drift.gearbox

└ 3. Car Part Type : Mechanics_Gearbox

6. Drift Setup (file : ks_toyota_supra_mkiv_setup_drift.tuningpart)

└ 1. Physics Tuning : Setup

└ 2. Setup

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\setup\supra_drift.carsetup

└ 3. Car Part Type : None

7. Drift Setup Limits (file : ks_toyota_supra_mkiv_setuplimits_drift.tuningpart)

└ 1. Physics Tuning : Setup Limits

└ 2. Setup Limits

└ └ 2a. Path :

content\cars\ks_toyota_supra_mkiv\data\setup\supra_mk_iv_limits_drift.carsetuplimits

└ 3. Car Part Type : None

Datsun 240z Fairlady

1. 5 Speed Gearbox (file : ks_datsun_240z_fairlady_5speed.tuningpart)

- | 1. Physics Tuning : Gearbox
- | 2. Gearbox
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_5speed.gearbox
- | 3. Car Part Type : Mechanics_Gearbox

2. Brake (file : ks_datsun_240z_fairlady_brake.tuningpart)

- | 1. Physics Tuning : Brake System
- | 2. Gearbox
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_brakes.brakesystem
- | 3. Car Part Type : Mechanics_CaliperFL

3. 5 Speed Clutch (file : ks_datsun_240z_fairlady_clutch_5speed.tuningpart)

- | 1. Physics Tuning : Clutch
- | 2. Clutch
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_clutch_5speed.clutch
- | 3. Car Part Type : Mechanics_Clutch

4. G Nose Coilovers (file : ks_datsun_240z_fairlady_g_nose_coilovers.tuningpart)

- | 1. Physics Tuning : Suspensions
- | 2. Suspensions
 - | 2a. Coilover Path Front :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_g_nose_front.coilover
 - | 2b. Coilover Path Rear :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_g_nose_rear.coilover
- | 3. Car Part Type : Mechanics_SuspensionCoilover_Front

5. Setup (file : ks_datsun_240z_fairlady_g_nose_setup.tuningpart)

- | 1. Physics Tuning : Setup
- | 2. Setup
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\setup\ks_datsun_240z_fairlady_g_nose.carsetup
- | 3. Car Part Type : None

6. Setup Limits (file : ks_datsun_240z_fairlady_g_nose_setuplimits.tuningpart)

- | 1. Physics Tuning : Setup Limits
- | 2. Setup Limits
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\setup\ks_datsun_240z_fairlady_g_nose.carsetuplimits
- | 3. Car Part Type : None

7. L28 Engine (file : ks_datsun_240z_fairlady_l28engine.tuningpart)

- | 1. Physics Tuning : Engine
- | 2. Engine
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_L28engine.carengine
- | 3. Car Part Type : Mechanics_Engine

8. Limited-Slip Differential (file : ks_datsun_240z_fairlady_lsd.tuningpart)

- | 1. Physics Tuning : Drivetrain
- | 2. Drivetrain
 - | 2a. Path :
content\cars\ks_datsun_240z_fairlady\data\tuning\ks_datsun_240z_fairlady_LSD.drivetrain
- | 3. Car Part Type : Mechanics_Drivetrain

Porsche 992 GT3 Cup

1. No ABS No TC (file : 992_gt3_no_abs_no_tc.tuningpart)

- | 1. Physics Tuning : <None>
- | 2. Car Part Type : Mechanics_Electronics

2. No ABS No TC Setup Limits (file : 992_gt3_no_abs_no_tc_setuplimits.tuningpart)

- | 1. Physics Tuning : Setup Limits
- | 2. Setup Limits
 - | 2a. Path :
content\cars\ks_porsche_992_gt3_cup\data\Setup\limitsporsche992cup_no_abs_no_tc.carsetuplimits
- | 3. Car Part Type : None

3. Only ABS (file : 992_gt3_only_abs.tuningpart)

- | 1. Physics Tuning : <None>
- | 2. Car Part Type : Mechanics_Electronics

4. Only ABS Setup Limits (file : 992_gt3_only_abs_setuplimits.tuningpart)

- | 1. Physics Tuning : Setup Limits
- | 2. Setup Limits
 - | 2a. Path :
content\cars\ks_porsche_992_gt3_cup\data\Setup\limitsporsche992cup_only_abs.carsetuplimits
- | 3. Car Part Type : None

9. Car Electronics [.carelectronics]

A. Description

Software layer for driver aids: traction control, ABS, optional electronic differential lock (EDL), and ESP. It reads wheel slip, yaw, and steering telemetry and intervenes on engine torque or brake pressure — it does not replace mechanical brakes, drivetrain, or tyre physics.

Car Setup only stores which map index is selected (TC 3, ABS 5...). The actual slip windows and cut severity live here. Tuning parts / setup limits can swap or lock entire .CARELECTRONICS files (Cup “No ABS No TC” vs “Only ABS”).

I. Role in the stack

Concern	Handled here	Handled elsewhere
TC / ABS / EDL / ESP map matrices	.CARELECTRONICS	—
Which map is active in the garage	—	.CARSETUP Electronics indices
Allowed TC/ABS range in UI	—	.CARSETUPLIMITS
Hydraulic brake torque and bias	—	.BRAKESYSTEM / .BRAKES
Engine torque the TC can cut	—	.CARENGINE
Swap / strip aids per variant	Path or None via .TUNINGPART	Companion setup limits

Car Data (or a tuning part) points at one electronics asset. Empty TC/ABS objects mean that aid is absent for that file — not “map 0 soft”.

II. What you are really tuning

- Traction control envelope** — Per-setting slip window (*Min / Max Slip Ratio*), lateral tolerance (*Ref Slip Angle Deg*, *Slip Angle Activation Deg*), and how hard the ECU chops torque (*Engine Cut Level*). Gains on angular acceleration and oversteer decide how early the system reacts to yaw, not only pure wheelspin.

Higher map numbers in GT examples usually tighten the slip band and raise intervention gains — wet / worn / safer. Setting 1 is often a hard off slot (zeros, or slip ratios at -1 for ABS).
- TC global clocks** — *Frequency Hz* (Huracan / MC20 at 333), *Min Speed Kmh* (often 40 — no TC crawling in pit lane), *Gear Change Time* (brief mute around shifts: Huracan 0.045 s, MC20 0.025 s), and *Min / Max Cut Level* clamp how soft or brutal cuts can get. *Has TC2* flags a second TC channel when present.
- ABS envelope** — Per-setting longitudinal slip targets, reference slip angle, *Cut Level* (how aggressively pressure is released), and *Max Torque Variation* (how fast brake torque may change — smoothing vs violent cycling). Setting 1 with min/max slip at -1 is the usual “ABS off” sentinel in race packs.
- ABS global clocks** — *Frequency* (examples at 200 Hz), *Channels* (4 = four-wheel independent), *Min Speed Kmh* (often 20). Faster sample rate → smoother pedal pulse if the rest of the map allows it.

5. **EDL and ESP** — *EDL* can apply brake torque under power/coast spin with deadzones and max-spin thresholds — electronic locker behaviour. Race examples here leave it *None*. *ESP* settings (gain, steer gains, over/understeer gains, brake percent) are present in schema but often disabled in motorsport files (*Frequency Hz : 0*, empty settings).

III. Architecture

1 - TRACTION CONTROL (SCHEMA 1)

Object *TC*: globals (*Has TC2*, frequency, min speed, gear-change mute, cut clamps) plus *Settings[x]* rows of slip / angle / cut / gain fields.

2 - ANTI-LOCK BRAKING (SCHEMA 2)

Object *ABS*: *Settings[x]* rows, then frequency, channel count, min speed.

3 - EDL (SCHEMA 3)

Optional object: *Active*, brake torque power/coast, deadzones, max spin power/coast, min speed. Entire block may be *None*.

4 - ESP (SCHEMA 4)

Frequency, min speed, and *Settings[x]* with gain / steer / slip / brake-percent fields. Often stubbed off on GT / Cup cars.

IV. How to read the examples

1 - LAMBORGHINI HURACAN ST EVO 2

Full pro multi-map suite: 10 TC settings (1 = null/off; 2→10 tighten slip from about 0.18–0.45 down to 0.05–0.15 and raise angular / oversteer gains). ABS has 12 settings (1 = off via -1 slips; live maps around 0.03–0.12 slip with *Max Torque Variation* 0.7). EDL *None*; ESP frequency 0. Classic GT3-style adjustable aids.

2 - MASERATI MC20 GT2

Similar layout, 13 TC maps and 12 ABS maps. Mid maps push oversteer gain hard (up to about 14) while late maps drop engine cut toward 0.1 — different authoring curve than Huracan, same idea: map index = intervention personality. EDL *None*; ESP empty.

3 - PORSCHE 992 GT3 CUP (TWO FILES)

No ABS No TC: *TC* and *ABS* both *None* — pure mechanical driving, electronics file is a stub.

Only ABS: *TC* : *None*, ABS populated (12 settings, off slot + progressive slip windows). Paired with tuning parts + setup limits so the garage cannot re-enable stripped aids. Regulation by swapping the whole asset, not by zeroing every map in place.

V. Practical notes

- Setup index N must exist as *Settings N* here; pointing setup at map 8 when only 5 rows exist is a content bug.
- Treat Setting 1 zeros / -1 slips as intentional Off, not as “very soft intervention”.
- Stripping aids for Cup / historic cars: prefer a dedicated .CARELECTRONICS with TC/ABS *None* (and matching limits) over leaving dead maps that the UI can still select.
- *Min Speed gates intervention* — low-speed spins may be intentional if the threshold is high.
- OCR spacing in the dump (*T C, A B S, Ref Slip Angledeg*) is cosmetic; field meaning is the slip/cut/gain set above.
- ESP/EDL stubs with frequency 0 or *None* mean “not used on this car”, not “map missing”.

VI. Related assets

- **5 / 6. Car Setup / Setup Limits** — select and clamp TC/ABS/ESC indices
- **8. Car Tuning Parts** — swap electronics files or tag Mechanics_Electronics variants
- **1 / 2. Brake System / Brakes** — hydraulic plant ABS modulates
- **4. Car Engine** — torque source TC cuts
- **3. Car Data** — default electronics path for the stock car

B. Schema

- ```
1. T C : object
 1a. Has T C2 : boolean
 1b. Frequency Hz : float
 1c. Min Speed Kmh : float
 1d. Gear Change Time : float
 1e. Min Cut Level : float
 1f. Max Cut Level : float
 1g. Settings [x] : object | can have multiple Settings
 1g1. Min Slip Ratio : float
 1g2. Max Slip Ratio : float
 1g3. Ref Slip Angle Deg : float
 1g4. Engine Cut Level : float
 1g5. Angular A C Cgain : float
 1g6. Oversteer Gain : float
 1g7. Slip Angle Activation Deg : float
2. A B S : object
 2a. Settings [x] : object | can have multiple Settings
 2a1. Min Slip Ratio : float
 2a2. Max Slip Ratio : float
 2a3. Ref Slip Angledeg : float
 2a4. Cut Level : float
 2a5. Max Torque Variation : float
 2b. Frequency : float
 2c. Channels : integer
 2d. Min Speed Kmh : float
```



- 3. E D L : **object**
  - 3a. Active : **boolean**
  - 3b. Braketorquepower : **float**
  - 3c. Braketorquecoast : **float**
  - 3d. Deadzonecoast : **float**
  - 3e. Deadzonepower : **float**
  - 3f. Maxspinpower : **float**
  - 3g. Maxspincoast : **float**
  - 3h. Minspeed : **float**
- 4. E S P : **object**
  - 4a. Frequency Hz : **float**
  - 4b. Min Speed Kmh : **float**
  - 4c. Settings [x] : **object** | can have multiple Settings
    - 4c1. Gain : **float**
    - 4c2. Steer Gain : **float**
    - 4c3. Min Steer Gain : **float**
    - 4c4. Steer Gain Max Speed : **float**
    - 4c5. Oversteer Gain : **float**
    - 4c6. Understeer Gain : **float**
    - 4c7. Max Slip Ratio : **float**
    - 4c8. Dead Zone : **float**
    - 4c9. Filter Gain : **float**
    - 4c10. Brake Perc : **float**
    - 4c11. Brake Perc Activation : **float**

### C. Measurement Units & Descriptions

| ID  | Name             | Unit of Measurement             | Description                                                                                      |
|-----|------------------|---------------------------------|--------------------------------------------------------------------------------------------------|
| 1.  | TC               | None ( Object )                 | Traction Control system block; processes wheel slip and yaw to modulate engine torque.           |
| 1a. | Has TC2          | None ( Boolean : True / False ) | Enables a secondary traction-control map (TC2) for dual-stage intervention logic.                |
| 1b. | Frequency Hz     | Hz ( Hertz )                    | Control loop update frequency of the traction-control system (e.g., 333 Hz in Huracán ST Evo 2). |
| 1c. | Min Speed Kmh    | km/h ( Kilometers per hour )    | Minimum vehicle speed below which traction control deactivates (e.g., 40 km/h).                  |
| 1d. | Gear Change Time | s ( Seconds )                   | Temporary TC suppression window duration during gear changes to avoid false interventions.       |
| 1e. | Min Cut Level    | Ratio ( 0.0 - 1.0 )             | Minimum engine torque cut level applied at the lowest TC intervention step.                      |
| 1f. | Max Cut Level    | Ratio ( 0.0 - 1.0 )             | Maximum engine torque cut level at the most aggressive TC step.                                  |

| ID          | Name                      | Unit of Measurement   | Description                                                                                   |
|-------------|---------------------------|-----------------------|-----------------------------------------------------------------------------------------------|
| <b>1g.</b>  | Settings                  | None ( Object array ) | Per-map TC settings block; multiple entries (Settings 1–12+) selectable from the cockpit.     |
| <b>1g1.</b> | Min Slip Ratio            | Dimensionless ratio   | Lower slip-ratio threshold at which TC begins monitoring wheel spin for this settings map.    |
| <b>1g2.</b> | Max Slip Ratio            | Dimensionless ratio   | Upper slip-ratio threshold triggering full TC intervention for this settings map.             |
| <b>1g3.</b> | Ref Slip Angle Deg        | deg ( Degrees )       | Reference rear slip angle used to scale TC aggression relative to yaw behavior.               |
| <b>1g4.</b> | Engine Cut Level          | Ratio ( 0.0 - 1.0 )   | Engine torque reduction factor applied when slip thresholds are exceeded.                     |
| <b>1g5.</b> | Angular ACC Gain          | Dimensionless gain    | Yaw-rate / angular acceleration gain for TC correction authority (schema: Angular A C Cgain). |
| <b>1g6.</b> | Oversteer Gain            | Dimensionless gain    | Additional TC sensitivity multiplier when oversteer is detected.                              |
| <b>1g7.</b> | Slip Angle Activation Deg | deg ( Degrees )       | Minimum slip angle required before this TC map becomes active.                                |
| <b>2.</b>   | ABS                       | None ( Object )       | Anti-lock Braking System block; modulates brake line pressure to prevent wheel lockup.        |
| <b>2a.</b>  | Settings                  | None ( Object array ) | Per-map ABS settings block; multiple entries (Settings 1–12+) for intervention intensity.     |
| <b>2a1.</b> | Min Slip Ratio            | Dimensionless ratio   | Lower wheel slip threshold for ABS monitoring on this settings map.                           |
| <b>2a2.</b> | Max Slip Ratio            | Dimensionless ratio   | Upper slip threshold triggering maximum ABS pressure modulation.                              |
| <b>2a3.</b> | Ref Slip Angle Deg        | deg ( Degrees )       | Reference slip angle for ABS yaw-stability cross-correlation (schema: Ref Slip Angledeg).     |
| <b>2a4.</b> | Cut Level                 | Ratio ( 0.0 - 1.0 )   | Brake pressure reduction factor applied during ABS pulsing.                                   |
| <b>2a5.</b> | Max Torque Variation      | Nm or ratio           | Maximum allowed brake torque fluctuation per ABS cycle on this map.                           |
| <b>2b.</b>  | Frequency                 | Hz ( Hertz )          | ABS control loop pulsing frequency (e.g., 200 Hz in Huracán ST Evo 2).                        |

| ID  | Name               | Unit of Measurement             | Description                                                                                                      |
|-----|--------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------|
| 2c. | Channels           | None ( Integer )                | Number of independent ABS control channels (e.g., 4 = one per wheel).                                            |
| 2d. | Min Speed Kmh      | km/h ( Kilometers per hour )    | Minimum speed below which ABS intervention is disabled (e.g., 20 km/h).                                          |
| 3.  | EDL                | None ( Object )                 | Electronic Differential Lock block; uses brake torque to emulate a locking differential (may be None/ disabled). |
| 3a. | Active             | None ( Boolean : True / False ) | Master enable for Electronic Differential Lock brake-based torque transfer.                                      |
| 3b. | Brake Torque Power | Nm ( Newton-meters )            | Brake torque applied to the spinning wheel under power to emulate diff lock (schema: Braketorquepower).          |
| 3c. | Brake Torque Coast | Nm ( Newton-meters )            | Brake torque applied during coasting to synchronize wheel speeds (schema: Braketorquecoast).                     |
| 3d. | Dead Zone Coast    | Ratio or rad/s                  | Speed-difference deadband below which coast EDL does not intervene (schema: Deadzonecoast).                      |
| 3e. | Dead Zone Power    | Ratio or rad/s                  | Speed-difference deadband below which power EDL does not intervene (schema: Deadzonepower).                      |
| 3f. | Max Spin Power     | rad/s or ratio                  | Maximum allowed wheel spin differential under power before EDL fully engages (schema: Maxspinpower).             |
| 3g. | Max Spin Coast     | rad/s or ratio                  | Maximum allowed wheel spin differential on coast before EDL engages (schema: Maxspincoast).                      |
| 3h. | Min Speed          | M/s or km/h ( Speed threshold   | Minimum vehicle speed required for EDL operation (schema: Minspeed).                                             |
| 4.  | ESP                | None ( Object )                 | Electronic Stability Program block; applies selective braking and torque reduction to control yaw.               |
| 4a. | Frequency Hz       | Hz ( Hertz )                    | ESP control loop update frequency (0 = disabled in some example configs).                                        |
| 4b. | Min Speed Kmh      | km/h ( Kilometers per hour )    | Minimum speed below which ESP deactivates.                                                                       |

| ID    | Name                  | Unit of Measurement                  | Description                                                                      |
|-------|-----------------------|--------------------------------------|----------------------------------------------------------------------------------|
| 4c.   | Settings              | None ( Object array )                | Per-map ESP settings block; multiple entries for intervention authority presets. |
| 4c1.  | Gain                  | Dimensionless gain                   | Global ESP intervention authority multiplier for this settings map.              |
| 4c2.  | Steer Gain            | Dimensionless gain                   | Steering-angle sensitivity gain for ESP yaw correction.                          |
| 4c3.  | Min Steer Gain        | Dimensionless gain                   | Minimum steering gain applied at low speeds within ESP logic.                    |
| 4c4.  | Steer Gain Max Speed  | km/h ( Kilometers per hour )         | Speed at which steering gain reaches its maximum ESP authority.                  |
| 4c5.  | Oversteer Gain        | Dimensionless gain                   | Corrective gain when rear slip (oversteer) is detected.                          |
| 4c6.  | Understeer Gain       | Dimensionless gain                   | Corrective gain when front slip (understeer) is detected.                        |
| 4c7.  | Max Slip Ratio        | Dimensionless ratio                  | Maximum wheel slip ratio before ESP applies full brake/engine intervention.      |
| 4c8.  | Dead Zone             | deg or ratio                         | Yaw/steer error deadband where ESP remains passive.                              |
| 4c9.  | Filter Gain           | Coefficient ( Smoothing multiplier ) | Low-pass filter on ESP sensor inputs to prevent oscillatory corrections.         |
| 4c10. | Brake Perc            | Ratio ( 0.0 - 1.0 )                  | Percentage of available brake pressure ESP may apply for stability correction.   |
| 4c11. | Brake Perc Activation | Ratio ( 0.0 - 1.0 )                  | Slip/yaw threshold fraction at which ESP begins applying brake intervention.     |

## D. Example data

### I. Chosen Cars for Example

- Lamborghini Huracan ST Evo 2 ( slug : ks\_lamborghini\_huracan\_st\_evo2 )
- Maserati MC20 GT2 ( slug : ks\_maserati\_mc20\_gt2 )
- Porsche 992 GT3 Cup ( slug : ks\_porsche\_992\_gt3\_cup ) [ 2 variants ]

### II. Example

#### Lamborghini Huracan ST Evo 2

```

| 1. T C
| | 1a. Has T C2 : false

```

- 1b. Frequency Hz : 333.00000
- 1c. Min Speed Kmh : 40.00000
- 1d. Gear Change Time : 0.04500
- 1e. Min Cut Level : 1000.00000
- 1f. Max Cut Level : 1.00000
- 1g. Settings 1
  - 1g1. Min Slip Ratio : 0.00000
  - 1g2. Max Slip Ratio : 0.00000
  - 1g3. Ref Slip Angle Deg : 0.00000
  - 1g4. Engine Cut Level : 0.00000
  - 1g5. Angular A C Cgain : 0.00000
  - 1g6. Oversteer Gain : 0.00000
  - 1g7. Slip Angle Activation Deg : 100.00000
- 1g. Settings 2
  - 1g1. Min Slip Ratio : 0.18000
  - 1g2. Max Slip Ratio : 0.45000
  - 1g3. Ref Slip Angle Deg : 10.00000
  - 1g4. Engine Cut Level : 1.50000
  - 1g5. Angular A C Cgain : 3.00000
  - 1g6. Oversteer Gain : 3.00000
  - 1g7. Slip Angle Activation Deg : 6.00000
- 1g. Settings 3
  - 1g1. Min Slip Ratio : 0.15000
  - 1g2. Max Slip Ratio : 0.40000
  - 1g3. Ref Slip Angle Deg : 9.50000
  - 1g4. Engine Cut Level : 1.50000
  - 1g5. Angular A C Cgain : 4.00000
  - 1g6. Oversteer Gain : 4.00000
  - 1g7. Slip Angle Activation Deg : 5.25000
- 1g. Settings 4
  - 1g1. Min Slip Ratio : 0.12000
  - 1g2. Max Slip Ratio : 0.35000
  - 1g3. Ref Slip Angle Deg : 9.00000
  - 1g4. Engine Cut Level : 1.50000
  - 1g5. Angular A C Cgain : 5.00000
  - 1g6. Oversteer Gain : 5.00000
  - 1g7. Slip Angle Activation Deg : 4.75000
- 1g. Settings 5
  - 1g1. Min Slip Ratio : 0.10000
  - 1g2. Max Slip Ratio : 0.30000
  - 1g3. Ref Slip Angle Deg : 9.00000
  - 1g4. Engine Cut Level : 1.50000
  - 1g5. Angular A C Cgain : 6.50000
  - 1g6. Oversteer Gain : 6.50000
  - 1g7. Slip Angle Activation Deg : 4.50000
- 1g. Settings 6
  - 1g1. Min Slip Ratio : 0.09000
  - 1g2. Max Slip Ratio : 0.25000
  - 1g3. Ref Slip Angle Deg : 8.50000
  - 1g4. Engine Cut Level : 1.50000
  - 1g5. Angular A C Cgain : 7.50000
  - 1g6. Oversteer Gain : 7.50000

- L 1g7. Slip Angle Activation Deg : 4.50000
- 1g. Settings 7
  - 1g1. Min Slip Ratio : 0.08000
  - 1g2. Max Slip Ratio : 0.25000
  - 1g3. Ref Slip Angle Deg : 8.00000
  - 1g4. Engine Cut Level : 1.25000
  - 1g5. Angular A C Cgain : 8.50000
  - 1g6. Oversteer Gain : 7.50000
  - L 1g7. Slip Angle Activation Deg : 4.50000
- 1g. Settings 8
  - 1g1. Min Slip Ratio : 0.07000
  - 1g2. Max Slip Ratio : 0.20000
  - 1g3. Ref Slip Angle Deg : 8.00000
  - 1g4. Engine Cut Level : 1.25000
  - 1g5. Angular A C Cgain : 8.50000
  - 1g6. Oversteer Gain : 7.50000
  - L 1g7. Slip Angle Activation Deg : 4.50000
- 1g. Settings 9
  - 1g1. Min Slip Ratio : 0.06000
  - 1g2. Max Slip Ratio : 0.20000
  - 1g3. Ref Slip Angle Deg : 7.50000
  - 1g4. Engine Cut Level : 1.25000
  - 1g5. Angular A C Cgain : 8.50000
  - 1g6. Oversteer Gain : 7.50000
  - L 1g7. Slip Angle Activation Deg : 4.00000
- 1g. Settings 10
  - 1g1. Min Slip Ratio : 0.05000
  - 1g2. Max Slip Ratio : 0.15000
  - 1g3. Ref Slip Angle Deg : 7.00000
  - 1g4. Engine Cut Level : 1.00000
  - 1g5. Angular A C Cgain : 9.00000
  - 1g6. Oversteer Gain : 7.50000
  - L 1g7. Slip Angle Activation Deg : 3.50000
- 2. A B S
  - 2a. Settings 1
    - 2a1. Min Slip Ratio : -1.00000
    - 2a2. Max Slip Ratio : -1.00000
    - 2a3. Ref Slip Angledeg : 0.00000
    - 2a4. Cut Level : 0.00000
    - L 2a5. Max Torque Variation : 0.00000
  - 2a. Settings 2
    - 2a1. Min Slip Ratio : 0.08000
    - 2a2. Max Slip Ratio : 0.12000
    - 2a3. Ref Slip Angledeg : 5.00000
    - 2a4. Cut Level : 0.10000
    - L 2a5. Max Torque Variation : 0.70000
  - 2a. Settings 3
    - 2a1. Min Slip Ratio : 0.07000
    - 2a2. Max Slip Ratio : 0.11000
    - 2a3. Ref Slip Angledeg : 5.00000
    - 2a4. Cut Level : 0.00000
    - L 2a5. Max Torque Variation : 0.70000
  - 2a. Settings 4
    - 2a1. Min Slip Ratio : 0.06000

- 2a2. Max Slip Ratio : 0.08000
- 2a3. Ref Slip Angledeg : 5.00000
- 2a4. Cut Level : 0.10000
- 2a5. Max Torque Variation : 0.70000
- 2a. Settings 5
  - 2a1. Min Slip Ratio : 0.05000
  - 2a2. Max Slip Ratio : 0.07000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 6
  - 2a1. Min Slip Ratio : 0.04000
  - 2a2. Max Slip Ratio : 0.06000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 7
  - 2a1. Min Slip Ratio : 0.03200
  - 2a2. Max Slip Ratio : 0.07500
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 8
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.07000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 9
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.07000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 10
  - 2a1. Min Slip Ratio : 0.02900
  - 2a2. Max Slip Ratio : 0.06500
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 11
  - 2a1. Min Slip Ratio : 0.02900
  - 2a2. Max Slip Ratio : 0.06500
  - 2a3. Ref Slip Angledeg : 4.50000
  - 2a4. Cut Level : 0.10000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 12
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.06500
  - 2a3. Ref Slip Angledeg : 4.25000
  - 2a4. Cut Level : 0.00000
  - 2a5. Max Torque Variation : 0.70000
- 2b. Frequency : 200.00000
- 2c. Channels : 4

- L 2d. Min Speed Kmh : 20.00000
- 3. E D L : None
- 4. E S P
  - 4a. Frequency Hz : 0.00000
  - 4b. Min Speed Kmh : 0.0

## Maserati MC20 GT2

- 1. T C
  - 1a. Has T C2 : false
  - 1b. Frequency Hz : 333.00000
  - 1c. Min Speed Kmh : 40.00000
  - 1d. Gear Change Time : 0.02500
  - 1e. Min Cut Level : 1.50000
  - 1f. Max Cut Level : 0.10000
  - 1g. Settings 1
    - 1g1. Min Slip Ratio : 0.00000
    - 1g2. Max Slip Ratio : 0.00000
    - 1g3. Ref Slip Angle Deg : 0.00000
    - 1g4. Engine Cut Level : 0.00000
    - 1g5. Angular A C Cgain : 0.20000
    - 1g6. Oversteer Gain : 0.50000
    - 1g7. Slip Angle Activation Deg : 0.20000
  - 1g. Settings 2
    - 1g1. Min Slip Ratio : 0.25000
    - 1g2. Max Slip Ratio : 0.50000
    - 1g3. Ref Slip Angle Deg : 18.00000
    - 1g4. Engine Cut Level : 1.50000
    - 1g5. Angular A C Cgain : 2.00000
    - 1g6. Oversteer Gain : 2.00000
    - 1g7. Slip Angle Activation Deg : 4.00000
  - 1g. Settings 3
    - 1g1. Min Slip Ratio : 0.23000
    - 1g2. Max Slip Ratio : 0.31000
    - 1g3. Ref Slip Angle Deg : 14.80000
    - 1g4. Engine Cut Level : 1.25000
    - 1g5. Angular A C Cgain : 3.50000
    - 1g6. Oversteer Gain : 4.00000
    - 1g7. Slip Angle Activation Deg : 3.50000
  - 1g. Settings 4
    - 1g1. Min Slip Ratio : 0.20000
    - 1g2. Max Slip Ratio : 0.30000
    - 1g3. Ref Slip Angle Deg : 13.00000
    - 1g4. Engine Cut Level : 1.10000
    - 1g5. Angular A C Cgain : 5.00000
    - 1g6. Oversteer Gain : 6.00000
    - 1g7. Slip Angle Activation Deg : 3.00000
  - 1g. Settings 5
    - 1g1. Min Slip Ratio : 0.18000
    - 1g2. Max Slip Ratio : 0.30000
    - 1g3. Ref Slip Angle Deg : 11.00000
    - 1g4. Engine Cut Level : 0.95000
    - 1g5. Angular A C Cgain : 4.00000
    - 1g6. Oversteer Gain : 10.00000



└ 1g7. Slip Angle Activation Deg : 2.50000

- 1g. Settings 6

└ 1g1. Min Slip Ratio : 0.17000

└ 1g2. Max Slip Ratio : 0.28000

└ 1g3. Ref Slip Angle Deg : 9.00000

└ 1g4. Engine Cut Level : 0.80000

└ 1g5. Angular A C Cgain : 13.00000

└ 1g6. Oversteer Gain : 11.00000

└ 1g7. Slip Angle Activation Deg : 2.25000

- 1g. Settings 7

└ 1g1. Min Slip Ratio : 0.15000

└ 1g2. Max Slip Ratio : 0.24000

└ 1g3. Ref Slip Angle Deg : 8.90000

└ 1g4. Engine Cut Level : 0.65000

└ 1g5. Angular A C Cgain : 20.00000

└ 1g6. Oversteer Gain : 14.00000

└ 1g7. Slip Angle Activation Deg : 2.00000

- 1g. Settings 8

└ 1g1. Min Slip Ratio : 0.15000

└ 1g2. Max Slip Ratio : 0.24000

└ 1g3. Ref Slip Angle Deg : 8.90000

└ 1g4. Engine Cut Level : 0.65000

└ 1g5. Angular A C Cgain : 20.00000

└ 1g6. Oversteer Gain : 14.00000

└ 1g7. Slip Angle Activation Deg : 1.80000

- 1g. Settings 9

└ 1g1. Min Slip Ratio : 0.12000

└ 1g2. Max Slip Ratio : 0.25000

└ 1g3. Ref Slip Angle Deg : 8.90000

└ 1g4. Engine Cut Level : 0.65000

└ 1g5. Angular A C Cgain : 20.00000

└ 1g6. Oversteer Gain : 14.00000

└ 1g7. Slip Angle Activation Deg : 1.80000

- 1g. Settings 10

└ 1g1. Min Slip Ratio : 0.10000

└ 1g2. Max Slip Ratio : 0.24000

└ 1g3. Ref Slip Angle Deg : 8.90000

└ 1g4. Engine Cut Level : 0.65000

└ 1g5. Angular A C Cgain : 20.00000

└ 1g6. Oversteer Gain : 14.00000

└ 1g7. Slip Angle Activation Deg : 1.70000

- 1g. Settings 11

└ 1g1. Min Slip Ratio : 0.08000

└ 1g2. Max Slip Ratio : 0.18000

└ 1g3. Ref Slip Angle Deg : 8.00000

└ 1g4. Engine Cut Level : 0.10000

└ 1g5. Angular A C Cgain : 0.00000

└ 1g6. Oversteer Gain : 1.00000

└ 1g7. Slip Angle Activation Deg : 1.60000

- 1g. Settings 12

└ 1g1. Min Slip Ratio : 0.08000

└ 1g2. Max Slip Ratio : 0.18000

└ 1g3. Ref Slip Angle Deg : 7.00000

└ 1g4. Engine Cut Level : 0.10000

- 1g5. Angular A C Cgain : 0.00000
- 1g6. Oversteer Gain : 1.00000
- 1g7. Slip Angle Activation Deg : 1.60000
- 1g. Settings 13
  - 1g1. Min Slip Ratio : 0.06000
  - 1g2. Max Slip Ratio : 0.18000
  - 1g3. Ref Slip Angle Deg : 5.00000
  - 1g4. Engine Cut Level : 0.10000
  - 1g5. Angular A C Cgain : 0.00000
  - 1g6. Oversteer Gain : 1.00000
  - 1g7. Slip Angle Activation Deg : 1.50000

## - 2. A B S

- 2a. Settings 1
  - 2a1. Min Slip Ratio : -1.00000
  - 2a2. Max Slip Ratio : -1.00000
  - 2a3. Ref Slip Angledeg : 0.00000
  - 2a4. Cut Level : 0.00000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 2
  - 2a1. Min Slip Ratio : 0.02000
  - 2a2. Max Slip Ratio : 0.30000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.60000
- 2a. Settings 3
  - 2a1. Min Slip Ratio : 0.03800
  - 2a2. Max Slip Ratio : 0.09000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 4
  - 2a1. Min Slip Ratio : 0.03300
  - 2a2. Max Slip Ratio : 0.08400
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 5
  - 2a1. Min Slip Ratio : 0.03100
  - 2a2. Max Slip Ratio : 0.08200
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 6
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.07900
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.80000
- 2a. Settings 7
  - 2a1. Min Slip Ratio : 0.02000
  - 2a2. Max Slip Ratio : 0.07500
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.80000

- 2a. Settings 8
  - 2a1. Min Slip Ratio : 0.04000
  - 2a2. Max Slip Ratio : 0.07000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 9
  - 2a1. Min Slip Ratio : 0.03700
  - 2a2. Max Slip Ratio : 0.07000
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 10
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.06500
  - 2a3. Ref Slip Angledeg : 5.00000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 11
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.06500
  - 2a3. Ref Slip Angledeg : 4.50000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.70000
- 2a. Settings 12
  - 2a1. Min Slip Ratio : 0.03000
  - 2a2. Max Slip Ratio : 0.06500
  - 2a3. Ref Slip Angledeg : 4.50000
  - 2a4. Cut Level : 0.15000
  - 2a5. Max Torque Variation : 0.70000
- 2b. Frequency : 200.00000
- 2c. Channels : 4
- 2d. Min Speed Kmh : 20.00000
- 3. E D L : None
- 4. E S P
  - 4a. Frequency Hz : 0.00000
  - 4b. Min Speed Kmh : 0.00000
- 4c. Settings : None

### Porsche 992 GT3 Cup

#### 1. No ABS No TC ( file : 992\_gt3\_no\_abs\_no\_tc.carelectronics )

- 1. T C : None
- 2. A B S : None
- 3. E D L : None
- 4. E S P
  - 4a. Frequency Hz : 0.00000
  - 4b. Min Speed Kmh : 0.00000
- 4c. Settings : None

#### 2. Only ABS ( file : 992\_gt3\_only\_abs.carelectronics )

- 1. T C : None

## 2. A B S

### 2a. Settings 1

- 2a1. Min Slip Ratio : -1.00000 | | 2a2. Max Slip Ratio : -1.00000
- 2a3. Ref Slip Angleddeg : 0.00000
- 2a4. Cut Level : 0.00000
- 2a5. Max Torque Variation : 0.00000

### 2a. Settings 2

- 2a1. Min Slip Ratio : 0.12000
- 2a2. Max Slip Ratio : 0.14000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 3

- 2a1. Min Slip Ratio : 0.11000
- 2a2. Max Slip Ratio : 0.14000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 4

- 2a1. Min Slip Ratio : 0.10000
- 2a2. Max Slip Ratio : 0.12000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 5

- 2a1. Min Slip Ratio : 0.08000
- 2a2. Max Slip Ratio : 0.10000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.02000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 6

- 2a1. Min Slip Ratio : 0.07000
- 2a2. Max Slip Ratio : 0.08000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 7

- 2a1. Min Slip Ratio : 0.06000
- 2a2. Max Slip Ratio : 0.08000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 8

- 2a1. Min Slip Ratio : 0.05000
- 2a2. Max Slip Ratio : 0.07000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 9

- 2a1. Min Slip Ratio : 0.05000
- 2a2. Max Slip Ratio : 0.06000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000

### 2a. Settings 10

- 2a1. Min Slip Ratio : 0.04000

- 2a2. Max Slip Ratio : 0.05000
- 2a3. Ref Slip Angleddeg : 7.00000
- 2a4. Cut Level : 0.20000
- 2a5. Max Torque Variation : 1.00000
- 2a. Settings 11
  - 2a1. Min Slip Ratio : 0.02500
  - 2a2. Max Slip Ratio : 0.03500
  - 2a3. Ref Slip Angleddeg : 7.00000
  - 2a4. Cut Level : 0.20000
  - 2a5. Max Torque Variation : 1.00000
- 2a. Settings 12
  - 2a1. Min Slip Ratio : 0.01000
  - 2a2. Max Slip Ratio : 0.02000
  - 2a3. Ref Slip Angleddeg : 7.00000
  - 2a4. Cut Level : 0.20000
  - 2a5. Max Torque Variation : 1.00000
- 2b. Frequency : 200.00000
- 2c. Channels : 4
- 2d. Min Speed Kmh : 20.00000
- 3. E D L : None
- 4. E S P
  - 4a. Frequency Hz : 0.00000
  - 4b. Min Speed Kmh : 0.00000
  - 4c. Settings : None

# 10. Clutch [ .clutch ]

## A. Description

Mechanical bridge between crankshaft and gearbox input: how much torque the plates can hold, how much inertia they add when spinning, and how automated clutch logic (anti-stall, shift assist, forced engagement) behaves when the driver is not working a three-pedal clutch by hand.

Gear ratios and shift times live in the gearbox; engine torque lives in `.CARENGINE`. This asset decides whether that torque couples cleanly, slips under load, or is handed off by an autoclutch during paddle / assisted shifts.

---

## I. Role in the stack

| Concern                                 | Handled here             | Handled elsewhere                                 |
|-----------------------------------------|--------------------------|---------------------------------------------------|
| Plate inertia and max clamp torque      | <code>.CLUTCH</code>     | —                                                 |
| Bite / engagement curve (optional)      | <i>Clutch Curve</i> path | External <code>.CURVE</code>                      |
| Up/down shift actuator curves           | Autoclutch profiles      | External <code>.curve</code>                      |
| Gear ratios, shift timing hardware      | —                        | <code>.GEARBOX</code>                             |
| Peak engine torque the clutch must hold | —                        | <code>.CARENGINE</code>                           |
| Which clutch file loads                 | —                        | <code>.CAR</code> / <code>.TUNINGPART</code> path |

Loaded from Car Data *Clutch Path*, or swapped by a tuning part (e.g. Datsun 5-speed clutch pack).

---

## II. What you are really tuning

- Holding capacity** — *Clutch Max Torque* (Nm) is the fuse: above it, the plates slip even at full clamp. Caterham 500 Nm, Golf GTI 450 Nm, F2004 700 Nm. Engine upgrades that exceed this number waste power as heat and slip instead of acceleration.
- Rotational inertia** — *Clutch Inertia* is the spinning mass of the assembly. Low values (F2004 0.005) let RPM rise and fall fast when decoupled — race response. Higher values (Caterham 0.017, Golf 0.010) store more kinetic energy and feel heavier / more road-car like on launch and blips.
- Autoclutch behaviour** — Forced On true (Golf GTI) means the sim owns the clutch — pedal binding is effectively obsolete. *false* (Caterham, F2004) leaves manual clutch feel available.

*Min Rpm*s / *Max Rpm*s bound anti-stall / launch assist windows (Caterham 1200–2000, Golf 1500–2400, F2004 a high 4300–4900 race idle band). Below min, the actuator can slip or open to save the engine.

*Use On Changes* true = autoclutch intervenes on gear changes (F2004, Golf). *false* (Caterham) = no automatic clutch work on shifts — pure driver friction zone.

4. **Shift profiles** — *Upshift Profile / Down Shift Profile* point at .CURVE files that shape engagement timing and damping during automated or assisted shifts. Caterham wires both; F2004 only a downshift profile in the dump; Golf leaves both None and relies on Forced On + RPM window instead.
5. **Clutch curve** — Optional path for a progressive bite map vs pedal / actuator travel. All three examples leave *Clutch Curve : None* — default linear or engine-side behaviour unless a curve is authored.

---

### III. Architecture

#### **1 - MECHANICAL CONSTANTS (SCHEMA 1-2)**

*Clutch Inertia, Clutch Max Torque*

#### **2 - AUTOCLUTCH OBJECT (SCHEMA 3)**

Paths for up/down profiles; *Forced On*; *Min / Max Rpm*s; *Use On Changes*.

#### **3 - ENGAGEMENT MAP (SCHEMA 4)**

*Clutch Curve* string path (or None).

---

### IV. How to read the examples

#### **1 - CATERHAM 485 CSR — ANALOG / HISTORIC**

Inertia 0.017, max torque 500 Nm. Both shift profiles present, but *Forced On* and *Use On Changes* false. Manual three-pedal philosophy with curve blueprints available if assist paths are ever enabled; no forced electronic safety net against stalls.

#### **2 - FERRARI F2004 — RACE SEQUENTIAL STYLE**

Very low inertia (0.005), high clamp (700 Nm). Downshift profile only; *Use On Changes* true; high RPM autoclutch window (4300–4900). Built for fast engine response and assisted shifts without full *Forced On* automation.

#### **3 - VOLKSWAGEN GOLF GTI MK8 — ROAD / ASSISTED**

Mid inertia (0.010), 450 Nm. No up/down curve paths; *Forced On* true and *Use On Changes* true with a 1500–2400 RPM band. Everyday drivability: software owns the clutch, profiles optional.

---

### V. Practical notes

- After an engine torque bump, re-check *Clutch Max Torque* or launches and WOT pulls will slip the plates.
- *Forced On : true* + empty profiles can still work if Min/Max RPM logic is enough; empty profiles + *Forced On* false usually means pure pedal.
- Tuning-part clutch swaps (alternate gearbox packages) should match the gearbox they ship with — 5-speed clutch with a different ratio set is a common content pairing.

- *Clutch Curve : None* is normal in the sample cars; do not assume a missing curve is a bug unless bite feel is wrong in-session.
- Schema dump spacing (*Down Shift Profile*) is cosmetic.

## VI. Related assets

- **3. Car Data** — Clutch Path
- **14. Gearbox** — ratios and shift events the clutch couples through
- **4. Car Engine** — torque the clutch must host
- **13. Drivetrain** — downstream of a closed clutch
- **8. Car Tuning Parts** — default electronics path for the stock car

## B. Schema

- 1. Clutch Inertia : float
- 2. Clutch Max Torque : float
- 3. Autoclutch : object
  - 3a. Upshift Profile : string - path
  - 3b. Down Shift Profile : string - path
  - 3c. Forced On : boolean
  - 3d. Min Rpms : float
  - 3e. Max Rpms : float
  - 3f. Use On Changes : boolean
- 4. Clutch Curve : string - path

## C. Measurement Units & Descriptions

| ID  | Name              | Unit of Measurement                          | Description                                                                                                                                                           |
|-----|-------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Clutch Intertia   | kg·m <sup>2</sup> ( Kilogram square meters ) | Rotational inertia of the clutch pressure plates and disc assembly; lower values allow faster RPM changes when decoupled (e.g., 0.005 F2004, 0.017 Caterham 485 CSR). |
| 2.  | Clutch Max Torque | Nm ( Newton-meters )                         | Maximum torque the friction plates can transmit when fully clamped before slip; acts as a powertrain fuse (e.g., 500 Nm Caterham, 700 Nm F2004).                      |
| 3.  | Autoclutch        | None ( Object )                              | Automated clutch assist profile block for shift timing, anti-stall, and launch logic.                                                                                 |
| 3a. | Upshift Profile   | None ( .curve file path )                    | Engagement/disengagement curve applied during upshifts (e.g., upShiftProfile.curve); may be None if unused.                                                           |



| ID  | Name               | Unit of Measurement             | Description                                                                                                                                     |
|-----|--------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 3b. | Down Shift Profile | None ( .curve file path )       | Engagement curve applied during downshifts (e.g., downShiftProfile.curve); may be None if unused.                                               |
| 3c. | Forced On          | None ( Boolean : True / False ) | When true, fully automates clutch operation (no manual pedal required; e.g., Golf GTI mk8).                                                     |
| 3d. | Min Rpms           | RPM ( Revolutions per minute )  | Lower RPM threshold for automated anti-stall logic; clutch slips or disengages below this speed (e.g., 1200 Caterham, 4300 F2004).              |
| 3e. | Max Rpms           | RPM ( Revolutions per minute )  | Upper RPM window bound for automated clutch/launch assist logic (e.g., 2000 Caterham, 4900 F2004).                                              |
| 3f. | Use On Changes     | None ( Boolean : True / False ) | When true, autoclutch intervenes during gear changes; when false, driver controls friction zone manually (Caterham: false).                     |
| 4.  | Clutch Curve       | None ( .curve file path )       | Look-up table mapping pedal travel or actuator position to clamping force / friction coefficient; often None when autoclutch profiles are used. |

## D. Example data

### I. Chosen Cars for Example

- Caterham 485 CSR ( slug : ks\_caterham\_485\_csr )
- Ferrari F2004 ( slug : ks\_ferrari\_f2004 )
- Volkswagen Golf GTI mk8 ( slug : ks\_volkswagen\_golf\_gti\_mk8 )

### II. Example

#### Caterham 485 CSR

```

| 1. Clutch Inertia : 0.01700
| 2. Clutch Max Torque : 500.00000
| 3. Autoclutch
| | 3a. Upshift Profile :
content\cars\ks_caterham_485_csr\data\upShiftProfile.curve
| | 3b. Down Shift Profile :
content\cars\ks_caterham_485_csr\data\downShiftProfile.curve
| | 3c. Forced On : false
| | 3d. Min Rpms : 1200.00000
| | 3e. Max Rpms : 2000.00000

```

- | L 3f. Use On Changes : false
- | 4. Clutch Curve : None

### **Ferrari F2004**

- | 1. Clutch Inertia : 0.00500
- | 2. Clutch Max Torque : 700.00000
- | 3. Autoclutch
  - | 3a. Upshift Profile : None
  - | 3b. Down Shift Profile :  
content\cars\ks\_ferrari\_f2004\data\downShiftProfile.curve
  - | 3c. Forced On : false
  - | 3d. Min Rpms : 4300.00000
  - | 3e. Max Rpms : 4900.00000
  - | 3f. Use On Changes : true
- | 4. Clutch Curve : None

### **Volkswagen Golf GTI mk8**

- | 1. Clutch Inertia : 0.01000
- | 2. Clutch Max Torque : 450.00000
- | 3. Autoclutch
  - | 3a. Upshift Profile : None
  - | 3b. Down Shift Profile : None
  - | 3c. Forced On : true
  - | 3d. Min Rpms : 1500.00000
  - | 3e. Max Rpms : 2400.00000
  - | 3f. Use On Changes : true
- | 4. Clutch Curve : None

# 11. Coilover [ .coilover ]

## A. Description

Per-axle (usually front and rear files) spring–damper unit: base spring rate, progressive rate, bump/rebound stops, collar position, slow/fast damper valves, optional damper LUT pack, helper spring, and optional rod controllers.

Car Setup overlays garage wheel rates, damper clicks, and ride-height collars on top of this hardware. Suspension kinematics (pickups, arms) live in .suspension. Alignment (camber, toe, pressure) is a setup concern — not fields in this schema, despite older prose that mixes the topics.

## I. Role in the stack

| Concern                                    | Handled here                    | Handled elsewhere                               |
|--------------------------------------------|---------------------------------|-------------------------------------------------|
| Base spring / progressive rate             | .COILOVER                       | Garage override in .CARSETUP                    |
| Bump stop up/down shape                    | .COILOVER                       | Setup may retarget ranges/rates                 |
| Slow/fast damper coefficients + thresholds | .COILOVER Damper                | Click overlay in setup; curves in .DAMPERCURVES |
| Damper force LUT pack                      | <i>Lut List</i> → .DAMPERCURVES | Shared <i>common_phsx</i> packs                 |
| Helper spring K / range                    | .COILOVER                       | Setup helpers when exposed                      |
| Active rod / heave logic                   | Rod Controllers stages          | Same stage pattern as other controllers         |
| Hardpoints / motion ratio geometry         | —                               | .SUSPENSION                                     |
| Which front/rear files load                | —                               | .CAR / .TUNINGPART paths                        |

One coilover asset ≈ one corner family (front file + rear file in examples). Tuning parts can swap both (Datsun G-nose pack).

## II. What you are really tuning

1. **Spring platform** — *Sprint Rate* in the dump is **Spring Rate** (OCR). Effective stiffness at the unit — Caterham about 14000 front / 16000 rear; Alpine 31000 / 40000; Dallara EXP 50000 / 80000. Higher = flatter platform, less mechanical compliance.

*Progressive Spring Rate* adds rate with travel when non-zero (Alpine front 30000, rear 40000). Zero (Caterham, Dallara) = linear main spring.

2. **Bump stops** — *Bump Stop Up / Down: Range, Reference, Force, Gamma, Length, Damping*. Up = compression packer behaviour; Down = extension / droop stop. Short ranges and high gamma (Dallara front Up range negative / tight reference 0.08) engage early under load — race aero platform. Caterham keeps longer ranges (about 0.04–0.07) for travel.

3. **Collar / ride reference** — *Collar Position* sets the threaded perch baseline that setup ride-height / collar adjustments build on (Caterham front about 0.05, rear about 0.07; Alpine and Dallara vary per axle).
4. **Damper valves** — *Damper.Slow / Fast* bump and rebound, plus *Fast Threshold* bump/rebound (velocity where “fast” map takes over). Caterham thresholds 0.15; Alpine 0.5 / 0.8; Dallara 0.8 / 0.8 — race packs separate chassis control from kerb hits more explicitly.

Thermal / fatigue fields (*Cooling Surface*, *Nominal Force*, stress, *Thermal Capacity*, *Heat Transfer Coef*) are present but zeroed in these examples.

5. **Damper LUT** — *Lut List* can point at a `.DAMPERCURVES` asset (Dallara → Penske pack in *common\_phsx*) with *Damper Lut Scale* (1.0). Caterham / Alpine leave *Lut None* and rely on the scalar slow/fast values alone.
6. **Helpers and rods** — *Helper K / Helper Range* for tender springs — unused (0) in all three samples. *Rod Controllers* stages (Input Var × Combinator × Lut × limits) enable active or linked rod behaviour; examples leave Name/Stages empty.

---

### III. Architecture

#### 1 - SPRINGS AND STOPS (SCHEMA 1-5)

*Spring rate, progressive rate, bump stop Up/Down objects, collar position.*

#### 2 - DAMPER BLOCK (SCHEMA 6)

Fast/Slow bump & rebound; thresholds; thermal/fatigue placeholders; optional Lut path + scale.

#### 3 - HELPER AND CONTROLLERS (SCHEMA 7-9)

*Helper K/range; Rod Controllers with staged Input Var / Combinator Mode enums (Brake, Gas, LatG, LonG, Steer, Speed, slip/angle metrics, ERS...; Add / Mult).*

---

### IV. How to read the examples

#### 1 - CATERHAM 485 CSR — COMPLIANT ROAD / HISTORIC

Soft springs (14k / 16k), progressive 0, generous bump-stop ranges, low fast thresholds (0.15), no damper LUT, no helpers/rods. Same damper scalars front and rear in the dump — balance comes more from spring rates and geometry elsewhere.

#### 2 - ALPINE A110S — STIFF ROAD / TRACK SPORT

Much higher rates (31k–40k) plus strong progressive springs. Shorter compression stop up front; denser slow rebound (7000–7500). Still no LUT — authored as scalar valves with higher fast thresholds than Caterham.

#### 3 - DALLARA EXP — HIGH-DOWNFORCE RACE

50k / 80k linear springs, aggressive bump-stop packing, high slow damper numbers (rear rebound 12800), Penske .dampercurves LUT at scale 1.0. Classic race pattern: stiff platform + shared professional damper map.

---

## V. Practical notes

- Schema label **Sprint Rate** = Spring Rate typo in the converted dump.
- Setup wheel rates / damper clicks must stay consistent with whether this car is SI ( $Ns/m$ ) or click-profile — see Setup Units.
- Negative bump-stop *Range* values appear in Alpine rear Down and Dallara front Up — treat as authored packer geometry, not necessarily a parse error, but verify in-game travel.
- Wiring a Lut List without a valid .DAMPERCURVES path breaks the valve model; scale 0 with a path is suspicious.
- Front/rear are separate files — a tuning part that only swaps one axle will create an asymmetric package on purpose or by mistake.
- PDF “alignments inside coilover” prose does not match this schema; use Car Setup for camber/toe/pressure.

---

## VI. Related assets

- **5 / 6 / 7. Car Setup / Limits / Units** — garage overlays and display language
- **12. Damper Curves** — LUT packs referenced by Lut List
- **17. Suspension** — kinematic hardpoints and motion
- **3. Car Data** — front/rear coilover paths
- **8. Car Tuning Parts** — coilover path redirects (G-nose, drift packs)

### B. Schema

- 1. Spring Rate : float
- 2. Progressive Spring Rate : float
- 3. Bump Stop Up : object
  - 3a. Range : float
  - 3b. Reference : float
  - 3c. Force : float
  - 3d. Gamma : float
  - 3e. Length : float
  - 3f. Damping : float
- 4. Bump Stop Down : object
  - 4a. Range : float
  - 4b. Reference : float
  - 4c. Force : float
  - 4d. Gamma : float
  - 4e. Length : float
  - 4f. Damping : float
- 5. Collar Position : float

```

6. Damper : object
├── 6a. Fast : object
│ ├── 6a1. Bump : float
│ └── 6a2. Rebound : float
├── 6b. Slow : object
│ ├── 6a1. Bump : float
│ └── 6a2. Rebound : float
├── 6c. Fast Threshold Bump : float
├── 6d. Fast Threshold Rebound : float
├── 6e. Cooling Surface : float
├── 6f. Nominal Force : float
├── 6g. Min Stress Fatigue : float
├── 6h. Max Stress Fatigue : float
├── 6i. Thermal Capacity : float
├── 6j. Heat Transfer Coef : float
├── 6k. Lut List : string – path
└── 6l. Damper Lut Scale : float
7. Helper K : float
8. Helper Range : float
9. Rod Controllers : object
├── 9a. Name : string
└── 9b. Stages [x] : object | can have multiple Stages
 ├── 9b1. Input Var : enum
 ├── 9b2. Combinator Mode : enum
 ├── 9b3. Lut : string – path
 ├── 9b4. Filter Gain : float
 ├── 9b5. Up Limit : float
 ├── 9b6. Down Limit : float
 ├── 9b7. Current Value : float
 └── 9b8. Const Value : float

```

### Enum - Car Coilover

|     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9b2 | Input Var       | UndefinedInput, Brake, Gas, LatG, LonG, Steer, Speed, Gear, SlipRatioFrontAVG, SlipRatioRearAVG, SlipRatioFrontMAX, SlipRatioRearMAX, SlipAngleFrontAVG, SlipAngleRearAVG, SlipAngleFrontMAX, SlipAngleRearMAX, OversteerFactor, RearSpeedRatio, SteerDEG, Const, RPMS, WheelSteerDEG, LoadSpreadLF, LoadSpreadRF, AvgTravelRear, SusTravelLR, SusTravelRR, SteerYawDeltaLeft, SteerYawDeltaRight, ErsChargeLevel, ErsCoastTorque |
| 9b2 | Combinator Mode | UndefinedMode, Add, Mult                                                                                                                                                                                                                                                                                                                                                                                                          |

### C. Measurement Units & Descriptions

| ID | Name        | Unit of Measurement       | Description                                                                                                                                        |
|----|-------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Spring Rate | N/m ( Newtons per meter ) | Primary linear spring stiffness at the wheel (schema label Sprint Rate; likely Spring Rate).<br>Caterham front: 14000 N/m, Alpine rear: 40000 N/m. |

| ID  | Name                    | Unit of Measurement                          | Description                                                                                                  |
|-----|-------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 2.  | Progressive Spring Rate | N/m <sup>2</sup> or N/m ( Progressive rate ) | Secondary progressive spring rate; stiffness rises non-linearly with compression (0 = disabled on Caterham). |
| 3.  | Bump Stop Up            | None ( Object )                              | Upper bump-stop block limiting compression travel beyond main spring range.                                  |
| 3a. | Range                   | m ( Meters )                                 | Travel range before upper bump-stop engages (Bump Stop Up).                                                  |
| 3b. | Reference               | m ( Meters )                                 | Reference suspension position where upper bump-stop force scaling begins.                                    |
| 3c. | Force                   | N ( Newtons )                                | Peak resistive force of the upper bump-stop at maximum compression.                                          |
| 3d. | Gamma                   | Dimensionless exponent                       | Non-linear shaping exponent for upper bump-stop force vs. compression.                                       |
| 3e. | Length                  | m ( Meters )                                 | Physical contact length of the upper bump-stop element.                                                      |
| 3f. | Damping                 | N·s/m ( Damping coefficient )                | Damping applied during upper bump-stop engagement.                                                           |
| 4.  | Bump Stop Down          | None ( Object )                              | Lower bump-stop block limiting droop/rebound travel.                                                         |
| 4a. | Range                   | m ( Meters )                                 | Travel range before lower bump-stop engages (Bump Stop Down).                                                |
| 4b. | Reference               | m ( Meters )                                 | Reference position where lower bump-stop force model activates.                                              |
| 4c. | Force                   | N ( Newtons )                                | Peak resistive force of the lower bump-stop at maximum extension.                                            |
| 4d. | Gamma                   | Dimensionless exponent                       | Non-linear shaping exponent for lower bump-stop force curve.                                                 |
| 4e. | Length                  | m ( Meters )                                 | Physical contact length of the lower bump-stop element.                                                      |
| 4f. | Damping                 | N·s/m ( Damping coefficient )                | Damping applied during lower bump-stop engagement on rebound.                                                |
| 5.  | Collar Position         | m ( Meters )                                 | Spring perch/collar position controlling static ride height and preload (e.g., 0.04961 m Caterham front).    |
| 6.  | Damper                  | None ( Object )                              | Hydraulic damper block: fast/slow circuits, thresholds, thermal model, and optional LUT.                     |

| ID   | Name                   | Unit of Measurement                   | Description                                                                                        |
|------|------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------|
| 6a.  | Fast                   | None ( Object )                       | Fast damper circuit sub-block for high-velocity wheel movements (kerbs, bumps).                    |
| 6a1. | Bump                   | N·s/m ( Damping rate )                | Fast-speed compression damping under 6a Fast (e.g., 700 Caterham, 1500 Alpine).                    |
| 6a2. | Rebound                | N·s/m ( Damping rate )                | Fast-speed extension damping under 6a Fast (e.g., 1700 Caterham front).                            |
| 6b.  | Slow                   | None ( Object )                       | Slow damper circuit sub-block for body roll, pitch, and weight transfer.                           |
| 6a1. | Bump                   | N·s/m ( Damping rate )                | Slow-speed compression damping under 6b Slow; schema reuses ID 6a1 (examples may label 6b1).       |
| 6a2. | Rebound                | N·s/m ( Damping rate )                | Slow-speed extension damping under 6b Slow; schema reuses ID 6a2 (examples may label 6b2).         |
| 6c.  | Fast Threshold Bump    | m/s ( Velocity threshold )            | Compression velocity above which the fast bump circuit engages (e.g., 0.15 Caterham, 0.50 Alpine). |
| 6d.  | Fast Threshold Rebound | m/s ( Velocity threshold )            | Extension velocity above which the fast rebound circuit engages.                                   |
| 6e.  | Cooling Surface        | m <sup>2</sup> ( Square meters )      | Exposed damper body surface area for thermal dissipation modeling.                                 |
| 6f.  | Nominal Force          | N ( Newtons )                         | Reference force level for damper nominal operating point and fatigue calculations.                 |
| 6g.  | Min Stress Fatigue     | Pa or N ( Stress threshold )          | Lower bound stress below which damper fatigue accumulation is negligible.                          |
| 6h.  | Max Stress Fatigue     | Pa or N ( Stress threshold )          | Upper bound stress at which damper fatigue reaches maximum degradation rate.                       |
| 6i.  | Thermal Capacity       | J/K or J/°C                           | Heat storage capacity of the damper assembly under repeated high-energy cycles.                    |
| 6j.  | Heat Transfer Coef     | W/(m <sup>2</sup> ·K) ( Coefficient ) | Convective heat transfer rate from damper body to ambient airflow.                                 |
| 6k.  | Lut List               | None ( File path )                    | Path to damper force-velocity LUT list; often None when using scalar rates only.                   |
| 6i.  | Damper Lut Scale       | Dimensionless multiplier              | Scaling factor applied to LUT damper output (0 = unused when Lut List is None).                    |



| ID   | Name            | Unit of Measurement                  | Description                                                                                   |
|------|-----------------|--------------------------------------|-----------------------------------------------------------------------------------------------|
| 7.   | Helper K        | N/m ( Spring Rate )                  | Stiffness of helper/tender spring assisting the main coil at low compression.                 |
| 8.   | Helper Range    | m ( Meters )                         | Travel range over which the helper spring is active before the main spring carries full load. |
| 9.   | Rod Controllers | None ( Object )                      | Dynamic suspension rod controller block; optional active/heave control pipelines.             |
| 9a.  | Name            | None ( String )                      | Internal identifier for the rod controller; often None when inactive.                         |
| 9b.  | Stages          | None ( Object array )                | Multi-stage controller pipeline; each stage maps telemetry input through a LUT (often None).  |
| 9b1. | Input Var       | None ( Telemetry enum )              | Telemetry input channel (Speed, SusTravelLR, SteerDEG, SlipRatio, ErsChargeLevel, etc.).      |
| 9b2. | Combinator Mode | None ( Math enum : Add / Mult )      | How stage output combines with prior stages: additive or multiplicative.                      |
| 9b3. | Lut             | None ( .curve file path )            | Look-up table mapping filtered input to rod/damper modifier output.                           |
| 9b4. | Filter Gain     | Coefficient ( Smoothing multiplier ) | Low-pass filter coefficient smoothing controller input spikes.                                |
| 9b5. | Up Limit        | Depends on input variable            | Upper clamp on processed input before LUT evaluation.                                         |
| 9b6. | Down Limit      | Depends on input variable            | Lower clamp on processed input before LUT evaluation.                                         |
| 9b7. | Current Value   | Depends on input variable            | Runtime controller stage output during simulation (telemetry/ debug).                         |
| 9b8. | Const Value     | Depends on input variable            | Constant fallback output when Input Var is Const or LUT is inactive.                          |

## D. Example data

### I. Chosen Cars for Example

- Caterham 485 CSR ( slug : ks\_caterham\_485\_csr )
- Alpine A110s ( slug : ks\_alpine\_a110\_s )
- Dallara EXP ( slug : ks\_dallara\_exp )

---

## II. Example

### Caterham 485 CSR

#### 1. Front Coilover ( file : caterham\_485\_csr\_fr\_coil.coilover )

- 1. Spring Rate : 14000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
  - 3a. Range : 0.04539
  - 3b. Reference : 0.01000
  - 3c. Force : 100.00000
  - 3d. Gamma : 2.00000
  - 3e. Length : 0.02000
  - 3f. Damping : 300.00000
- 4. Bump Stop Down : object
  - 4a. Range : 0.06961
  - 4b. Reference : 0.01000
  - 4c. Force : 80.00000
  - 4d. Gamma : 2.00000
  - 4e. Length : 0.02000
  - 4f. Damping : 100.00000
- 5. Collar Position : 0.04961
- 6. Damper : object
  - 6a. Fast : object
    - 6a1. Bump : 700.00000
    - 6a2. Rebound : 1700.00000
  - 6b. Slow : object
    - 6b1. Bump : 2000.00000
    - 6b2. Rebound : 3000.00000
  - 6c. Fast Threshold Bump : 0.15000
  - 6d. Fast Threshold Rebound : 0.15000
  - 6e. Cooling Surface : 0.00000
  - 6f. Nominal Force : 0.00000
  - 6g. Min Stress Fatigue : 0.00000
  - 6h. Max Stress Fatigue : 0.00000
  - 6i. Thermal Capacity : 0.00000
  - 6j. Heat Transfer Coef : 0.00000
  - 6k. Lut List : None
  - 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
  - 9a. Name : None
  - 9b. Stages [x] : None

#### 2. Rear Coilover ( file : caterham\_485\_csr\_r\_coil.coilover )

- 1. Spring Rate : 16000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
  - 3a. Range : 0.04031
  - 3b. Reference : 0.01000
  - 3c. Force : 100.00000

- 3d. Gamma : 2.00000
- 3e. Length : 0.02000
- 3f. Damping : 300.00000
- 4. Bump Stop Down : object
  - 4a. Range : 0.04469
  - 4b. Reference : 0.01000
  - 4c. Force : 80.00000
  - 4d. Gamma : 2.00000
  - 4e. Length : 0.02000
  - 4f. Damping : 100.00000
- 5. Collar Position : 0.06969
- 6. Damper : object
  - 6a. Fast : object
    - 6a1. Bump : 700.00000
    - 6a2. Rebound : 1700.00000
  - 6b. Slow : object
    - 6b1. Bump : 2000.00000
    - 6b2. Rebound : 3000.00000
  - 6c. Fast Threshold Bump : 0.15000
  - 6d. Fast Threshold Rebound : 0.15000
  - 6e. Cooling Surface : 0.00000
  - 6f. Nominal Force : 0.00000
  - 6g. Min Stress Fatigue : 0.00000
  - 6h. Max Stress Fatigue : 0.00000
  - 6i. Thermal Capacity : 0.00000
  - 6j. Heat Transfer Coef : 0.00000
  - 6k. Lut List : None
  - 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
  - 9a. Name : None
  - 9b. Stages [x] : None

## Alpine A110s

### 1. Front Coilover ( file : ks\_alpine\_a110\_s\_front.coilover )

- 1. Spring Rate : 31000.00000
- 2. Progressive Spring Rate : 30000.00000
- 3. Bump Stop Up : object
  - 3a. Range : 0.01616
  - 3b. Reference : 0.00600
  - 3c. Force : 300.00000
  - 3d. Gamma : 3.00000
  - 3e. Length : 0.02000
  - 3f. Damping : 400.00000
- 4. Bump Stop Down : object
  - 4a. Range : 0.05184
  - 4b. Reference : 0.01000
  - 4c. Force : 300.00000
  - 4d. Gamma : 1.50000
  - 4e. Length : 0.04500
  - 4f. Damping : 400.00000
- 5. Collar Position : 0.07184

- 6. Damper : object
  - 6a. Fast : object
    - 6a1. Bump : 1500.00000
    - 6a2. Rebound : 3500.00000
  - 6b. Slow : object
    - 6b1. Bump : 3000.00000
    - 6b2. Rebound : 7000.00000
  - 6c. Fast Threshold Bump : 0.50000
  - 6d. Fast Threshold Rebound : 0.80000
  - 6e. Cooling Surface : 0.00000
  - 6f. Nominal Force : 0.00000
  - 6g. Min Stress Fatigue : 0.00000
  - 6h. Max Stress Fatigue : 0.00000
  - 6i. Thermal Capacity : 0.00000
  - 6j. Heat Transfer Coef : 0.00000
  - 6k. Lut List : None
  - 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
  - 9a. Name : None
  - 9b. Stages [x] : None

## 2. Rear Coilover ( file : ks\_alpine\_a110\_s\_rear.coilover )

- 1. Spring Rate : 40000.00000
- 2. Progressive Spring Rate : 40000.00000
- 3. Bump Stop Up : object
  - 3a. Range : 0.06087
  - 3b. Reference : 0.00600
  - 3c. Force : 400.00000
  - 3d. Gamma : 2.00000
  - 3e. Length : 0.02000
  - 3f. Damping : 500.00000
- 4. Bump Stop Down : object
  - 4a. Range : -0.00887
  - 4b. Reference : 0.01000
  - 4c. Force : 400.00000
  - 4d. Gamma : 1.50000
  - 4e. Length : 0.02000
  - 4f. Damping : 500.00000
- 5. Collar Position : 0.00913
- 6. Damper : object
  - 6a. Fast : object
    - 6a1. Bump : 1500.00000
    - 6a2. Rebound : 2300.00000
  - 6b. Slow : object
    - 6b1. Bump : 4200.00000
    - 6b2. Rebound : 7500.00000
  - 6c. Fast Threshold Bump : 0.50000
  - 6d. Fast Threshold Rebound : 0.80000
  - 6e. Cooling Surface : 0.00000
  - 6f. Nominal Force : 0.00000
  - 6g. Min Stress Fatigue : 0.00000
  - 6h. Max Stress Fatigue : 0.00000

- | 6i. Thermal Capacity : 0.00000
- | 6j. Heat Transfer Coef : 0.00000
- | 6k. Lut List : None
- | 6l. Damper Lut Scale : 0.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
  - | 9a. Name : None
  - | 9b. Stages [x] : None

## Dallara EXP

### 1. Front Coilover ( file : ks\_dallara\_exp\_front\_coil.coilover )

- 1. Spring Rate : 50000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
  - | 3a. Range : -0.01135
  - | 3b. Reference : 0.08000
  - | 3c. Force : 350.00000
  - | 3d. Gamma : 3.00000
  - | 3e. Length : 0.01500
  - | 3f. Damping : 1000.00000
- 4. Bump Stop Down : object
  - | 4a. Range : 0.05635
  - | 4b. Reference : 0.01000
  - | 4c. Force : 350.00000
  - | 4d. Gamma : 1.50000
  - | 4e. Length : 0.02000
  - | 4f. Damping : 500.00000
- 5. Collar Position : 0.05135
- 6. Damper : object
  - | 6a. Fast : object
    - | 6a1. Bump : 1234.00000
    - | 6a2. Rebound : 2889.00000
  - | 6b. Slow : object
    - | 6b1. Bump : 5500.00000
    - | 6b2. Rebound : 8200.00000
  - | 6c. Fast Threshold Bump : 0.80000
  - | 6d. Fast Threshold Rebound : 0.80000
  - | 6e. Cooling Surface : 0.00000
  - | 6f. Nominal Force : 0.00000
  - | 6g. Min Stress Fatigue : 0.00000
  - | 6h. Max Stress Fatigue : 0.00000
  - | 6i. Thermal Capacity : 0.00000
  - | 6j. Heat Transfer Coef : 0.00000
  - | 6k. Lut List :
    - content\cars\common\_phsx\dampers\penske\damper\_penske.dampercurves
    - | 6l. Damper Lut Scale : 1.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
  - | 9a. Name : None
  - | 9b. Stages [x] : None

## 2. Rear Coilover ( file : ks\_dallara\_exp\_rear\_coil.coilover )

- 1. Spring Rate : 80000.00000
- 2. Progressive Spring Rate : 0.00000
- 3. Bump Stop Up : object
  - 3a. Range : 0.01310
  - 3b. Reference : 0.08000
  - 3c. Force : 400.00000
  - 3d. Gamma : 3.00000
  - 3e. Length : 0.01500
  - 3f. Damping : 1000.00000
- 4. Bump Stop Down : object
  - 4a. Range : 0.03490
  - 4b. Reference : 0.01000
  - 4c. Force : 400.00000
  - 4d. Gamma : 1.50000
  - 4e. Length : 0.02000
  - 4f. Damping : 1000.00000
- 5. Collar Position : 0.02690
- 6. Damper : object
  - 6a. Fast : object
    - 6a1. Bump : 2042.00000
    - 6a2. Rebound : 6889.00000
  - 6b. Slow : object
    - 6b1. Bump : 7056.00000
    - 6b2. Rebound : 12800.00000
  - 6c. Fast Threshold Bump : 0.80000
  - 6d. Fast Threshold Rebound : 0.80000
  - 6e. Cooling Surface : 0.00000
  - 6f. Nominal Force : 0.00000
  - 6g. Min Stress Fatigue : 0.00000
  - 6h. Max Stress Fatigue : 0.00000
  - 6i. Thermal Capacity : 0.00000
  - 6j. Heat Transfer Coef : 0.00000
  - 6k. Lut List :  
content\cars\common\_phsx\dampers\penske\damper\_penske.dampercurves
    - 6l. Damper Lut Scale : 1.00000
- 7. Helper K : 0.00000
- 8. Helper Range : 0.00000
- 9. Rod Controllers : object
  - 9a. Name : None
  - 9b. Stages [x] : None

## 12. Damper Curves [ .dampercurves ]

### A. Description

Indexable bank of damper force LUTs: an ordered list of paths to .CURVE files. Coilover assets reference one of these packs via *Damper.Lut List*, then scale or select into it with *Damper Lut Scale* and garage damper settings.

The official prose block in the source dump is unfinished placeholder text (copied spring/alignment headings). Behaviour below is inferred from the schema and from how coilovers actually wire these packs (e.g. Dallara EXP → Penske).

---

### I. Role in the stack

| Concern                             | Handled here  | Handled elsewhere                 |
|-------------------------------------|---------------|-----------------------------------|
| Ordered list of damper LUT curves   | .DAMPERCURVES | Each entry → .CURVE               |
| Which pack a corner uses            | —             | .COILOVER <i>Lut List</i>         |
| Scale / blend of that pack          | —             | .COILOVER <i>Damper Lut Scale</i> |
| Slow/fast scalar valves when no LUT | —             | .COILOVER <i>Damper Fast/Slow</i> |
| Garage damper clicks / rates        | —             | .CARSETUP (+ limits / units )     |

Almost always shipped under *content\cars\common\_phsx\dampers\...* and shared across many cars — not duplicated per vehicle folder.

---

### II. What you are really tuning

- LUT catalogue, not a single curve** — The asset is only *Damper Lut [x]* paths. Index 1, 2, 3... are alternate valve maps (click positions, bump/rebound families, or manufacturer presets). The coilover / setup layer picks which entry matters in session.
- Front vs rear packs** — Some brands ship separate .DAMPERCURVES files per axle (Ford GT3 front/rear, Cayman GT4 front/rear) so F and R can use different curve sets. Others ship one shared pack (Penske) referenced by both front and rear coilovers.
- Depth of the click rack** — Catalogue size varies by hardware philosophy:

| Pack                            | Approx. LUT count | Pattern                         |
|---------------------------------|-------------------|---------------------------------|
| Ford GT3 front / rear           | 5 + 5             | Compact race click set per axle |
| Porsche Cayman GT4 front / rear | 12 + 12           | Wider per-axle rack             |
| Penske (shared)                 | 60                | Dense shared library            |

More entries = finer garage resolution if setup/limits expose that many steps and the coilover scale maps onto them cleanly.

4. **Relationship to scalar dampers** — Caterham / Alpine coilovers in the coilover examples leave *Lut List* : *None* and use Slow/Fast bump-rebound floats only. Dallara sets *Lut List* to *Penske* and *Damper Lut Scale* : *1.0* — the curve bank becomes the real force shape; scalars and scale still matter as multipliers / baselines depending on authoring.

---

## III. Architecture

### 1 - DAMPER CURVES LIST EDIT (SCHEMA 1)

Single repeating field: *Damper Lut [x] : string* (path). Can have many entries; no other fields in the dump.

### 2 - REFERENCE PAYLOAD

Each path points at a *.curve* (force vs velocity or equivalent LUT). Editing damper feel means editing those curves or swapping which *.dampercurves* pack the coilover points at — not adding sibling fields inside this asset.

---

## IV. How to read the examples

### 1 - FORD GT3 DAMPERS

Two files: *ford\_gt3\_front\_damper.dampercurves* and rear counterpart. Each lists five LUTs (*ford\_damper\_front\_1...5.curve* / *rear\_1...5*). Axle-specific, short click banks — typical GT3 shared physics.

### 2 - PENSKE (COMMON PHSX)

One pack with sixty LUTs (*damper\_1.curve* ... *damper\_60.curve*). Same file Dallara EXP coilovers reference. Broad manufacturer library reused by multiple race cars.

### 3 - PORSCHE CAYMAN GT4 DAMPERS

Front and rear packs under ...*\porsche\cayman\*, twelve curves each (*front1...12*, *rear1...12*). Mid-size per-axle racks between Ford's five and Penske's sixty.

---

## V. Practical notes

- Source Description section is placeholder (Xxxxxxx) — do not trust its spring/alignment headings for this asset.
- Broken or missing *.CURVE* paths inside the list fail at the LUT the garage actually selects, which can look like “random click does nothing”.
- Changing only the *.DAMPERCURVES* file while coilovers still point at another pack does nothing until *Lut List* is updated.
- *Damper Lut Scale* : *0* with a non-None *Lut List* (or scale 1 with empty curves) is a content smell.



- Prefer editing shared *common\_phsx* packs carefully — one change hits every car that references them.
- Setup damper indices must not exceed the LUT count in the referenced pack.

---

## VI. Related assets

- **11. Coilover** — Lut List / Damper Lut Scale consumers
- **5 / 6 / 7. Car Setup / Limits / Units** — which click/rate the driver selects and how it is labelled
- **External .CURVE files** — actual force–velocity (or equivalent) maps listed here
- **3. Car Data** — does not hold dampercurvews directly; reaches them through coilover paths

## B. Schema

[ Data in a “Damper Curves List Edit” object ]

- └ 1. Damper Lut [x] : string – path | can have multiple Damper Lut

## C. Measurement Units & Descriptions

| ID | Name       | Unit of Measurement       | Description                                                                                                                                                                                                                                                                                                     |
|----|------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Damper Lut | None ( .curve file path ) | Indexed list of damper force–velocity LUT curves (.curve); each entry defines non-linear bump/rebound damping for a discrete damper click or valve map. Referenced from Coilover <b>6k Lut List</b> (.DAMPERCURVES asset). Ford GT3: 5 curves per axle; Penske kit: 60 curves; Porsche Cayman GT4: 12 per axle. |

## D. Example data

---

### I. Chosen Cars for Example

- Ford - GT3 Dampers ( common\_phsx ) [ 2 dampers ]
- Penske Dampers ( common\_phsx )
- Porsche Cayman Dampers ( common\_phsx ) [ 2 dampers ]

---

## II. Example

### Ford - GT3 Dampers

1. Front Damper ( file : ford\_gt3\_front\_damper.dampercurves )

└ 1. Damper Lut 1 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_front\_1.curve  
└ 1. Damper Lut 2 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_front\_2.curve  
└ 1. Damper Lut 3 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_front\_3.curve  
└ 1. Damper Lut 4 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_front\_4.curve  
└ 1. Damper Lut 5 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_front\_5.curve

## **2. Rear Damper ( file : ford\_gt3\_rear\_damper.dampercurves )**

└ 1. Damper Lut 1 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_rear\_1.curve  
└ 1. Damper Lut 2 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_rear\_2.curve  
└ 1. Damper Lut 3 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_rear\_3.curve  
└ 1. Damper Lut 4 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_rear\_4.curve  
└ 1. Damper Lut 5 :  
content\cars\common\_phsx\dampers\ford\ford\_damper\_rear\_5.curve

## **Penske**

└ 1. Damper Lut 1 :  
content\cars\common\_phsx\dampers\penske\damper\_1.curve  
└ 1. Damper Lut 2 :  
content\cars\common\_phsx\dampers\penske\damper\_2.curve  
└ 1. Damper Lut 3 :  
content\cars\common\_phsx\dampers\penske\damper\_3.curve  
└ 1. Damper Lut 4 :  
content\cars\common\_phsx\dampers\penske\damper\_4.curve  
└ 1. Damper Lut 5 :  
content\cars\common\_phsx\dampers\penske\damper\_5.curve  
└ 1. Damper Lut 6 :  
content\cars\common\_phsx\dampers\penske\damper\_6.curve  
└ 1. Damper Lut 7 :  
content\cars\common\_phsx\dampers\penske\damper\_7.curve  
└ 1. Damper Lut 8 :  
content\cars\common\_phsx\dampers\penske\damper\_8.curve  
└ 1. Damper Lut 9 :  
content\cars\common\_phsx\dampers\penske\damper\_9.curve  
└ 1. Damper Lut 10 :  
content\cars\common\_phsx\dampers\penske\damper\_10.curve  
└ 1. Damper Lut 11 :  
content\cars\common\_phsx\dampers\penske\damper\_11.curve  
└ 1. Damper Lut 12 :  
content\cars\common\_phsx\dampers\penske\damper\_12.curve  
└ 1. Damper Lut 13 :  
content\cars\common\_phsx\dampers\penske\damper\_13.curve  
└ 1. Damper Lut 14 :  
content\cars\common\_phsx\dampers\penske\damper\_14.curve

└ 1. Damper Lut 15 :  
content\cars\common\_phsx\dampers\penske\damper\_15.curve  
└ 1. Damper Lut 16 :  
content\cars\common\_phsx\dampers\penske\damper\_16.curve  
└ 1. Damper Lut 17 :  
content\cars\common\_phsx\dampers\penske\damper\_17.curve  
└ 1. Damper Lut 18 :  
content\cars\common\_phsx\dampers\penske\damper\_18.curve  
└ 1. Damper Lut 19 :  
content\cars\common\_phsx\dampers\penske\damper\_19.curve  
└ 1. Damper Lut 20 :  
content\cars\common\_phsx\dampers\penske\damper\_20.curve  
└ 1. Damper Lut 21 :  
content\cars\common\_phsx\dampers\penske\damper\_21.curve  
└ 1. Damper Lut 22 :  
content\cars\common\_phsx\dampers\penske\damper\_22.curve  
└ 1. Damper Lut 23 :  
content\cars\common\_phsx\dampers\penske\damper\_23.curve  
└ 1. Damper Lut 24 :  
content\cars\common\_phsx\dampers\penske\damper\_24.curve  
└ 1. Damper Lut 25 :  
content\cars\common\_phsx\dampers\penske\damper\_25.curve  
└ 1. Damper Lut 26 :  
content\cars\common\_phsx\dampers\penske\damper\_26.curve  
└ 1. Damper Lut 27 :  
content\cars\common\_phsx\dampers\penske\damper\_27.curve  
└ 1. Damper Lut 28 :  
content\cars\common\_phsx\dampers\penske\damper\_28.curve  
└ 1. Damper Lut 29 :  
content\cars\common\_phsx\dampers\penske\damper\_29.curve  
└ 1. Damper Lut 30 :  
content\cars\common\_phsx\dampers\penske\damper\_30.curve  
└ 1. Damper Lut 31 :  
content\cars\common\_phsx\dampers\penske\damper\_31.curve  
└ 1. Damper Lut 32 :  
content\cars\common\_phsx\dampers\penske\damper\_32.curve  
└ 1. Damper Lut 33 :  
content\cars\common\_phsx\dampers\penske\damper\_33.curve  
└ 1. Damper Lut 34 :  
content\cars\common\_phsx\dampers\penske\damper\_34.curve  
└ 1. Damper Lut 35 :  
content\cars\common\_phsx\dampers\penske\damper\_35.curve  
└ 1. Damper Lut 36 :  
content\cars\common\_phsx\dampers\penske\damper\_36.curve  
└ 1. Damper Lut 37 :  
content\cars\common\_phsx\dampers\penske\damper\_37.curve  
└ 1. Damper Lut 38 :  
content\cars\common\_phsx\dampers\penske\damper\_38.curve  
└ 1. Damper Lut 39 :  
content\cars\common\_phsx\dampers\penske\damper\_39.curve  
└ 1. Damper Lut 40 :  
content\cars\common\_phsx\dampers\penske\damper\_40.curve  
└ 1. Damper Lut 41 :  
content\cars\common\_phsx\dampers\penske\damper\_41.curve  
└ 1. Damper Lut 42 :  
content\cars\common\_phsx\dampers\penske\damper\_42.curve

└ 1. Damper Lut 43 :  
content\cars\common\_phsx\dampers\penske\damper\_43.curve  
└ 1. Damper Lut 44 :  
content\cars\common\_phsx\dampers\penske\damper\_44.curve  
└ 1. Damper Lut 45 :  
content\cars\common\_phsx\dampers\penske\damper\_45.curve  
└ 1. Damper Lut 46 :  
content\cars\common\_phsx\dampers\penske\damper\_46.curve  
└ 1. Damper Lut 47 :  
content\cars\common\_phsx\dampers\penske\damper\_47.curve  
└ 1. Damper Lut 48 :  
content\cars\common\_phsx\dampers\penske\damper\_48.curve  
└ 1. Damper Lut 49 :  
content\cars\common\_phsx\dampers\penske\damper\_49.curve  
└ 1. Damper Lut 50 :  
content\cars\common\_phsx\dampers\penske\damper\_50.curve  
└ 1. Damper Lut 51 :  
content\cars\common\_phsx\dampers\penske\damper\_51.curve  
└ 1. Damper Lut 52 :  
content\cars\common\_phsx\dampers\penske\damper\_52.curve  
└ 1. Damper Lut 53 :  
content\cars\common\_phsx\dampers\penske\damper\_53.curve  
└ 1. Damper Lut 54 :  
content\cars\common\_phsx\dampers\penske\damper\_54.curve  
└ 1. Damper Lut 55 :  
content\cars\common\_phsx\dampers\penske\damper\_55.curve  
└ 1. Damper Lut 56 :  
content\cars\common\_phsx\dampers\penske\damper\_56.curve  
└ 1. Damper Lut 57 :  
content\cars\common\_phsx\dampers\penske\damper\_57.curve  
└ 1. Damper Lut 58 :  
content\cars\common\_phsx\dampers\penske\damper\_58.curve  
└ 1. Damper Lut 59 :  
content\cars\common\_phsx\dampers\penske\damper\_59.curve  
└ 1. Damper Lut 60 :  
content\cars\common\_phsx\dampers\penske\damper\_60.curve

## **Porsche Cayman Dampers**

### **1. Front Damper ( file : cayman\_gt4\_front.dampercurves )**

└ 1. Damper Lut 1 :  
content\cars\common\_phsx\dampers\porsche\cayman\front1.curve  
└ 1. Damper Lut 2 :  
content\cars\common\_phsx\dampers\porsche\cayman\front2.curve  
└ 1. Damper Lut 3 :  
content\cars\common\_phsx\dampers\porsche\cayman\front3.curve  
└ 1. Damper Lut 4 :  
content\cars\common\_phsx\dampers\porsche\cayman\front4.curve  
└ 1. Damper Lut 5 :  
content\cars\common\_phsx\dampers\porsche\cayman\front5.curve  
└ 1. Damper Lut 6 :  
content\cars\common\_phsx\dampers\porsche\cayman\front6.curve

└ 1. Damper Lut 7 :  
content\cars\common\_phsx\dampers\porsche\cayman\front7.curve  
└ 1. Damper Lut 8 :  
content\cars\common\_phsx\dampers\porsche\cayman\front8.curve  
└ 1. Damper Lut 9 :  
content\cars\common\_phsx\dampers\porsche\cayman\front9.curve  
└ 1. Damper Lut 10 :  
content\cars\common\_phsx\dampers\porsche\cayman\front10.curve  
└ 1. Damper Lut 11 :  
content\cars\common\_phsx\dampers\porsche\cayman\front11.curve  
└ 1. Damper Lut 12 :  
content\cars\common\_phsx\dampers\porsche\cayman\front12.curve

## **2. Rear Damper ( file : cayman\_gt4\_rear.dampercurves )**

└ 1. Damper Lut 1 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear1.curve  
└ 1. Damper Lut 2 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear2.curve  
└ 1. Damper Lut 3 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear3.curve  
└ 1. Damper Lut 4 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear4.curve  
└ 1. Damper Lut 5 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear5.curve  
└ 1. Damper Lut 6 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear6.curve  
└ 1. Damper Lut 7 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear7.curve  
└ 1. Damper Lut 8 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear8.curve  
└ 1. Damper Lut 9 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear9.curve  
└ 1. Damper Lut 10 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear10.curve  
└ 1. Damper Lut 11 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear11.curve  
└ 1. Damper Lut 12 :  
content\cars\common\_phsx\dampers\porsche\cayman\rear12.curve

# 13. Drivetrain [ .drivetrain ]

## A. Description

How engine torque reaches the wheels: driven-axle layout, differential type and lock (power / coast / preload), optional multi-diff AWD tree, halfshaft / driveline torsion, and staged lock or AWD-clutch controllers.

Gear ratios live in .GEARBOX. Garage power/coast/preload clicks live in .CARSETUP. This asset is the mechanical baseline those layers sit on — and where full AWD clutch logic is authored.

Official Description prose in the source dump is unfinished placeholder text (spring/damper headings). Content below follows the schema and examples.

---

## I. Role in the stack

| Concern                           | Handled here                 | Handled elsewhere        |
|-----------------------------------|------------------------------|--------------------------|
| FWD / RWD / AWD layout            | <i>Traction Type</i>         | —                        |
| Primary diff type & locks         | <i>Differential Data</i>     | Setup may override locks |
| Front / center / rear diffs (AWD) | <i>Four WD Differentials</i> | —                        |
| Driveline stiffness / torsion     | Stiffness fields             | —                        |
| Dynamic lock / eLSD maps          | Lock Controllers stages      | .CURVE LUTs              |
| AWD clutch packs                  | <i>Awd Clutches [x]</i>      | Controllers + curves     |
| Ratios / shift                    | —                            | .GEARBOX                 |
| Which file loads                  | —                            | .CAR / .TUNINGPART path  |

---

## II. What you are really tuning

1. **Traction layout** — *Traction Type* selects the driven scheme. Examples: *AWDF* (Audi RS3 — AWD with front-biased architecture), *RWD* (F40 LM), *FWD* (Abarth 695). Wrong type with filled AWD blocks is a content bug.
2. **Primary differential** — *Differential Data: Type* (LSD, Spool, Torsen, EpicyclicTorsen, EpicyclicLSD, TorqueVectoring — same family as Car Data performance-mode enums), plus *Power / Coast / Preload*, optional *Front Share*, torque-bias ratios, and thermal / wear / friction-vs-temperature fields.

On simple FWD/RWD cars this block is the real diff (Abarth: LSD, power/coast 0.25, preload 20). On the RS3 AWD example the primary block is zeroed — locks live under Four WD Differentials instead.

3. **Four-wheel-drive tree** — *Four WD Differentials* → Front / Center / Rear, each with the same lock/thermal shape as Differential Data. RS3: front LSD (0.5 / 0.1 / preload 30), center EpicyclicLSD with *Front Share* 0.40, rear Spool with torque-bias ratios (power 2.5 / coast 2.0). RWD/FWD cars still show the tree in the dump but with zeroed stubs.

4. **Driveline torsion** — *Stiffness, Stiffness Mult, Damping Ratio, Max Torsion Deg, Non Linear Model* model shaft wind-up between gearbox and wheels. F40 LM fills these in (stiffness 4500, mult 2, damping 0.05, max torsion 40°, nonlinear true). RS3 / Abarth leave them at 0 / false — rigid coupling assumption.
5. **Flags and lock controllers** — *Max Between Lsd And Elsd, Has Cockpit Controls*. Front / Center / Rear / Left / Right **Lock Controllers** reuse the staged Input × Combinator × Lut pipeline (Gas, LatG, slip, oversteer...). Empty in these three samples.
6. **AWD clutches** — *Awd Clutches[x]: Position, Preload*, nested Controllers. RS3 ships two clutches (positions 2 and 3) with five stages each — Gas and front slip (Add), then LatG, OversteerFactor, and Speed (Mult) — classic Haldex-style transfer logic. F40 / Abarth: None.

---

### III. Architecture

#### **1 - LAYOUT AND PRIMARY DIFF (SCHEMA 1-2)**

*Traction Type; Differential Data* object.

#### **2 - AWD DIFFERENTIALS (SCHEMA 3)**

Front / Center / Rear diff objects (same leaf fields as 2).

#### **3 - TORSION AND FLAGS (SCHEMA 4-10)**

Stiffness stack; nonlinear flag; LSD/eLSD max flag; cockpit controls flag.

#### **4 - CONTROLLERS AND CLUTCHES (SCHEMA 11-16)**

Five lock-controller slots; repeating AWD clutch entries with their own stage lists.

---

### IV. How to read the examples

#### **1 - AUDI RS3 SPORTBACK — CONTROLLED AWD**

*AWDF*. Real behaviour in Four WD diffs + two AWD clutches driven by Gas / slip / LatG / oversteer / speed curves. Primary Differential Data empty. Shows where modern quattro-style logic lives: not a single power/coast pair, but center share + clutch maps.

#### **2 - FERRARI F40 LM — RWD WITH SHAFT TORSION**

*RWD*, LSD type declared, lock numbers zero in the dump (expect setup overlays or open baseline until garage sets them). Four WD / AWD clutches unused. Driveline torsion fully authored — race RWD that can wind up and unload.

#### **3 - ABARTH 695 BIPOSTO — SIMPLE FWD LSD**

*FWD* with Differential Data LSD at 0.25 / 0.25 / preload 20. Four WD stubs and AWD clutches empty; torsion off. Compact hot-hatch pattern: one mechanical LSD, no transfer case.

---

## V. Practical notes

- Source Description section is placeholder — ignore its spring/alignment headings.
- For AWD, always read **Four WD Differentials** and **Awd Clutches** before trusting Differential Data (may be a zero stub).
- Setup power/coast/preload must stay consistent with the active diff node (primary vs front/rear).
- OCR quirks: duplicated *2i* labels, *LSF* on F40 front stub, nested controller IDs reused as *11a/11b* under clutch 16.
- Empty lock controllers + nonzero AWD clutch stages is normal — clutch pack owns the dynamics.
- Tuning-part drivetrain swaps (Supra drift, Datsun LSD) replace this whole file; pair with matching setup/limits.

---

## VI. Related assets

- **3. Car Data** — Drivetrain Path
- **14 / 10 / 4. Gearbox / Clutch / Engine** — upstream torque path
- **5 / 6. Car Setup / Limits** — garage diff locks
- **8. Car Tuning Parts** — alternate .DRIVETRAIN redirects
- **9. Car Electronics** — EDL / TC intervene on top of mechanical lock, they do not replace this asset

## B. Schema

```
1. Traction Type : enum
2. Differential Data : object
 2a. Type : enum
 2b. Power : float
 2c. Coast : float
 2d. Preload : float
 2e. Front Share : float
 2f. Torque Bias Ratio Power : float
 2g. Torque Bias Ratio Coast : float
 2h. Thermal Capacity : float
 2i. Surface : float
 2j. Heat Transfer Coeff : float
 2k. Wear Factor : float
 2l. Friction Reduction With T : float
 2m. Friction Ref T : float
3. Four W D Differentials : object
 3a. Front Diff : object
 2a. Type : float
 2b. Power : float
 2c. Coast : float
 2d. Preload : float
 2e. Front Share : float
 2f. Torque Bias Ratio Power : float
 2g. Torque Bias Ratio Coast : float
 2h. Thermal Capacity : float
```



- 2i. Surface : float
- 2j. Heat Transfer Coeff : float
- 2k. Wear Factor : float
- 2l. Friction Reduction With T : float
- 2m. Friction Ref T : float
- 3b. Center Diff : object
  - 2a. Type : enum
  - 2b. Power : float
  - 2c. Coast : float
  - 2d. Preload : float
  - 2e. Front Share : float
  - 2f. Torque Bias Ratio Power : float
  - 2g. Torque Bias Ratio Coast : float
  - 2h. Thermal Capacity : float
  - 2i. Surface : float
  - 2j. Heat Transfer Coeff : float
  - 2k. Wear Factor : float
  - 2l. Friction Reduction With T : float
  - 2m. Friction Ref T : float
- 3c. Rear Diff : object
  - 2a. Type : enum
  - 2b. Power : float
  - 2c. Coast : float
  - 2d. Preload : float
  - 2e. Front Share : float
  - 2f. Torque Bias Ratio Power : float
  - 2g. Torque Bias Ratio Coast : float
  - 2h. Thermal Capacity : float
  - 2i. Surface : float
  - 2j. Heat Transfer Coeff : float
  - 2k. Wear Factor : float
  - 2l. Friction Reduction With T : float
  - 2m. Friction Ref T : float
- 4. Stiffness : float
- 5. Stiffness Mult : float
- 6. Damping Ratio : float
- 7. Max Torsion Deg : float
- 8. Non Linear Model : boolean
- 9. Max Between Lsd And Elsd : boolean
- 10. Has Cockpit Controls : boolean
- 11. Front Lock Controllers : object
  - 11a. Name : string
  - 11b. Stages [x] : object | can have multiple Stages
    - 11b1. Input Var : enum
    - 11b2. Combinator Mode : enum
    - 11b3. Lut : string – path
    - 11b4. Filter Gain : float
    - 11b5. Up Limit : float
    - 11b6. Down Limit : float
    - 11b7. Current Value : float
    - 11b8. Const Value : float
- 12. Center Lock Controllers : object
  - 11a. Name : string
  - 11b. Stages [x] : object | can have multiple Stages
    - 11b1. Input Var : enum
    - 11b2. Combinator Mode : enum

- 11b3. Lut : string - path
  - 11b4. Filter Gain : float
  - 11b5. Up Limit : float
  - 11b6. Down Limit : float
  - 11b7. Current Value : float
  - 11b8. Const Value : float
- 13. Rear Lock Controllers : object
  - 11a. Name : string
  - 11b. Stages [x] : object | can have multiple Stages
    - 11b1. Input Var : enum
    - 11b2. Combinator Mode : enum
    - 11b3. Lut : string - path
    - 11b4. Filter Gain : float
    - 11b5. Up Limit : float
    - 11b6. Down Limit : float
    - 11b7. Current Value : float
    - 11b8. Const Value : float
- 14. Left Lock Controllers : object
  - 11a. Name : string
  - 11b. Stages [x] : object | can have multiple Stages
    - 11b1. Input Var : enum
    - 11b2. Combinator Mode : enum
    - 11b3. Lut : string - path
    - 11b4. Filter Gain : float
    - 11b5. Up Limit : float
    - 11b6. Down Limit : float
    - 11b7. Current Value : float
    - 11b8. Const Value : float
- 15. Right Lock Controllers : object
  - 11a. Name : string
  - 11b. Stages [x] : object | can have multiple Stages
    - 11b1. Input Var : enum
    - 11b2. Combinator Mode : enum
    - 11b3. Lut : string - path
    - 11b4. Filter Gain : float
    - 11b5. Up Limit : float
    - 11b6. Down Limit : float
    - 11b7. Current Value : float
    - 11b8. Const Value : float
- 16. Awd Clutches [x] : object | can have multiple Awd Clutches
  - 16a. Position : integer
  - 16b. Preload : float
  - 16c. Controllers : object
    - 11a. Name : string
    - 11b. Stages [x] : object | can have multiple Stages
      - 11b1. Input Var : enum
      - 11b2. Combinator Mode : enum
      - 11b3. Lut : string - path
      - 11b4. Filter Gain : float
      - 11b5. Up Limit : float
      - 11b6. Down Limit : float
      - 11b7. Current Value : float
      - 11b8. Const Value : float

## C. Measurement Units & Descriptions

| ID  | Name                      | Unit of Measurement                                                                     | Description                                                                                                          |
|-----|---------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1.  | Traction Type             | None ( Enum : FWD / RWD / AWDF / ... )                                                  | Global driveline layout enum (e.g., AWDF Audi RS3, RWD Ferrari F40 LM, FWD Abarth 695).                              |
| 2.  | Differential Data         | None ( Object )                                                                         | Primary differential parameter block for single-diff or default axle configuration.                                  |
| 2a. | Type                      | None ( Enum : LSD / Spool / Torsen / EpicyclicTorsen / EpicyclicLSD / TorqueVectoring ) | Differential mechanism type governing torque split and lock behavior.                                                |
| 2b. | Power                     | Nm or ratio                                                                             | Lock/acceleration sensitivity under power (on-throttle); higher values increase drive-wheel coupling on corner exit. |
| 2c. | Coast                     | Nm or ratio                                                                             | Lock sensitivity on coast/off-throttle; stabilizes the axle under braking and lift-off.                              |
| 2d. | Preload                   | Nm ( Newton-meters )                                                                    | Static preload torque before differential plates begin slipping.                                                     |
| 2e. | Front Share               | Ratio ( 0.0 - 1.0 )                                                                     | Fraction of total driveline torque directed to the front axle (AWD/ center diff).                                    |
| 2f. | Torque Bias Ratio Power   | Ratio ( e.g., 2.0:1 )                                                                   | Torque bias ratio between sides or axles under power.                                                                |
| 2g. | Torque Bias Ratio Coast   | Ratio                                                                                   | Torque bias ratio between sides or axles on coast.                                                                   |
| 2h. | Thermal Capacity          | J/K or J/°C                                                                             | Heat storage capacity of differential friction components for thermal wear modeling.                                 |
| 2i. | Surface                   | m <sup>2</sup> ( Square meters )                                                        | Exposed friction surface area of the differential housing for cooling calculations.                                  |
| 2j. | Heat Transfer Coeff       | W/(m <sup>2</sup> ·K) ( Coefficient )                                                   | Convective heat dissipation rate from differential to ambient/oil.                                                   |
| 2k. | Wear Factor               | Dimensionless coefficient                                                               | Rate at which differential friction surfaces degrade under load and temperature.                                     |
| 2l. | Friction Reduction With T | Ratio/°C                                                                                | Linear friction reduction per degree of differential temperature rise.                                               |
| 2m. | Friction Ref T            | °C ( Degrees Celsius )                                                                  | Reference temperature at which nominal differential friction is defined.                                             |

| ID   | Name                     | Unit of Measurement                 | Description                                                                                                                                                                        |
|------|--------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.   | Four W D Differentials   | None ( Object )                     | AWD/4WD triple-differential container: front, center, and rear axle blocks.                                                                                                        |
| 3a.  | Front Diff               | None ( Object )                     | Front Diff sub-block within Four W D Differentials (3).                                                                                                                            |
| 3b.  | Center Diff              | None ( Object )                     | Center Diff sub-block within Four W D Differentials (3).                                                                                                                           |
| 3c.  | Rear Diff                | None ( Object )                     | Rear Diff sub-block within Four W D Differentials (3).                                                                                                                             |
| 4.   | Stiffness                | Nm/deg ( Newton-meters per degree ) | Torsional stiffness of the differential coupling model (e.g., 4500 Nm/deg Ferrari F40 LM).                                                                                         |
| 5.   | Stiffness Mult           | Dimensionless multiplier            | Scalar multiplier applied to base stiffness for non-linear or epicyclic models (e.g., 2.0 F40 LM).                                                                                 |
| 6.   | Damping Ratio            | Dimensionless ratio ( $\zeta$ )     | Torsional damping ratio of the differential dynamics model (e.g., 0.05 F40 LM).                                                                                                    |
| 7.   | Max Torsion Deg          | deg ( Degrees )                     | Maximum torsion angle before the differential model saturates (e.g., 40° F40 LM).                                                                                                  |
| 8.   | Non Linear Model         | None ( Boolean : True / False )     | When true, enables non-linear torsional differential physics (true on F40 LM).                                                                                                     |
| 9.   | Max Between Lsd And Elsa | None ( Boolean : True / False )     | When true, caps combined LSD and eLSD lock contribution to a maximum blended value.                                                                                                |
| 10.  | Has Cockpit Controls     | None ( Boolean : True / False )     | When true, exposes in-cockpit differential lock or bias adjustment to the driver.                                                                                                  |
| 11.  | Front Lock Controllers   | None ( Object )                     | Dynamic controller modulating front lock controllers under telemetry conditions.                                                                                                   |
| 11a. | Name                     | None ( String )                     | Internal identifier for the lock controller; often None when inactive. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.                                  |
| 11b. | Stages                   | None ( Object array )               | Multi-stage controller pipeline mapping telemetry input through LUTs (often None on simple LSD setups). Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8. |

| ID    | Name                    | Unit of Measurement                  | Description                                                                                                                                                     |
|-------|-------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11b1. | Input Var               | None ( Telemetry enum )              | Telemetry input channel (Gas, SlipRatioFrontAVG, LatG, Speed, ErsChargeLevel, etc.). Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8. |
| 11b2. | Combinator Mode         | None ( Math enum : Add / Mult )      | How stage output combines with prior stages: additive or multiplicative. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.             |
| 11b3. | Lut                     | None ( .curve file path )            | Look-up table mapping filtered input to differential/clutch lock modifier. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.           |
| 11b4. | Filter Gain             | Coefficient ( Smoothing multiplier ) | Low-pass filter coefficient smoothing controller input spikes. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.                       |
| 11b5. | Up Limit                | Depends on input variable            | Upper clamp on processed input before LUT evaluation. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.                                |
| 11b6. | Down Limit              | Depends on input variable            | Lower clamp on processed input before LUT evaluation. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.                                |
| 11b7. | Current Value           | Depends on input variable            | Runtime controller stage output during simulation (telemetry/ debug). Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.                |
| 11b8. | Const Value             | Depends on input variable            | Constant fallback output when Input Var is Const or LUT is inactive. Under <b>Front Lock Controllers</b> ; schema reuses IDs 11a/11b/11b1–11b8.                 |
| 12.   | Center Lock Controllers | None ( Object )                      | Dynamic controller modulating center lock controllers under telemetry conditions.                                                                               |
| 13.   | Rear Lock Controllers   | None ( Object )                      | Dynamic controller modulating rear lock controllers under telemetry conditions.                                                                                 |
| 14.   | Left Lock Controllers   | None ( Object )                      | Dynamic controller modulating left lock controllers under telemetry conditions.                                                                                 |
| 15.   | Right Lock Controllers  | None ( Object )                      | Dynamic controller modulating right lock controllers under telemetry conditions.                                                                                |

| ID   | Name         | Unit of Measurement    | Description                                                                                              |
|------|--------------|------------------------|----------------------------------------------------------------------------------------------------------|
| 16.  | Awd Clutches | None ( Object array )  | AWD clutch pack array; each entry couples axles with preload and optional active controllers (Audi RS3). |
| 16a. | Position     | None ( Integer index ) | Clutch pack index/position in the AWD coupling matrix (e.g., 2 on Audi RS3).                             |
| 16b. | Preload      | Nm ( Newton-meters )   | Static preload torque on this AWD clutch before slip occurs.                                             |
| 16c. | Controllers  | None ( Object )        | Nested lock controller block for this AWD clutch pack.                                                   |

## D. Example data

### I. Chosen Cars for Example

- Audi RS3 Sportback ( slug : ks\_audi\_rs\_3\_sportback )
- Ferrari F40 LM ( slug : ks\_ferrari\_f40\_lm )
- Abarth 695 Biposto ( slug : ks\_abarth\_695\_biposto )

### II. Example

#### Audi RS3 Sportback

```

1. Traction Type : AWDF
2. Differential Data
 2a. Type : LSD
 2b. Power : 0.00000
 2c. Coast : 0.00000
 2d. Preload : 0.00000
 2e. Front Share : 0.00000
 2f. Torque Bias Ratio Power : 0.00000
 2g. Torque Bias Ratio Coast : 0.00000
 2h. Thermal Capacity : 0.00000
 2i. Surface : 0.00000
 2j. Heat Transfer Coeff : 0.00000
 2k. Wear Factor : 0.00000
 2l. Friction Reduction With T : 0.00000
 2m. Friction Ref T : 0.00000
3. Four W D Differentials
 3a. Front Diff
 2a. Type : LSD
 2b. Power : 0.50000
 2c. Coast : 0.10000
 2d. Preload : 30.00000
 2e. Front Share : 0.00000
 2f. Torque Bias Ratio Power : 0.00000
 2g. Torque Bias Ratio Coast : 0.00000

```

- | 2h. Thermal Capacity : 0.00000
- | 2i. Surface : 0.00000
- | 2j. Heat Transfer Coeff : 0.00000
- | 2k. Wear Factor : 0.00000
- | 2l. Friction Reduction With T : 0.00000
- | 2m. Friction Ref T : 0.00000
- 3b. Center Diff
  - | 2a. Type : EpicyclicLSD
  - | 2b. Power : 0.03000
  - | 2c. Coast : 0.03000
  - | 2d. Preload : 1.00000
  - | 2e. Front Share : 0.40000
  - | 2f. Torque Bias Ratio Power : 0.00000
  - | 2g. Torque Bias Ratio Coast : 0.00000
  - | 2h. Thermal Capacity : 0.00000
  - | 2i. Surface : 0.00000
  - | 2j. Heat Transfer Coeff : 0.00000
  - | 2k. Wear Factor : 0.00000
  - | 2l. Friction Reduction With T : 0.00000
  - | 2m. Friction Ref T : 0.00000
- 3c. Rear Diff
  - | 2a. Type : Spool
  - | 2b. Power : 0.03000
  - | 2c. Coast : 0.03000
  - | 2d. Preload : 0.00000
  - | 2e. Front Share : 0.00000
  - | 2f. Torque Bias Ratio Power : 2.50000
  - | 2g. Torque Bias Ratio Coast : 2.00000
  - | 2h. Thermal Capacity : 0.00000
  - | 2i. Surface : 0.00000
  - | 2j. Heat Transfer Coeff : 0.00000
  - | 2k. Wear Factor : 0.00000
  - | 2l. Friction Reduction With T : 0.00000
  - | 2m. Friction Ref T : 0.00000
- 4. Stiffness : 0.00000
- 5. Stiffness Mult : 0.00000
- 6. Damping Ratio : 0.00000
- 7. Max Torsion Deg : 0.00000
- 8. Non Linear Model : false
- 9. Max Between Lsd And Elsd : false
- 10. Has Cockpit Controls : false
- 11. Front Lock Controllers
  - | 11a. Name : None
  - | 11b. Stages : None
- 12. Center Lock Controllers
  - | 11a. Name : None
  - | 11b. Stages : None
- 13. Rear Lock Controllers
  - | 11a. Name : None
  - | 11b. Stages : None
- 14. Left Lock Controllers
  - | 11a. Name : None
  - | 11b. Stages : None
- 15. Right Lock Controllers
  - | 11a. Name : None
  - | 11b. Stages : None

```

- 16. Awd Clutches 1
- - 16a. Position : 2
- - 16b. Preload : 0.000
- - 16c. Controllers
- - 11a. Name : None
- - 11b. Stages 1
- - 11b1. Input Var : Gas
- - 11b2. Combinator Mode : Add
- - 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_gas_add.curve
- - 11b4. Filter Gain : 0.00000
- - 11b5. Up Limit : 3000.00000
- - 11b6. Down Limit : 0.00000
- - 11b7. Current Value : 0.00000
- - 11b8. Const Value : 0.00000
- - 11b. Stages 2
- - 11b1. Input Var : SlipRatioFrontAVG
- - 11b2. Combinator Mode : Add
- - 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_SRavg.curve
- - 11b4. Filter Gain : 0.00000
- - 11b5. Up Limit : 3000.00000
- - 11b6. Down Limit : 0.00000
- - 11b7. Current Value : 0.00000
- - 11b8. Const Value : 0.00000
- - 11b. Stages 3
- - 11b1. Input Var : LatG
- - 11b2. Combinator Mode : Mult
- - 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_steer_right_clutch_Perf1.curve
- - 11b4. Filter Gain : 0.00000
- - 11b5. Up Limit : 3000.00000
- - 11b6. Down Limit : 0.00000
- - 11b7. Current Value : 0.00000
- - 11b8. Const Value : 0.00000
- - 11b. Stages 4
- - 11b1. Input Var : OversteerFactor
- - 11b2. Combinator Mode : Mult
- - 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_clutch_oversteer.curve
- - 11b4. Filter Gain : 0.00000
- - 11b5. Up Limit : 3000.00000
- - 11b6. Down Limit : 0.00000
- - 11b7. Current Value : 0.00000
- - 11b8. Const Value : 0.00000
- - 11b. Stages 5
- - 11b1. Input Var : Speed
- - 11b2. Combinator Mode : Mult
- - 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_speed_power.curve
- - 11b4. Filter Gain : 0.00000

```



```

| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000
- 16. Awd Clutches 2
| | | | 16a. Position : 3
| | | | 16b. Preload : 0.000
| | | | 16c. Controllers
| | | | | 11a. Name : None
| | | | | 11b. Stages 1
| | | | | | 11b1. Input Var : Gas
| | | | | | 11b2. Combinator Mode : Add
| | | | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_gas_add.curve
| | | | | 11b4. Filter Gain : 0.00000
| | | | | 11b5. Up Limit : 3000.00000
| | | | | 11b6. Down Limit : 0.00000
| | | | | 11b7. Current Value : 0.00000
| | | | | 11b8. Const Value : 0.00000
| | | | | 11b. Stages 2
| | | | | | 11b1. Input Var : SlipRatioFrontAVG
| | | | | | 11b2. Combinator Mode : Add
| | | | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_SRavg.curve
| | | | | 11b4. Filter Gain : 0.00000
| | | | | 11b5. Up Limit : 3000.00000
| | | | | 11b6. Down Limit : 0.00000
| | | | | 11b7. Current Value : 0.00000
| | | | | 11b8. Const Value : 0.00000
| | | | | 11b. Stages 3
| | | | | | 11b1. Input Var : LatG
| | | | | | 11b2. Combinator Mode : Mult
| | | | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_steer_right_clutch_Perf1.curve
| | | | | 11b4. Filter Gain : 0.00000
| | | | | 11b5. Up Limit : 3000.00000
| | | | | 11b6. Down Limit : 0.00000
| | | | | 11b7. Current Value : 0.00000
| | | | | 11b8. Const Value : 0.00000
| | | | | 11b. Stages 4
| | | | | | 11b1. Input Var : OversteerFactor
| | | | | | 11b2. Combinator Mode : Mult
| | | | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_sportback_awd_clutch_oversteer.curve
| | | | | 11b4. Filter Gain : 0.00000
| | | | | 11b5. Up Limit : 3000.00000
| | | | | 11b6. Down Limit : 0.00000
| | | | | 11b7. Current Value : 0.00000
| | | | | 11b8. Const Value : 0.00000
| | | | | 11b. Stages 5
| | | | | | 11b1. Input Var : Speed
| | | | | | 11b2. Combinator Mode : Mult

```

```

| | | | 11b3. Lut :
content\cars\ks_audi_rs_3_sportback\data\controller\new_awd\ks_audi_rs_3
_awd_clutch_speed_power.curve
| | | | 11b4. Filter Gain : 0.00000
| | | | 11b5. Up Limit : 3000.00000
| | | | 11b6. Down Limit : 0.00000
| | | | 11b7. Current Value : 0.00000
| | | | 11b8. Const Value : 0.00000

```

## Ferrari F40 LM

```

| 1. Traction Type : RWD
| 2. Differential Data
| | 2a. Type : LSD
| | 2b. Power : 0.00000
| | 2c. Coast : 0.00000
| | 2d. Preload : 0.00000
| | 2e. Front Share : 0.00000
| | 2f. Torque Bias Ratio Power : 0.00000
| | 2g. Torque Bias Ratio Coast : 0.00000
| | 2h. Thermal Capacity : 0.00000
| | 2i. Surface : 0.00000
| | 2j. Heat Transfer Coeff : 0.00000
| | 2k. Wear Factor : 0.00000
| | 2l. Friction Reduction With T : 0.00000
| | 2m. Friction Ref T : 0.00000
| 3. Four W D Differentials
| | 3a. Front Diff
| | | 2a. Type : LSF
| | | 2b. Power : 0.00000
| | | 2c. Coast : 0.00000
| | | 2d. Preload : 0.00000
| | | 2e. Front Share : 0.00000
| | | 2f. Torque Bias Ratio Power : 0.00000
| | | 2g. Torque Bias Ratio Coast : 0.00000
| | | 2h. Thermal Capacity : 0.00000
| | | 2i. Surface : 0.00000
| | | 2j. Heat Transfer Coeff : 0.00000
| | | 2k. Wear Factor : 0.00000
| | | 2l. Friction Reduction With T : 0.00000
| | | 2m. Friction Ref T : 0.00000
| | 3b. Center Diff
| | | 2a. Type : LSD
| | | 2b. Power : 0.00000
| | | 2c. Coast : 0.00000
| | | 2d. Preload : 0.00000
| | | 2e. Front Share : 0.00000
| | | 2f. Torque Bias Ratio Power : 0.00000
| | | 2g. Torque Bias Ratio Coast : 0.00000
| | | 2h. Thermal Capacity : 0.00000
| | | 2i. Surface : 0.00000
| | | 2j. Heat Transfer Coeff : 0.00000
| | | 2k. Wear Factor : 0.00000
| | | 2l. Friction Reduction With T : 0.00000

```

- 2m. Friction Ref T : 0.00000
  - 3c. Rear Diff
    - 2a. Type : LSD
    - 2b. Power : 0.00000
    - 2c. Coast : 0.00000
    - 2d. Preload : 0.00000
    - 2e. Front Share : 0.00000
    - 2f. Torque Bias Ratio Power : 0.00000
    - 2g. Torque Bias Ratio Coast : 0.00000
    - 2h. Thermal Capacity : 0.00000
    - 2i. Surface : 0.00000
    - 2j. Heat Transfer Coeff : 0.00000
    - 2k. Wear Factor : 0.00000
    - 2l. Friction Reduction With T : 0.00000
    - 2m. Friction Ref T : 0.00000
- 4. Stiffness : 4500.00000
- 5. Stiffness Mult : 2.00000
- 6. Damping Ratio : 0.05000
- 7. Max Torsion Deg : 40.00000
- 8. Non Linear Model : true
- 9. Max Between Lsd And Elsd : false
- 10. Has Cockpit Controls : false
- 11. Front Lock Controllers
  - 11a. Name : None
  - 11b. Stages : None
- 12. Center Lock Controllers
  - 11a. Name : None
  - 11b. Stages : None
- 13. Rear Lock Controllers
  - 11a. Name : None
  - 11b. Stages : None
- 14. Left Lock Controllers
  - 11a. Name : None
  - 11b. Stages : None
- 15. Right Lock Controllers
  - 11a. Name : None
  - 11b. Stages : None
- 16. Awd Clutches : None

### Abarth 695 Biposto

- 1. Traction Type : FWD
- 2. Differential Data
  - 2a. Type : LSD
  - 2b. Power : 0.25000
  - 2c. Coast : 0.25000
  - 2d. Preload : 20.00000
  - 2e. Front Share : 0.00000
  - 2f. Torque Bias Ratio Power : 0.00000
  - 2g. Torque Bias Ratio Coast : 0.00000
  - 2h. Thermal Capacity : 0.00000
  - 2i. Surface : 0.00000
  - 2j. Heat Transfer Coeff : 0.00000
  - 2k. Wear Factor : 0.00000

- 2l. Friction Reduction With T : 0.00000
- 2m. Friction Ref T : 0.00000
- 3. Four W D Differentials
  - 3a. Front Diff
    - 2a. Type : LSD
    - 2b. Power : 0.00000
    - 2c. Coast : 0.00000
    - 2d. Preload : 0.00000
    - 2e. Front Share : 0.00000
    - 2f. Torque Bias Ratio Power : 0.00000
    - 2g. Torque Bias Ratio Coast : 0.00000
    - 2h. Thermal Capacity : 0.00000
    - 2i. Surface : 0.00000
    - 2j. Heat Transfer Coeff : 0.00000
    - 2k. Wear Factor : 0.00000
    - 2l. Friction Reduction With T : 0.00000
    - 2m. Friction Ref T : 0.00000
  - 3b. Center Diff
    - 2a. Type : LSD
    - 2b. Power : 0.00000
    - 2c. Coast : 0.00000
    - 2d. Preload : 0.00000
    - 2e. Front Share : 0.00000
    - 2f. Torque Bias Ratio Power : 0.00000
    - 2g. Torque Bias Ratio Coast : 0.00000
    - 2h. Thermal Capacity : 0.00000
    - 2i. Surface : 0.00000
    - 2j. Heat Transfer Coeff : 0.00000
    - 2k. Wear Factor : 0.00000
    - 2l. Friction Reduction With T : 0.00000
    - 2m. Friction Ref T : 0.00000
  - 3c. Rear Diff
    - 2a. Type : LSD
    - 2b. Power : 0.00000
    - 2c. Coast : 0.00000
    - 2d. Preload : 0.00000
    - 2e. Front Share : 0.00000
    - 2f. Torque Bias Ratio Power : 0.00000
    - 2g. Torque Bias Ratio Coast : 0.00000
    - 2h. Thermal Capacity : 0.00000
    - 2i. Surface : 0.00000
    - 2j. Heat Transfer Coeff : 0.00000
    - 2k. Wear Factor : 0.00000
    - 2l. Friction Reduction With T : 0.00000
    - 2m. Friction Ref T : 0.00000
- 4. Stiffness : 0.00000
- 5. Stiffness Mult : 0.00000
- 6. Damping Ratio : 0.00000
- 7. Max Torsion Deg : 0.00000
- 8. Non Linear Model : false
- 9. Max Between Lsd And Elsd : false
- 10. Has Cockpit Controls : false
- 11. Front Lock Controllers
  - 11a. Name : None
  - 11b. Stages : None
- 12. Center Lock Controllers

```
| 11a. Name : None
| 11b. Stages : None
- 13. Rear Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 14. Left Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 15. Right Lock Controllers
| 11a. Name : None
| 11b. Stages : None
- 16. Awd Clutches : None
```

## 14. Gearbox [ .gearbox ]

### A. Description

Transmission ratios and shift behaviour: gear list (including reverse and neutral), final drive, up/down shift times, throttle cut during shifts, dual-clutch / H-pattern flags, downshift protection, autoblip, autoshifter thresholds, gearbox inertia, and optional gear fatigue stress parameters.

Clutch engagement curves live in .CLUTCH. Diff and driven axles live in .DRIVETRAIN. This asset decides what ratio sits between engine and final drive, and how long / how violently a gear change interrupts torque.

Official Description prose in the source dump is unfinished placeholder text. Content below follows the schema and examples.

---

### I. Role in the stack

| Concern                             | Handled here                          | Handled elsewhere                 |
|-------------------------------------|---------------------------------------|-----------------------------------|
| Pear-gear ratios (+ R / N)          | <i>Gears[x]</i>                       | —                                 |
| Final drive                         | <i>Final Ratio</i>                    | —                                 |
| Shift duration / throttle cut       | Up/Dn time, Auto Cut Off, Gas Cut Off | Clutch autoclutch profiles        |
| Dual clutch / H-shifter support     | Flags                                 | Controls bindings                 |
| Overrev / neutral lock on downshift | Downshift Protection                  | —                                 |
| Autoblip / auto shift assists       | Autoblip, Autoshifter                 | .CURVE profiles                   |
| Spinning gearbox inertia            | <i>Gearbox Inertia</i>                | Engine / clutch inertia elsewhere |
| Which file loads                    | —                                     | .CAR / .TUNINGPART path           |

---

### II. What you are really tuning

- Ratio stack** — *Gear Count* is the number of forward gears. *Gears[x]* always lists named slots with ratios — examples include *R* (negative), *N* (0), then 1...*n*. Cayman GT4 CS MR: 6-speed with *R* –3.55 through 6th 0.881, final 3.72. Alpine A290 b: *Gear Count* : 1 with only *R* / *N* / 1 and final 1.0 — single-ratio / EV-style reduction. R5: classic 5-speed, final 3.73.
- Shift timing and cut** — *Gear Up Time* / *Gear Dn Time* (Cayman 30 / 80, Alpine 10 / 16, R5 250 / 270 — race DCT vs slow manual). *Auto Cut Off Time* and *Gas Cut Off Level* shape how much and how long throttle is pulled during automated cuts (Cayman cut level 0.7; Alpine/R5 level 0).
- Hardware personality** — *Has Dual Clutch* (Cayman true; Alpine/R5 false). *Is Shifter Supported* (R5 true for H-pattern; paddle cars false). Together with shift times they separate sequential/DCT race boxes from three-pedal manuals.

4. **Downshift protection** — *Is Active*, *Overrev* margin, *Lock N*. Cayman/Alpine active with overrev 300 / 100 and Lock N true — blocks suicidal downshifts. R5 inactive (0 / Lock N false) — period-correct “you can grenade it”.
5. **Shift windows and inertia** — *Damage Rpm Window*, *Valid Shift Rpm Window*, *Controls Window Gain* gate when a shift request is accepted vs damaging. *Gearbox Inertia* (Cayman 0.012, Alpine 0.008, R5 0.017) adds rotating mass in the box.
6. **Autoblip and auto shifter** — *Autoblip Profile* path + *Is Electronic* (true on Cayman/Alpine, false on R5 — mechanical vs ECU blip). Autosifter: *Up RPM*, down threshold, *Slip Threshold*, *Gas Cutoff Time* for assisted / AI-style shifting (Cayman up 7700, Alpine 6850, R5 6000).
7. **Fatigue (optional)** — *Gears Fatigue Log10 A*, nominal torque, max/min stress — present but zeroed in all three examples.

---

### III. Architecture

#### **1 - RATIOS (SCHEMA 1-3)**

*Gear count; repeating Name/Ratio gears; final ratio.*

#### **2 - SHIFT ACTUATORS AND FLAGS (SCHEMA 4-9)**

Up/down times; auto cut-off time/level; dual clutch; shifter supported.

#### **3 - PROTECTION AND WINDOWS (SCHEMA 10-14)**

Downshift Protection object; damage / valid shift windows; controls gain; gearbox inertia.

#### **4 - ASSISTS AND FATIGUE (SCHEMA 15-20)**

Autoblip (profile + electronic flag — dump reuses 15a for both); Autosifter; four fatigue floats.

---

### IV. How to read the examples

#### **1 - PORSCHE 718 CAYMAN GT4 CS MR — RACE DCT STYLE**

6 forward gears, dual clutch, fast upshifts (30), slower downs (80), electronic autoblip, downshift protection on, shifter not supported. Autosift up around 7700 RPM. Modern cup/GT transmission feel.

#### **2 - ALPINE A290B — SINGLE-SPEED / EV LAYOUT**

*Gear Count* : 1, tall  $\pm 9.0$  R/1 ratios, final 1.0, very short shift times, no dual clutch, electronic autoblip still wired. Shows the schema working for non-H multi-gear ICE boxes.

#### **3 - RENAULT 5 GT TURBO — ANALOG MANUAL**

5-speed, long 250/270 shift times, dual clutch off, H-shifter supported, downshift protection off, non-electronic autoblip curve, higher gearbox inertia (0.017). Classic hot-hatch: slow shifts, driver can overrev.

---

## V. Practical notes

- Source Description section is placeholder — ignore its spring/alignment headings.
- *Gear Count* counts forward gears; array length is larger because R and N are included.
- Shift time units look like milliseconds in race cars (30–80) vs hundreds for manuals — keep consistency within a car, do not mix blindly across titles.
- Schema OCR duplicates Autoblip field id *15a* for both Profile path and *Is Electronic*.
- Tuning-part gearbox swaps (Datsun 5-speed, Supra drift) must stay paired with a compatible clutch file.
- Fatigue zeros mean “unused”, not “indestructible by design” unless the rest of the damage model agrees.

---

## VI. Related assets

- **3. Car Data** — *Gearbox Path*
- **10. Clutch** — couples/decouples during shifts; autoclutch profiles
- **4. Car Engine** — RPM band the ratios must match
- **13. Drivetrain** — after the final drive
- **8. Car Tuning Parts** — alternate .GEARBOX redirects

### B. Schema

- 1. Gear Count : integer
- 2. Gears [x] : object | can have multiple Gears
  - 2a. Name : string
  - 2b. Ratio : float
- 3. Final Ratio : float
- 4. Gear Up Time : float
- 5. Gear Dn Time : float
- 6. Auto Cut Off Time : float
- 7. Gas Cut Off Level : float
- 8. Has Dual Clutch : boolean
- 9. Is Shifter Supported : boolean
- 10. Downshift Protection : object
  - 10a. Is Active : boolean
  - 10b. Is Debug : boolean
  - 10c. Overrev : integer
  - 10d. Lock N : boolean
- 11. Damage Rpm Window : float
- 12. Valid Shift Rpm Window : float
- 13. Controls Window Gain : float
- 14. Gearbox Inertia : float
- 15. Autoblip : object
  - 15a. Profile : float
  - 15b. Is Electronic : boolean
- 16. Autosifter : object
  - 16a. Up : integer
  - 16b. Down Rpm Threshold : integer



- 16c. Slip Threshold : float
- 16d. Gas Cutoff Time : float
- 17. Gears Fatigue Log10 A : float
- 18. Gears Fatigue Nominal Torque : float
- 19. Gears Fatigue Max Stress : float
- 20. Gears Fatigue Min Stress : float

### C. Measurement Units & Descriptions

| ID  | Name                 | Unit of Measurement             | Description                                                                                                             |
|-----|----------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1.  | Gear Count           | None ( Integer count )          | Number of forward/reverse gear ratios configured (excludes R/N slots in Gear Count; Porsche GT4: 6).                    |
| 2.  | Gears                | None ( Object array )           | Per-gear entries including reverse (R), neutral (N), and numbered forward ratios; each slot repeats IDs 2a/2b.          |
| 2a. | Name                 | None ( String )                 | Gear label (R, N, 1, 2, ...); identifies reverse, neutral, and forward gears in the ratio table.                        |
| 2b. | Ratio                | Ratio ( dimensionless )         | Internal gear ratio; negative for reverse (e.g., -3.55 R), 0 for neutral, positive for forward (3.909 1st Porsche GT4). |
| 3.  | Final Ratio          | Ratio ( dimensionless )         | Final-drive/differential ring-and-pinion multiplier applied after the selected gear (e.g., 3.72461 Porsche GT4).        |
| 4.  | Gear Up Time         | ms ( Milliseconds )             | Upshift engagement duration; lower values yield faster shifts (30 ms DCT Porsche, 250 ms Renault 5 manual).             |
| 5.  | Gear Dn Time         | ms ( Milliseconds )             | Downshift engagement duration; often longer than upshift for rev-matching (80 ms Porsche, 16 ms Alpine).                |
| 6.  | Auto Cut Off Time    | ms ( Milliseconds )             | Engine fuel/torque cut duration during automated or semi-automated shifts.                                              |
| 7.  | Gas Cut Off Level    | Ratio ( 0.0 - 1.0 )             | Fraction of throttle cut during shift fuel interruption (0.7 Porsche DCT; 0 on single-speed Alpine).                    |
| 8.  | Has Dual Clutch      | None ( Boolean : True / False ) | When true, gearbox uses a dual-clutch layout with overlapping shift phases (Porsche GT4: true).                         |
| 9.  | Is Shifter Supported | None ( Boolean : True / False ) | When true, H-pattern or sequential shifter hardware input is supported for manual shifts.                               |

| ID   | Name                   | Unit of Measurement                          | Description                                                                                                |
|------|------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 10.  | Downshift Protection   | None ( Object )                              | Anti-over-rev downshift protection block preventing destructive downshift RPM excursions.                  |
| 10a. | Is Active              | None ( Boolean : True / False )              | Master enable for downshift over-rev protection logic.                                                     |
| 10b. | Is Debug               | None ( Boolean : True / False )              | When true, exposes debug telemetry for downshift protection triggers.                                      |
| 10c. | Overrev                | RPM ( Revolutions per minute )               | Allowed RPM overshoot margin before downshift is blocked (300 Porsche, 100 Alpine).                        |
| 10d. | Lock N                 | None ( Boolean : True / False )              | When true, prevents accidental departure from neutral without clutch/shift validation.                     |
| 11.  | Damage Rpm Window      | RPM ( Revolutions per minute )               | RPM band width used when evaluating gearbox/drivetrain damage from over-rev events (100 in examples).      |
| 12.  | Valid Shift Rpm Window | RPM ( Revolutions per minute )               | RPM tolerance window within which a shift request is considered valid (800 in examples).                   |
| 13.  | Controls Window Gain   | Dimensionless coefficient                    | Gain scaling driver shift-input timing tolerance (0.4 across example cars).                                |
| 14.  | Gearbox Inertia        | kg·m <sup>2</sup> ( Kilogram square meters ) | Rotational inertia of gears and shafts; affects rev hang and shift feel (0.012 Porsche, 0.017 Renault 5).  |
| 15.  | Autoblip               | None ( Object )                              | Throttle-blip profile block for downshift rev-matching on manual/semi-auto setups.                         |
| 15a. | Profile                | None ( .curve file path )                    | Autoblip throttle envelope curve (e.g., 718_gt4_autoblip.curve); schema types float but examples use path. |
| 15b. | Is Electronic          | None ( Boolean : True / False )              | When true, autoblip is applied electronically via ECU/throttle actuation rather than pedal coaching.       |
| 16.  | Autoshifter            | None ( Object )                              | Automated upshift/downshift assist thresholds for sequential or auto modes.                                |
| 16a. | Up                     | RPM ( Revolutions per minute )               | Engine RPM threshold triggering upshift in autoshifter mode (7700 Porsche, 6850 Alpine).                   |
| 16b. | Down Rpm Threshold     | RPM ( Revolutions per minute )               | RPM threshold below which downshift is requested (0 = disabled in examples).                               |

| ID   | Name                         | Unit of Measurement                             | Description                                                                                      |
|------|------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 16c. | Slip Threshold               | Ratio ( 0.0 - 1.0 )                             | Wheel-slip ratio threshold influencing autoshifter shift decisions (0.96 Porsche, 0.95 Alpine).  |
| 16d. | Gas Cutoff Time              | s ( Seconds )                                   | Duration of throttle cut during autoshifter shifts (0.063 s Porsche, 0.15 s Alpine).             |
| 17.  | Gears Fatigue Log10 A        | Dimensionless coefficient ( log10 coefficient ) | Log10 fatigue curve coefficient A for gear tooth stress accumulation (0 = disabled in examples). |
| 18.  | Gears Fatigue Nominal Torque | Nm ( Newton-meters )                            | Reference torque level for gear fatigue life calculations.                                       |
| 19.  | Gears Fatigue Max Stress     | Pa or N ( Stress threshold )                    | Upper stress bound for gear fatigue degradation modeling.                                        |
| 20.  | Gears Fatigue Min Stress     | Pa or N ( Stress threshold )                    | Lower stress bound below which gear fatigue accumulation is negligible.                          |

## D. Example data

### I. Chosen Cars for Example

- Porsche 718 Cayman GT4 CS MR ( slug : ks\_porsche\_718\_cayman\_gt4\_cs\_mr )
- Alpine A290 b ( slug : ks\_alpine\_a290\_b )
- Renault 5 GT Turbo ( slug : ks\_renault\_5\_gt\_turbo )

### II. Example

#### Porsche 718 Cayman GT4 CS MR

```

1. Gear Count : 6
2. Gears 1
 | 2a. Name : R
 | 2b. Ratio : -3.55000
2. Gears 2
 | 2a. Name : N
 | 2b. Ratio : 0.00000
2. Gears 3
 | 2a. Name : 1
 | 2b. Ratio : 3.90900
2. Gears 4
 | 2a. Name : 2
 | 2b. Ratio : 2.29167
2. Gears 5
 | 2a. Name : 3
 | 2b. Ratio : 1.65384
2. Gears 6
 | 2a. Name : 4

```

- └ 2b. Ratio : 1.30303
- 2. Gears 7
  - └ 2a. Name : 5
  - └ 2b. Ratio : 1.08108
- 2. Gears 8
  - └ 2a. Name : 6
  - └ 2b. Ratio : 0.88095
- 3. Final Ratio : 3.72461
- 4. Gear Up Time : 30.00000
- 5. Gear Dn Time : 80.00000
- 6. Auto Cut Off Time : 30.00000
- 7. Gas Cut Off Level : 0.70000
- 8. Has Dual Clutch : true
- 9. Is Shifter Supported : false
- 10. Downshift Protection
  - └ 10a. Is Active : true
  - └ 10b. Is Debug : false
  - └ 10c. Overrev : 300
  - └ 10d. Lock N : true
- 11. Damage Rpm Window : 100.00000
- 12. Valid Shift Rpm Window : 800.00000
- 13. Controls Window Gain : 0.40000
- 14. Gearbox Inertia : 0.01200
- 15. Autoblip
  - └ 15a. Profile :
   
content\cars\ks\_porsche\_718\_cayman\_gt4\_cs\_mr\data\curves\718\_gt4\_autobli
   
p.curve
    - └ 15b. Is Electronic : true
- 16. Autoshifter
  - └ 16a. Up : 7700
  - └ 16b. Down Rpm Threshold : 0
  - └ 16c. Slip Threshold : 0.96000
  - └ 16d. Gas Cutoff Time : 0.06300
- 17. Gears Fatigue Log10 A : 0.00000
- 18. Gears Fatigue Nominal Torque : 0.00000
- 19. Gears Fatigue Max Stress : 0.00000
- 20. Gears Fatigue Min Stress : 0.00000

### Alpine A290 b

- 1. Gear Count : 1
- 2. Gears 1
  - └ 2a. Name : R
  - └ 2b. Ratio : -9.00000
- 2. Gears 2
  - └ 2a. Name : N
  - └ 2b. Ratio : 0.00000
- 2. Gears 3
  - └ 2a. Name : 1
  - └ 2b. Ratio : 9.00000
- 3. Final Ratio : 1.00000
- 4. Gear Up Time : 10.00000
- 5. Gear Dn Time : 16.00000
- 6. Auto Cut Off Time : 11.00000

- 7. Gas Cut Off Level : 0.00000
- 8. Has Dual Clutch : false
- 9. Is Shifter Supported : false
- 10. Downshift Protection
  - 10a. Is Active : true
  - 10b. Is Debug : false
  - 10c. Overrev : 100
  - 10d. Lock N : true
- 11. Damage Rpm Window : 100.00000
- 12. Valid Shift Rpm Window : 800.00000
- 13. Controls Window Gain : 0.40000
- 14. Gearbox Inertia : 0.00800
- 15. Autoblip
  - 15a. Profile :  
content\cars\ks\_alpine\_a290\_b\data\autoBlipProfile.curve
  - 15b. Is Electronic : true
- 16. Autoshifter
  - 16a. Up : 6850
  - 16b. Down Rpm Threshold : 0
  - 16c. Slip Threshold : 0.95000
  - 16d. Gas Cutoff Time : 0.15000
- 17. Gears Fatigue Log10 A : 0.00000
- 18. Gears Fatigue Nominal Torque : 0.00000
- 19. Gears Fatigue Max Stress : 0.00000
- 20. Gears Fatigue Min Stress : 0.00000

### Renault 5 GT Turbo

- 1. Gear Count : 5
- 2. Gears 1
  - 2a. Name : R
  - 2b. Ratio : -3.54500
- 2. Gears 2
  - 2a. Name : N
  - 2b. Ratio : 0.00000
- 2. Gears 3
  - 2a. Name : 1
  - 2b. Ratio : 3.09100
- 2. Gears 4
  - 2a. Name : 2
  - 2b. Ratio : 1.84200
- 2. Gears 5
  - 2a. Name : 3
  - 2b. Ratio : 1.32100
- 2. Gears 6
  - 2a. Name : 4
  - 2b. Ratio : 0.96700
- 2. Gears 7
  - 2a. Name : 5
  - 2b. Ratio : 0.75800
- 3. Final Ratio : 3.73300
- 4. Gear Up Time : 250.00000
- 5. Gear Dn Time : 270.00000
- 6. Auto Cut Off Time : 0.00000

- 7. Gas Cut Off Level : 0.00000
- 8. Has Dual Clutch : false
- 9. Is Shifter Supported : true
- 10. Downshift Protection
  - 10a. Is Active : false
  - 10b. Is Debug : false
  - 10c. Overrev : 0
  - 10d. Lock N : false
- 11. Damage Rpm Window : 100.00000
- 12. Valid Shift Rpm Window : 1000.00000
- 13. Controls Window Gain : 0.40000
- 14. Gearbox Inertia : 0.01700
- 15. Autoblip
  - 15a. Profile :  
content\cars\ks\_renault\_5\_gt\_turbo\data\autoBlipProfile.curve
  - 15b. Is Electronic : false
- 16. Autoshifter
  - 16a. Up : 6000
  - 16b. Down Rpm Threshold : 0
  - 16c. Slip Threshold : 0.95000
  - 16d. Gas Cutoff Time : 0.30000
- 17. Gears Fatigue Log10 A : 0.00000
- 18. Gears Fatigue Nominal Torque : 0.00000
- 19. Gears Fatigue Max Stress : 0.00000
- 20. Gears Fatigue Min Stress : 0.00000

# 15. General [ .generalcar ]

## A. Description

Standalone “chassis identity” block: display name, total mass, fuel tank and fuel economy, body box, pickup heights, ride-height rule flags, torsional stiffness/damping, and optional body-mesh offset.

It is the same field set as the **General** object embedded inside **3. Car Data** (.CAR), exposed as its own asset type so a car can optionally load it via *General Path* instead of (or as an override for) the inline General object.

Official Description prose in the source dump is unfinished placeholder text. The example section states explicitly that **no shipped car or common asset currently uses a .GENERALCAR file** — all practical values live inline in Car Data with *General Path* : *None*.

---

## I. Role in the stack

| Concern                             | Handled here       | Handled elsewhere                |
|-------------------------------------|--------------------|----------------------------------|
| Screen name, mass, fuel, tank       | .GENERALCAR fields | .CAR → General object            |
| Body box / pickup heights           | .GENERALCAR        | Same inline General              |
| Check rules / minimum height        | .GENERALCAR        | Same inline General              |
| Chassis torsion pair                | .GENERALCAR        | Same inline General              |
| Body mesh offset                    | .GENERALCAR        | Same inline General              |
| Which general block loads           | —                  | .CAR <i>General Path</i>         |
| Wheelbase, tracks, CG, module paths | —                  | Rest of .CAR (not in this asset) |

Think of .GENERALCAR as an extractable shared fragment — useful for tooling or multi-car inheritance — not as a second physics domain.

---

## II. What you are really tuning

- Identity and mass** — *Screen Name* for UI. *Total Mass* is the  $F=ma$  baseline (in Car Data examples: R5 about 910 kg, R8 GT3 about 1355 kg, 296 GTB about 1750 kg).
- Fuel payload** — *Tank Position* (x,y,z), *Fuel / Max Fuel*, *Efficiency*, *Kg Per Liter*. Changing fuel changes mass and CG over a stint — same behaviour whether authored inline or via this asset.
- Body envelope and pickups** — *Body Box Sizes* (x,y,z), *Pickup Front Height / Pickup Rear Height* — collision / jack / reference heights tied to the body, not suspension hardpoints.
- Regulatory floor** — *Check Rules + Minimum Height* enable ride-height legality checks when true (race cars in Car Data examples); road cars often leave checks off.
- Chassis beam** — *Torsional Stiffness / Torsional Damping* — soft shells twist (R5 about 11000 stiffness in Car Data); stiff tubs stay flatter (R8 about 40000).

6. **Mesh alignment** — Body Mesh Offset → Position / Rotation / Scale — visual/collider mesh placement relative to the physics body.

---

### III. Architecture

#### 1 - CORE SCALARS (SCHEMA 1-14)

*Screen name; mass; tank and fuel block; body box; pickup heights; check rules; minimum height; torsional stiffness and damping.*

#### 2 - MESH OFFSET (SCHEMA 15)

Nested Position / Rotation / Scale vectors.

Field order matches Car Data *General* leaves (2a...2o), flattened to top-level 1...15 in this asset type.

---

### IV. How to read the examples

#### 1 - SOURCE STATEMENT

No cars or common assets use an asset of the "generalcar" type.

There is no vehicle dump to walk through for this file. Treat the schema as the contract; treat **Car Data** → **General** dumps as the living examples of the same numbers.

#### 2 - PRACTICAL READING VIA CAR DATA

| Car Data example       | What the General block shows                                          |
|------------------------|-----------------------------------------------------------------------|
| Ferrari 296 GTB        | Mass 1750, torsional stiffness 30000, large tank, General Path : None |
| Audi R8 LMS GT3 Evo II | Mass 1355, stiffness 40000, rules/min height used, path None          |
| Renault 5 GT Turbo     | Mass 910, soft shell about 11000, path None                           |

If a future car sets *General Path* to a .GENERALCAR file, expect those fields to supply (or replace) the inline General object — verify in-session which layer wins.

---

### V. Practical notes

- Source Description section is placeholder — ignore spring/alignment headings.
- Do not invent a .GENERALCAR dependency for current ACE cars; shipped content keeps everything in .car.
- Duplicating mass/fuel here **and** inline without a clear Path policy will confuse authors — pick one source of truth per vehicle.



- Extension name in the doc title is `.GENERALCAR`; Car Data only exposes *General Path* (string). Match whatever on-disk extension the toolchain actually writes.
- Editing torsion or mass for handling work still means editing Car Data General today.

---

## VI. Related assets

- **3. Car Data** — owns inline General today; optional *General Path* consumer
- **5. Car Setup** — session fuel load overlays Max Fuel capacity defined here / in General
- **11 / 17. Coilover / Suspension** — ride and kinematics; not substitutes for torsional stiffness

### B. Schema

- ```

1. Screen Name : string
2. Total Mass : float
3. Tank Position : x, y, z float
4. Fuel : float
5. Max Fuel : float
6. Efficiency : float
7. Kg Per Liter : float
8. Body Box Sizes : x, y, z float
9. Pickup Front Height : float
10. Pickup Rear Height : float
11. Check Rules : boolean
12. Minimum Height : float
13. Torsional Stiffness : float
14. Torsional Damping : float
15. Body Mesh Offset : object
    | 15a. Position : x, y, z float
    | 15b. Rotation : x, y, z float
    | 15c. Scale : x, y, z float

```

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Screen Name	None (String)	Human-readable label for this <code>.GENERALCAR</code> profile in editor or UI tooling; does not affect physics.
2.	Total Mass	Kg (Kilograms)	Baseline vehicle mass including driver and fluids; fundamental input for weight transfer and inertia (Car Data 2b).
3.	Tank Position	m (Meters, X / Y / Z)	3D coordinates of the fuel tank centroid relative to the vehicle reference frame (Car Data 2c).
4.	Fuel	L (Liters)	Starting fuel volume at session load; reduces mass and shifts CG as fuel burns (Car Data 2d).

ID	Name	Unit of Measurement	Description
5.	Max Fuel	L (Liters)	Maximum fuel tank capacity; upper bound for fuel load and endurance calculations (Car Data 2e).
6.	Efficiency	Dimensionless ratio	Fuel consumption efficiency modifier applied to engine burn calculations (Car Data 2f).
7.	Kg Per Liter	kg/L (Kilograms per liter)	Fuel density factor converting volumetric fuel into mass for physics updates (Car Data 2g).
8.	Body Box Sizes	m (Meters, X / Y / Z)	Axis-aligned bounding box dimensions of the chassis body envelope for regulatory/collision checks (Car Data 2h).
9.	Pickup Front Height	m (Meters)	Vertical pickup/reference height at the front axle; aligns suspension hardpoints (Car Data 2i).
10.	Pickup Rear Height	m (Meters)	Vertical pickup/reference height at the rear axle (Car Data 2j).
11.	Check Rules	None (Boolean : True / False)	When true, enables automated regulatory validation such as minimum ride height (Car Data 2k).
12.	Minimum Height	m (Meters)	Regulatory minimum ride-height threshold enforced when Check Rules is active (Car Data 2l).
13.	Torsional Stiffness	Nm/rad or N/m (Torsional spring rate)	Chassis torsional spring rate resisting twist between front and rear axles (Car Data 2m).
14.	Torsional Damping	Nm-s/rad (Torsional damping)	Chassis torsional damping coefficient dissipating body twist oscillations (Car Data 2n).
15.	Body Mesh Offset	None (Object)	Visual/collision body mesh alignment block relative to the physics origin (Car Data 2o).
15a.	Position	m (Meters, X / Y / Z)	Translation offset of the body mesh from the physics origin (Car Data 2o1).
15b.	Rotation	deg or rad (X / Y / Z)	Euler rotation offset applied to the body mesh for model alignment (Car Data 2o2).
15c.	Scale	Dimensionless (X / Y / Z)	Non-uniform scale factor on the body mesh offset block (Car Data 2o3).

D. Example data

I. Chosen Cars for Example

No cars or common assets use an asset of the “GENERALCAR” type.

16. Surface 3D [.surface3d]

A. Description

Transmission ratios and shift behaviour: gear list (including reverse and neutral), final drive, up/down shift times, throttle cut during shifts, dual-clutch / H-pattern flags, downshift protection, autoblip, autoshifter thresholds, gearbox inertia, and optional gear fatigue stress parameters.

Clutch engagement curves live in `.CLUTCH`. Diff and driven axles live in `.DRIVETRAIN`. This asset decides what ratio sits between engine and final drive, and how long / how violently a gear change interrupts torque.

Official Description prose in the source dump is unfinished placeholder text. Content below follows the schema and examples.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Cd / Cl (or CX / CZ)	Interpolation table	—
Axis span and grid density	min/max + size front/rear	—
Optional secondary height LUTs	<i>Downforce h/dh</i> mm paths	Usually None in samples
Denser runtime interp cache	<i>Interp Map</i> when done	—
CSV authoring import/export	Import/export paths	Tooling only
Which maps the car uses	—	<code>.CAR</code> Aero paths / tuning parts
Wing angles / element gains	—	<code>.WING</code> , <code>setup aero</code>
Session ride heights	—	<code>.CARSETUP</code> Aero targets

Typical triplet per car: one drag (CX) map + front lift/downforce (CZ) + rear lift/downforce (CZ).

II. What you are really tuning

- The lookup axes** — *min / max* front and rear plus *size front / size rear* define the grid. Table headers are the sampled axis values (Mercedes drag about 65–160 on both axes; R8 CX/CZ front 30–80 with 6 samples, rear 30–100 with 8). Outside the span the runtime must clamp or extrapolate — keep setup ride heights inside the map.
- Coefficient cells** — Each cell is the aero coefficient at that front/rear pairing. Reading a row = fixed front state while rear varies (columns). R8 rear CZ rises strongly as front height drops (more front rake / lower nose → big rear load in that map). Drag maps often rise when both ends sit high (Mercedes high-high corner about 0.67 vs diagonal mid about 0.46).
- Naming conventions** — Same schema, different filenames/roles:

Role	Example filenames
Front downforce / CZ	front_downforce, frontczmap, front_lift_map
Rear downforce / CZ	rear_downforce, rearczmap, rear_lift_map

“Lift” vs “downforce” in the filename is authoring vocabulary — sign and magnitude live in the cell values.

4. **Interpolation bookkeeping** — *interpolation done* true + filled *Interp Map* (min/max val, resolution front/rear) marks a denser baked interp (Mercedes front/rear DF and all R8 maps: resolution 18×18, val span −0.5...2.0). Raw grids with *interpolation done* : false and *Interp Map* : None (Mercedes drag, Dallara maps) rely on the coarse table alone.
5. **Tooling fields** — *Downforce h mm / dh mm* paths, CSV import/export, *Import front / Import range x y* — authoring hooks. All examples leave paths None and import flags false.

III. Architecture

1 - OPTIONAL LINKED CURVES (SCHEMA 1-2)

Paths for height / delta-height downforce helpers (unused in samples).

2 - GRID DEFINITION (SCHEMA 3-8)

Integer sizes (0–30); float min/max per axis.

3 - TABLE AND INTERP (SCHEMA 9-11)

Interpolation table of size rear × size front; interpolation done; optional Interp Map object.

4 - CSV / IMPORT (SCHEMA 12-15)

Booleans and paths for bulk edit workflows.

IV. How to read the examples

1 - MERCEDES AMG GT2 — SYMMETRIC 8X8 PACK

Three files: drag, front downforce, rear downforce. Drag left at coarse interp; DF maps baked to 18×18. Drag diagonal soft mid values; high-high corner climbs. Rear DF axis spans much taller (about 2–288) than front (about 20–154) — rear map covers a wider operating window.

2 - AUDI R8 LMS GT3 EVO 2 — CX / CZ RACE MAPS

Wired from Car Data as *Drag* → *cxmap*, *Front Lift* → *frontczmap*, *Rear Lift* → *rearczmap*. Shared 6×8 ride-height window (F 30–80, R 30–100). CX stays about 1.04–1.16; front CZ collapses toward 0 as front height rises; rear CZ sits about 1.8–2.6. Classic GT3 platform sensitivity to ride height.

3 - DALLARA STRADALE COUPE — COARSER ROAD/TRACK PACK

7 rear × 4 front, axes 60–120. Drag and front lift tables are numerically identical in the dump (likely shared/placeholder front map — verify before trusting). Rear lift is distinct and higher (about 1.4–2.1). All three leave interp undone.

V. Practical notes

- Source Description section is placeholder — ignore spring/alignment headings.
- Table size must match *size front* × *size rear*; a mismatched CSV import is a silent footgun.
- Keep garage ride-height limits inside min/max or the car runs off-map.
- Editing one map without the sibling CX/CZ pair breaks aero balance even if numbers “look fine” in isolation.
- Filename “lift” does not imply positive lift — read cell signs against how Car Data gains multiply them.
- OCR can corrupt edge cells (e.g. truncated decimals); prefer re-export from source assets when numbers look broken.

VI. Related assets

- **3. Car Data** — Aero Front Lift / Rear Lift / Drag paths
- **20. Wing** — discrete wing elements referenced beside these maps
- **5 / 6. Car Setup / Limits** — ride height and wing angles that feed the axes
- **8. Car Tuning Parts** — aero package redirects that can swap these maps

B. Schema

```
1. Downforce h mm : string - path
2. Downforce dh mm : string - path
3. size rear : range integer ( 0 - 30 )
4. size front : range integer ( 0 - 30 )
5. min rear : float
6. max rear : float
7. min front : float
8. max front : float
(9). Interpolation Table : table of size "3. size rear" * "4. size
front"
10. interpolation done : boolean
11. Interp Map : object
  11a. min val : float
  11b. max val : float
  11c. resolution front : range integer ( 3 - 90 )
  11d. resolution rear : range integer ( 3 - 90 )
12. Import front : boolean
13. Import range x y : boolean
14. CSV import path : string - path
15. CSV export name : string - path
```

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Downforce h mm	None (File path)	Optional source height map defining absolute ride-height reference for aero lookup (often None; Mercedes drag map).
2.	Downforce dh mm	None (File path)	Optional delta ride-height map layered on Downforce h mm for dynamic aero offset (often None).
3.	Size rear	Integer count (0 - 30)	Rear-axis grid resolution of the 2D interpolation table (rows; e.g., 8 on Mercedes/Audi GT3 maps).
4.	Size front	Integer count (0 - 30)	Front-axis grid resolution of the interpolation table (columns; e.g., 6–8).
5.	Min rear	mm (Millimeters)	Minimum rear ride-height sample bound for table axis (e.g., 65.338 mm Mercedes drag, 30 mm Audi Cx).
6.	Max rear	mm (Millimeters)	Maximum rear ride-height sample bound for table axis (e.g., 160.227 mm Mercedes drag).
7.	Min front	mm (Millimeters)	Minimum front ride-height sample bound for table axis.
8.	Max front	mm (Millimeters)	Maximum front ride-height sample bound for table axis.
9.	Interpolation Table	Dimensionless coefficient (2D matrix)	size rear × size front lookup of aero coefficients (Cd, Cl, Cz) vs. front/rear ride height; schema ID shown as (9). in examples.
10.	Interpolation done	None (Boolean : True / False)	When true, raw table data has been processed into the Interp Map representation (true on downforce maps).
11.	Interp Map	None (Object)	Baked interpolation map metadata block derived from the table (None when using raw matrix only).
11a.	Min val	Dimensionless coefficient	Lower clamp of interpolated aero coefficient output (e.g., -0.500).
11b.	Max val	Dimensionless coefficient	Upper clamp of interpolated aero coefficient output (e.g., 2.000).
11c.	Resolution front	Integer count (3 - 90)	Front-axis resolution of the baked interpolation map (18 on processed GT3/GT2 maps).
11d.	Resolution rear	Integer count (3 - 90)	Rear-axis resolution of the baked interpolation map (18 in examples).

ID	Name	Unit of Measurement	Description
12.	Import front	None (Boolean : True / False)	When true, front-axis data is imported from an external source rather than edited inline.
13.	Import range x y	None (Boolean : True / False)	When true, imports axis range bounds from external CSV/tooling rather than manual min/max fields.
14.	CSV Import path	None (File path)	Path to a CSV file for bulk table import; often None when table is authored in-editor.
15.	CSV export name	None (File path or string)	Target path or filename for CSV export of the interpolation table; often None.

D. Example data

I. Chosen Cars for Example

- Mercedes AMG GT2 (slug : ks_mercedes_amg_gt2) [3 surface 3d]
- Audi R8 LMS GT3 Evo 2 (slug : ks_audi_r8_lms_gt3_evo_2) [3 surface 3d]
- Dallara Stradale Coupe (slug : ks_dallara_stradale_coupe) [3 surface 3d]

II. Example

Mercedes AMG GT2

1. Drag map (file : drag_map.surface3d)

```

1. Downforce h mm : None
2. Downforce dh mm : None
3. size rear : 8
4. size front : 8
5. min rear : 65.338
6. max rear : 160.227
7. min front : 65.338
8. max front : 160.227
(9). Interpolation Table

```

Front v / Rear	65.3	78.9	92.4	106.0	119.6	133.1	146.7	160.2
65.3	0.454	0.475	0.487	0.498	0.509	0.539	0.579	0.671
78.9	0.487	0.456	0.476	0.489	0.506	0.522	0.545	0.582
92.4	0.515	0.489	0.457	0.478	0.497	0.519	0.528	0.548
106.0	0.540	0.516	0.490	0.459	0.486	0.510	0.525	0.531
119.6	0.556	0.541	0.518	0.492	0.467	0.499	0.516	0.528

Front v / Rear	65.3	78.9	92.4	106.0	119.6	133.1	146.7	160.2
133.1	0.572	0.557	0.542	0.520	0.500	0.480	0.505	0.519
146.7	0.597	0.573	0.558	0.545	0.528	0.513	0.486	0.508
160.2	0.642	0.599	0.574	0.561	0.553	0.541	0.519	0.489

- 10. interpolation done : false
- 11. Interp Map : None
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

2. Front downforce (file : front_downforce.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 8
- 5. min rear : 20.039
- 6. max rear : 154.236
- 7. min front : 20.039
- 8. max front : 154.236
- (9). Interpolation Table

Front v / Rear	20.0	39.2	58.4	77.6	96.7	115.9	135.1	154.2
20.0	0.210	0.273	0.288	0.282	0.272	0.260	0.255	0.255
39.2	0.176	0.211	0.273	0.288	0.283	0.272	0.261	0.256
58.4	0.172	0.177	0.212	0.275	0.289	0.284	0.273	0.262
77.6	0.175	0.175	0.180	0.215	0.277	0.292	0.287	0.276
96.7	0.181	0.175	0.176	0.181	0.215	0.278	0.293	0.287
115.9	0.184	0.174	0.169	0.169	0.174	0.209	0.271	0.286
135.1	0.189	0.180	0.170	0.164	0.164	0.169	0.204	0.266
154.2	0.187	0.187	0.178	0.168	0.162	0.163	0.167	0.202

- 10. interpolation done : true
- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

3. Rear downforce (file : rear_downforce.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 8
- 5. min rear : 1.732
- 6. max rear : 288.196
- 7. min front : 1.732
- 8. max front : 288.196
- (9). Interpolation Table

Front v / Rear	1.7	42.7	83.6	124.5	165.4	206.3	247.3	288.2
1.7	0.444	0.431	0.453	0.439	0.440	0.422	0.384	0.346
42.7	0.445	0.477	0.452	0.439	0.440	0.422	0.384	0.346
83.6	0.443	0.477	0.499	0.438	0.440	0.422	0.384	0.346
124.5	0.443	0.475	0.499	0.484	0.439	0.422	0.384	0.346
165.4	0.443	0.475	0.497	0.484	0.485	0.421	0.384	0.346
206.3	0.443	0.475	0.497	0.482	0.486	0.467	0.383	0.346
247.3	0.443	0.475	0.497	0.482	0.484	0.467	0.429	0.45
288.2	0.443	0.475	0.497	0.482	0.484	0.465	0.430	0.391

- 10. interpolation done : true
- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

Audi R8 LMS GT3 Evo 2

1. CX Map (file : cxmap.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 6
- 5. min rear : 30.000
- 6. max rear : 100.000
- 7. min front : 30.000

```

| 8. max front : 80.000
| (9). Interpolation Table

```

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
30.0	1.041	1.049	1.051	1.054	1.057	1.061	1.065	1.068
40.0	1.076	1.080	1.080	1.078	1.080	1.085	1.089	1.092
50.0	1.092	1.104	1.108	1.113	1.121	1.119	1.123	1.127
60.0	1.098	1.110	1.119	1.127	1.129	1.137	1.141	1.145
70.0	1.110	1.123	1.131	1.140	1.141	1.145	1.152	1.153
80.0	1.120	1.134	1.141	1.146	1.149	1.152	1.158	1.164

```

| 10. interpolation done : true
| 11. Interp Map
|   | 11a. min val : -0.500
|   | 11b. max val : 2.000
|   | 11c. resolution front : 18
|   | 11d. resolution rear : 18
| 12. Import front : false
| 13. Import range x y : false
| 14. CSV import path : None
| 15. CSV export name : None

```

2. Front CZ Map (file : frontczmap.surface3d)

```

| 1. Downforce h mm : None
| 2. Downforce dh mm : None
| 3. size rear : 8
| 4. size front : 6
| 5. min rear : 30.000
| 6. max rear : 100.000
| 7. min front : 30.000
| 8. max front : 80.000
| (9). Interpolation Table

```

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
30.0	0.913	1.056	1.197	1.323	1.428	1.516	1.581	1.633
40.0	0.784	0.911	1.022	1.120	1.215	1.300	1.384	1.455
50.0	0.616	0.726	0.836	0.946	1.051	1.138	1.214	1.281
60.0	0.415	0.508	0.618	0.740	0.854	0.954	1.035	1.100
70.0	0.214	0.311	0.415	0.519	0.621	0.721	0.816	0.898
80.0	0.011	0.116	0.223	0.333	0.431	0.516	0.605	0.689

- 10. interpolation done : true
- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

3. Rear CZ Map (file : rearczmap.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 8
- 4. size front : 6
- 5. min rear : 30.000
- 6. max rear : 100.000
- 7. min front : 30.000
- 8. max front : 80.000
- (9). Interpolation Table

Front v / Rear	30.0	40.0	50.0	60.0	70.0	80.0	90.0	100.0
30.0	1.927	2.011	2.026	1.953	1.905	1.892	1.859	1.823
40.0	2.085	2.122	2.125	2.074	2.015	1.965	1.917	1.871
50.0	2.267	2.262	2.240	2.191	2.124	2.053	1.992	1.934
60.0	2.428	2.407	2.365	2.299	2.227	2.151	2.080	2.005
70.0	2.542	2.513	2.464	2.395	2.316	2.232	2.156	2.079
80.0	2.633	2.591	2.539	2.481	2.401	2.305	2.217	2.138

- 10. interpolation done : true
- 11. Interp Map
 - 11a. min val : -0.500
 - 11b. max val : 2.000
 - 11c. resolution front : 18
 - 11d. resolution rear : 18
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

Dallara Stradale Coupe

1. Drag map (file : drag_map.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 7
- 4. size front : 4
- 5. min rear : 60.000
- 6. max rear : 120.000
- 7. min front : 60.000
- 8. max front : 120.000
- (9). Interpolation Table

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
60.0	0.930	0.943	0.950	0.960	0.986	0.994	1.010
80.0	0.945	0.960	0.965	0.980	1.000	1.016	1.040
100.0	0.960	0.970	0.974	0.990	1.001	1.015	1.023
120.0	0.970	0.972	0.976	0.995	1.003	1.010	1.022

- 10. interpolation done : false
- 11. Interp Map : None
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None
- 15. CSV export name : None

2. Front lift map (file : front_lift_map.surface3d)

- 1. Downforce h mm : None
- 2. Downforce dh mm : None
- 3. size rear : 7
- 4. size front : 4
- 5. min rear : 60.000
- 6. max rear : 120.000
- 7. min front : 60.000
- 8. max front : 120.000
- (9). Interpolation Table

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
60.0	0.930	0.943	0.950	0.960	0.986	0.994	1.010
80.0	0.945	0.960	0.965	0.980	1.000	1.016	1.040
100.0	0.960	0.970	0.974	0.990	1.001	1.015	1.023
120.0	0.970	0.972	0.976	0.995	1.003	1.010	1.022

- 10. interpolation done : false
- 11. Interp Map : None
- 12. Import front : false
- 13. Import range x y : false
- 14. CSV import path : None

L 15. CSV export name : None

3. *Rear lift map (file : rear_lift_map.surface3d)*

| 1. Downforce h mm : None
| 2. Downforce dh mm : None
| 3. size rear : 7
| 4. size front : 4
| 5. min rear : 60.000
| 6. max rear : 120.000
| 7. min front : 60.000
| 8. max front : 120.000
| (9). Interpolation Table

Front v / Rear	60.0	70.0	80.0	90.0	100.0	110.0	120.0
60.0	1.619	1.562	1.482	1.414	1.402	1.365	1.396
80.0	1.766	1.751	1.664	1.575	1.517	1.474	1.428
100.0	1.716	1.739	1.754	2.096	1.645	1.551	1.534
120.0	1.833	1.837	1.835	1.754	1.711	1.613	1.597

| 10. interpolation done : false
| 11. Interp Map : None
| 12. Import front : false
| 13. Import range x y : false
| 14. CSV import path : None
| 15. CSV export name : None

17. Suspension [.suspension]

A. Description

Kinematic hardpoint package for one axle (or corner family): hub mass, baseline camber/toe linear, rim offset, plus exactly one filled geometry topology (strut, double wishbone, trailing arm, multilink, axle, etc.).

Coilover rates and dampers live in `.COILOVER`. Garage camber/toe/pressure live in `.CARSETUP`. This file decides how the upright moves through travel and steer — motion ratios, camber gain, bump steer — via pickup coordinates.

Official Description prose in the source dump is unfinished placeholder text. Content below follows the schema and examples.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Hub / upright mass	<i>Basic Data</i>	—
Baseline static camber / toe-out linear	<i>Basic Data</i>	Setup alignment overlays
Rim offset	<i>Basic Data</i>	—
Wishbone / strut / trailing / multilink / pickups	Topology objects 2-9	—
Coilover attach points (when in geometry)	Dw Coil / Multi Link New	Spring rates in <code>.COILOVER</code>
Wheelbase, track, CG	—	<code>.CAR</code> Suspensions hub
Which front/rear files load	—	<code>.CAR</code> / <code>.TUNINGPART</code> paths

One `.SUSPENSION` file per axle side of the architecture (front file + rear file; optional drift variants).

II. What you are really tuning

- Basic hub identity** — Always filled: *Hub Mass*, *Toe Out Linear*, *Static Camber*, *Rim Offset*. Golf front hub 50 kg / camber -0.3° ; S2000 front 37 kg / -0.4° with rim offset 0.045; Porsche GT3 R front 51 kg / camber -5.7° — race baseline geometry, not a mild road setup.
- One topology wins** — Only one of the geometry blocks should be non-None. The schema lists every layout; authors fill the matching family:

Block	Typical layout
D W Data	Double wishbone (top/bottom car+tyre + steer)
Strut	MacPherson strut + lower wishbone + steer
Strut MI	Strut multilink (adds thrust balls/arm)

Block	Typical layout
Axle	Live/beam axle links, leak K, torque reaction
Multi Link Data	Older joint-car / joint-tyre arrays
Trailing Arm Data	Trailing arm hinges + upright + steer
Multi Link New Data	Named links/arms, coilover flags, toe link
Dw Coil Data	Double wishbone + explicit car/bottom coilover points

3. **Coordinate pickups** — All hardpoints are x, y, z in the car/tyre frames used by the toolchain (car-side vs tyre-side pairs). Moving a pickup changes instantaneous centre, roll centre, camber gain, and bump steer — more sensitive than tweaking a spring rate.
4. **Coilover attachment in geometry** — *Dw Coil Data* and *Multi Link New* expose *Car Coilover / Bottom Coilover* (and per-link *Has Coilover Attached*). That is where the damper unit sits in space relative to the links — motion ratio comes from this geometry + coilover rates.
5. **Variant packs** — Same Basic Data with relocated DW pickups = alternate kinematics without a new spring file (S2000 stock front vs front drift: top mounts lowered in Y, steer/tyre points shifted). Tuning parts can redirect geometry paths the same way.

III. Architecture

1 - BASIC DATA (SCHEMA 1)

Hub mass, toe-out linear, static camber, rim offset.

2 - LEGACY / CLASSIC TOPOLOGIES (SCHEMA 2-7)

DW Data; Strut; Strut MI; Axle; Multi Link Data; Trailing Arm Data.

3 - MODERN MULTILINK AND DW+COIL (SCHEMA 8-9)

Multi Link New: Links (ball car/tyre, coilover flag, Is Toe) and Arms (two car balls + tyre ball); coilover car/bottom points. Dw Coil Data: full DW set plus coilover pickups.

IV. How to read the examples

1 - VOLKSWAGEN GOLF GTI MK1 — STRUT FRONT / TRAILING REAR

Front fills **Strut** (classic Mk1 MacPherson). Rear fills **Trailing Arm Data** (Steer Link To Body : false). All other topology blocks None. Period road-car architecture, mild static cambers.

2 - HONDA S2000 AP1 — DOUBLE WISHBONE + DRIFT FRONT

Stock front and rear use **D W Data**. Extra *front_drift* file keeps the same hub/camber/rim basics but relocates upper mounts and steer points — geometry-only drift package. OCR note: the rear dump header incorrectly reuses the drift front filename; treat content (hub 44.5, camber -1.6, DW points) as the rear axle.

3 - PORSCHE 992 GT3 R RENNSPORT — DW COIL FRONT / MULTILINK REAR

Front uses **Dw Coil Data** (wishbones + coilover car/bottom). Rear uses **Multi Link New Data** with five links (one named *steer* with *Is Toe : true*, one with coilover attached) and explicit coilover points. Aggressive static camber (−5.7 / −4.4). Modern GT race pattern.

V. Practical notes

- Source Description section is placeholder — ignore spring/alignment headings.
- Filling two topology blocks in one file is a content bug; leave unused families *None*.
- Setup camber/toe adjust on top of *Static Camber / Toe Out Linear* — do not double-count when comparing cars.
- Drift / geometry tuning parts must point at real `.SUSPENSION` files; swapping only coilovers will not change roll centres.
- OCR can garble coordinates (e.g. *-010540*); prefer re-reading source assets when a point looks impossible.
- Rim offset changes scrub/track feel without touching Car Data track width — keep both consistent.

VI. Related assets

- **3. Car Data** — front/rear suspension paths, wheelbase/track/CG
- **11. Coilover** — spring/damper unit hanging on these pickups
- **5 / 6. Car Setup / Limits** — alignment and ride overlays
- **8. Car Tuning Parts** — Suspensions Geometry path redirects
- **19. Tyre** — contact patch that kinematics aim at the ground

B. Schema

- 1. Basic Data : **object**
 - 1a. Hub Mass : **float**
 - 1b. Toe Out Linear : **float**
 - 1c. Static Camber : **float**
 - 1d. Rim Offset : **float**
- 2. D W Data : **object**
 - 2a. Car Top Front : **x, y, z float**
 - 2b. Car Top Rear : **x, y, z float**
 - 2c. Tyre Top : **x, y, z float**
 - 2d. Car Bottom Front : **x, y, z float**
 - 2e. Car Bottom Rear : **x, y, z float**
 - 2f. Tyre Bottom : **x, y, z float**
 - 2g. Car Steer : **x, y, z float**
 - 2h. Tyre Steer : **x, y, z float**
- 3. Strut : **object**
 - 3a. Car Strut : **x, y, z float**

- 3b. Tyre Strut : `x, y, z float`
- 3c. Car Bottom W B F : `x, y, z float`
- 3d. Car Bottom W B R : `x, y, z float`
- 3e. Tyre Bottom W B : `x, y, z float`
- 3f. Car Steer : `x, y, z float`
- 3g. Tyre Steer : `x, y, z float`
- 4. Strut Ml : `object`
 - 4a. Car Strut : `x, y, z float`
 - 4b. Tyre Strut : `x, y, z float`
 - 4c. Car Bottom W B F : `x, y, z float`
 - 4d. Car Bottom W B R : `x, y, z float`
 - 4e. Tyre Bottom W B : `x, y, z float`
 - 4f. Car Thrust Ball1 : `x, y, z float`
 - 4g. Car Thrust Ball2 : `x, y, z float`
 - 4h. Tyre Thrust Arm : `x, y, z float`
 - 4i. Car Steer : `x, y, z float`
 - 4j. Tyre Steer : `x, y, z float`
- 5. Axle : `object`
 - 5a. Attach Relative Pos : `float`
 - 5b. Link Count : `integer`
 - 5c. Car Side `[x]` : `x, y, z float` | `can have multiple Car Side`
 - 5d. Axle Side `[x]` : `x, y, z float` | `can have multiple Axle Side`
 - 5e. Hub Mass : `float`
 - 5f. Leaf Spring Lat K : `float`
 - 5g. Torquereaction : `float`
- 6. Multi Link Data : `object`
 - 6a. Joint Car `[x]` : `x, y, z float` | `can have multiple Joint Car`
 - 6b. Joint Type `[x]` : `x, y, z float` | `can have multiple Joint Type`
- 7. Trailing Arm Data : `object`
 - 7a. Car Hinge Int : `x, y, z float`
 - 7b. Car Hinge Ext : `x, y, z float`
 - 7c. Tyre Top : `x, y, z float`
 - 7d. Tyre Bottom : `x, y, z float`
 - 7e. Car Steer : `x, y, z float`
 - 7f. Tyre Steer : `x, y, z float`
 - 7g. Steer Link To Body : `boolean`
- 8. Multi Link New Data : `object`
 - 8a. Links `[x]` : `object` | `can have multiple Links`
 - 8a1. Name : `string`
 - 8a2. Ball Car : `x, y, z float`
 - 8a3. Ball Tyre : `x, y, z float`
 - 8a4. Has Coilover Attached : `boolean`
 - 8a5. Is Toe : `boolean`
 - 8b. Arms `[x]` : `object` | `can have multiple Arms`
 - 8b1. Name : `string`
 - 8b2. Ball Car 1 : `x, y, z float`
 - 8b3. Ball Car 2 : `x, y, z float`
 - 8b4. Ball Tyre : `x, y, z float`
 - 8b5. Has Coilover Attached : `boolean`
 - 8c. Car Coilover : `x, y, z float`
 - 8d. Bottom Coilover : `x, y, z float`
- 9. Dw Coil Data : `object`
 - 9a. Car Top Front : `x, y, z float`
 - 9b. Car Top Rear : `x, y, z float`
 - 9c. Tyre Top : `x, y, z float`
 - 9d. Car Bottom Front : `x, y, z float`

- 9e. Car Bottom Rear : x, y, z float
- 9f. Tyre Bottom : x, y, z float
- 9g. Car Steer : x, y, z float
- 9h. Tyre Steer : x, y, z float
- 9i. Car Coilover : x, y, z float
- 9j. Bottom Coilover : x, y, z float

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Basic Data	None (Object)	Per-corner suspension asset root; holds unsprung mass, static alignment, and exactly one kinematic block (2–9). Referenced from Car Data front/rear suspension paths (.suspension).
1a.	Hub Mass	kg (Kilograms)	Unsprung mass at the hub: wheel, tire, brake rotor/caliper, hub, and upright share. Typical range ~37–51 kg in examples (S2000 light front 37 kg, Golf 50 kg, GT3 R 51 kg). Lower mass improves bump compliance and tire contact response.
1b.	Toe Out Linear	rad/m (Radians per meter)	Linear kinematic toe gradient vs. suspension displacement or stroke (aligned with Car Setup 5f). Small signed values (e.g. ± 0.0002 – 0.0011) model bump-steer / compliance toe; positive often means increasing toe-out with travel depending on axis convention.
1c.	Static Camber	deg (Degrees)	Fixed geometric camber at reference ride height before live setup offsets. Negative = top of wheel leans inward (-0.3° street Golf, -1.6° S2000 rear, -5.7° GT3 R race front). Sets baseline contact patch and camber gain reference.
1d.	Rim Offset	m (Meters)	Wheel rim lateral offset (ET equivalent) from hub mounting face; shifts scrub radius and track (e.g. 0.045 m S2000 front, 0.037 m rear). Affects steering feel, kingpin offset, and fender clearance.
2.	D W Data	None (Object)	Double-wishbone (DWB) kinematic hardpoints; used when Strut / Axle / Trailing Arm / Multi Link blocks are None (e.g. Honda S2000 front). Defines upper/lower arm pivots and steer tie-rod points in the car body frame.

ID	Name	Unit of Measurement	Description
2a.	Car Top Front	m (Meters, X / Y / Z)	Chassis-side pivot of the upper wishbone, forward inner bushing (subframe / body mount). With 2b defines upper arm axis and camber / caster kinematics during jounce and rebound.
2b.	Car Top Rear	m (Meters, X / Y / Z)	Chassis-side pivot of the upper wishbone, rear inner bushing. Pair with 2a sets upper control arm length, inclination, and roll-center contribution.
2c.	Tyre Top	m (Meters, X / Y / Z)	Upright-side ball joint for the upper wishbone (outer upper pivot). Primary camber-change link as the wheel moves vertically and steers.
2d.	Car Bottom Front	m (Meters, X / Y / Z)	Chassis-side pivot of the lower wishbone, forward inner bushing. Sets anti-dive geometry with 2e and influences roll center height.
2e.	Car Bottom Rear	m (Meters, X / Y / Z)	Chassis-side pivot of the lower wishbone, rear inner bushing. With 2d defines lower arm plane and longitudinal compliance.
2f.	Tyre Bottom	m (Meters, X / Y / Z)	Upright-side ball joint for the lower wishbone (outer lower pivot). Carries vertical load into the lower arm; strongly affects scrub radius and kingpin axis.
2g.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner attachment on rack or steering arm (body side). With 2h defines bump-steer and Ackermann geometry for DWB layouts.
2h.	Tyre Steer	m (Meters, X / Y / Z)	Steering tie-rod outer ball joint on the upright/knuckle. Motion relative to 2g produces steer angle change through travel.
3.	Strut	None (Object)	McPherson strut kinematic set; lower arm + telescopic strut define wheel motion (e.g. Golf GTI Mk1 front). Mutually exclusive with other geometry blocks when set to None elsewhere.
3a.	Car Strut	m (Meters, X / Y / Z)	Strut top mount on the body (shock tower / dome). Defines primary vertical load path and strut axis inclination (caster/camber offset from strut angle).

ID	Name	Unit of Measurement	Description
3b.	Tyre Strut	m (Meters, X / Y / Z)	Strut lower joint at the steering knuckle (wheel-side strut pin). Sliding joint along strut axis; sets kingpin inclination with 3a.
3c.	Car Bottom W B F	m (Meters, X / Y / Z)	Lower control arm (wishbone) inner forward bushing on the subframe. "W B" = wishbone; front pivot of the LCA.
3d.	Car Bottom W B R	m (Meters, X / Y / Z)	Lower control arm inner rear bushing on the subframe. With 3c sets LCA swing axis, anti-roll, and longitudinal location.
3e.	Tyre Bottom W B	m (Meters, X / Y / Z)	Lower control arm outer ball joint at the knuckle. Vertical wheel motion rotates about the LCA line 3c-3d-3e.
3f.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner point (rack or steering arm on body) for McPherson layout.
3g.	Tyre Steer	m (Meters, X / Y / Z)	Tie-rod outer joint on the knuckle; combined with 3f defines steer angle and McPherson bump-steer.
4.	Strut MI	None (Object)	McPherson-strut variant with additional locating links (multi-link McPherson / strut-plus-links). Adds thrust links and extra steer points beyond block 3.
4a.	Car Strut	m (Meters, X / Y / Z)	Strut top mount on the body for Strut MI layout (same role as 3a).
4b.	Tyre Strut	m (Meters, X / Y / Z)	Strut-to-knuckle lower joint for Strut MI (same role as 3b).
4c.	Car Bottom W B F	m (Meters, X / Y / Z)	Lower wishbone forward inner pivot (Strut MI).
4d.	Car Bottom W B R	m (Meters, X / Y / Z)	Lower wishbone rear inner pivot (Strut MI).
4e.	Tyre Bottom W B	m (Meters, X / Y / Z)	Lower wishbone outer ball joint at the upright (Strut MI).
4f.	Car Thrust Ball1	m (Meters, X / Y / Z)	First chassis-side pivot of a thrust / radius rod limiting knuckle rotation about the strut axis; improves camber control under lateral load.
4g.	Car Thrust Ball2	m (Meters, X / Y / Z)	Second chassis-side pivot paired with 4f (dual-link or split bush axis) for the thrust linkage.

ID	Name	Unit of Measurement	Description
4h.	Tyre Thrust Arm	m (Meters, X / Y / Z)	Upright-side joint for the thrust / radius arm connecting to 4f/4g; resists fore-aft knuckle rotation and brake steer.
4i.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner attachment (Strut MI geometry).
4j.	Tyre Steer	m (Meters, X / Y / Z)	Steering tie-rod outer joint on the upright (Strut MI).
5.	Axle	None (Object)	Live / solid axle kinematic block for dependent rear (or front) suspension: locating links, leaf spring, and torque reaction. Used on trucks and classic live-axle cars when DW/Strut blocks are None
5a.	Attach Relative Pos	m (Meters) or dimensionless ratio (0 -1)	Longitudinal or normalized position along the axle/leaf where the spring or link attaches relative to axle center; sets load transfer and wrap-up geometry.
5b.	Link Count	Integer count	Number of axle locating links modeled (Panhard rod, trailing links, Watt's link, etc.); pairs with indexed 5c/5d entries.
5c.	Car Side	m (Meters, X / Y / Z)	Chassis-side attachment point for an axle locating link (indexed [x] per link). Defines roll center and lateral axle location.
5d.	Axle Side	m (Meters, X / Y / Z)	Axle-side attachment for the corresponding locating link [x]. With 5c sets link length and lateral constraint stiffness.
5e.	Hub Mass	kg (Kilograms)	Unsprung mass per corner for the axle layout (same physical meaning as 1a but authored inside the Axle block for live-axle assets).
5f.	Leaf Spring Lat K	N/m (Newtons per meter)	Lateral stiffness of the leaf spring stack resisting sideways axle shift (leaf-as-lateral-locator). Higher values reduce axle walk under cornering and braking.
5g.	Torquereaction	Nm/rad (Newton-meters per radian)	Axle torque-reaction stiffness against driveline/brake wrap-up (rotation of axle about its lateral axis). Models anti-squat / anti-lift from link geometry and compliant mounts.

ID	Name	Unit of Measurement	Description
6.	Multi Link Data	None (Object)	Legacy multi-link hardpoint list (indexed joints) predating Multi Link New Data ; simpler joint-car enumeration for older ACE suspension templates.
6a.	Joint Car	m (Meters, X / Y / Z)	Chassis-side pivot for link index [x] in the legacy multi-link solver (one point per link).
6b.	Joint Type	m (Meters, X / Y / Z)	Per-link constraint data for index [x]; schema types as x,y,z float (likely encoded joint class / axis flags rather than a literal position—treat as solver metadata vector in meters or normalized codes).
7.	Trailing Arm Data	None (Object)	Semi-trailing or trailing-arm rear geometry (e.g. Golf GTI Mk1 rear). Arm hinges on body and carries hub at outer end.
7a.	Car Hinge Int	m (Meters, X / Y / Z)	Inner trailing-arm bushing on the body (inboard pivot). Defines primary arm axis with 7b.
7b.	Car Hinge Ext	m (Meters, X / Y / Z)	Secondary inner hinge point on the body (outboard bushing of the arm inner pair). Two-point inner mount sets bushing compliance axis and toe change under load.
7c.	Tyre Top	m (Meters, X / Y / Z)	Hub/upright attachment at the upper outer end of the trailing arm.
7d.	Tyre Bottom	m (Meters, X / Y / Z)	Hub/upright attachment at the lower outer end; with 7c fixes wheel plane orientation on the arm.
7e.	Car Steer	m (Meters, X / Y / Z)	Steering or toe-link inner point on body (often inactive on driven rear arms).
7f.	Tyre Steer	m (Meters, X / Y / Z)	Steering / toe-link outer point on the hub for rear steer or toe compliance.
7g.	Steer Link To Body	None (Boolean : True / False)	When true, the steer/toe link inner mount is fixed to the body; when false, it may attach to the arm, changing toe compliance under traction/braking.
8.	Multi Link New Data	None (Object)	Explicit graph-based multi-link: indexed Links (two-point struts) and Arms (three-point control arms) plus optional coilover mounts (e.g. Porsche 992 GT3 R rear).

ID	Name	Unit of Measurement	Description
8a.	Links	None (Object array [x])	Repeatable link elements (tie rods, toe links, strut rods); each index defines a rigid link between Ball Car and Ball Tyre.
8a1.	Name	None (String)	Optional label for the link (e.g. “steer” on GT3 R toe link); aids authoring/debug, not used in physics directly.
8a2.	Ball Car	m (Meters, X / Y / Z)	Chassis-side spherical joint for link [x].
8a3.	Ball Tyre	m (Meters, X / Y / Z)	Upright-side spherical joint for link [x]; distance to 8a2 sets link length and constraint direction.
8a4.	Has Coilover Attached	None (Boolean : True / False)	When true, the coilover damper unit (.coilover asset) acts on this link (force along link or at pickup on link geometry).
8a5.	Is Toe	None (Boolean : True / False)	When true, this link is flagged as the primary toe-control member (toe link / steering compliance arm); GT3 R example sets true on link named steer.
8b.	Arms	None (Object array [x])	Repeatable control arms with two body pivots and one upright pivot (classic multi-link arm).
8b1.	Name	None (String)	Optional arm identifier for authoring (camper arm, upper lateral, etc.).
8b2.	Ball Car 1	m (Meters, X / Y / Z)	First chassis-side pivot of arm [x] (inner bushing pair or front link point).
8b3.	Ball Car 2	m (Meters, X / Y / Z)	Second chassis-side pivot of arm [x]; with 8b2 defines arm plane and bush axis.
8b4.	Ball Tyre	m (Meters, X / Y / Z)	Upright-side ball joint for arm [x]; primary load path from wheel into the arm.
8b5.	Has Coilover Attached	None (Boolean : True / False)	When true, coilover spring/damper attaches to this arm (common on lower or virtual swing arm).
8c.	Car Coilover	m (Meters, X / Y / Z)	Coilover upper mount on the body when not carried by a flagged link/ arm (shock tower pickup).
8d.	Bottom Coilover	m (Meters, X / Y / Z)	Coilover lower mount on the upright or link; with 8c defines damper line and motion ratio.

ID	Name	Unit of Measurement	Description
9.	Dw Coil Data	None (Object)	Double wishbone geometry with integrated coilover pick-ups (e.g. Porsche 992 GT3 R front). Combines DW hardpoints with explicit damper mount points 9i/9j.
9a.	Car Top Front	m (Meters, X / Y / Z)	Upper wishbone forward inner pivot (Dw Coil layout; same role as 2a).
9b.	Car Top Rear	m (Meters, X / Y / Z)	Upper wishbone rear inner pivot (Dw Coil; same role as 2b).
9c.	Tyre Top	m (Meters, X / Y / Z)	Upper wishbone outer ball joint on upright (Dw Coil; same role as 2c).
9d.	Car Bottom Front	m (Meters, X / Y / Z)	Lower wishbone forward inner pivot (Dw Coil; same role as 2d).
9e.	Car Bottom Rear	m (Meters, X / Y / Z)	Lower wishbone rear inner pivot (Dw Coil; same role as 2e).
9f.	Tyre Bottom	m (Meters, X / Y / Z)	Lower wishbone outer ball joint (Dw Coil; same role as 2f).
9g.	Car Steer	m (Meters, X / Y / Z)	Steering tie-rod inner point for Dw Coil geometry.
9h.	Tyre Steer	m (Meters, X / Y / Z)	Steering tie-rod outer joint for Dw Coil geometry.
9i.	Car Coilover	m (Meters, X / Y / Z)	Coilover upper mount on chassis/ subframe; defines damper axis relative to DW kinematics.
9j.	Bottom Coilover	m (Meters, X / Y / Z)	Coilover lower mount on upright or lower arm; with 9i sets spring motion ratio and anti-dive line.

D. Example data

I. Chosen Cars for Example

- Volkswagen Golf GTI Mk1 (slug : ks_volkswagen_golf_gti_mk1) [2 suspensions]
- Honda S2000 AP1 (slug : ks_honda_s2000_ap1) [3 suspensions]
- Porsche 992 GT3 R Rennsport (slug : ks_porsche_992_gt3_r_rennsport) [2 suspensions]

II. Example

Volkswagen Golf GTI Mk1

1. Front Suspension (file : ks_volkswagen_golf_gti_mk1_front.suspension)

```

1. Basic Data
| 1a. Hub Mass : 50.00000

```

- 1b. Toe Out Linear : 0.00030
- 1c. Static Camber : -0.30000
- 1d. Rim Offset : 0.00000
- 2. D W Data : None
- 3. Strut
 - 3a. Car Strut : 0.13200, 0.38800, -0.01050
 - 3b. Tyre Strut : 0.05200, -0.05000, 0.01500
 - 3c. Car Bottom W B F : 0.41000, -0.05700, 0.32500
 - 3d. Car Bottom W B R : 0.39000, -0.05000, -0.04000
 - 3e. Tyre Bottom W B : 0.05200, -0.05000, 0.01500
 - 3f. Car Steer : 0.39000, -0.05250, 0.06000
 - 3g. Tyre Steer : 0.05200, -0.05000, 0.15000
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

2. Rear Suspension (file : ks_volkswagen_golf_gti_mk1_rear.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 45.00000
 - 1b. Toe Out Linear : 0.00025
 - 1c. Static Camber : -1.00000
 - 1d. Rim Offset : 0.00000
- 2. D W Data : None
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data
 - 7a. Car Hinge Int : 1.40000, 0.00000, 0.35000
 - 7b. Car Hinge Ext : 1.30000, 0.00000, 0.40000
 - 7c. Tyre Top : 0.09622, 0.49070, -0.01650
 - 7d. Tyre Bottom : 0.04602, 0.03294, 0.00870
 - 7e. Car Steer : 0.48002, -0.07906, -0.10540
 - 7f. Tyre Steer : 0.04602, -0.07506, -0.04540
 - 7g. Steer Link To Body : false
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

Honda S2000 AP1

1. Front Suspension (file : ks_honda_s2000_ap1_front.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 37.00000
 - 1b. Toe Out Linear : -0.00020
 - 1c. Static Camber : -0.40000
 - 1d. Rim Offset : 0.04500
- 2. D W Data
 - 2a. Car Top Front : 0.34203, 0.12311, 0.08000
 - 2b. Car Top Rear : 0.34302, 0.12451, -0.08000

- 2c. Tyre Top : 0.12479, 0.10951, -0.02210
- 2d. Car Bottom Front : 0.45520, -0.06960, -0.26000
- 2e. Car Bottom Rear : 0.42590, -0.08110, 0.03000
- 2f. Tyre Bottom : 0.09050, -0.13502, 0.01250
- 2g. Car Steer : 0.45520, -0.02000, 0.05890
- 2h. Tyre Steer : 0.09480, -0.06012, 0.11691
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

2. Front Drift Suspension (file : ks_honda_s2000_ap1_front_drift.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 37.00000
 - 1b. Toe Out Linear : -0.00020
 - 1c. Static Camber : -0.40000
 - 1d. Rim Offset : 0.04500
- 2. D W Data
 - 2a. Car Top Front : 0.34302, 0.08311, 0.08000
 - 2b. Car Top Rear : 0.34302, 0.08451, -0.08000
 - 2c. Tyre Top : 0.11479, 0.10951, -0.02210
 - 2d. Car Bottom Front : 0.45520, -0.06960, -0.26000
 - 2e. Car Bottom Rear : 0.42590, -0.08110, 0.03000
 - 2f. Tyre Bottom : 0.08050, -0.10502, 0.01250
 - 2g. Car Steer : 0.45520, -0.05000, 0.05890
 - 2h. Tyre Steer : 0.09480, -0.06012, 0.11691
- 3. Strut : None
- 4. Strut Ml : None
- 5. Axle : None
- 6. Multi Link Data : None
- 7. Trailing Arm Data : None
- 8. Multi Link New Data : None
- 9. Dw Coil Data : None

3. Rear Suspension (file : ks_honda_s2000_ap1_front_drift.suspension)

- 1. Basic Data
 - 1a. Hub Mass : 44.50000
 - 1b. Toe Out Linear : -0.00030
 - 1c. Static Camber : -1.60000
 - 1d. Rim Offset : 0.03700
- 2. D W Data
 - 2a. Car Top Front : 0.39022, 0.08701, 0.04129
 - 2b. Car Top Rear : 0.42022, 0.09469, -0.24111
 - 2c. Tyre Top : 0.11000, 0.11220, 0.00000
 - 2d. Car Bottom Front : 0.44100, -0.07010, 0.22113
 - 2e. Car Bottom Rear : 0.44042, -0.08989, -0.10935
 - 2f. Tyre Bottom : 0.09130, -0.10511, -0.01000
 - 2g. Car Steer : 0.38990, -0.04540, 0.16020
 - 2h. Tyre Steer : 0.09130, -0.05342, 0.14988
- 3. Strut : None

- | 4. Strut Ml : None
- | 5. Axle : None
- | 6. Multi Link Data : None
- | 7. Trailing Arm Data : None
- | 8. Multi Link New Data : None
- | 9. Dw Coil Data : None

Porsche 992 GT3 R Rennport

1. Front Suspension (file : ks_porsche_992_gt3_r_rennsport_front.suspension)

- | 1. Basic Data
 - | 1a. Hub Mass : 51.00000
 - | 1b. Toe Out Linear : 0.00044
 - | 1c. Static Camber : -5.70000
 - | 1d. Rim Offset : 0.00000
- | 2. D W Data : None
- | 3. Strut : None
- | 4. Strut Ml : None
- | 5. Axle : None
- | 6. Multi Link Data : None
- | 7. Trailing Arm Data : None
- | 8. Multi Link New Data : None
- | 9. Dw Coil Data
 - | 9a. Car Top Front : 0.36128, 0.06671, 0.10834
 - | 9b. Car Top Rear : 0.36128, 0.03131, -0.14089
 - | 9c. Tyre Top : 0.08582, 0.10517, -0.01868
 - | 9d. Car Bottom Front : 0.45261, -0.15879, 0.21226
 - | 9e. Car Bottom Rear : 0.45095, -0.14676, -0.08504
 - | 9f. Tyre Bottom : 0.03941, -0.11288, 0.01898
 - | 9g. Car Steer : 0.49910, -0.11521, 0.13421
 - | 9h. Tyre Steer : 0.03338, -0.06370, 0.13886
 - | 9i. Car Coilover : 0.40708, 0.21593, -0.04583
 - | 9j. Bottom Coilover : 0.10508, -0.11831, 0.00237

2. Rear Suspension (file : ks_porsche_992_gt3_r_rennsport_rear.suspension)

- | 1. Basic Data
 - | 1a. Hub Mass : 46.00000
 - | 1b. Toe Out Linear : -0.00110
 - | 1c. Static Camber : -4.40000
 - | 1d. Rim Offset : 0.00000
- | 2. D W Data : None
- | 3. Strut : None
- | 4. Strut Ml : None
- | 5. Axle : None
- | 6. Multi Link Data : None
- | 7. Trailing Arm Data : None
- | 8. Multi Link New Data
 - | 8a. Links 1
 - | 8a1. Name : None
 - | 8a2. Ball Car : 0.45733, -0.21779, -0.03274
 - | 8a3. Ball Tyre : 0.05940, -0.15917, -0.05147
 - | 8a4. Has Coilover Attached : true

- └ 8a5. Is Toe : false
- 8a. Links 2
 - └ 8a1. Name : None
 - └ 8a2. Ball Car : 0.33837, -0.13867, 0.26058
 - └ 8a3. Ball Tyre : 0.07196, -0.16195, 0.00295
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : false
- 8a. Links 3
 - └ 8a1. Name : None
 - └ 8a2. Ball Car : 0.29399, -0.00924, 0.18603
 - └ 8a3. Ball Tyre : 0.08416, 0.09013, 0.15594
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : false
- 8a. Links 4
 - └ 8a1. Name : None
 - └ 8a2. Ball Car : 0.31754, 0.04240, 0.18574
 - └ 8a3. Ball Tyre : 0.06361, 0.16582, -0.00746
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : false
- 8a. Links 5
 - └ 8a1. Name : steer
 - └ 8a2. Ball Car : 0.32123, -0.00156, -0.12150
 - └ 8a3. Ball Tyre : 0.05699, 0.10790, -0.12715
 - └ 8a4. Has Coilover Attached : false
 - └ 8a5. Is Toe : true
- 8b. Arms : None
- 8c. Car Coilover : 0.32123, -0.00156, -0.12150
- 8d. Bottom Coilover : 0.05699, 0.10790, -0.12715
- 9. Dw Coil Data : None

18. Turbo [.turbo]

A. Description

Per-charger boost behaviour: peak boost, spool lag up/down, RPM reference, RPM/throttle gamma shaping, wastegate ceiling, whether boost is garage-adjustable, and an index into the engine's turbo list.

Loaded from Car Engine Turbos To Load (and related controller / max-boost logic). Controllers and maps on the engine decide *when* boost is requested; this asset decides *how* the charger builds and caps pressure.

Official Description prose in the source dump is unfinished placeholder text. Content below follows the schema and examples.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Max boost / wastegate	<i>Max Boost, Watergate</i>	Engine Max Turbo Boost / controllers
Spool / decay lag	<i>Lag UP / Lag DN</i>	Throttle lag on <code>.CARENGINE</code>
RPM / throttle response shape	<i>Rpm Ref, gammas</i>	Engine turbo controllers LUTs
Garage-adjustable boost	<i>Is Adjustable</i>	Setup turbo boost level
Which turbo file loads	—	Engine <i>Turbos To Load</i> paths
Base torque curve	—	<code>.CARENGINE</code> power maps

One engine can load several `.TURBO` files (205 T16 pair, Supra twin / drift twins).

II. What you are really tuning

- Boost ceiling** — *Max Boost* is the pressure target/cap for this charger (205 soft stage 0.7, hard 1.2; Camaro 1.0; Supra drift secondary 2.0). Stock Supra examples sit at 0.0 here — boost may be driven entirely from engine controllers / maps with this file only supplying lag shape.
- Wastegate** — Watergate in the dump is **wastegate** (OCR). Often mirrors or sits just under max boost (205 0.7 / 1.2; Camaro 0.9 vs max 1.0; drift Supra 0.3 / 1.5 vs max 0.3 / 2.0). Caps how much pressure the gate will hold.
- Lag factors** — *Lag UP / Lag DN* near 1.0 (0.99x) = slow exponential spool/bleed typical of exhaust turbos. Camaro compressor uses **0.0 / 0.0** — effectively no turbo lag (supercharger-style instant response). Drift Supra stages tweak lag slightly (0.9975 / 0.988 vs stock 0.995 / 0.990).
- RPM and gamma shaping** — *Rpm Ref* anchors where the charger “comes alive” (205 2300 vs 4100 for the two stages; Camaro 6000; Supra stock 3500 / 4000). *Gamma Rpm / Gamma Gas* steepen or soften build vs RPM and throttle (205 both 2.5; Camaro gamma gas 0; drift hard stage gamma RPM 3.0).

5. **Adjustability and wiring** — *Is Adjustable* true (Supra drift) exposes boost to garage / cockpit control; false locks the stage. *CarData* path is optional linkage (None in samples). *Turbo no* indexes the unit in the engine's turbo list (0 in all dumps here — filename / load order still distinguishes twins).

III. Architecture

Flat schema (1–10):

- 1. *Max Boost*
- 2-3. *Lag UP / Lag DN*
- 4. *Rpm Ref*
- 5-6. *Gamma Rpm / Gamma Gas*
- 7. *Wategate* (wastegate)
- 8. *Is Adjustable*
- 9. *CarData* path
- 10. *Turbo no*

No nested controllers inside this asset — those live on `.CARENGINE`.

IV. How to read the examples

1 - PEUGEOT 205 T16 — TWO TURBO STAGES

Turbo 0: max 0.7, Rpm Ref 2300, wastegate 0.7. Turbo 1: max 1.2, Rpm Ref 4100, wastegate 1.2. Same lag (0.995 / 0.990) and gammas (2.5). Classic sequential / staged Group B style: low-RPM soft charger + high-RPM hard charger. Not adjustable.

2 - CHEVROLET CAMARO ZL1 — COMPRESSOR

Max 1.0, wastegate 0.9, **zero lag**, Gamma Gas 0, Rpm Ref 6000. Labelled compressor in the source list — behaviour matches a blower: pressure without exhaust spool delay. Engine example loads this via *Turbos To Load*.

3 - TOYOTA SUPRA MKIV — STOCK VS DRIFT WINS

Stock turbo 0/1: Max Boost and wastegate 0, lag present, soft gammas 0.5 — lag skeleton with boost authored elsewhere. Drift turbo 0/1: real boost 0.3 then 2.0, wastegates 0.3 / 1.5, *Is Adjustable* : *true*, stronger gammas on the big stage. Tuning-part drift package swaps the charger personality with the rest of the car.

V. Practical notes

- Source Description section is placeholder — ignore spring/alignment headings.
- Always read **engine** turbo controllers + *Max Turbo Boost* together with this file; zeros here do not always mean NA.
- *Wategate* = wastegate typo in the converted dump.
- Lag 0/0 is a feature (compressor), not a missing value — do not “fix” it to 0.99 unless you want turbo lag.

- Multi-turbo engines: match *Turbo no* / load order to the stage you intend; duplicate *Turbo no : 0* on every file in the dump means identity is path-based.
- Filename OCR (*t15* vs *t16*) appears in the Peugeot examples — trust the car slug, not the mistyped stem.

VI. Related assets

- **4. Car Engine** — *Turbos To Load*, turbo/wastegate controllers, max boost, BOV
- **5 / 6. Car Setup / Limits** — turbo boost level when *Is Adjustable*
- **8. Car Tuning Parts** — drift / upgrade packs that retarget .TURBO files

B. Schema

- 1. Max Boost : float
- 2. Lag U P : float
- 3. Lag D N : float
- 4. Rpm Ref : float
- 5. Gamma Rpm : float
- 6. Gamma Gas : float
- 7. Wastegate : float
- 8. Is Adjustable : boolean
- 9. CarData : string – path
- 10. Turbo no : integer

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Max Boost	bar (Bar gauge pressure)	Peak compressor boost relative to ambient (gauge). Typical street/ rally ~0.7–1.2 bar (205 T16), race/ drift up to ~2.0 bar; 0.0 disables forced-induction output on that stage. Aligns with Car Engine §18 Max Turbo Boost ceiling.
2.	Lag U P	Dimensionless lag coefficient (0 - 1)	Exponential spool-up filter toward target boost under rising demand (UP = up). Values near 1.0 (e.g. 0.995–0.9975) mean heavy turbo lag; 0.0 = instantaneous rise (Camaro ZL1 compressor / supercharger-style response).
3.	Lag D N	Dimensionless lag coefficient (0 - 1)	Exponential decay filter when boost demand falls (DN = down): throttle lift, gear change, or RPM drop. Slightly below Lag UP in examples (0.988–0.990) so boost bleeds off a bit faster than it builds; 0.0 with Lag UP = no lag (ZL1).

ID	Name	Unit of Measurement	Description
4.	Rpm Ref	RPM (Revolutions per minute)	Reference engine speed for the RPM-shaped boost curve (with Gamma Rpm). Below this region spool is weak; around/above it boost builds toward Max Boost. Examples: 2300/4100 (205 staged twin), 6000 (ZL1), 3200–4400 (Supra drift).
5.	Gamma Rpm	Dimensionless exponent (power-law shape)	Exponent shaping how boost scales with RPM relative to Rpm Ref. Higher gamma (>1, e.g. 2.5–3.0) = abrupt high-RPM spool; lower (0.5–1.0) = progressive/early response (Supra stock / ZL1).
6.	Gamma Gas	Dimensionless exponent (power-law shape)	Exponent shaping boost vs. throttle/gas pedal. Higher = needs more pedal to spool; 0.0 = boost independent of throttle position (ZL1 compressor model); ~0.5–2.5 typical turbo response.
7.	Wastegate	bar (Bar gauge pressure)	Wastegate setpoint. Caps regulated boost; often equals Max Boost (205 T16) or slightly below (ZL1 0.9 vs 1.0; Supra drift 1.5 vs 2.0 Max). Models mechanical/ electronic wastegate opening pressure.
8.	Is Adjustable	None (Boolean : True / False)	When true, boost level is exposed to setup / performance-mode adjustment (Turbo Boost Lv); false = fixed factory map (205 T16, ZL1, Supra stock). Supra drift turbos set true.
9.	CarData	None (File path)	Optional path linking this .turbo asset to a Car Data context; None in all documented examples (boost controlled via Car Engine Turbos To Load + controllers).
10.	Turbo no	Integer index (0-based)	Ordinal index of this turbocharger in a multi-turbo / sequential stack (Turbo 0 = primary/low-RPM, Turbo 1 = secondary). Examples list file Turbo 0/1 but store index 0 in-asset—verify against loaded stage order.

D. Example data

I. Chosen Cars for Example

- Peugeot 205 T16 (slug : ks_peugeot_205_t16) [2 turbos]

- Chevrolet Camaro ZL1 (slug : ks_chevrolet_camaro_zl1) [compressor]
- Toyota Supra MKIV (slug : ks_toyota_supra_mkiv) [2 turbos / 2 types]

II. Example

Peugeot 205 T16

1. Turbo 0 (file : ks_peugeot_205_t15.turbo)

- 1. Max Boost : 0.70000
- 2. Lag U P : 0.99500
- 3. Lag D N : 0.99000
- 4. Rpm Ref : 2300.00000
- 5. Gamma Rpm : 2.50000
- 6. Gamma Gas : 2.50000
- 7. Wastegate : 0.70000
- 8. Is Adjustable : false
- 9. CarData : None
- 10. Turbo no : 0

2. Turbo 1 (file : ks_peugeot_205_t15_turbo0.turbo)

- 1. Max Boost : 1.20000
- 2. Lag U P : 0.99500
- 3. Lag D N : 0.99000
- 4. Rpm Ref : 4100.00000
- 5. Gamma Rpm : 2.50000
- 6. Gamma Gas : 2.50000
- 7. Wastegate : 1.20000
- 8. Is Adjustable : false
- 9. CarData : None
- 10. Turbo no : 0

Chevrolet Camaro ZL1

- 1. Max Boost : 1.00000
- 2. Lag U P : 0.00000
- 3. Lag D N : 0.00000
- 4. Rpm Ref : 6000.00000
- 5. Gamma Rpm : 1.00000
- 6. Gamma Gas : 0.00000
- 7. Wastegate : 0.90000
- 8. Is Adjustable : false
- 9. CarData : None
- 10. Turbo no : 0

Toyota Supra MKIV

1. Turbo 0 (file : ks_toyota_supra_mkiv0.turbo)

- 1. Max Boost : 0.00000

- | 2. Lag U P : 0.99500
- | 3. Lag D N : 0.99000
- | 4. Rpm Ref : 3500.00000
- | 5. Gamma Rpm : 0.50000
- | 6. Gamma Gas : 0.50000
- | 7. Wastegate : 0.00000
- | 8. Is Adjustable : false
- | 9. CarData : None
- | 10. Turbo no : 0

2. Turbo 1 (file : ks_toyota_supra_mkiv1.turbo)

- | 1. Max Boost : 0.00000
- | 2. Lag U P : 0.99100
- | 3. Lag D N : 0.98800
- | 4. Rpm Ref : 4000.00000
- | 5. Gamma Rpm : 0.50000
- | 6. Gamma Gas : 0.50000
- | 7. Wastegate : 0.00000
- | 8. Is Adjustable : false
- | 9. CarData : None
- | 10. Turbo no : 0

3. Drift Turbo 0 (file : ks_toyota_supra_mkiv_drift0.turbo)

- | 1. Max Boost : 0.30000
- | 2. Lag U P : 0.99750
- | 3. Lag D N : 0.98800
- | 4. Rpm Ref : 3200.00000
- | 5. Gamma Rpm : 1.00000
- | 6. Gamma Gas : 1.00000
- | 7. Wastegate : 0.30000
- | 8. Is Adjustable : true
- | 9. CarData : None
- | 10. Turbo no : 0

4. Drift Turbo 1 (file : ks_toyota_supra_mkiv_drift1.turbo)

- | 1. Max Boost : 2.00000
- | 2. Lag U P : 0.99650
- | 3. Lag D N : 0.99700
- | 4. Rpm Ref : 4400.00000
- | 5. Gamma Rpm : 3.00000
- | 6. Gamma Gas : 1.00000
- | 7. Wastegate : 1.50000
- | 8. Is Adjustable : true
- | 9. CarData : None
- | 10. Turbo no : 0

19. Tyre [.tyre]

A. Description

Full tyre compound physics: carcass size and stiffness, brush/grip model, thermal mass and degradation (grain/blister), pressure and camber sensitivity, speed-dependent μ , rolling resistance, and groove factors.

Car Data only lists which `.TYRE` paths sit on front/rear. Setup picks pressure, camber, and compound index. This file is where grip, heat, and wear actually live — usually under `common_phsx\tyres\...` and shared across many cars.

Official Description prose in the source dump is unfinished placeholder text. Content below follows the schema and examples.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Size, rate, flex, inertia, tread	<i>Tyre Data</i>	—
Peak grip / slip shape / Mz / wear curve	<i>Model Data</i>	Wear <code>.CURVE</code>
Heat, grain, blister, $\mu(T)$ curve	<i>Thermal Data</i>	Thermal perf <code>.CURVE</code>
Ideal pressure / flex vs P	<i>Pressure Data + Pressure</i>	Setup cold pressures
Camber sensitivity	<i>Camber Data</i>	Setup camber
μ vs speed (lat/long)	<i>Speed Sensitivity</i>	—
Rolling resistance	<i>Rolling Resistance</i>	—
Wet / groove behaviour	<i>Groove Data</i>	—
Which compounds a car offers	—	<code>.CAR</code> Front/Rear Tyre Compounds
Compound index in garage	—	<code>.CARSETUP</code> Alignements

II. What you are really tuning

1. **Identity** — Name / Short Name for UI. Tyre Compound enum tags the family: Eco, Road, SuperCar, HyperCar, Slick_Medium, Wet, Racing_Vintage, F1_Soft/Medium/Hard/Wet/Intermediate, ...
2. **Carcass (Tyre Data)** — Width and radius (metres), vertical *Rate* / progressive rate, damping (+ *Damping Mode*: Simple / hystereticMaxwell / hystereticNando), angular inertia, rim radius, tread height and consumption, mass, lateral/longitudinal flex K/C, explosion / blanket temperatures, flat-spot K, contact patch camber/flex helpers.

Eco 165 mm / radius 0.25; Vintage 195 / 0.31; F1 wet 305 / 0.36. F1 wet tread height 3 mm vs 8 mm on Eco/Vintage. Vintage uses large negative flex K values in the dump — treat as authored offsets relative to the model's baseline, not “negative stiffness” in plain language.

3. **Grip model (Model Data)** — $Dy0$ / $Dx0$ lateral/longitudinal grip scales; load sensitivity exponents; $Fz0$ reference load; friction limit angle; flex gain; combined / slip factors; wear curve path; relaxation lengths; self-aligning moment (Mz) tweaks; brush exponent.

Eco $Dy0$ about 0.96; Vintage about 1.11; F1 wet about 1.47 (with $Dx0$ about 1.56). Peak μ rises hard from economy rubber to open-wheel wet.

4. **Thermal world** — Density, specific heat, surface/core mass ratio, cool factors (dry + rain), road conduction, conductivity, rolling heat factor, grain/blister gains and slip-angle thresholds, brake heat transfer into the tyre, thermal performance curve path, friction-limit angle vs temperature.

Eco cools faster dry (*Cool Factor* 1.4) than Vintage (0.7) or F1 wet (0.9). F1 wet pushes grain gains much higher and heat partition 1.4 — aggressive thermal personality. Blister gains are 0 in all three samples.

5. **Pressure and camber** — Ideal vs reference pressure (Eco 31 / 30; Vintage 31 / 28; F1 wet 26 / 17), pressure spring gain, curb puncture thresholds. Camber gain and vertical-K vs camber range shape how camber buys grip vs kills the patch.
6. **Speed, RR, grooves** — Separate lateral and longitudinal μ -speed / μ -temp blocks ($Mu0$ T , Ref $Speed$, sensitivities, $Tref$). Rolling resistance $Rr0/Rr1$ /slip/wear. *Groove Factor* about 0.49 on Eco/Vintage vs **68** on F1 wet — wet siping / drainage strength, not a typo scale shared with road tyres.
7. **Init Data** — Human tyre-size metadata: width, aspect ratio, diameter (inch), load index, pressure — Eco 165-60-12, Vintage 195-60-15, F1 wet 305-720-18 style sizing in the examples.

III. Architecture

1 - HEADER (SCHEMA 1-3)

Name, short name, compound enum.

2 - STRUCTURE AND MODEL (SCHEMA 4-5)

Tyre Data carcass; Model Data grip/wear/ Mz .

3 - ENVIRONMENT COUPLING (SCHEMA 6-8)

Thermal Data; Pressure Data; Camber Data.

4 - SPEED, RR, GROOVE, COLD FILL (SCHEMA 9-13)

Speed Sensitivity; Rolling Resistance; Groove Data; scalar *Pressure*; Init Data size card.

IV. How to read the examples

1 - ECO (165-60-12) — ECONOMY ROAD

Small, light carcass, $Dy0$ under 1.0, ideal pressure about 31 psi, mild groove factor, eco wear + $tcurve$ under *common_phsx\tyres\eco*. Soft contact flex vs race tyres. Baseline “efficient road” compound.

2 - RACING VINTAGE (195-60-15) — PERIOD RACE

Wider/taller, higher Dy0/Dx0 than Eco, cooler dry cool factor (0.7), much stronger grain gains, vintage wear/tcurve paths. Flex K authored negative in the dump. Classic historic race rubber personality on a road-ish size.

3 - F1 2025 WET (305-720-18) — OPEN-WHEEL WET

Wide, high inertia, short tread, very high Dy0/Dx0, low reference pressure (17) vs ideal 26, huge groove factor (68), elevated grain and heat partition, F1 wet curves. Built for standing water and high peak load — not a road compound with a wet label slapped on.

V. Practical notes

- Source Description section is placeholder — ignore spring/alignment headings.
- Almost all live tyres sit in *common_phsx*; editing one file hits every car that lists it.
- Setup pressure should track *Ideal Pressure / Pressure Reference*; running far off rewrites flex and heat via Pressure Data gains.
- Wear and thermal curves are mandatory personality — swapping only Dy0 without curves gives inconsistent fade.
- Negative flex K on vintage is an authoring convention in this dump; verify in tooling before “correcting” it to positive.
- Compound enum must match how Car Data / UI filter lists; wrong enum can hide a tyre from selectors.
- Brake heat couples through *Brake Transfer Factor* (small but nonzero in samples) — pad temp and tyre temp are not fully isolated.

VI. Related assets

- **3. Car Data** — Front/Rear Tyre Compounds path lists
- **5 / 6 / 7. Car Setup / Limits / Units** — pressures, camber, compound index, PSI labels
- **2. Brakes** — thermal neighbour via brake transfer
- **17. Suspension** — camber/toe kinematics feeding the contact patch
- **External .CURVE** — wear and thermal performance maps referenced here

B. Schema

- 1. Name : **string**
- 2. Short Name : **string**
- 3. Tyre Compound : **enum**
- 4. Tyre Data : **object**
 - 4a. Width : **float**
 - 4b. Radius : **float**
 - 4c. Rate : **float**
 - 4d. Progressive Rate : **float**
 - 4e. Damping : **float**
 - 4f. Angular Inertia : **float**
 - 4g. Rim Radius : **float**

- 4h. Radius Raise K : float
- 4i. Tread Height M M : float
- 4j. Tread Consumption K : float
- 4k. Mass : float
- 4l. Lateral Flex K : float
- 4m. Lateral Flex C : float
- 4n. Longitudinal Flex K : float
- 4o. Longitudinal Flex C : float
- 4p. Explosion Temperature : float
- 4q. Blanket Temperature : float
- 4r. Flat Spot K : float
- 4s. Normal To Flex Ratio : float
- 4t. Contact Camber : float
- 4u. Contact Flex : float
- 4v. Contact Vertical Flex : float
- 4w. Damping Mode : enum
- 4x. Maxwell Damping Peak Frequency : float
- 4y. Maxwell Stiffening Percent : float
- 4z. Damping Threshold Speed Ms : float
- 4aa. Speed Damping Factor : float
- 4ab. Deflection Damping Factor : float
- 5. Model Data : object
 - 5a. Dy0 : float
 - 5b. Dx0 : float
 - 5c. Ls Exp Y : float
 - 5d. Ls Exp X : float
 - 5e. Fz0 : float
 - 5f. Friction Limit Angle : float
 - 5g. Flex Gain : float
 - 5h. Cf Xmult : float
 - 5i. Brake D X Mod : float
 - 5j. Combined Factor : float
 - 5k. Grip Slip Factor : float
 - 5l. Wear Curve : string – path
 - 5m. Grain Factor : float
 - 5n. Contact Wear I M 0 : float
 - 5o. Relaxation Length Y : float
 - 5p. Relaxation Length X : float
 - 5q. Mz Tweak Mult : float
 - 5r. Mz Scale : float
 - 5s. Mz Trail Nd Slip Reduction : float
 - 5t. Mz Trail Remap : float
 - 5u. Brush Exponent : float
- 6. Thermal Data : object
 - 6a. Vertical Spring : float
 - 6b. Vertical Damp K : float
 - 6c. Wear Mult : float
 - 6d. Density : float
 - 6e. Specific Heat : float
 - 6f. Surface Core Mass Ratio : float
 - 6g. Cool Factor Rain : float
 - 6h. Cool Factor : float
 - 6i. Heat Partition Coeff : float
 - 6j. Road Conduction : float
 - 6k. Thermal Conductivity : float
 - 6l. Rolling Factor : float

- 6m. Grain Gain : float
- 6n. Grain Gamma : float
- 6o. Grain Slip Angle Gain : float
- 6p. Grain Slip Angle Gamma : float
- 6q. Grain Slip Angle Threshold : float
- 6r. Blister Gain : float
- 6s. Blister Gamma : float
- 6t. Practical Temp Source : float
- 6u. Brake Transfer Factor : float
- 6v. Thermal Performance Curve : string - path
- 6w. Contact I M 0 : float
- 6x. Tref Friction Limit Angle : float
- 6y. Friction Limit Angle T Sensitivity : float
- 7. Pressure Data : object
 - 7a. Pressure Flex Gain : float
 - 7b. Rolling Resistance Gain : float
 - 7c. Rolling Heat Gain : float
 - 7d. Gain D : float
 - 7e. Ideal Pressure : float
 - 7f. Pressure Reference : float
 - 7g. Pressure Spring Curve : string - path
 - 7h. Pressure Spring Gain : float
 - 7i. Curb Pressure Loss Must : float
 - 7j. Curb Pressure Loss Threshold : float
- 8. Camber Data : object
 - 8a. Camber Gain : float
 - 8b. Camber Vertical K Range Deg : float
 - 8c. Camber Vertical K Gain : float
 - 8d. Dcamber0 : float
 - 8e. Dcamber1 : float
- 9. Speed Sensitivity : object
 - 9a. Mu0 T : float
 - 9b. Ref Speed : float
 - 9c. Mu Speed Sensitivity : float
 - 9d. Mu T Sensitivity : float
 - 9e. Tref Mu T : float
 - 9f. Mu0 T X : float
 - 9g. Ref Speed X : float
 - 9h. Mu Speed Sensitivity X : float
 - 9i. Mu T Sensitivity X : float
 - 9j. Tref Mu T X : float
- 10. Rolling Resistance : object
 - 10a. Rr0 : float
 - 10b. Rr1 : float
 - 10c. Rr Slip : float
 - 10d. Rr Wear Mult : float
- 11. Groove Data : object
 - 11a. Groove Factor : float
 - 11b. Groove S A Factor : float
 - 11c. Groove S R Factor : float
- 12. Pressure : float
- 13. Init Data : object
 - 13a. Width : float
 - 13b. Aspect Ratio : float
 - 13c. Diameter : float
 - 13d. Load Index : float

L L 13e. Pressure : float

Enum - Car Tyre

3	Tyre Compound	Eco, Road, SuperCar, HyperCar, Slick_Medium, Wet, Racing_Vintage, F1_Soft, F1_Medium, F1_Hard, F1_Wet, F1_Intermediate
4w	Damping Mode	Simple, hystereticMaxwell, hystereticNando

C. Measurement Units & Descriptions

ID	Name	Unit of Measurement	Description
1.	Name	None (String)	Full display name of the compound (e.g. Eco (E), Racing Vintage (RV), Wet (W)).
2.	Short Name	None (String)	Abbreviated HUD / telemetry label (E, RV, W).
3.	Tyre Compound	None (Enum : Eco / Road / SuperCar / HyperCar / Slick_Medium / Wet / Racing_Vintage / F1_Soft / F1_Medium / F1_Hard / F1_Wet / F1_Intermediate)	Compound class selecting grip, thermal, and wear baselines (Eco street, Racing_Vintage, F1_Wet, etc.).
4.	Tyre Data	None (Object)	Structural carcass block: geometry, vertical spring/damper, flex, temperatures, and contact-patch parameters.
4a.	Width	m (Meters)	Contact-section width in SI (0.165 Eco = 165 mm, 0.305 F1 wet = 305 mm). Distinct from Init Data width in mm.
4b.	Radius	m (Meters)	Unloaded overall tyre radius (Eco 0.251 m, Vintage 0.308 m, F1 wet 0.360 m). Sets rolling circumference and gearing.
4c.	Rate	N/m (Newtons per meter)	Vertical carcass spring rate (~263–307 kN/m in examples). Primary tyre contribution to wheel rate and ride frequency.
4d.	Progressive Rate	N/m ² or N/m ³ (Progressive stiffening)	Additional rate that rises with deflection (progressive carcass). 0.0 in examples = linear spring.
4e.	Damping	N·s/m (Newton-seconds per meter)	Vertical viscous damping of the tyre spring (Eco ~549, F1 wet ~846). Controls carcass oscillation and curb impact.

ID	Name	Unit of Measurement	Description
4f.	Angular Inertia	kg·m ² (Kilogram square meters)	Rotational inertia of tyre + rim about spin axis (Eco 0.91, Vintage 2.27, F1 wet 4.67). Affects acceleration, braking, and ABS feel.
4g.	Rim Radius	m (Meters)	Wheel rim outer radius (12" → 0.1524 m, 15" → 0.1905 m, 18" → 0.2286 m). Used with tyre radius for sidewall height.
4h.	Radius Raise K	m/N or dimensionless gain	Radius growth coefficient under load/centrifugal effects (0.001 in examples); slight diameter increase with speed/load.
4i.	Tread Height M M	mm (Millimeters)	Usable tread depth (Eco/Vintage 8 mm, F1 wet 3 mm). Depletes via wear; wet grooves need remaining depth for water evacuation.
4j.	Tread Consumption K	Dimensionless wear rate	Scalar converting slip/energy work into tread-height loss (6.0 across examples).
4k.	Mass	kg (Kilograms)	Tyre (and often rim share) mass for unsprung dynamics (6 kg in examples — simplified vs. real F1 ~9–10 kg).
4l.	Lateral Flex K	N/m (Newtons per meter)	Lateral carcass stiffness (Eco 28 kN/m, F1 wet 30 kN/m). Negative values (Vintage –10 kN/m) can encode special flex models; governs sidewall lean under Fy.
4m.	Lateral Flex C	Dimensionless damping ratio or N·s/m	Lateral flex damping (typically ~2.0). Limits oscillation of the contact patch relative to the rim.
4n.	Longitudinal Flex K	N/m (Newtons per meter)	Fore-aft carcass stiffness under Fx (Eco 18 kN/m, F1 wet 20 kN/m). Affects traction/braking compliance feel.
4o.	Longitudinal Flex C	Dimensionless damping ratio or N·s/m	Longitudinal flex damping (~2.0–2.5).
4p.	Explosion Temperature	°C (Degrees Celsius)	Catastrophic overheat threshold (450 °C in examples) above which the tyre is considered failed/ exploded.
4q.	Blanket Temperature	°C (Degrees Celsius)	Tyre-warmer / pit-blanket target start temperature (40 °C in examples).
4r.	Flat Spot K	Dimensionless gain	Sensitivity of flat-spot damage under locked-wheel scrub (0.1 in examples).

ID	Name	Unit of Measurement	Description
4s.	Normal To Flex Ratio	Dimensionless ratio	Coupling of vertical load into carcass flex (0.0 = no extra coupling in examples).
4t.	Contact Camber	deg (Degrees) or model coefficient	Contact-patch camber sensitivity / effective patch camber shaping (Eco ~2.8, F1 wet ~11).
4u.	Contact Flex	m (Meters) or dimensionless	Lateral contact-patch flex magnitude under load (Eco 0.023, F1 wet 0.345).
4v.	Contact Vertical Flex	Dimensionless or mm-scale	Vertical contact-patch compliance shaping (Eco ~3.6, F1 wet ~16.6).
4w.	Damping Mode	None (Enum : Simple / hystereticMaxwell / hystereticNando)	Carcass damping model: Simple viscous, Maxwell hysteretic, or Nando hysteretic (examples use hystereticNando).
4x.	Maxwell Damping Peak Frequency	Hz (Hertz)	Peak frequency for Maxwell hysteretic damper (0 when unused / Nando mode).
4y.	Maxwell Stiffening Percent	% (Percent)	Extra dynamic stiffening in Maxwell mode (0 when unused).
4z.	Damping Threshold Speed Ms	m/s (Meters per second)	Vertical deflection-speed threshold where speed-dependent damping engages (1.0 m/s in examples).
4aa.	Speed Damping Factor	Dimensionless gain	Multiplier on damping above the speed threshold (0.5 in examples).
4ab.	Deflection Damping Factor	Dimensionless gain	Multiplier scaling damping with deflection amplitude (3.0 in examples).
5.	Model Data	None (Object)	Brush / friction model: peak μ , load sensitivity, slip limits, wear, relaxation lengths, and self-aligning torque (Mz).
5a.	Dy0	Dimensionless friction coefficient (μ_y)	Peak lateral friction coefficient at reference load Fz0 (Eco 0.96, Vintage 1.11, F1 wet 1.47).
5b.	Dx0	Dimensionless friction coefficient (μ_x)	Peak longitudinal friction coefficient at Fz0 (Eco 1.00, F1 wet 1.56). Braking/traction grip baseline.
5c.	Ls Exp Y	Dimensionless exponent	Lateral load-sensitivity exponent: how μ_y falls as Fz rises above Fz0 (~0.82–0.87).
5d.	Ls Exp X	Dimensionless exponent	Longitudinal load-sensitivity exponent for μ_x (~0.83–1.07).
5e.	Fz0	N (Newtons)	Reference vertical load for Dy0/ Dx0 (~3840–4230 N $\approx$ 390–430 kgf per tyre).

ID	Name	Unit of Measurement	Description
5f.	Friction Limit Angle	deg (Degrees)	Slip angle (or slip equivalent) at which peak friction is reached (~7.8–8.6°). Beyond this, grip falls toward sliding.
5g.	Flex Gain	Dimensionless gain	Scales carcass flex contribution into the friction model (0.0207 in examples).
5h.	Cf Xmult	Dimensionless multiplier	Multiplier on longitudinal cornering/stiffness factor Cf_x (1.8–2.0).
5i.	Brake D X Mod	Dimensionless gain	Modifier reducing Dx under heavy braking (Eco 0.05, F1 wet 0.00). Models brake-lock μ drop.
5j.	Combined Factor	Dimensionless coefficient	Combined-slip ellipse shaping (Pacejka-like): how Fx and Fy share friction budget (1.8–2.0).
5k.	Grip Slip Factor	Dimensionless coefficient	Post-peak sliding grip retention vs. peak (0.76–0.82). Higher = more forgiving slide.
5l.	Wear Curve	None (.curve file path)	LUT of wear state → grip/thermal modifiers (wear_eco.curve, wear_f12025_wet.curve, ...).
5m.	Grain Factor	Dimensionless gain	Base graining susceptibility (0.001). Raised in thermal grain gains for soft compounds.
5n.	Contact Wear I M O	Dimensionless I/M/O weights	Inner / middle / outer contact-wear distribution weight (4.0 in examples).
5o.	Relaxation Length Y	m (Meters)	Lateral force lag distance (~0.10 m). Shorter = snappier turn-in; longer = delayed Fy build-up.
5p.	Relaxation Length X	m (Meters)	Longitudinal force relaxation length (~0.10 m). Affects ABS and throttle traction transient.
5q.	Mz Tweak Mult	Dimensionless multiplier	Global scale on self-aligning torque Mz feel (0.6).
5r.	Mz Scale	Dimensionless multiplier	Base Mz magnitude scale (0.12). Steering weight / trail feedback.
5s.	Mz Trail Nd Slip Reduction	Dimensionless coefficient	How pneumatic trail (Mz) shrinks with normalized slip (0.5).
5t.	Mz Trail Remap	Dimensionless coefficient	Remapping curve intensity for trail vs. slip (0.2).
5u.	Brush Exponent	Dimensionless exponent	Brush-model stress distribution exponent (3.0). Shapes the force build-up before peak slip.

ID	Name	Unit of Measurement	Description
6.	Thermal Data	None (Object)	Heat capacity, cooling, conduction, grain/blister, brake heat soak, and temperature→grip curve.
6a.	Vertical Spring	N/m or thermal model scale	Thermal-node coupling spring (40 in examples) between surface/core temperature layers.
6b.	Vertical Damp K	Thermal damping coefficient	Damping of surface↔core temperature exchange (2.0).
6c.	Wear Mult	Dimensionless multiplier	Temperature-linked wear rate multiplier (0.2).
6d.	Density	kg/m ³ (Kilograms per cubic meter)	Rubber density (1100 kg/m ³ — typical elastomer). Used with volume for thermal mass.
6e.	Specific Heat	J/(kg·K) (Joules per kilogram-kelvin)	Specific heat capacity of the tread rubber (1600 J/(kg·K) in examples).
6f.	Surface Core Mass Ratio	Dimensionless ratio (0 -1)	Fraction of thermal mass in the surface skin vs. core (0.15). Low = surface heats/cools fast.
6g.	Cool Factor Rain	Dimensionless gain	Extra convective cooling in wet/ rain conditions (3.0).
6h.	Cool Factor	Dimensionless gain	Base air/road cooling rate (Eco 1.4, Vintage 0.7, F1 wet 0.9).
6i.	Heat Partition Coeff	Dimensionless coefficient	Split of generated heat into surface vs. bulk (0.5–1.4).
6j.	Road Conduction	W/(m ² ·K) or model units	Conductive heat transfer to asphalt (1000 in examples).
6k.	Thermal Conductivity	W/(m·K) (Watts per meter-kelvin)	Rubber conductivity (Eco 0.20, Vintage 0.28, F1 wet 0.35 W/(m·K)).
6l.	Rolling Factor	Dimensionless gain	Heat generation from rolling hysteresis (Eco 0.6, F1 wet 1.6).
6m.	Grain Gain	Dimensionless gain	Rate of surface graining under thermal/slip stress (Eco 0.017, F1 wet 0.47).
6n.	Grain Gamma	Dimensionless exponent	Non-linear shaping of grain growth vs. drivers (matches Grain Gain in examples).
6o.	Grain Slip Angle Gain	Dimensionless gain	Extra grain rate from slip angle (F1 wet 0.565 vs Eco 0.165).
6p.	Grain Slip Angle Gamma	Dimensionless exponent	Exponent on slip-angle grain contribution.
6q.	Grain Slip Angle Threshold	deg (Degrees)	Slip angle above which grain acceleration engages (~10.5–11.3°).

ID	Name	Unit of Measurement	Description
6r.	Blister Gain	Dimensionless gain	Blister formation rate (0 in these compounds; used on soft slicks).
6s.	Blister Gamma	Dimensionless exponent	Non-linear blister growth shaping (0 when blister unused).
6t.	Practical Temp Source	Dimensionless blend (0 - 1)	Blend between surface and core temperature for “practical” grip temp (0.8).
6u.	Brake Transfer Factor	Dimensionless gain	Fraction of brake rotor heat conducted into the tyre (0.0007).
6v.	Thermal Performance Curve	None (.curve file path)	LUT temperature (°C) → friction efficiency (tcurve_eco.curve, tcurve_f12025_wet.curve).
6w.	Contact I M O	Dimensionless I/M/O weights	Inner/middle/outer contact thermal weighting (1.6).
6x.	Tref Friction Limit Angle	°C (Degrees Celsius)	Reference temperature for friction-limit-angle thermal correction (75 °C).
6y.	Friction Limit Angle T Sensitivity	deg/°C or gain	How Friction Limit Angle shifts with temperature (0.06).
7.	Pressure Data	None (Object)	Inflation effects on flex, rolling resistance, heat, spring rate, and curb puncture loss.
7a.	Pressure Flex Gain	Dimensionless gain	How carcass flex changes with pressure offset from reference (0.25).
7b.	Rolling Resistance Gain	Dimensionless gain	Pressure→rolling-resistance coupling (1.2). Under-inflation raises RR.
7c.	Rolling Heat Gain	Dimensionless gain	Extra heat from pressure-related hysteresis (0.1).
7d.	Gain D	Dimensionless gain	Pressure influence on longitudinal grip/damping term (~0.0025–0.0026).
7e.	Ideal Pressure	psi (Pounds per square inch)	Target hot pressure for peak performance (Eco/Vintage 31 psi, F1 wet 26 psi).
7f.	Pressure Reference	psi (Pounds per square inch)	Cold/reference pressure for spring and flex tables (Eco 30, Vintage 28, F1 wet 17 psi).
7g.	Pressure Spring Curve	None (.curve file path)	Optional LUT of pressure → vertical spring rate; None when using Pressure Spring Gain only.
7h.	Pressure Spring Gain	N/m per psi (or N/m)	Vertical rate contribution from inflation (~9.9–11.1 kN/m scale in examples).

ID	Name	Unit of Measurement	Description
7i.	Curb Pressure Loss Must	psi or loss amount	Pressure drop applied when curb impact exceeds threshold (1.0).
7j.	Curb Pressure Loss Threshold	m/s or impact units	Impact severity before curb puncture loss (Eco/Vintage 10, F1 wet 30 — tougher wet carcass).
8.	Camber Data	None (Object)	Camber sensitivity of grip and vertical stiffness.
8a.	Camber Gain	Dimensionless gain	Grip change per degree of camber (0.2).
8b.	Camber Vertical K Range Deg	deg (Degrees)	Camber window over which vertical rate gain applies ($\pm 5^\circ$).
8c.	Camber Vertical K Gain	Dimensionless gain	Vertical stiffness change with camber (-0.1).
8d.	Dcamber0	Dimensionless coefficient	Camber-thrust / μ offset coefficient at small camber (0.6).
8e.	Dcamber1	Dimensionless coefficient	Higher-order camber sensitivity (-11). Shapes camber thrust curve.
9.	Speed Sensitivity	None (Object)	Speed- and temperature-dependent friction for lateral (T) and longitudinal (X) axes.
9a.	Mu0 T	Dimensionless friction coefficient (μ)	Base lateral μ at reference speed/temp (Eco 0.78, F1 wet 0.85).
9b.	Ref Speed	m/s (Meters per second)	Reference slip/ground speed for lateral μ speed law (4.5 m/s).
9c.	Mu Speed Sensitivity	Dimensionless gain	Lateral μ change with speed relative to Ref Speed (Eco 0.78, F1 wet 1.2).
9d.	Mu T Sensitivity	1/°C or gain	Lateral μ temperature sensitivity (0.025–0.035).
9e.	Tref Mu T	°C (Degrees Celsius)	Reference temperature for lateral μ thermal correction (70–80 °C).
9f.	Mu0 T X	Dimensionless friction coefficient (μ_x)	Base longitudinal μ at reference conditions (Eco 0.70, F1 wet 0.68).
9g.	Ref Speed X	m/s (Meters per second)	Reference speed for longitudinal μ law (2.0–3.5 m/s).
9h.	Mu Speed Sensitivity	Dimensionless gain	Longitudinal μ speed sensitivity (Eco 0.62, F1 wet 1.8 — wet more speed-critical).
9i.	Mu T Sensitivity X	1/°C or gain	Longitudinal μ temperature sensitivity (0.035–0.095).
9j.	Tref Mu T X	°C (Degrees Celsius)	Reference temperature for longitudinal μ thermal correction (70–80 °C).

ID	Name	Unit of Measurement	Description
10.	Rolling Resistance	None (Object)	Steady and slip-dependent rolling drag opposing wheel rotation.
10a.	Rr0	N (Newtons)	Constant rolling-resistance force (~17.8–20.1 N). Dominant at low speed.
10b.	Rr1	N·s/m or N per (m/s)	Speed-proportional RR term (0.002). Small linear rise with velocity.
10c.	Rr Slip	N or model units	Extra RR under slip/scrub (646). Raises drag when sliding.
10d.	Rr Wear Mult	Dimensionless multiplier	How wear state scales rolling resistance (0.14).
11.	Groove Data	None (Object)	Tread groove / void effects on wet grip and aquaplaning (critical for F1_Wet).
11a.	Groove Factor	Dimensionless or groove index	Overall groove/void intensity (street Eco 0.49 vs F1 wet 68 — deep rain channels).
11b.	Groove S A Factor	Dimensionless coefficient	Groove effect on slip-angle grip (0.45). SA = slip angle.
11c.	Groove S R Factor	Dimensionless coefficient	Groove effect on slip-ratio / longitudinal grip (0.45). SR = slip ratio.
12.	Pressure	psi (Pounds per square inch)	Default cold inflation for this compound asset (Eco 30, Vintage 28, F1 wet 17 psi). Overridden by setup pressures.
13.	Init Data	None (Object)	Human-readable tyre size designation (width–aspect–rim) and load/pressure ratings.
13a.	Width	mm (Millimeters)	Nominal section width (165 / 195 / 305). Matches commercial size marking.
13d.	Aspect Ratio	% (Percent of width)	Sidewall height as % of width (60 street, 43 F1 wet ≈ low-profile race).
13c.	Diameter	in (Inches)	Rim diameter (12 / 15 / 18). Must match Rim Radius = Diameter × 0.0254 / 2.
13d.	Load Index	None (Integer load index)	ISO load index (Eco 99, Vintage 88) — max load capacity class.
13e.	Pressure	psi (Pounds per square inch)	Rated / nominal inflation for the size designation (mirrors field 12 / Pressure Reference).

D. Example data

I. Chosen Tyres for Example

- Eco (slug : eco_165_60_12)
- Vintage (slug : vintage_195_60_15)
- F1 2025 Wet (slug : f12025_wet_305_720_18)

II. Example

Eco | Size : 165 - 60 - 12

- 1. Name : Eco (E)
- 2. Short Name : E
- 3. Tyre Compound : Eco
- 4. Tyre Data
 - 4a. Width : 0.16500
 - 4b. Radius : 0.25140
 - 4c. Rate : 306900.00000
 - 4d. Progressive Rate : 0.00000
 - 4e. Damping : 549.08398
 - 4f. Angular Inertia : 0.91034
 - 4g. Rim Radius : 0.15240
 - 4h. Radius Raise K : 0.00100
 - 4i. Tread Height M M : 8.00000
 - 4j. Tread Consumption K : 6.00000
 - 4k. Mass : 6.00000
 - 4l. Lateral Flex K : 28000.00000
 - 4m. Lateral Flex C : 2.00000
 - 4n. Longitudinal Flex K : 18000.00000
 - 4o. Longitudinal Flex C : 2.50000
 - 4p. Explosion Temperature : 450.00000
 - 4q. Blanket Temperature : 40.00000
 - 4r. Flat Spot K : 0.10000
 - 4s. Normal To Flex Ratio : 0.00000
 - 4t. Contact Camber : 2.79367
 - 4u. Contact Flex : 0.02293
 - 4v. Contact Vertical Flex : 3.58828
 - 4w. Damping Mode : hystereticNando
 - 4x. Maxwell Damping Peak Frequency : 0.00000
 - 4y. Maxwell Stiffening Percent : 0.00000
 - 4z. Damping Threshold Speed Ms : 1.00000
 - 4aa. Speed Damping Factor : 0.50000
 - 4ab. Deflection Damping Factor : 3.00000
- 5. Model Data
 - 5a. Dy0 : 0.96160
 - 5b. Dx0 : 1.00483
 - 5c. Ls Exp Y : 0.82088
 - 5d. Ls Exp X : 1.07088
 - 5e. Fz0 : 4170.00000
 - 5f. Friction Limit Angle : 8.56818
 - 5g. Flex Gain : 0.02070
 - 5h. Cf Xmult : 2.00000

- 5i. Brake D X Mod : 0.05000
- 5j. Combined Factor : 1.90000
- 5k. Grip Slip Factor : 0.78000
- 5l. Wear Curve : content\cars\common_phsx\tyres\eco\wear_eco.curve
- 5m. Grain Factor : 0.00100
- 5n. Contact Wear I M 0 : 4.00000
- 5o. Relaxation Length Y : 0.10051
- 5p. Relaxation Length X : 0.10051
- 5q. Mz Tweak Mult : 0.60000
- 5r. Mz Scale : 0.12000
- 5s. Mz Trail Nd Slip Reduction : 0.50000
- 5t. Mz Trail Remap : 0.20000
- 5u. Brush Exponent : 3.00000

6. Thermal Data

- 6a. Vertical Spring : 40.00000
- 6b. Vertical Damp K : 2.00000
- 6c. Wear Mult : 0.20000
- 6d. Density : 1100.00000
- 6e. Specific Heat : 1600.00000
- 6f. Surface Core Mass Ratio : 0.15000
- 6g. Cool Factor Rain : 3.00000
- 6h. Cool Factor : 1.40000
- 6i. Heat Partition Coeff : 0.50000
- 6j. Road Conduction : 1000.00000
- 6k. Thermal Conductivity : 0.20000
- 6l. Rolling Factor : 0.60000
- 6m. Grain Gain : 0.01700
- 6n. Grain Gamma : 0.01700
- 6o. Grain Slip Angle Gain : 0.16500
- 6p. Grain Slip Angle Gamma : 0.24500
- 6q. Grain Slip Angle Threshold : 11.26818
- 6r. Blister Gain : 0.00000
- 6s. Blister Gamma : 0.00000
- 6t. Practical Temp Source : 0.80000
- 6u. Brake Transfer Factor : 0.00070
- 6v. Thermal Performance Curve :

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- 6w. Contact I M 0 : 1.60000
- 6x. Tref Friction Limit Angle : 75.00000
- 6y. Friction Limit Angle T Sensitivity : 0.06000

7. Pressure Data

- 7a. Pressure Flex Gain : 0.25000
- 7b. Rolling Resistance Gain : 1.20000
- 7c. Rolling Heat Gain : 0.10000
- 7d. Gain D : 0.00260
- 7e. Ideal Pressure : 31.00000
- 7f. Pressure Reference : 30.00000
- 7g. Pressure Spring Curve : None
- 7h. Pressure Spring Gain : 10890.00000
- 7i. Curb Pressure Loss Must : 1.00000
- 7j. Curb Pressure Loss Threshold : 10.00000

8. Camber Data

- 8a. Camber Gain : 0.20000
- 8b. Camber Vertical K Range Deg : 5.00000
- 8c. Camber Vertical K Gain : -0.10000
- 8d. Dcamber0 : 0.60000

- L 8e. Dcamber1 : -11.00000
- 9. Speed Sensitivity
 - 9a. Mu0 T : 0.78000
 - 9b. Ref Speed : 4.50000
 - 9c. Mu Speed Sensitivity : 0.78000
 - 9d. Mu T Sensitivity : 0.02500
 - 9e. Tref Mu T : 70.00000
 - 9f. Mu0 T X : 0.70000
 - 9g. Ref Speed X : 3.50000
 - 9h. Mu Speed Sensitivity X : 0.62000
 - 9i. Mu T Sensitivity X : 0.09500
 - 9j. Tref Mu T X : 70.00000
- 10. Rolling Resistance
 - 10a. Rr0 : 17.79780
 - 10b. Rr1 : 0.00200
 - 10c. Rr Slip : 646.00000
 - 10d. Rr Wear Mult : 0.14000
- 11. Groove Data
 - 11a. Groove Factor : 0.49000
 - 11b. Groove S A Factor : 0.45000
 - 11c. Groove S R Factor : 0.45000
- 12. Pressure : 30.00000
- 13. Init Data
 - 13a. Width : 165.00000
 - 13b. Aspect Ratio : 60.00000
 - 13c. Diameter : 12.00000
 - 13d. Load Index : 99.00000
 - 13e. Pressure : 30.00000

Vintage | Size : 195 - 60 - 15

- 1. Name : Racing Vintage (RV)
- 2. Short Name : RV
- 3. Tyre Compound : Racing_Vintage
- 4. Tyre Data
 - 4a. Width : 0.19500
 - 4b. Radius : 0.30750
 - 4c. Rate : 272800.00000
 - 4d. Progressive Rate : 0.00000
 - 4e. Damping : 488.07468
 - 4f. Angular Inertia : 2.26550
 - 4g. Rim Radius : 0.19050
 - 4h. Radius Raise K : 0.00100
 - 4i. Tread Height M M : 8.00000
 - 4j. Tread Consumption K : 6.00000
 - 4k. Mass : 6.00000
 - 4l. Lateral Flex K : -10000.00000
 - 4m. Lateral Flex C : 2.00000
 - 4n. Longitudinal Flex K : -1800.00000
 - 4o. Longitudinal Flex C : 2.00000
 - 4p. Explosion Temperature : 450.00000
 - 4q. Blanket Temperature : 40.00000
 - 4r. Flat Spot K : 0.10000

- 4s. Normal To Flex Ratio : 0.00000
- 4t. Contact Camber : 9.33809
- 4u. Contact Flex : 0.16605
- 4v. Contact Vertical Flex : 14.45922
- 4w. Damping Mode : hystereticNando
- 4x. Maxwell Damping Peak Frequency : 0.00000
- 4y. Maxwell Stiffening Percent : 0.00000
- 4z. Damping Threshold Speed Ms : 1.00000
- 4aa. Speed Damping Factor : 0.50000
- 4ab. Deflection Damping Factor : 3.00000

- 5. Model Data

- 5a. Dy0 : 1.10844
- 5b. Dx0 : 1.14557
- 5c. Ls Exp Y : 0.85581
- 5d. Ls Exp X : 0.88381
- 5e. Fz0 : 3840.00000
- 5f. Friction Limit Angle : 8.30538
- 5g. Flex Gain : 0.02070
- 5h. Cf Xmult : 2.00000
- 5i. Brake D X Mod : 0.05000
- 5j. Combined Factor : 1.80000
- 5k. Grip Slip Factor : 0.76000
- 5l. Wear Curve :

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- 5m. Grain Factor : 0.00100
- 5n. Contact Wear I M 0 : 4.00000
- 5o. Relaxation Length Y : 0.10051
- 5p. Relaxation Length X : 0.10051
- 5q. Mz Tweak Mult : 0.60000
- 5r. Mz Scale : 0.12000
- 5s. Mz Trail Nd Slip Reduction : 0.50000
- 5t. Mz Trail Remap : 0.20000
- 5u. Brush Exponent : 3.00000

- 6. Thermal Data

- 6a. Vertical Spring : 40.00000
- 6b. Vertical Damp K : 2.00000
- 6c. Wear Mult : 0.20000
- 6d. Density : 1100.00000
- 6e. Specific Heat : 1600.00000
- 6f. Surface Core Mass Ratio : 0.15000
- 6g. Cool Factor Rain : 3.00000
- 6h. Cool Factor : 0.70000
- 6i. Heat Partition Coeff : 0.70000
- 6j. Road Conduction : 1000.00000
- 6k. Thermal Conductivity : 0.28000
- 6l. Rolling Factor : 1.00000
- 6m. Grain Gain : 0.27000
- 6n. Grain Gamma : 0.27000
- 6o. Grain Slip Angle Gain : 0.26500
- 6p. Grain Slip Angle Gamma : 0.34500
- 6q. Grain Slip Angle Threshold : 11.00538
- 6r. Blister Gain : 0.00000
- 6s. Blister Gamma : 0.00000
- 6t. Practical Temp Source : 0.80000
- 6u. Brake Transfer Factor : 0.00070

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| | 6v. Thermal Performance Curve :
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| | 6w. Contact I M 0 : 1.60000
| | 6x. Tref Friction Limit Angle : 75.00000
| | 6y. Friction Limit Angle T Sensitivity : 0.06000
- 7. Pressure Data
| | 7a. Pressure Flex Gain : 0.25000
| | 7b. Rolling Resistance Gain : 1.20000
| | 7c. Rolling Heat Gain : 0.10000
| | 7d. Gain D : 0.00250
| | 7e. Ideal Pressure : 31.00000
| | 7f. Pressure Reference : 28.00000
| | 7g. Pressure Spring Curve : None
| | 7h. Pressure Spring Gain : 9860.00000
| | 7i. Curb Pressure Loss Must : 1.00000
| | 7j. Curb Pressure Loss Threshold : 10.00000
- 8. Camber Data
| | 8a. Camber Gain : 0.20000
| | 8b. Camber Vertical K Range Deg : 5.00000
| | 8c. Camber Vertical K Gain : -0.10000
| | 8d. Dcamber0 : 0.60000
| | 8e. Dcamber1 : -11.00000
- 9. Speed Sensitivity
| | 9a. Mu0 T : 0.78000
| | 9b. Ref Speed : 4.50000
| | 9c. Mu Speed Sensitivity : 0.78000
| | 9d. Mu T Sensitivity : 0.02500
| | 9e. Tref Mu T : 70.00000
| | 9f. Mu0 T X : 0.80000
| | 9g. Ref Speed X : 2.00000
| | 9h. Mu Speed Sensitivity X : 0.82000
| | 9i. Mu T Sensitivity X : 0.08500
| | 9j. Tref Mu T X : 70.00000
- 10. Rolling Resistance
| | 10a. Rr0 : 18.36000
| | 10b. Rr1 : 0.00200
| | 10c. Rr Slip : 646.00000
| | 10d. Rr Wear Mult : 0.14000
- 11. Groove Data
| | 11a. Groove Factor : 0.49000
| | 11b. Groove S A Factor : 0.45000
| | 11c. Groove S R Factor : 0.45000
- 12. Pressure : 28.00000
- 13. Init Data
| | 13a. Width : 195.00000
| | 13b. Aspect Ratio : 60.00000
| | 13c. Diameter : 15.00000
| | 13d. Load Index : 88.00000
| | 13e. Pressure : 28.00000

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F1 2025 [Wet] | Size : 305 - 720 - 18

- 1. Name : Wet (W)
- 2. Short Name : W

- 3. Tyre Compound : F1_Wet
- 4. Tyre Data
 - 4a. Width : 0.30500
 - 4b. Radius : 0.35975
 - 4c. Rate : 262600.00000
 - 4d. Progressive Rate : 0.00000
 - 4e. Damping : 845.68597
 - 4f. Angular Inertia : 4.66531
 - 4g. Rim Radius : 0.22860
 - 4h. Radius Raise K : 0.00100
 - 4i. Tread Height M M : 3.00000
 - 4j. Tread Consumption K : 6.00000
 - 4k. Mass : 6.00000
 - 4l. Lateral Flex K : 30000.00000
 - 4m. Lateral Flex C : 2.00000
 - 4n. Longitudinal Flex K : 20000.00000
 - 4o. Longitudinal Flex C : 2.50000
 - 4p. Explosion Temperature : 450.00000
 - 4q. Blanket Temperature : 40.00000
 - 4r. Flat Spot K : 0.10000
 - 4s. Normal To Flex Ratio : 0.00000
 - 4t. Contact Camber : 10.98116
 - 4u. Contact Flex : 0.34464
 - 4v. Contact Vertical Flex : 16.61791
 - 4w. Damping Mode : hystereticNando
 - 4x. Maxwell Damping Peak Frequency : 0.00000
 - 4y. Maxwell Stiffening Percent : 0.00000
 - 4z. Damping Threshold Speed Ms : 1.00000
 - 4aa. Speed Damping Factor : 0.500000
 - 4ab. Deflection Damping Factor : 3.00000
- 5. Model Data
 - 5a. Dy0 : 1.46825
 - 5b. Dx0 : 1.56499
 - 5c. Ls Exp Y : 0.86528
 - 5d. Ls Exp X : 0.83328
 - 5e. Fz0 : 4230.00000
 - 5f. Friction Limit Angle : 7.81311
 - 5g. Flex Gain : 0.02070
 - 5h. Cf Xmult : 1.80000
 - 5i. Brake D X Mod : 0.00000
 - 5j. Combined Factor : 2.00000
 - 5k. Grip Slip Factor : 0.82000
 - 5l. Wear Curve :
 - content\cars\common_phsx\tyres\f1_2025\wear_f12025_wet.curve
 - 5m. Grain Factor : 0.00100
 - 5n. Contact Wear I M 0 : 4.00000
 - 5o. Relaxation Length Y : 0.10051
 - 5p. Relaxation Length X : 0.10051
 - 5q. Mz Tweak Mult : 0.60000
 - 5r. Mz Scale : 0.12000
 - 5s. Mz Trail Nd Slip Reduction : 0.50000
 - 5t. Mz Trail Remap : 0.20000
 - 5u. Brush Exponent : 3.00000
- 6. Thermal Data
 - 6a. Vertical Spring : 40.00000
 - 6b. Vertical Damp K : 2.00000

- 6c. Wear Mult : 0.20000
- 6d. Density : 1100.00000
- 6e. Specific Heat : 1600.00000
- 6f. Surface Core Mass Ratio : 0.15000
- 6g. Cool Factor Rain : 3.00000
- 6h. Cool Factor : 0.90000
- 6i. Heat Partition Coeff : 1.40000
- 6j. Road Conduction : 1000.00000
- 6k. Thermal Conductivity : 0.35000
- 6l. Rolling Factor : 1.60000
- 6m. Grain Gain : 0.47000
- 6n. Grain Gamma : 0.47000
- 6o. Grain Slip Angle Gain : 0.56500
- 6p. Grain Slip Angle Gamma : 0.84500
- 6q. Grain Slip Angle Threshold : 10.51311
- 6r. Blister Gain : 0.00000
- 6s. Blister Gamma : 0.00000
- 6t. Practical Temp Source : 0.80000
- 6u. Brake Transfer Factor : 0.00070
- 6v. Thermal Performance Curve :

content\cars\common_phsx\tyres\f1_2025\tcurve_f12025_wet.curve

- 6w. Contact I M 0 : 1.60000
- 6x. Tref Friction Limit Angle : 75.00000
- 6y. Friction Limit Angle T Sensitivity : 0.06000

- 7. Pressure Data

- 7a. Pressure Flex Gain : 0.25000
- 7b. Rolling Resistance Gain : 1.20000
- 7c. Rolling Heat Gain : 0.10000
- 7d. Gain D : 0.00250
- 7e. Ideal Pressure : 26.00000
- 7f. Pressure Reference : 17.00000
- 7g. Pressure Spring Curve : None
- 7h. Pressure Spring Gain : 11110.00000
- 7i. Curb Pressure Loss Must : 1.00000
- 7j. Curb Pressure Loss Threshold : 30.00000

- 8. Camber Data

- 8a. Camber Gain : 0.20000
- 8b. Camber Vertical K Range Deg : 5.00000
- 8c. Camber Vertical K Gain : -0.10000
- 8d. Dcamber0 : 0.60000
- 8e. Dcamber1 : -11.00000

- 9. Speed Sensitivity

- 9a. Mu0 T : 0.85000
- 9b. Ref Speed : 4.50000
- 9c. Mu Speed Sensitivity : 1.20000
- 9d. Mu T Sensitivity : 0.03500
- 9e. Tref Mu T : 80.00000
- 9f. Mu0 T X : 0.68000
- 9g. Ref Speed X : 2.60000
- 9h. Mu Speed Sensitivity X : 1.80000
- 9i. Mu T Sensitivity X : 0.03500
- 9j. Tref Mu T X : 80.00000

- 10. Rolling Resistance

- 10a. Rr0 : 20.11450
- 10b. Rr1 : 0.00200
- 10c. Rr Slip : 646.00000

- | L 10d. Rr Wear Mult : 0.14000
- 11. Groove Data
 - | L 11a. Groove Factor : 68.00000
 - | L 11b. Groove S A Factor : 0.45000
 - | L 11c. Groove S R Factor : 0.45000
- 12. Pressure : 17.00000
- 13. Init Data
 - | L 13a. Width : 305.00000
 - | L 13b. Aspect Ratio : 43.00000
 - | L 13c. Diameter : 18.00000
 - | L 13d. Load Index : 101.00000
 - | L 13e. Pressure : 17.00000

20. Wing [.wing]

A. Description

Discrete aero element: a named lifting/drag surface with chord, span, world position, angle-of-attack CL/CD curves, optional ground-height multipliers, Cl/Cd gains, yaw sensitivity, damage stages, and optional wing controllers (active / linked aero).

Surface 3D maps handle the platform CX/CZ vs ride height. Wings are the individual BODY / FRONT / REAR / DIFFUSER pieces Car Data lists under Wings Path. Setup wing angles overlay the baseline *Angle* on adjustable elements.

Official Description prose in the source dump is unfinished placeholder text. Content below follows the schema and examples.

I. Role in the stack

Concern	Handled here	Handled elsewhere
Element size and mount point	Chord, Span, Position	—
CL/CD vs AOA	<i>Lut AOA CL/CD</i>	.CURVE files
Ground-effect multipliers	<i>Lut GH CL/CD Mult</i>	Height .CURVE
Gain, default angle, yaw	Cl/Cd Gain, Angle, Yaw Gain	Setup wing angles
Damage sensitivity	Damage CD/CL arrays	Collision / wear systems
Active angle logic	<i>Wing Controllers</i>	Input enums + LUTs
Which elements load	—	.CAR Aero Wings Path
Platform maps	—	.SURFACE3D Front/Rear Lift / Drag

II. What you are really tuning

- Element identity** — *Name* labels the role (*BODY*, *FRONT*, *REAR*, *DIFFUSER_F*, ...). *Vertical* flags orientation (all samples *false* — horizontal aero planes). Chord × Span set reference area; Position (x,y,z) places the force application point (front bumper vs rear wing vs underfloor).
- AOA polar** — *Lut AOA CL* / *Lut AOA CD* are required personality curves. Changing angle (setup or baseline *Angle*) walks these LUTs. BODY elements often carry drag with *Cl Gain* : 0 and nonzero *Cd Gain* (RS6 body Cd 1.6; Emira body Cd 1.0).
- Ground height** — Optional *Lut GH CL Mult* / *Lut GH CD Mult* scale lift/drag with ride height. RS6 front and SF-25 front wings / diffusers wire height curves; many rear wings leave them *None* (angle-dominated, less ground-effect in the authoring).
- Gains, angle, yaw** — *Cl Gain* / *Cd Gain* multiply the LUT outputs. *Angle* is the default incidence (Emira front/rear 2°; SF-25 front 6°, rear 12°, body/diffuser 3.5°). *Yaw Gain* (often negative on race wings) bleeds or shifts aero in yaw — SF-25 rear −1.5, front elements about −0.57.

5. **Cooling and fans** — *Drag Per Cool Transfer* couples cooling drag (0 in samples). *Has Fan / Fan Speed* for active blown elements — unused here.
6. **Damage stages** — *Damage CD[x] / Damage CL[x]* (typically four slots) raise drag or kill lift as the element takes hits. Front wings often damage slot 1; rear slot 2; body uses mid/high slots — progressive aero degradation.
7. **Controllers** — Staged pipeline: Combinator (Add / Must — OCR for Mult), Input (Brake, Gas, Yaw, LatG, LonG, Steer, Speed, suspension travel...), Filter, limits, Lut. Empty in these examples; used when DRS-like or dynamic wing logic lives on the element itself (Car Data may also own DRS hooks separately).

III. Architecture

1 - GEOMETRY AND POLARS (SCHEMA 1-9)

Vertical flag; name; chord/span; position; AOA CL/CD paths; optional GH CL/CD mult paths.

2 - GAINS AND EXTRAS (SCHEMA 10-16)

Cd/Cl gain; angle; yaw gain; cool-transfer drag; fan flags.

3 - DAMAGE AND CONTROL (SCHEMA 17-19)

Repeating damage CD/CL floats; Wing Controllers array.

IV. How to read the examples

1 - AUDI RS6 AVANT — ROAD TRIAD

Three wings, shared span 2.41: **BODY** (Cd only), **FRONT** (Cl 1.0, angle 2°, height CL/CD curves), **REAR** (Cl 1.0, no height LUTs). Classic road-car aero split: body drag + front height-sensitive + rear wing.

2 - LOTUS EMIRA — SPORT TRIAD

Same BODY/FRONT/REAR pattern, narrower span (1.945). Higher Cl gains (front 2.0, rear 2.2), both ends at 2° angle, no height multipliers. More “wing-driven” than ground-effect front.

3 - FERRARI SF-25 — OPEN-WHEEL PACKAGE

Many elements: BODY (Cl+Cd, yaw -0.2), **two FRONT** wings mirrored at $x \pm 0.56$ (span 0.42 each, Cl gain 3.9, angle 6°, height LUTs, strong yaw), REAR (chord 0.45, span 2.0, angle 12°, Cd 1.5 / Cl 0.7, yaw -1.5), plus DIFFUSER_F and DIFFUSER_RL (and further diffuser pieces in the dump) with height curves. Shows wings as a full aero assembly, not just a rear flap.

V. Practical notes

- Source Description section is placeholder; example header wrongly says “Chosen Tyres”.

- Setup wing angle only matters if Car Data lists the corresponding .WING and Limits allow adjustment.
- Editing AOA curves without updating Cl/Cd gains (or vice versa) double-scales the element.
- Dual front wings (SF-25) must stay symmetric unless intentional aero balance asymmetry.
- OCR: Combinator *Must* $\approx$ Mult; input *SusTravelLR* $\approx$ suspension travel L/R.
- Surface 3D maps and wings stack — changing only one layer leaves the other fighting your balance.

VI. Related assets

- **3. Car Data** — Aero Wings Path list, DRS hooks
- **16. Surface 3D** — platform CX/CZ vs ride height
- **5 / 6. Car Setup / Limits** — wing angles and legal windows
- **8. Car Tuning Parts** — Wing / Aero Package redirects
- **External .CURVE** — AOA and ground-height LUTs referenced here

B. Schema

- ```

1. Vertical : boolean
2. Name : string
3. Chord : float
4. Span : float
5. Position : x, y, z float
6. Lut A O A C L : string - path
7. Lut A O A C D : string - path
8. Lut G H C L Mult : string - path
9. Lut G H C D Mult : string - path
10. Cd Gain : float
11. Cl Gain : float
12. Angle : float
13. Yaw Gain : float
14. Drag Per Cool Transfer : float
15. Has Fan : boolean
16. Fan Speed : float
17. Damage C D [x] : float | can have multiple Damage C D
18. Damage C L [x] : float | can have multiple Damage C L
19. Wing Controllers [x] : object | can have multiple Wing Controllers
 19a. Combinator Mode : enum
 19b. Input : enum
 19c. Filter : float
 19d. Up Limit : float
 19e. Down Limit : float
 19f. Lut : string - path

```

### Enum - Car Wing

|            |                 |                                                                                     |
|------------|-----------------|-------------------------------------------------------------------------------------|
| <b>19a</b> | Combinator Mode | UndefinedMode, Add, Mult                                                            |
| <b>19b</b> | Input           | UndefinedInput, Brake, Gas, Yaw, LatG, LonG, Steer, Speed, SusTravelLR, SusTravelRR |

## C. Measurement Units & Descriptions

| ID | Name             | Unit of Measurement             | Description                                                                                                                                                                                                                                                 |
|----|------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Vertical         | None ( Boolean : True / False ) | When true, the element is a vertical aero surface (endplate / fin) producing mainly side force; false = horizontal wing, body, or diffuser plane (all documented examples are false).                                                                       |
| 2. | Name             | None ( String )                 | Element label used for authoring and DRS/setup mapping (BODY, FRONT, REAR, DIFFUSER_F, DIFFUSER_RL/RR).                                                                                                                                                     |
| 3. | Chord            | m ( Meters )                    | Streamwise reference length of the aero element. Often 1.0 as a unit chord for LUT scaling; physical chords appear on race cars (SF-25 rear wing 0.45 m, front diffuser 2.5 m). With Span sets effective area $A \approx \text{Chord} \times \text{Span}$ . |
| 4. | Span             | m ( Meters )                    | Lateral width of the element (RS6/Emira body ~1.95–2.41 m; SF-25 split front wings 0.42 m each, rear wing 2.0 m, half-diffusers 0.683 m).                                                                                                                   |
| 5. | Position         | m ( Meters, X / Y / Z )         | Force application point in the car body frame (m). Sets pitch/heave moment arms: front wings at +Z, rear at -Z, body near CG (e.g. RS6 FRONT z=1.9, REAR z=-2.1).                                                                                           |
| 6. | Lut A O A C L    | None ( .curve file path )       | Look-up table: angle of attack (AoA, deg) → lift coefficient $C_l$ (wing_body/front/rear_AOA_CL.curve). Core polar for downforce vs. incidence.                                                                                                             |
| 7. | Lut A O A C D    | None ( .curve file path )       | Look-up table: AoA → drag coefficient $C_d$ (..._AOA_CD.curve). Paired with $C_l$ LUT for L/D trade-off.                                                                                                                                                    |
| 8. | Lut G H C L Mult | None ( .curve file path )       | Ground-height (ride-height) multiplier LUT on $C_l$ (height_front_CL.curve, height_diffuser_CL.curve). Models ground effect / diffuser sensitivity; often None on body/rear wing.                                                                           |
| 9. | Lut G H C D Mult | None ( .curve file path )       | Ground-height multiplier LUT on $C_d$ . Used with GH $C_l$ Mult for front wings and underbody elements.                                                                                                                                                     |

| ID   | Name                   | Unit of Measurement                         | Description                                                                                                                                                                 |
|------|------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.  | Cd Gain                | Dimensionless coefficient                   | Scalar on Cd from LUTs (aligned with Car Data §17c3). Body often carries most drag (RS6 BODY 1.6, Emira BODY 1.0); wings may use 0–1.5.                                     |
| 11.  | Cl Gain                | Dimensionless coefficient                   | Scalar on Cl from LUTs (aligned with Car Data §17c2). BODY often 0 (pure drag blob); FRONT/REAR 1–2 (street) up to ~3.9 (SF-25 front).                                      |
| 12.  | Angle                  | deg ( Degrees )                             | Static geometric incidence / flap angle relative to the body (AoA offset into CL/CD LUTs). Setup-adjustable on race cars (Emira FRONT 2°, SF-25 FRONT 6°, REAR 12°).        |
| 13.  | Yaw Gain               | Dimensionless coefficient                   | Sensitivity of aero forces to vehicle yaw (aligned with Car Data §17c4). Negative values on F1 (SF-25 –0.2 to –1.5) model yaw-induced Cl/Cd change and side-force coupling. |
| 14.  | Drag Per Cool Transfer | Dimensionless coefficient                   | Drag penalty coupled to cooling-air / duct mass-flow (aligned with Car Data §17c5). 0.0 in all documented examples.                                                         |
| 15.  | Has Fan                | None ( Boolean : True / False )             | Enables an active suction fan on this element (e.g. fan-car / blown diffuser). False on RS6, Emira, and SF-25 examples.                                                     |
| 16.  | Fan Speed              | RPM or rad/s ( Fan rotational speed )       | Commanded fan speed when Has Fan is true; 0 when unused.                                                                                                                    |
| 17.  | Damage C D             | Dimensionless $\Delta C_d$ per damage stage | Indexed [x] damage stages increasing Cd after aero damage (e.g. 0.005–0.015). Stages map to progressive wing/body damage levels.                                            |
| 18.  | Damage C L             | Dimensionless $\Delta C_d$ per damage stage | Indexed [x] damage stages reducing (or shifting) Cl after damage; often matched to Damage CD on the impacted side (front crash → stage 1, rear → stage 2).                  |
| 19.  | Wing Controllers       | None ( Object array [x] )                   | Dynamic aero controllers (DRS, brake-duct, speed-sensitive flaps). None in documented examples; schema exposes per-controller fields 19a–19f.                               |
| 19a. | Combinator Mode        | None ( Enum : UndefinedMode / Add / Mult )  | How this controller's LUT output combines with base Angle/gains (Add = offset; Must likely Mult/override — schema spelling).                                                |

| ID   | Name       | Unit of Measurement                                                                                          | Description                                                                                                  |
|------|------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 19b. | Input      | None ( Enum : UndefinedInput / Brake / Gas / Yaw / LatG / LonG / Steer / Speed / SusTravelLR / SusTravelRR ) | Telemetry channel driving the controller (brake for DRS-off, speed, lateral G, suspension travel L/R, etc.). |
| 19c. | Filter     | Dimensionless filter coefficient ( 0 - 1 )                                                                   | Low-pass smoothing on the controller input to avoid aero chatter.                                            |
| 19d. | Up Limit   | deg or gain ( Depends on controller output )                                                                 | Upper clamp on controller output (max flap angle / gain).                                                    |
| 19e. | Down Limit | deg or gain ( Depends on controller output )                                                                 | Lower clamp on controller output (min flap angle / gain).                                                    |
| 19f. | Lut        | None ( .curve file path )                                                                                    | Look-up table mapping Input → angle or Cl/Cd modifier for this wing controller.                              |

## D. Example data

### I. Chosen Tyres for Example

- Audi RS6 Avant ( slug : ks\_audi\_rs\_6\_avant ) [ 3 wings ]
- Lotus Emira ( slug : ks\_lotus\_emira )
- Ferrari SF-25 ( slug : ks\_ferrari\_sf\_25 )

### II. Example

#### Audi RS6 Avant

##### 1. Wing 0 [ Body ] ( file : ks\_audi\_rs\_6\_avant0.wing )

```

1. Vertical : false
2. Name : BODY
3. Chord : 1.00000
4. Span : 2.41000
5. Position : 0.00000, 0.18000, -0.10000
6. Lut A 0 A C L :
content\cars\ks_audi_rs_6_avant\data\wing_body_A0A_CL.curve
7. Lut A 0 A C D :
content\cars\ks_audi_rs_6_avant\data\wing_body_A0A_CD.curve
8. Lut G H C L Mult : None
9. Lut G H C D Mult : None
10. Cd Gain : 1.60000
11. Cl Gain : 0.00000
12. Angle : 0.00000
13. Yaw Gain : 0.00000
14. Drag Per Cool Transfer : 0.00000
15. Has Fan : false
16. Fan Speed : 0.00000
17. Damage C D 1 : 0.00500

```

```
| 17. Damage C D 2 : 0.00500
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

#### **1. Wing 1 [ Front ] ( file : ks\_audi\_rs\_6\_avant1.wing )**

```
| 1. Vertical : false
| 2. Name : FRONT
| 3. Chord : 1.00000
| 4. Span : 2.41000
| 5. Position : 0.00000, -0.34000, 1.90000
| 6. Lut A O A C L :
content\cars\ks_audi_rs_6_avant\data\wing_front_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_audi_rs_6_avant\data\wing_front_AOA_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_audi_rs_6_avant\data\height_front_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_audi_rs_6_avant\data\height_front_CD.curve
| 10. Cd Gain : 0.00000
| 11. Cl Gain : 1.00000
| 12. Angle : 2.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01500
| 17. Damage C D 2 : 0.00500
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.01500
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

#### **1. Wing 2 [ Rear ] ( file : ks\_audi\_rs\_6\_avant2.wing )**

```
| 1. Vertical : false
| 2. Name : REAR
| 3. Chord : 1.00000
| 4. Span : 2.41000
| 5. Position : 0.00000, 0.80000, -2.10000
| 6. Lut A O A C L :
content\cars\ks_audi_rs_6_avant\data\wing_rear_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_audi_rs_6_avant\data\wing_rear_AOA_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 0.00000
```

- | 11. Cl Gain : 1.00000
- | 12. Angle : 0.00000
- | 13. Yaw Gain : 0.00000
- | 14. Drag Per Cool Transfer : 0.00000
- | 15. Has Fan : false
- | 16. Fan Speed : 0.00000
- | 17. Damage C D 1 : 0.00000
- | 17. Damage C D 2 : 0.01500
- | 17. Damage C D 3 : 0.00000
- | 17. Damage C D 4 : 0.00000
- | 18. Damage C L 1 : 0.00000
- | 18. Damage C L 2 : 0.01000
- | 18. Damage C L 3 : 0.00000
- | 18. Damage C L 4 : 0.00000
- | 19. Wing Controllers : None

## Lotus Emira

### 1. Wing 0 [ Body ] ( file : ks\_lotus\_emira0.wing )

- | 1. Vertical : false
- | 2. Name : BODY
- | 3. Chord : 1.00000
- | 4. Span : 1.94500
- | 5. Position : 0.00000, 0.20000, 0.50000
- | 6. Lut A O A C L :  
content\cars\ks\_lotus\_emira\data\wing\_body\_A0A\_CL.curve
- | 7. Lut A O A C D :  
content\cars\ks\_lotus\_emira\data\wing\_body\_A0A\_CD.curve
- | 8. Lut G H C L Mult : None
- | 9. Lut G H C D Mult : None
- | 10. Cd Gain : 1.00000
- | 11. Cl Gain : 0.00000
- | 12. Angle : 0.00000
- | 13. Yaw Gain : 0.00000
- | 14. Drag Per Cool Transfer : 0.00000
- | 15. Has Fan : false
- | 16. Fan Speed : 0.00000
- | 17. Damage C D 1 : 0.00500
- | 17. Damage C D 2 : 0.00500
- | 17. Damage C D 3 : 0.01000
- | 17. Damage C D 4 : 0.01000
- | 18. Damage C L 1 : 0.00000
- | 18. Damage C L 2 : 0.00000
- | 18. Damage C L 3 : 0.00000
- | 18. Damage C L 4 : 0.00000
- | 19. Wing Controllers : None

### 1. Wing 1 [ Front ] ( file : ks\_lotus\_emira1.wing )

- | 1. Vertical : false
- | 2. Name : FRONT
- | 3. Chord : 1.00000
- | 4. Span : 1.94500



```

| 5. Position : 0.00000, -0.30000, 1.48700
| 6. Lut A O A C L :
content\cars\ks_lotus_emira\data\wing_front_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_lotus_emira\data\wing_front_AOA_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 0.00000
| 11. Cl Gain : 2.00000
| 12. Angle : 2.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01500
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.01500
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

```

#### **1. Wing 2 [ Rear ] ( file : ks\_lotus\_emira2.wing )**

```

| 1. Vertical : false
| 2. Name : REAR
| 3. Chord : 1.00000
| 4. Span : 1.94500
| 5. Position : 0.00000, 0.10000, -1.00000
| 6. Lut A O A C L :
content\cars\ks_lotus_emira\data\wing_rear_AOA_CL.curve
| 7. Lut A O A C D :
content\cars\ks_lotus_emira\data\wing_rear_AOA_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 0.00000
| 11. Cl Gain : 2.20000
| 12. Angle : 2.00000
| 13. Yaw Gain : 0.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01500
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None

```

## Ferrari SF-25

### 1. Wing 0 [ Body ] ( file : ks\_ferrari\_sf\_250.wing )

```
| 1. Vertical : false
| 2. Name : BODY
| 3. Chord : 1.00000
| 4. Span : 1.36600
| 5. Position : 0.00000, 0.15000, -0.08000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_body_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_body_A0A_CD.curve
| 8. Lut G H C L Mult : None
| 9. Lut G H C D Mult : None
| 10. Cd Gain : 0.50000
| 11. Cl Gain : 1.00000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.20000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

### 2. Wing 1 [ Front ] ( file : ks\_ferrari\_sf\_251.wing )

```
| 1. Vertical : false
| 2. Name : FRONT
| 3. Chord : 1.00000
| 4. Span : 0.42000
| 5. Position : -0.56000, -0.09000, 2.90000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_front_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_front_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_frontwing_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_frontwing_CD.curve
| 10. Cd Gain : 1.25000
| 11. Cl Gain : 3.90000
| 12. Angle : 6.00000
| 13. Yaw Gain : -0.57100
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
```

- 17. Damage C D 2 : 0.00000
- 17. Damage C D 3 : 0.00000
- 17. Damage C D 4 : 0.00000
- 18. Damage C L 1 : 0.01000
- 18. Damage C L 2 : 0.00000
- 18. Damage C L 3 : 0.00000
- 18. Damage C L 4 : 0.00000
- 19. Wing Controllers : None

### 3. Wing 2 [ Front ] ( file : ks\_ferrari\_sf\_252.wing )

- 1. Vertical : false
- 2. Name : FRONT
- 3. Chord : 1.00000
- 4. Span : 0.42000
- 5. Position : 0.56000, -0.09000, 2.90000
- 6. Lut A 0 A C L :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_front\_A0A\_CL.curve
- 7. Lut A 0 A C D :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_front\_A0A\_CD.curve
- 8. Lut G H C L Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_frontwing\_CL.curve
- 9. Lut G H C D Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_frontwing\_CD.curve
- 10. Cd Gain : 1.25000
- 11. Cl Gain : 3.90000
- 12. Angle : 6.00000
- 13. Yaw Gain : -0.57100
- 14. Drag Per Cool Transfer : 0.00000
- 15. Has Fan : false
- 16. Fan Speed : 0.00000
- 17. Damage C D 1 : 0.01000
- 17. Damage C D 2 : 0.00000
- 17. Damage C D 3 : 0.00000
- 17. Damage C D 4 : 0.00000
- 18. Damage C L 1 : 0.01000
- 18. Damage C L 2 : 0.00000
- 18. Damage C L 3 : 0.00000
- 18. Damage C L 4 : 0.00000
- 19. Wing Controllers : None

### 4. Wing 3 [ Rear ] ( file : ks\_ferrari\_sf\_253.wing )

- 1. Vertical : false
- 2. Name : REAR
- 3. Chord : 0.45000
- 4. Span : 2.00000
- 5. Position : 0.00000, 0.55000, -1.70000
- 6. Lut A 0 A C L :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_rear\_A0A\_CL.curve
- 7. Lut A 0 A C D :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_rear\_A0A\_CD.curve
- 8. Lut G H C L Mult : None
- 9. Lut G H C D Mult : None
- 10. Cd Gain : 1.50000

```
| 11. Cl Gain : 0.70000
| 12. Angle : 12.00000
| 13. Yaw Gain : -1.50000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.00000
| 17. Damage C D 4 : 0.00000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.00000
| 18. Damage C L 4 : 0.00000
| 19. Wing Controllers : None
```

#### **5. Wing 4 [ Diffuser Front ] ( file : ks\_ferrari\_sf\_254.wing )**

```
| 1. Vertical : false
| 2. Name : DIFFUSER_F
| 3. Chord : 2.50000
| 4. Span : 1.36600
| 5. Position : 0.00000, -0.16000, 0.00000
| 6. Lut A O A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A O A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 1.00000
| 11. Cl Gain : 0.80000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None
```

#### **6. Wing 5 [ Diffuser Rear Left ] ( file : ks\_ferrari\_sf\_255.wing )**

```
| 1. Vertical : false
| 2. Name : DIFFUSER_RL
| 3. Chord : 1.00000
| 4. Span : 0.68300
| 5. Position : -0.34230, -0.17500, -1.57000
```

```

| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CD.curve
| 10. Cd Gain : 0.55000
| 11. Cl Gain : 0.71000
| 12. Angle : 3.50000
| 13. Yaw Gain : -1.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None

```

## 7. Wing 6 [ Diffuser Rear Right ] ( file : ks\_ferrari\_sf\_256.wing )

```

| 1. Vertical : false
| 2. Name : DIFFUSER_RR
| 3. Chord : 1.00000
| 4. Span : 0.68300
| 5. Position : 0.34230, -0.17500, -1.57000
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_rear_CD.curve
| 10. Cd Gain : 0.55000
| 11. Cl Gain : 0.71000
| 12. Angle : 3.50000
| 13. Yaw Gain : -1.00000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.00000
| 17. Damage C D 2 : 0.01000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.00000
| 18. Damage C L 2 : 0.01000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000

```

| 19. Wing Controllers : None

#### 8. Wing 7 [ DRS ] ( file : ks\_ferrari\_sf\_257.wing )

| 1. Vertical : false

| 2. Name : DRS

| 3. Chord : 1.00000

| 4. Span : 1.36600

| 5. Position : 0.00000, 0.59000, -1.80000

| 6. Lut A 0 A C L :

content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_rear2\_A0A\_CL.curve

| 7. Lut A 0 A C D :

content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_rear2\_A0A\_CD.curve

| 8. Lut G H C L Mult : None

| 9. Lut G H C D Mult : None

| 10. Cd Gain : 1.70000

| 11. Cl Gain : 2.00000

| 12. Angle : 17.00000

| 13. Yaw Gain : -1.00000

| 14. Drag Per Cool Transfer : 0.00000

| 15. Has Fan : false

| 16. Fan Speed : 0.00000

| 17. Damage C D 1 : 0.00000

| 17. Damage C D 2 : 0.01000

| 17. Damage C D 3 : 0.00000

| 17. Damage C D 4 : 0.00000

| 18. Damage C L 1 : 0.00000

| 18. Damage C L 2 : 0.01000

| 18. Damage C L 3 : 0.00000

| 18. Damage C L 4 : 0.00000

| 19. Wing Controllers : None

#### 9. Wing 8 [ Monkeyseat ] ( file : ks\_ferrari\_sf\_258.wing )

| 1. Vertical : false

| 2. Name : DRS

| 3. Chord : 1.00000

| 4. Span : 1.36600

| 5. Position : 0.00000, -0.10000, -1.80000

| 6. Lut A 0 A C L :

content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_monkey\_A0A\_CL.curve

| 7. Lut A 0 A C D :

content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_monkey\_A0A\_CD.curve

| 8. Lut G H C L Mult : None

| 9. Lut G H C D Mult : None

| 10. Cd Gain : 1.00000

| 11. Cl Gain : 1.00000

| 12. Angle : 10.00000

| 13. Yaw Gain : -0.15000

| 14. Drag Per Cool Transfer : 0.00000

| 15. Has Fan : false

| 16. Fan Speed : 0.00000

| 17. Damage C D 1 : 0.00000

| 17. Damage C D 2 : 0.01000

| 17. Damage C D 3 : 0.00000

| 17. Damage C D 4 : 0.00000  
| 18. Damage C L 1 : 0.00000  
| 18. Damage C L 2 : 0.01000  
| 18. Damage C L 3 : 0.00000  
| 18. Damage C L 4 : 0.00000  
| 19. Wing Controllers : None

**10. Wing 9 [ Diffuser Front Left ] ( file : ks\_ferrari\_sf\_diff\_l.wing )**

| 1. Vertical : false  
| 2. Name : DIFFUSER\_FL  
| 3. Chord : 2.50000  
| 4. Span : 0.35000  
| 5. Position : 0.31000, -0.16000, 0.00000  
| 6. Lut A 0 A C L :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_diffuser\_A0A\_CL.curve  
| 7. Lut A 0 A C D :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_diffuser\_A0A\_CD.curve  
| 8. Lut G H C L Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_diffuser\_CL.curve  
| 9. Lut G H C D Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_diffuser\_CD.curve  
| 10. Cd Gain : 0.83000  
| 11. Cl Gain : 1.70000  
| 12. Angle : 3.50000  
| 13. Yaw Gain : -0.07000  
| 14. Drag Per Cool Transfer : 0.00000  
| 15. Has Fan : false  
| 16. Fan Speed : 0.00000  
| 17. Damage C D 1 : 0.01000  
| 17. Damage C D 2 : 0.00000  
| 17. Damage C D 3 : 0.01000  
| 17. Damage C D 4 : 0.01000  
| 18. Damage C L 1 : 0.01000  
| 18. Damage C L 2 : 0.00000  
| 18. Damage C L 3 : 0.01000  
| 18. Damage C L 4 : 0.01000  
| 19. Wing Controllers : None

**11. Wing 10 [ Diffuser Front Left L ] ( file : ks\_ferrari\_sf\_diff\_ll.wing )**

| 1. Vertical : false  
| 2. Name : DIFFUSER\_FLL  
| 3. Chord : 2.50000  
| 4. Span : 0.35000  
| 5. Position : 0.66000, -0.16000, 0.00000  
| 6. Lut A 0 A C L :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_diffuser\_A0A\_CL.curve  
| 7. Lut A 0 A C D :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_diffuser\_A0A\_CD.curve  
| 8. Lut G H C L Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_diffuser\_CL.curve  
| 9. Lut G H C D Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_diffuser\_CD.curve  
| 10. Cd Gain : 0.83000

| 11. Cl Gain : 1.10000  
| 12. Angle : 3.50000  
| 13. Yaw Gain : -0.07000  
| 14. Drag Per Cool Transfer : 0.00000  
| 15. Has Fan : false  
| 16. Fan Speed : 0.00000  
| 17. Damage C D 1 : 0.01000  
| 17. Damage C D 2 : 0.00000  
| 17. Damage C D 3 : 0.01000  
| 17. Damage C D 4 : 0.01000  
| 18. Damage C L 1 : 0.01000  
| 18. Damage C L 2 : 0.00000  
| 18. Damage C L 3 : 0.01000  
| 18. Damage C L 4 : 0.01000  
| 19. Wing Controllers : None

**12. Wing 11 [ Diffuser Front Right ] ( file : ks\_ferrari\_sf\_diff\_r.wing )**

| 1. Vertical : false  
| 2. Name : DIFFUSER\_FR  
| 3. Chord : 2.50000  
| 4. Span : 0.35000  
| 5. Position : -0.31000, -0.16000, 0.00000  
| 6. Lut A O A C L :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_diffuser\_A0A\_CL.curve  
| 7. Lut A O A C D :  
content\cars\ks\_ferrari\_sf\_25\data\aero\wing\_diffuser\_A0A\_CD.curve  
| 8. Lut G H C L Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_diffuser\_CL.curve  
| 9. Lut G H C D Mult :  
content\cars\ks\_ferrari\_sf\_25\data\aero\height\_diffuser\_CD.curve  
| 10. Cd Gain : 0.83000  
| 11. Cl Gain : 1.70000  
| 12. Angle : 3.50000  
| 13. Yaw Gain : -0.07000  
| 14. Drag Per Cool Transfer : 0.00000  
| 15. Has Fan : false  
| 16. Fan Speed : 0.00000  
| 17. Damage C D 1 : 0.01000  
| 17. Damage C D 2 : 0.00000  
| 17. Damage C D 3 : 0.01000  
| 17. Damage C D 4 : 0.01000  
| 18. Damage C L 1 : 0.01000  
| 18. Damage C L 2 : 0.00000  
| 18. Damage C L 3 : 0.01000  
| 18. Damage C L 4 : 0.01000  
| 19. Wing Controllers : None

**13. Wing 12 [ Diffuser Front Right R ] ( file : ks\_ferrari\_sf\_diff\_rr.wing )**

| 1. Vertical : false  
| 2. Name : DIFFUSER\_FRR  
| 3. Chord : 2.50000  
| 4. Span : 0.35000  
| 5. Position : -0.66000, -0.16000, 0.00000



```
| 6. Lut A 0 A C L :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CL.curve
| 7. Lut A 0 A C D :
content\cars\ks_ferrari_sf_25\data\aero\wing_diffuser_A0A_CD.curve
| 8. Lut G H C L Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CL.curve
| 9. Lut G H C D Mult :
content\cars\ks_ferrari_sf_25\data\aero\height_diffuser_CD.curve
| 10. Cd Gain : 0.83000
| 11. Cl Gain : 1.10000
| 12. Angle : 3.50000
| 13. Yaw Gain : -0.07000
| 14. Drag Per Cool Transfer : 0.00000
| 15. Has Fan : false
| 16. Fan Speed : 0.00000
| 17. Damage C D 1 : 0.01000
| 17. Damage C D 2 : 0.00000
| 17. Damage C D 3 : 0.01000
| 17. Damage C D 4 : 0.01000
| 18. Damage C L 1 : 0.01000
| 18. Damage C L 2 : 0.00000
| 18. Damage C L 3 : 0.01000
| 18. Damage C L 4 : 0.01000
| 19. Wing Controllers : None
```